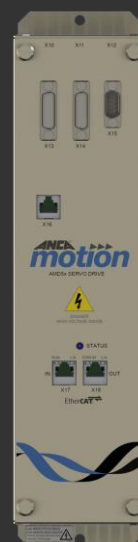




AMD Servo Drive - SoE v3.0 Configuration Guide

AMDOC-000192 Rev 02



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AMD Servo Drive - SoE v3.0 Configuration Guide

Some Important Links

Related Manuals and Brochures

[*Related Documentation*](#)

Sales and Support Contact Information

[*Product, Sales and Service Enquiries*](#)

For the latest copy of the manual visit us online

[*Manuals*](#)

**For the latest version of the ANCA MotionBench
Software visit**

[*Software*](#)

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1	Safety	General Product safety information
2	Introduction	Target Audience, model applicability, help in reading the manual and related manuals/brochures
3	Product Identification	How to Interpret the product label to ensure manual applies to the model of drive being configured
4	Start-up	Installing and using ANCA Motion Bench to configure and enable the drive
5	Basic Configuration	Guidance on configuration of the most commonly used features
6	Operation Modes	Describes the various motion control modes available
7	Advanced Configuration	Additional configuration parameters for the advanced user
8	Communication	Interfacing over EtherCAT
9	Fault Tracing	Indicators, drive states and diagnostics
10	Additional Information	How to contact ANCA Motion with your enquiries

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1 Safety



Warning: To prevent possible accidents or injury, ensure you read and understand this manual before commencing installation or commissioning work on the AMD servo drives.



DANGER HIGH VOLTAGE - The working DC bus is live at all times when power is on. The Main Isolator feeding the drive power supply must be switched to the off position at least 15 minutes before any work is commenced on the unit. The operator must check the bus voltage with a tested working voltage measuring instrument prior to disconnecting any connectors or opening the DC Bus terminal cover. The red LED indicator on the front of the drive which indicates that there is charge remaining in the drive is only to be used as an aid to visual troubleshooting. It should not be relied on as a means of safety.

Rotating permanent magnet motors can produce large voltages. Please ensure that the motors have stopped rotating before commencing work.

This manual and the warnings attached to the AMD servo drive only highlight hazards that can be predicted by ANCA Motion. Be aware they do not cover all possible hazards.

ANCA Motion shall not be responsible for any accidents caused by the misuse or abuse of the device by the operator.

Safe operation of these devices is your own responsibility. By taking note of the safety precautions, tips and warnings in this manual you can help to ensure your own safety and the safety of those around you. The AMD servo drive is equipped with safety features to protect the operator and equipment. Never operate the equipment if you are in doubt about how these safety features work.

1.1 General Safety

The following points must be understood and adhered to at all times:

- Equipment operators must read the relevant [AMD2000 Series D21xx Servo Drive - User Guide](#) or [AMD5x Passive Infeed Unit and AMD5x Servo Drive - User Guide](#) carefully and make sure of the correct procedure before operating the AMD servo drive.
- Memorize the locations of the power and drive isolator switches so that you can activate them immediately at any time if required.
- If two or more persons are working together, establish signals so that they can communicate to confirm safety before proceeding to another step.
- Always make sure there are no obstacles or people near the devices during installation and or operation. Be aware of your environment and what is around you.
- Take precautions to ensure that your clothing, hair or personal effects (such as jewellery) cannot become entangled in the equipment.
- Do not remove the covers to access the inside of the AMD servo drive unless authorized
- Do not turn on any of the equipment without all safety features in place and known to be functioning correctly. Never remove any covers or guards unless instructed by the procedures described in this manual.
- Never touch any exposed wiring, connections or fittings while the equipment is in operation.
- Visually check all switches on the operator panel before operating them.
- Do not apply any mechanical force to the AMD servo drive, which may cause malfunction or failure.
- Before removing equipment covers, be sure to turn OFF the power supply at the isolator. (Refer to the [AMD2000 Series D21xx Servo Drive - User Guide](#) or [AMD5x Passive Infeed Unit and AMD5x Servo Drive - User Guide](#)).

- Do not operate equipment with the covers removed.
- Keep the vicinity of the AMD servo drive clean and tidy.
- Never attempt cleaning or inspection during machine operation.
- Only suitably qualified personnel should install, operate, repair and/or replace this equipment.
- Be aware of the closest First Aid station.
- Ensure all external wiring is clearly labelled. This will assist you and your colleagues in identifying possible electrical safety hazards.
- Clean or inspect the equipment only after isolating all power sources.
- Use cables with the minimum cross sectional area as recommended or greater.
- Install cables according to local legislation and regulations as applicable.
- Insulation resistance testers (sometimes known as a 'megger' or hi-pot tester) are not to be used on the drive, as a false resistance reading and/or damage to the tester may result

2 Introduction

2.1 What this Chapter Contains

This chapter introduces reader to the manual, the target audience and some useful information with regards to comprehending the content.

2.2 Purpose

This manual provides the required information for commissioning and operation of the AMD servo drive using Servo over EtherCAT. It has been written specifically to meet the needs of qualified engineers, tradespersons, technicians and operators. It should be read in conjunction with the [AMD2000 Series D21xx Servo Drive - User Guide](#) or [AMD5x Passive Infeed Unit and AMD5x Servo Drive - User Guide](#) which will cover the planning and installation aspects of the drive.

Every effort has been made to simplify the procedures and processes applicable to the AMD servo drive in this Manual. However, given the sometimes complex nature of the information, some prior knowledge of associated units, their configuration and or programming has to be assumed.

2.3 About the AMD Servo Drive

The AMD servo drives are capable of motion control for applications that may vary from precise control of movement and angular position of permanent magnet synchronous motors through to less rigorous applications such as simple speed control of induction motors. In many of these applications the rotational control of the motor is converted to motion using mechanical means such as ball screws and belts.

Motion control is performed by the drive controller which accepts position feedback from motor encoders and/or separate linear scales. The drive utilizes state-of-the-art current-regulated, pulse-width-modulated voltage-source inverter technology that manages motor performance. In general, the drive control receives motion control commands via a higher level controller, which is based on an Ethernet-based field-bus interface. In certain applications the drive is capable of executing pre-defined moves that are stored in local memory, without the use of a motion controller. The AMD servo drives also supports position, velocity and torque control modes. Please refer to [AMD2000 Series D21xx Servo Drive - User Guide](#) or [AMD5x Passive Infeed Unit and AMD5x Servo Drive - User Guide](#) for more details of features available.

2.4 Drive Model Applicability

This manual is applicable to the following variants of the ANCA Motion AMD Servo Drives:

Product	Product variant	Product Number
AMD2000 Series Servo Drive	3A rms	D2103-2S2-A
	9A rms	D2109-2S2-A
AMD5x Series Servo Drive	12A rms	AMD5-11200-BA00
	20A rms	AMD5-12000-BA00
	35A rms	AMD5-13500-BA00

2.5 Related Documents

AMD2000 Series Servo Drive – CoE Configuration Guide

AMD2000 Series D21xx Servo Drive – User Guide

AMD5x Passive Infeed Unit and AMD5x Servo Drive - User Guide

ANCA Motion MotionBench – User Guide

Digital Servo Drive CoE/SoE Parameter Reference – Included with firmware bundle

Digital Servo Drive Error Code Reference – Included with firmware bundle

2.6 Terms and Abbreviations

DSD	Digital Servo Drive
EMC	Electromagnetic Compatibility
IEC	International Electrotechnical Commission
I/O	Bidirectional Input / Output
O	Output
AIN	Analogue Input
AOUT	Analogue Output
DI	Digital Input
DO	Digital Output
W.R.T.	With Respect To
GND	Ground
rms	root mean square
V / mV	Volt / millivolt
A / mA	Ampere / milliampere
Φ / Ø	phase
Ω	ohms
AC / DC	Alternating Current / Direct Current
Hz	Hertz
ms	millisecond
CNC	Computer Numerical Control
DCH	Drive-Controlled Homing
DCM	Drive-Controlled Moves
PMSM	Permanent Magnet Synchronous Motor
PMAC	Permanent Magnet Alternating Current
STO	Safe Torque Off
SoE	Servo Drive Profile over EtherCAT®
CoE	CANopen over EtherCAT® Profile
NC	Numeric Control
C1D	Class 1 Diagnostic
CUCH	Control Unit Controlled Homing
DCH	Drive Controlled Homing
LSB	Least Significant Bit

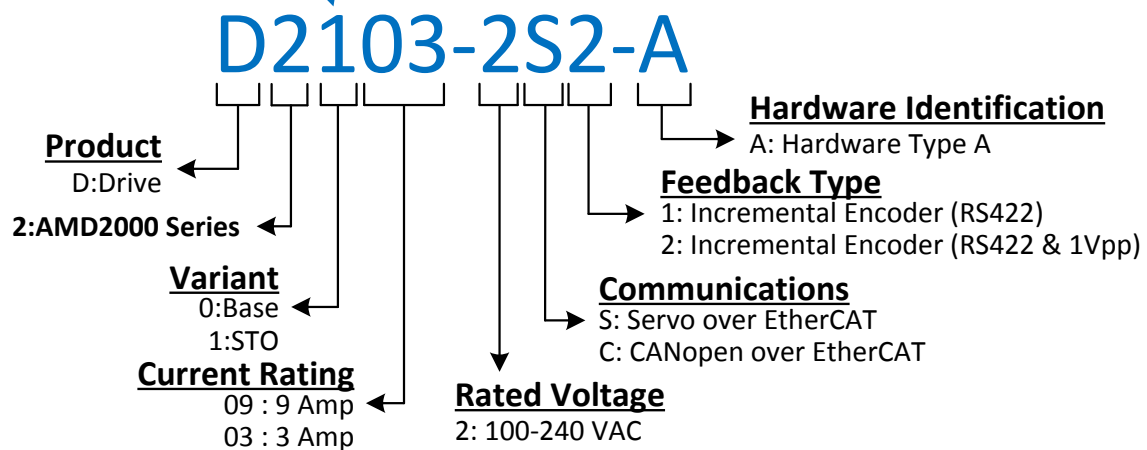
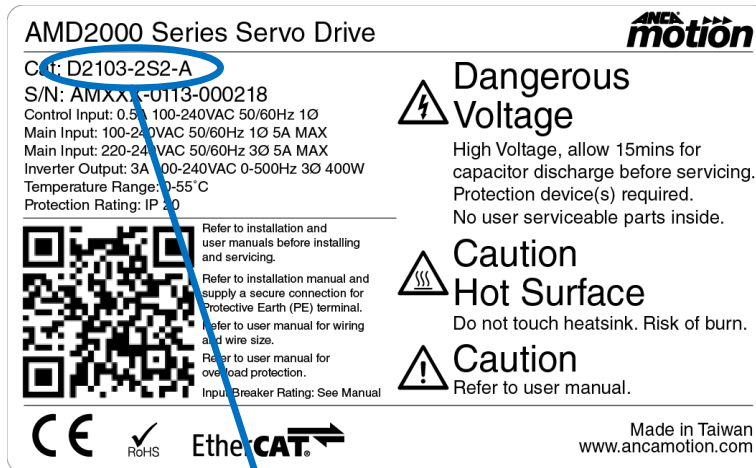
2.7 Trademarks

EtherCAT® is a registered trademark and patented technology, licensed by Beckhoff Automation GmbH, Germany.

3 Product Identification

3.1 AMD2000 Series Servo Drive Catalogue Number Interpretation

AMD2000 drives are marked with an identification label. The catalogue number is explained as follows:



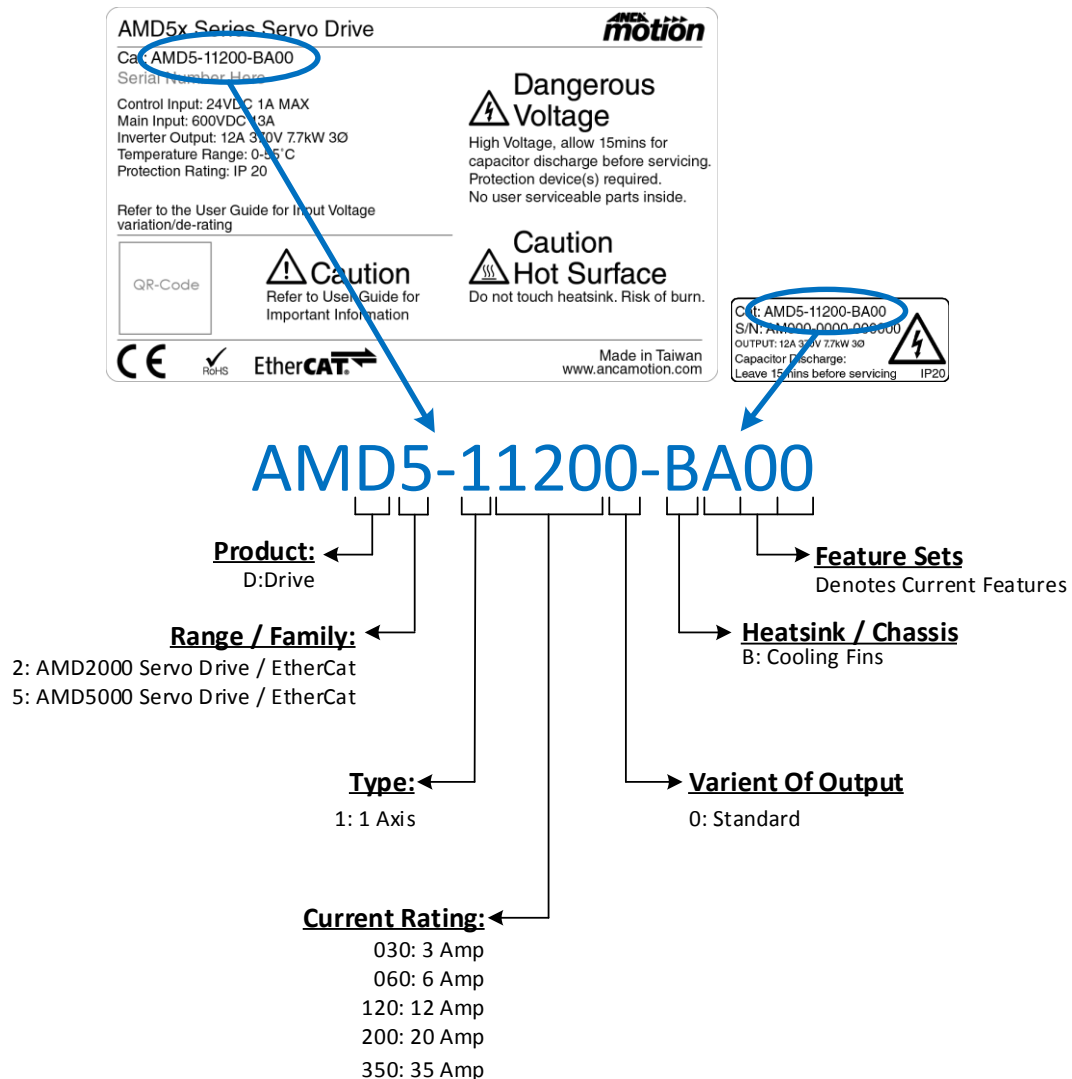
For any warranty work to be undertaken these labels must be readable and undamaged. Care should be taken to record these numbers in a separate register in the event of damage or loss.



NOTE: Do not under any circumstances tamper with these labels. Your warranty may be void if the labels are damaged.

3.2 AMD5x Series Servo Drive Catalogue Number Interpretation

AMD5x drives are marked with an identification label. The Catalogue number is explained as follows:



For any warranty work to be undertaken these labels must be readable and undamaged. Care should be taken to record these numbers in a separate register in the event of damage or loss.



NOTE: Do not under any circumstances tamper with these labels. Your warranty may be void if the labels are damaged.

4 Start-up

4.1 What this Chapter Contains

This chapter contains information related to the ANCA MotionBench that will guide the user in setting up and configuring the AMD servo drive:

- ANCA MotionBench Software installation and requirements
- Starting the drive using ANCA MotionBench
- Configuring and Commissioning of the drive
- Additional information on the ANCA MotionBench

4.2 Safe Start-Up and Operation

Please refer to sections [Installation Checklist](#) and section [Power-On Checks](#) sections of [AMD2000 Series D21xx Servo Drive - User Guide](#) or [AMD5x Passive Infeed Unit and AMD5x Servo Drive - User Guide](#) for information on how to start up the AMD servo drive safely.

4.3 Introduction to MotionBench

ANCA MotionBench is an application for inspecting and commissioning AMD servo drives. The MotionBench provides a connection wizard to aid in connecting directly to drive and a quick start-up wizard to get a motor moving quickly. Further panels are provided that give a functional overview and guided access to key drive functions, as well as an interface to access the entire list of parameters. MotionBench includes powerful real-time signal logging and graphing capability.

4.4 PC minimum specifications

The minimum PC requirements for MotionBench are:

- 1GB Memory (minimum)
- 2GB Free Disk Space (minimum)
- 1024 x 768 Screen Resolution 32-bit colour (recommended)
- Mouse or similar pointing device
- Microsoft .Net Framework 4
- Supported Operating System
- Supported Wired Network Adapter that is currently unused

Supported Operating Systems for MotionBench are:

- Windows XP
- Windows Vista
- Windows 7



NOTE: Both 32 and 64-bit versions of Windows are supported. However, only EN-AU and EN-US are guaranteed to work. Windows for other languages are known to cause problems with MotionBench.

Supported Wired Network Adapters for MotionBench are:

- Intel 82577LM Gigabit
- Broadcom NetXtream 57xx Gigabit
- Broadcom 57765-B0 PCI
- Marvell Yukon 88E8053 Gigabit
- ASIX AX88772A (USB2.0 to Ethernet dongle)
- Realtek RTL8139-810X
- Realtek PCIe GBE Family Controller
- Realtek PCIe FE Family Controller



NOTE: Most other wired network adaptors should be reliable but those listed are known to work. At this stage there are no wired network adapters which are known to be unreliable.

4.5 Configuring the Network Adapter



WARNING: To connect the AMD servo drive to a Laptop or PC requires the alteration of the Ethernet adapter configuration. This may affect the computer's office Ethernet connection. Installing a second Ethernet adaptor which is dedicated for use with the AMD servo drive will prevent this possible limitation. If you are uncomfortable about making changes to your Ethernet adapter configuration, or do not have the required user permission levels, then please consult with your IT administrator.

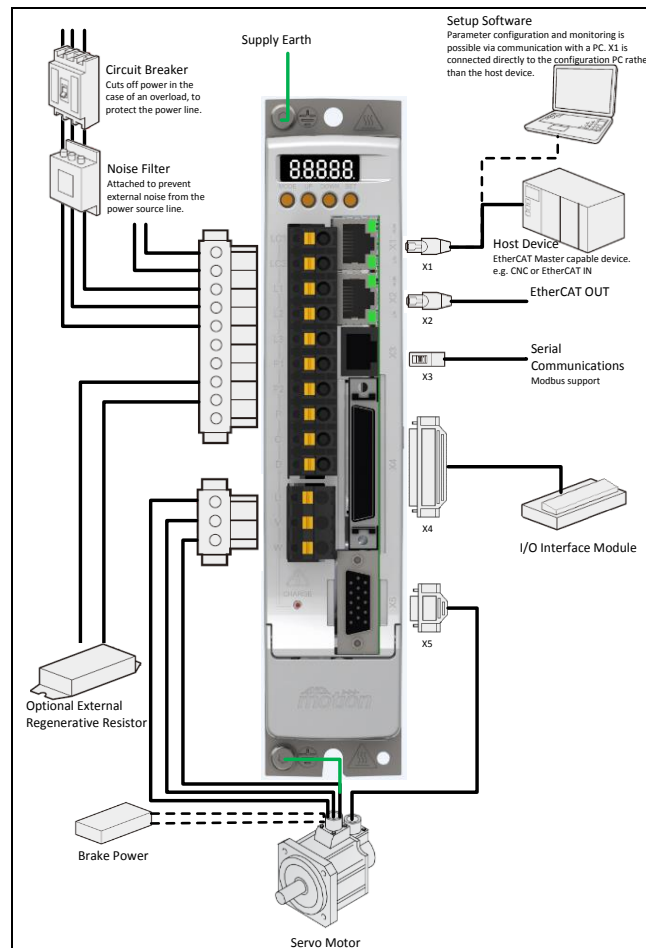
Note that the AMD servo drive must be directly connected to the Laptop or PC, it cannot be connected via an intermediate network.

1. **Windows 7:** Click Start menu, then Control Panel, then Network and Sharing Center.
Windows XP: Click Start menu, then Settings, then Control Panel, then Network Connections.
2. **Windows 7:** Click Change adapter settings. On the Local Area Connection for the required Ethernet adapter right-click, and then select Properties.
Windows XP: Right-click the Local Area Connection entry for the required Ethernet adapter and choose Properties.
3. **Windows 7:** Select the Internet Protocol Version 4 (TCP/IPv4) entry and click Properties.
Windows XP: Select Internet Protocol Version (TCP/IP) and click Properties.
4. On the General tab, make a note of the existing settings. Click Advanced, and make a note of any existing settings. Click Cancel and then click the Alternate Configuration tab (if one exists) and make a note of any existing settings.
5. On the General tab, choose the "Use the following IP" address option.
6. In the IP address box, enter the following IP address: 192.168.100.1. This is the IP address that will be assigned to the Ethernet adapter.
7. In the Subnet mask box, enter 255.255.255.0, no DNS settings required and Gateway IP address should be cleared and click OK.
8. Click Close to close the Local Connection Properties dialogue.

4.6 Connecting the AMD Servo Drive to a PC

Connect the supplied Ethernet cable between the PC network port and X1 of the AMD servo drive

	X1	EtherCAT IN
	X2	EtherCAT OUT



4.7 Installing the ANCA MotionBench

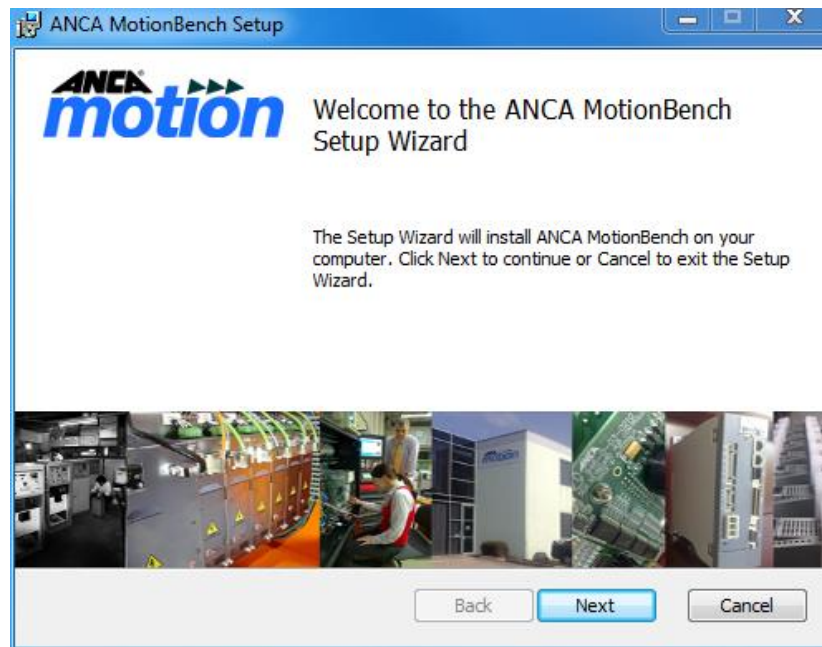


NOTE: For more information on ANCA MotionBench, refer to the [ANCA MotionBench User Guide](#).

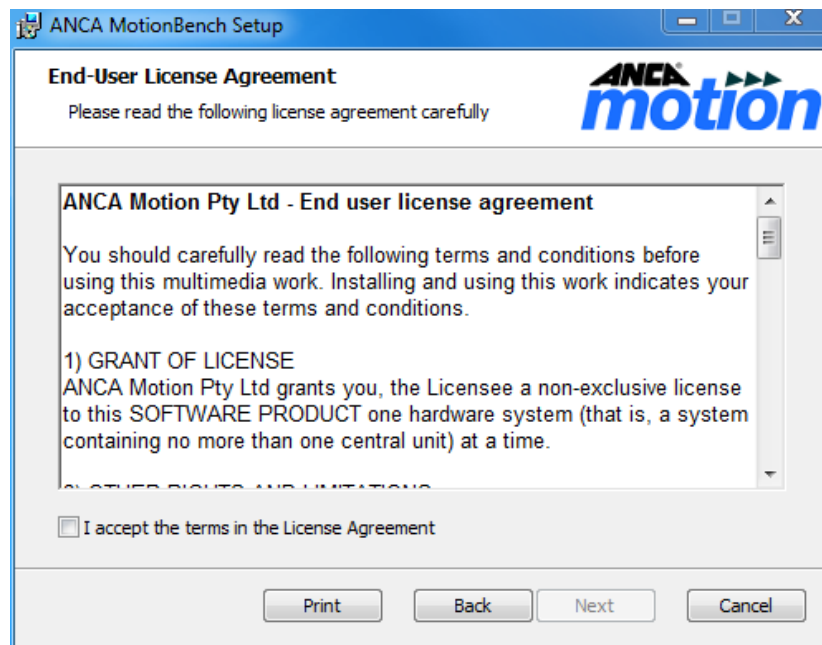
This section will guide you through the process to install ANCA MotionBench on your PC.

1. Ensure [4.2 Safe Start-Up and Operation](#) has been completed.
2. Double-click on the ANCA MotionBench .msi file. Latest .msi file can be downloaded from the ANCA Motion website.

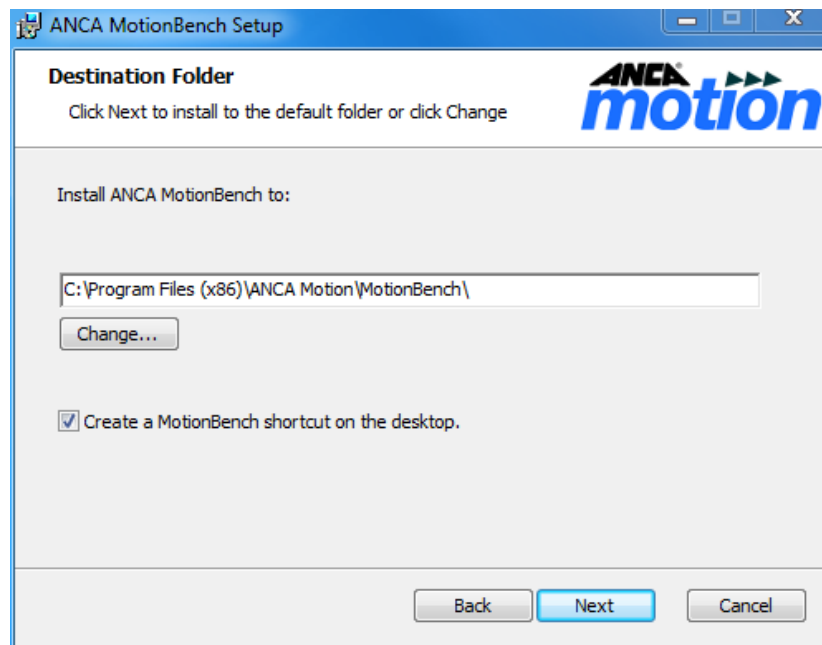
3. You will then be presented with the welcome screen shown below:



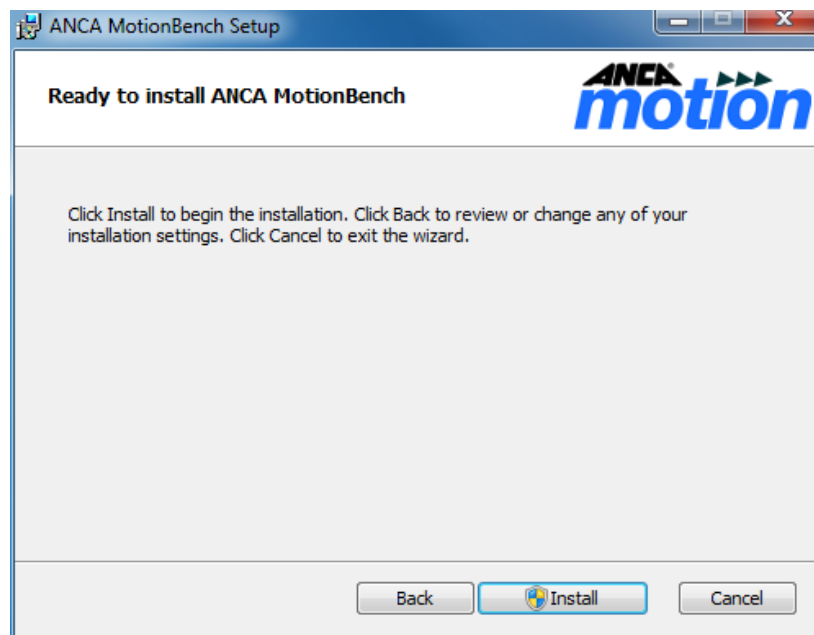
4. Click Next. You will then be presented with the *End-User License Agreement* shown below:



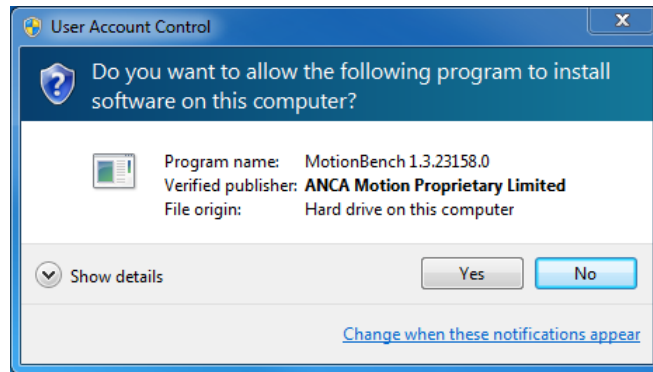
5. Please read the License Agreement and tick the "I accept the terms in the License Agreement" check box. Click Next. You will then be presented with the *Destination Folder* dialogue shown below:



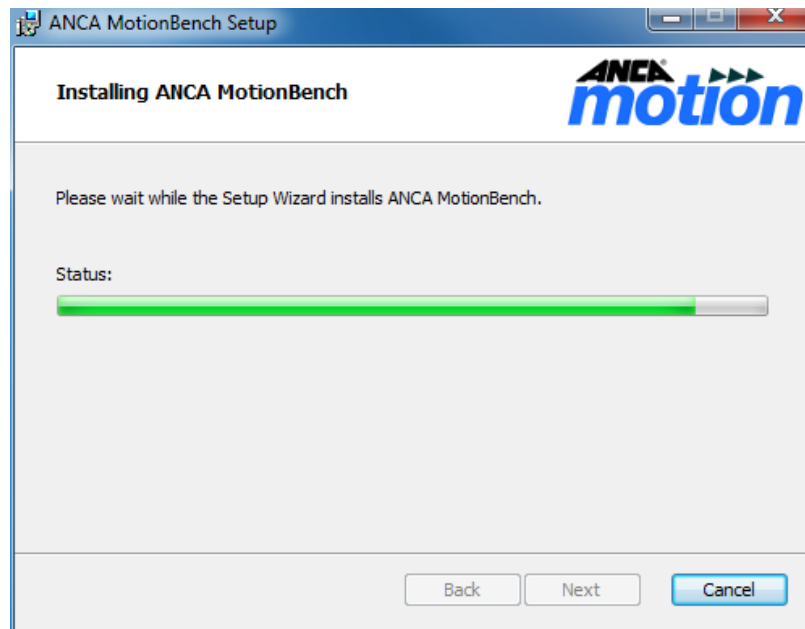
6. If you are happy with the default destination folder, simply click Next. Alternatively use the Change button to navigate to an alternative location. By default, a shortcut icon for launching MotionBench will be added to the Desktop. If you do not wish for an icon to be added to the desktop untick the “Create a MotionBench shortcut on the desktop” check box. Click Next. You will then be presented with *Install MotionBench* dialogue shown below:



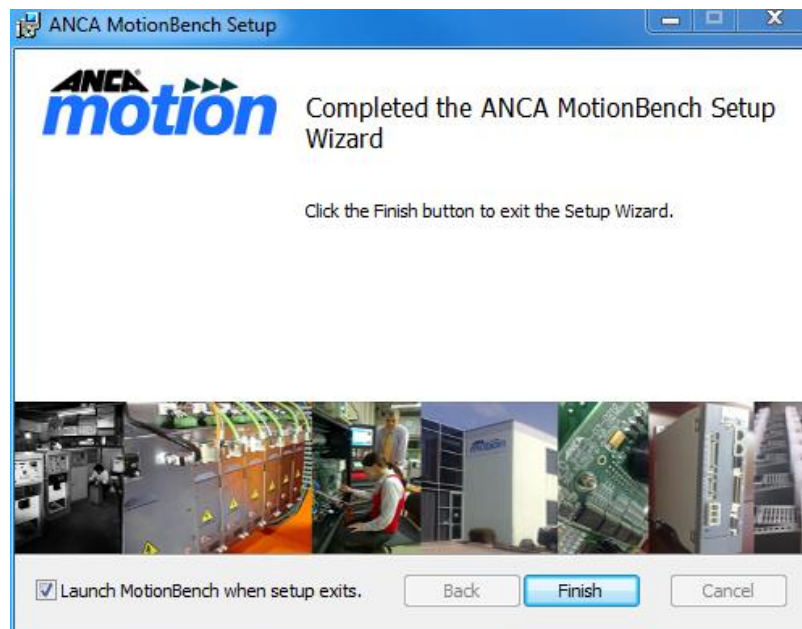
7. Click Install. You may be presented with the following dialogue:



8. Click Yes. MotionBench will then start installing on your PC. The dialogue shown below you indicate the status of the installation process.



9. When the installation process has completed the following dialogue will be shown.

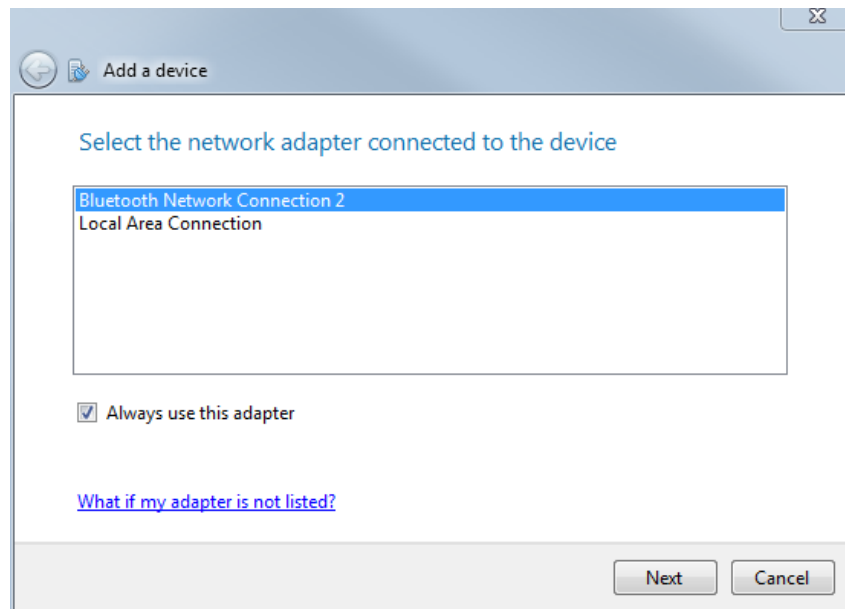


10. By default the MotionBench application will launch immediately after you click the Finish button. If you do not wish for the application to launch immediately, untick the "Launch MotionBench when setup exits" check box. Click Finish.

4.8 Configuring the AMD Servo Drive

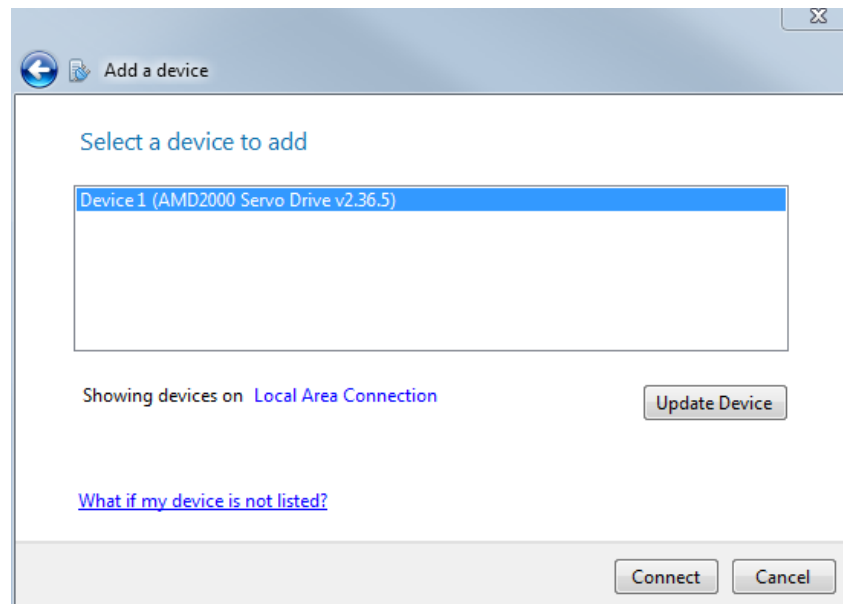
4.8.1 Launching ANCA MotionBench

1. Ensure the previous sections in [4 Start-up](#) have been completed.
11. Ensure the drive is powered-on.
12. Launch Motion Bench via the start menu or desktop icon.
13. You will be presented with the Add a device wizard.



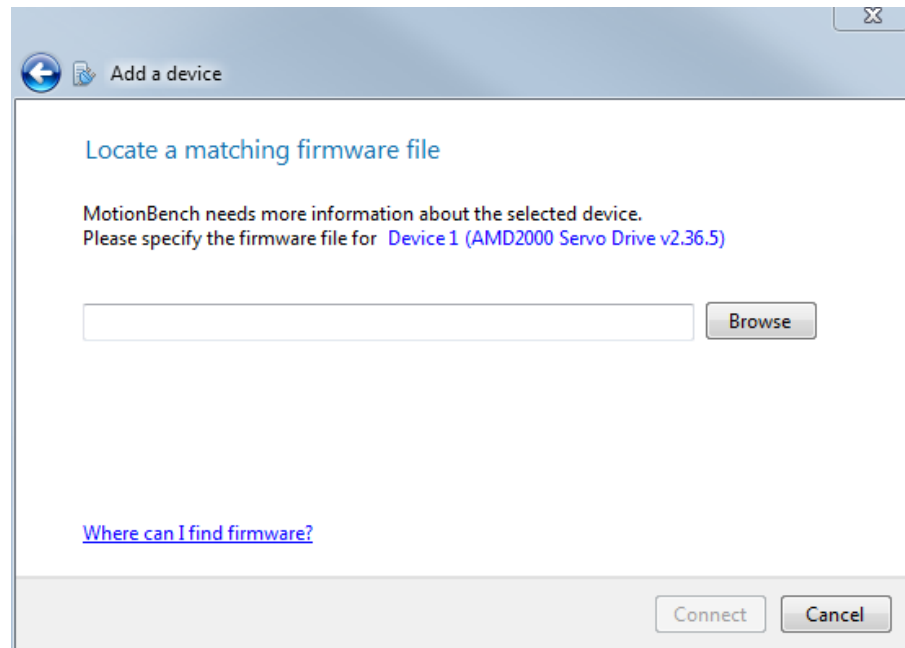
- a) Select the network adaptor connected to the device. This is the adapter that was configured in [4.5 Configuring the Network Adapter](#)
- b) The check box “Always use this adapter” will ensure that this adapter is used for future ANCA MotionBench session (recommended).
- c) Select Next

- d) Select a device to add.



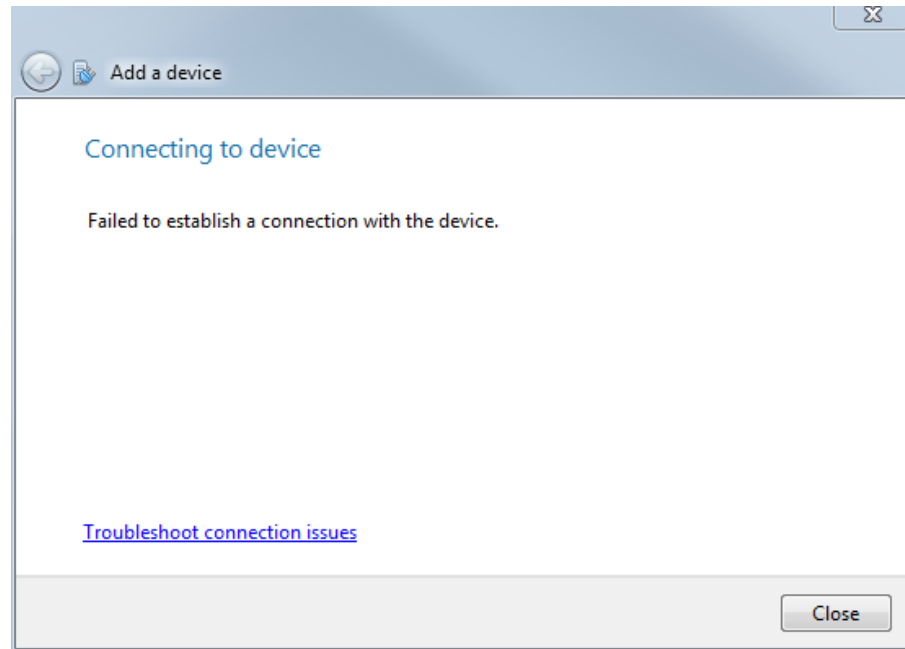
- e) If the device is not listed, there are a few possible things to check:
- Incorrect network adaptor selected in the previous dialogue. Use the back button to confirm this setting.
 - AMD servo drive is not powered on. Power-on the AMD servo drive.
 - Ethernet cable between the AMD servo drive and the PC is not connected. Refer to section [4.6 Connecting the AMD Servo Drive to a PC](#) for details.
 - Network adaptor configuration is incorrect. Refer to Section [4.5 Configuring the Network Adapter](#) for details.
 - Ensure that there is no Firewall blocking connections to the AMD servo drive. If unsure, please consult with your IT administrator.
 - Try closing the MotionBench application and then restarting.
 - Try power cycling the AMD servo drive.
 - Try rebooting the PC.
- f) The version of the firmware currently installed on the drive is indicated. If required, this can be updated using the Update Device button. Refer to [4.8.2.5 Firmware Update Wizard](#) below for details.

- g) If MotionBench cannot locate the .amf file which matches the firmware currently installed on the AMD servo drive the following screen is presented. Browse to the .amf file.



- h) Firmware can be downloaded from the ANCA Motion website.
i) Select Connect.

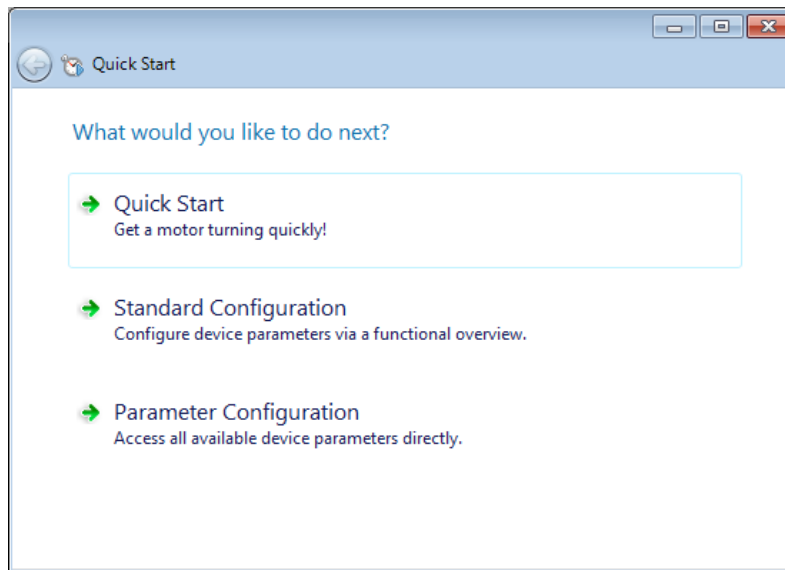
- j) If MotionBench fails to connect the following screen will be shown.



- k) If the device does not connect, there are a few possible things to check:
- i. AMD servo drive is not powered on. Power-on the AMD servo drive.
 - ii. Ethernet cable between the AMD servo drive and the PC is not connected. Refer to Section [4.6 Connecting the AMD Servo Drive to a PC](#) for details.
 - iii. Network adaptor configuration is incorrect. Refer to Section [4.5 Configuring the Network Adapter](#) for details.
 - iv. Ensure that there is no Firewall blocking connections to the AMD servo drive. If unsure, please consult with your IT administrator.
 - v. Try closing the MotionBench application and then restarting.
 - vi. Try power cycling the AMD servo drive.
 - vii. Try rebooting the PC.
- l) If MotionBench successfully connects to the drive, the Quick Start wizard will launch.

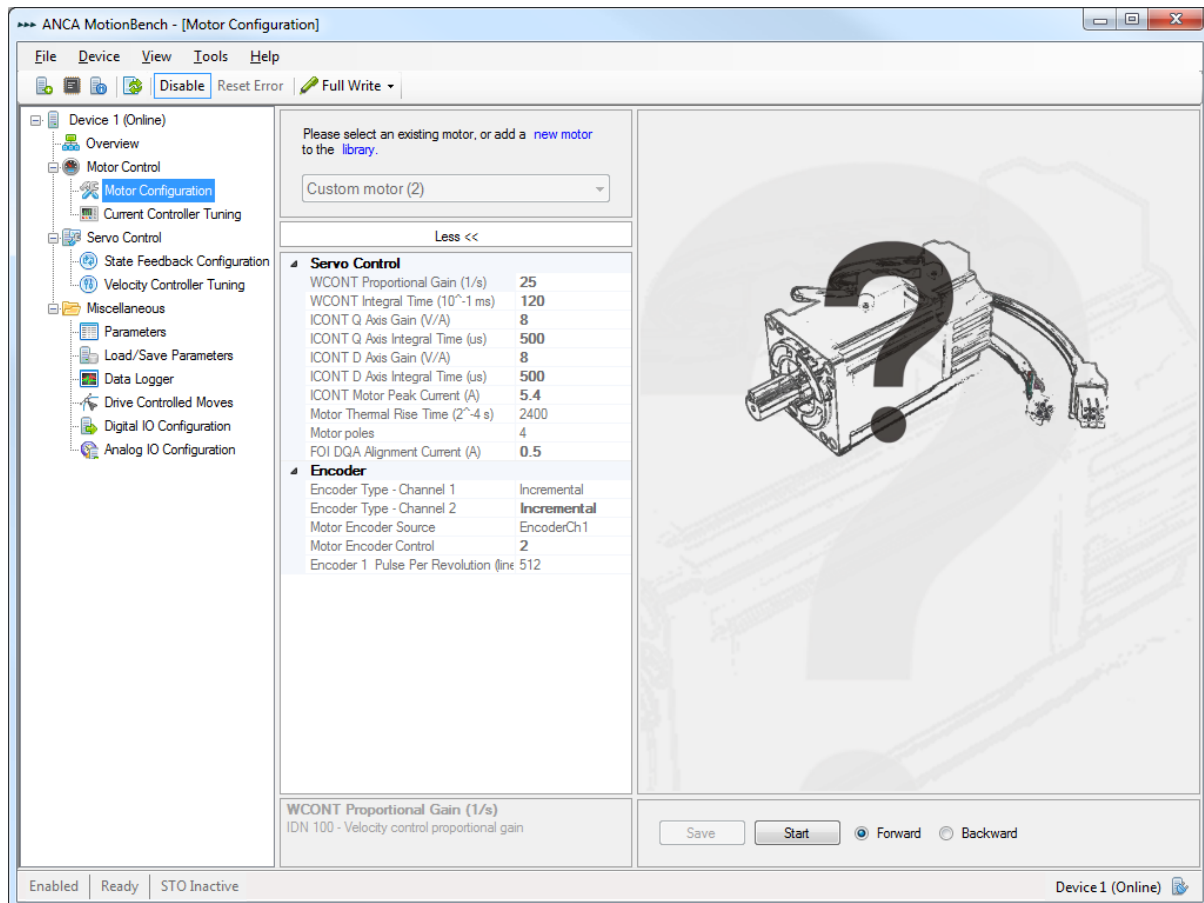
4.8.2 Quick Wizard

When the Quick Start wizard starts you will be given three options: Quick Start, Standard Configuration, and Parameter Configuration



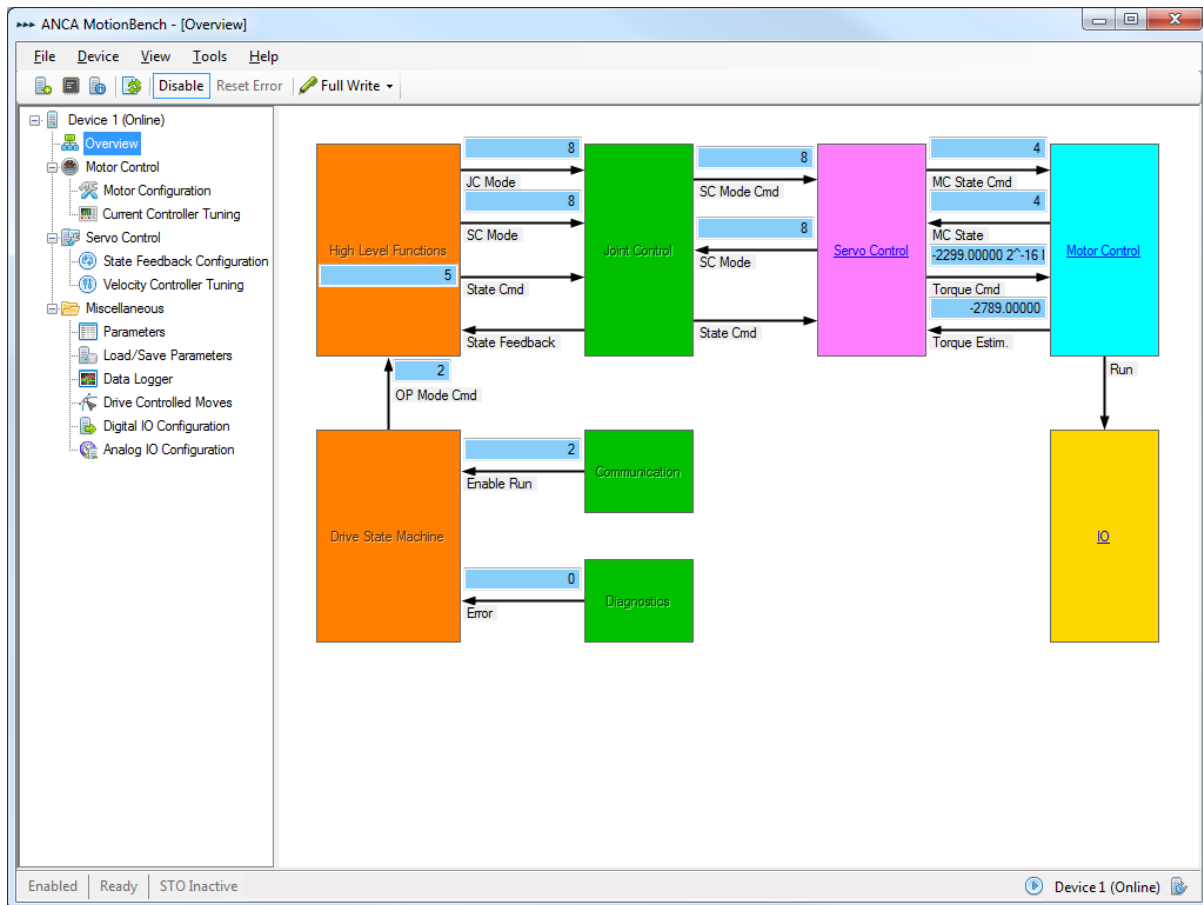
4.8.2.1 Quick Start

The *Quick Start* interface on the Quick Start Wizard takes you to the ANCA MotionBench – Motor Configuration page, which allows you to easily configure the drive parameters to suit a motor from the ANCA Motion range. Simply select your motor from the drop down box and click Save. Once the download process is complete you can then enable the drive by clicking on the *Enable* button in the toolbar. The drive will execute the Field Orientation Initialisation (FOI) procedure and be ready to accept motion commands. Click on the *Run Forward* or *Run Backward* buttons to jog the axis. Click on the *Stop* to pause the motion. Click on the *Disable* button (in the toolbar) to disable the drive and remove torque from the motor



4.8.2.2 Standard Configuration

The *Standard Configuration* interface from the Quick Start Wizard takes you to the ANCA MotionBench – [Overview] Page which allows you to easily configure the drive. Here you get a high level view of the current operating state of the drive. As well as access to other drive modules, for example, the Velocity Controller Tuning interface, Data Logger interface, etc.



4.8.2.3 Parameter Configuration

The Parameter Configuration interface from the Quick start Menu will take you to the ANCA MotionBench – [Parameter Access] table where all variables in the drive profile can be accessed.

ANCA MotionBench - [Parameters]

File Device View Tools Help Device

Enable Reset Error Full Write

Device 1 (Online)

- Overview
- Motor Control
 - Motor Configuration
 - Current Controller Tuning
- Servo Control
 - State Feedback Configuration
 - Velocity Controller Tuning
- Modes of Operation
 - Drive controlled homing
 - Profile position mode
 - Profile velocity mode
 - Drive Controlled Moves
- Miscellaneous
 - Parameters**
 - Load/Save Parameters
 - Digital IO Configuration
 - Analog IO Configuration
 - Scaling

Object ID	Name	Description	Value	Units
0x26F2.06	Units Message	Canopen current scaling units message	0	
0x26F2.07	Conversion Factor...	Canopen current conversion factor mantissa	1	
0x26F2.08	Conversion Factor...	Holds the conversion factor exponent for CoE current scaling	-3	
0x6040.00	Control Word		0	
0x6041.00	Status Word		624	
0x6042.00	vl Target Velocity	This command value is only active when the drive is in Velocity control mode (see IDN00032)	0	mm/s
0x6043.00	vl Velocity Demand		0	mm/s
0x6044.00	vl Velocity Actual	This is the estimated joint velocity.	0	mm/s
0x6046.01	vl Velocity Min Am...	Minimum acceptable value of the Target Velocity in CIA402 "Velocity Mode".	0	mm/s
0x6046.02	vl Velocity Max A...	Maximum acceptable value of the Target Velocity in CIA402 "Velocity Mode".	0	mm/s
0x6048.01	Delta Speed	Used in CIA402 "Velocity Mode".	0	mm/s
0x6048.02	Delta Time	Used in CIA402 "Velocity Mode".	0	s
0x6049.01	Delta Speed	Used in CIA402 "Velocity Mode".	0	mm/s
0x6049.02	Delta Time	Used in CIA402 "Velocity Mode".	0	s
0x6060.00	Modes of Operation		NoModeAssigned	
0x6061.00	Modes of Operatio...		NoModeAssigned	
0x6064.00	Position Actual Va...	The is the internal estimate of the joint position.	0.002	mm
0x606C.00	Velocity Actual Va...	This is the internal estimate of the joint velocity.	0	mm/s
0x606D.00	Velocity Window		8	mm/s
0x606F.00	Velocity Threshold		0.4	mm/s
0x607A.00	Target Position	This parameter is utilized in the drive position control operational mode, where the value spe...	0	mm
0x607F.00	Max Profile Velocity		400	mm/s
0x6081.00	Profile Velocity		40	mm/s
0x6083.00	Profile Acceleration		0.16	m/s/s
0x6084.00	Profile Deceleration		0.16	m/s/s
0x60C5.00	Max Acceleration		1.6	m/s/s
0x60C6.00	Max Deceleration		1.6	m/s/s
0x60FF.00	Target Velocity	This command value is only active when the drive is in Velocity control mode (see IDN00032)	0	mm/s
0x6410.01	Motor Continuous ...	This parameter, in conjunction with the motor thermal time constant, is use as the threshold f...	9	A
0x6410.02	Motor Overload T...	Motor I2T Load Limit in [A]	10	A
0x6410.03	Motor Load Limit	Load limit of the motor.	10000	
0x6410.04	Motor Thermal Ris...	Thermal rise time of the motor.	2400	
0x6502.00	Supported Drive ...	Canopen Supported drive modes	196997	

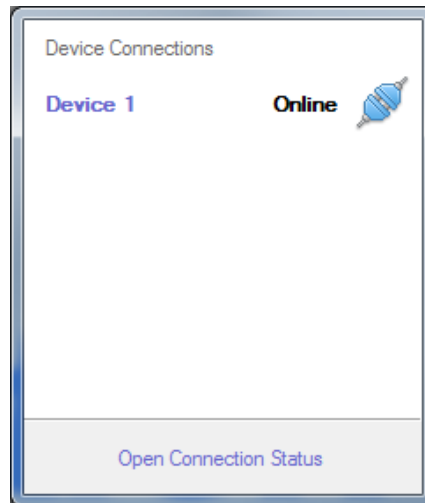
Filter

Disabled Ready STO Inactive

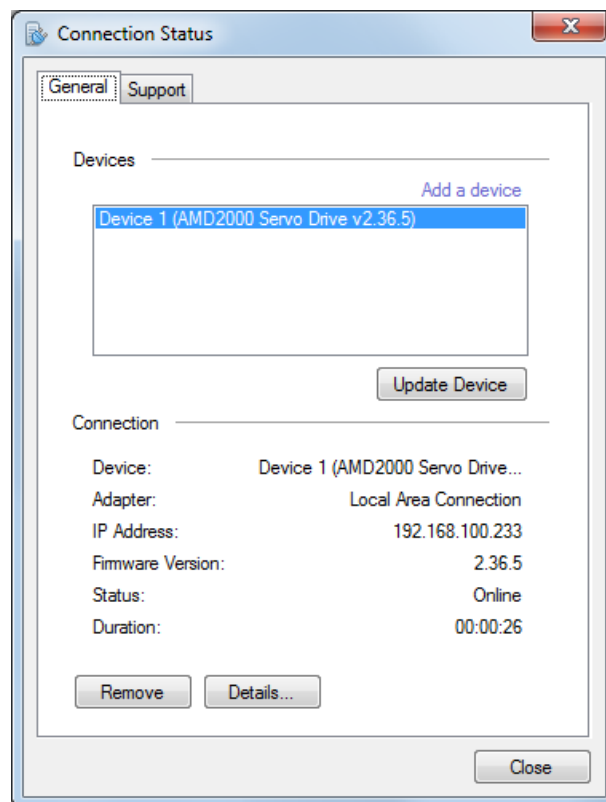
Device 1 (Online)

4.8.2.4 Connection Status Window

1. Clicking on the Device icon in the Status Bar of MotionBench will open the dialogue shown below. This interface shows the status of the devices connected to MotionBench.

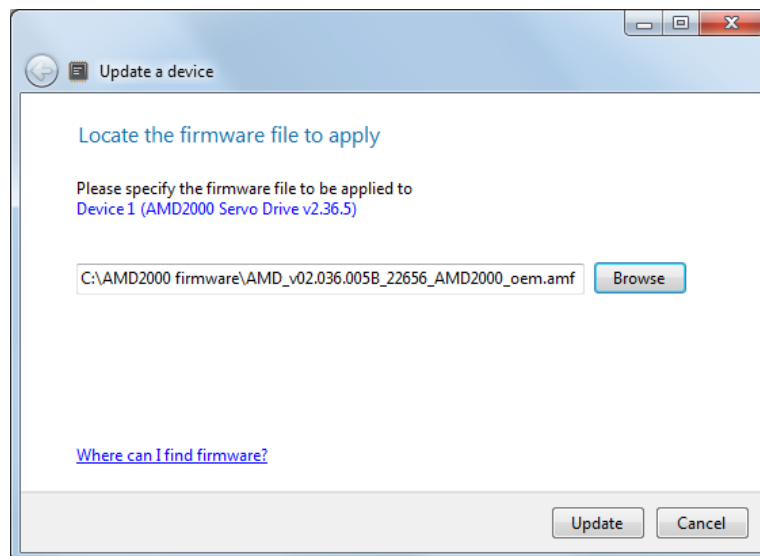


14. Clicking on Open Connection Status will open the dialogue shown below. Here additional information regarding the status of the device can be viewed. As well, it provides an interface to update the device firmware via the Update Device button. See Section [4.8.2.5 Firmware Update Wizard](#).

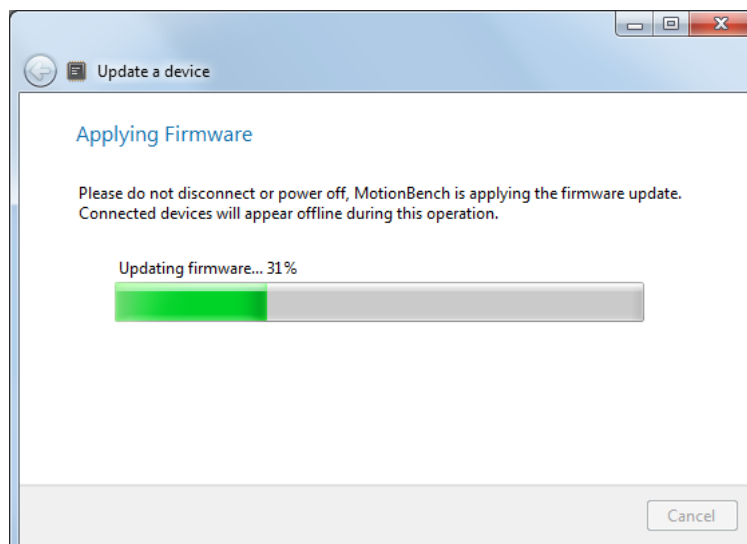


4.8.2.5 Firmware Update Wizard

1. Select Update Device button as part of the Add a device wizard (see Section [4.8.1 Launching ANCA MotionBench](#)), or from the Connection Status window (see Section [4.8.2.4 Connection Status Window](#)).
15. The Update a device dialogue shown below will open.

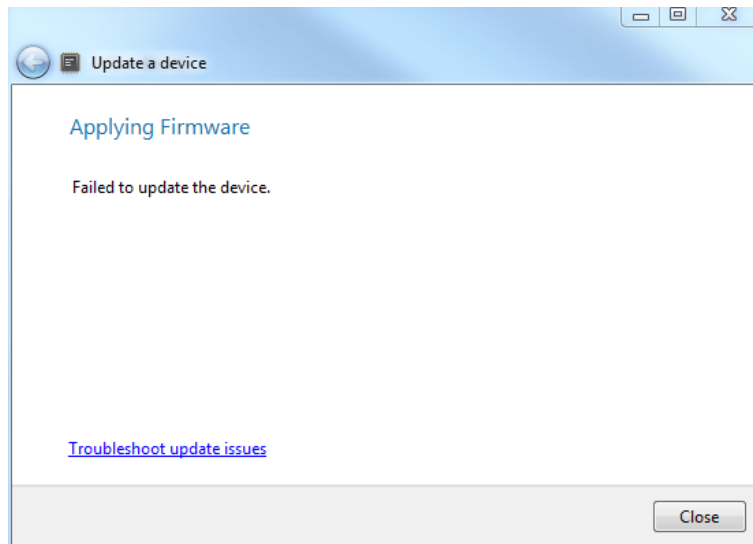


16. Browse to the amf file. Note that the Update button will only become available if a valid firmware file is selected. Click Update.



17. Firmware can be downloaded from the ANCA Motion website.

18. If MotionBench fails to update the firmware on the AMD servo drive, then the following screen will be shown.



19. If the firmware fails to update, there are a few possible things to check:
- a) AMD servo drive is not powered on. Power-on the AMD servo drive.
 - b) Ethernet cable between the AMD servo drive and the PC is not connected. Refer to Section [4.6 Connecting the AMD Servo Drive to a PC](#) for details.
 - c) Network adaptor configuration is incorrect. Refer to Section [4.5 Configuring the Network Adapter](#) for details.
 - d) Try closing the MotionBench application and then restarting.
 - e) Try power cycling the AMD servo drive.
 - f) Try rebooting the PC.

5 Basic Configuration

5.1 What this Chapter Contains

The following sections illustrate the configuration of most commonly used features of the AMD SoE servo drives.

5.2 Motor Configuration

5.2.1 Standard AM Motor

When using a standard ANCA Motion motor and MotionBench, basic motor setup can be done by following the instructions in [4.8.2.1 Quick Start](#). This provides a good starting point from which the axis can be configured. If the motor being used is not in the standard range provided by MotionBench then the motor can be configured as described in [5.2.2 Custom Motor](#) and then optionally saved into the MotionBench motor library for future use if required.

5.2.2 Custom Motor

If using a motor that is not stored in MotionBench by default, then a number of parameters need to be configured as a starting point to enable the motor. This section will assume a Permanent Magnet Synchronous Motor (PMSM) servo is used. For details on configuring the AMD servo drive for an Induction Motor, refer to [7.15.4 Induction Motor V/F Control](#).

5.2.2.1 Current Ratings

The motor manufacturer's specification sheet should list the following current ratings that should be configured in the AMD servo drive.

5.2.2.2 Related IDNs

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0109 / 109	Motor Peak Current Limit	Signed Integer (4 bytes)	Standard Current	0A	0A
S-0-0111 / 111	Motor Continuous Current Rating	Signed Integer (4 bytes)	Standard Current	1.25A	1.25A



WARNING: Selecting a peak current limit greater than specified by the motor manufacturer could result in damage to the motor.



NOTE: IDN S-0-0111 / 111 Motor Continuous Current Rating is defined in peak current units rather than RMS. If the continuous current rating of the motor used is specified in RMS then multiply this value by $\sqrt{2}$.



NOTE: While the maximum value the motor peak current limit should be set to what is specified in the motor data sheet.

5.2.2.3 I2R Overload Protection

5.2.2.3.1 Description

I2R Overload Protection is designed to detect a build-up of residual heat in the motor that can lead to permanent damage. Often a motor's peak current rating is larger than its continuous current rating, which means that it can operate at currents above its continuous rating, but not indefinitely.

5.2.2.3.2 General Operation

The formula used to estimate the residual heat in the motor is:

$$I^2R(k) = \left(\frac{T_s}{(\text{Cont. Current Rating})^2 \times (\text{Thermal Rise Time})} \right) \times (i_d^2 + i_q^2) + \left(1 - \frac{T_s}{(\text{Thermal Rise Time})} \right) \times I^2R(k-1)$$

which is updated every $T_s = 0.004$ seconds. If the value of I^2R exceeds one, then Error 325 will be triggered. Figure 5-1 shows an example of how the residual heat in the motor is estimated.

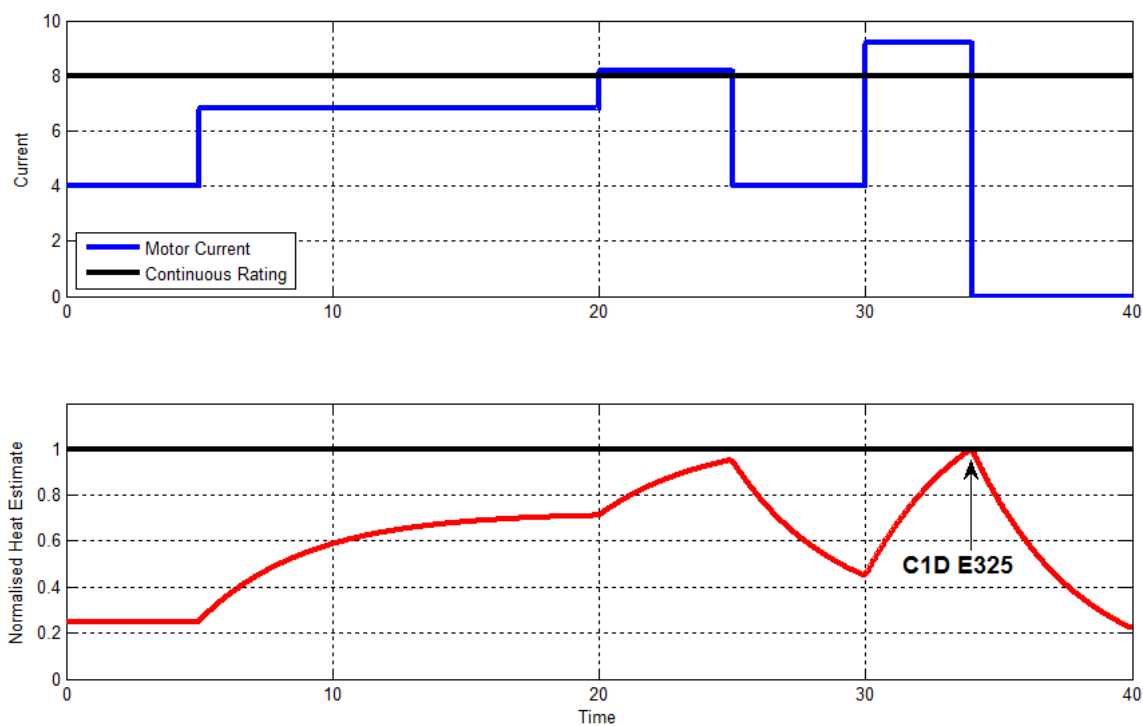


Figure 5-1 Motor I2R Overload Protection



Note: There is a similar protection mechanism for the AMD servo drive amplifier. If the current from the drive exceeds its continuous current rating for an extended period of time, then fault E309 will be triggered.

Please refer to the motor manufacturer's specification sheet for the continuous current rating and thermal rise time of the motor.

5.2.2.3.3 Related IDNs

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0111 / 111	Motor Continuous Current Rating	Signed Integer (4 bytes)	Standard Current	1.25A	1.25A
P-0-1232 / 34000	Motor Thermal Rise Time	Signed Integer (2 bytes)	2^{-4} seconds	2400	2400

P-0-0510 / 33278	Motor Control Tuning Procedure Command	Unsigned Integer (2 bytes)	Procedure Command	0	0
------------------	--	----------------------------	-------------------	---	---



NOTE: IDN S-0-0111 / 111 Motor Continuous Current Rating is defined in peak current units rather than RMS. If the continuous current rating of the motor used is specified in RMS then multiply this value by $\sqrt{2}$.

5.2.2.4 Field Orientation Initialisation

5.2.2.4.1 Description

If using a rotary motor, the motor manufacturer's specification sheet should contain the number of motor poles (2 x the number of pole pairs) of the permanent magnet motor to be entered into *IDN P-0-0006 / 32774*.

If using a linear motor, the motor manufacturer's specification sheet should contain the pole pitch (N-N distance) of the motor to be entered into *IDN S-0-0123 / 123*.

Using the default DQA as the Field Orientation Initialisation (FOI) method also requires a current to be configured. The current required depends on the combined friction of the motor and load but should be high enough to reliably move the rotor and low enough to safely use this amount of current for approximately 10 seconds. In most cases, setting this to the rated current of the motor is sufficient.

DQA will work for a wide range of motors in a wide range of applications; however, it causes motor movement during the first post power up initialisation. As such, it is not appropriate for braked axes. Refer [7.11 Field Orientation Initialisation](#) for more information on configuration options.

5.2.2.4.2 Related IDNs

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0123 / 123	Feed Constant	Signed Integer (4 bytes)	Standard Position	5mm	5mm
P-0-0006 / 32774	Motor Poles	Unsigned Integer (2 bytes)	Poles / rev	4	4
P-0-0301 / 33069	FOI DQA Alignment Current	Signed Integer (4 bytes)	Standard Current	4A	4A



WARNING: Selecting a FOI DQA alignment current limit greater than what the motor is capable of for that period of time could result in damage to the motor.

5.2.2.5 Encoder Configuration

The drive needs to be configured to determine which channels are connected to an encoder. The user can set the following parameters to either of the following two values:

- 0: No encoder connected
- 10: Incremental encoder connected

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-1432 / 34200	Encoder Type Connected - Channel 1	Unsigned Integer (2 bytes)	N/A	10	10
P-0-1433 / 34201	Encoder Type Connected - Channel 2	Unsigned Integer (2 bytes)	N/A	0	0

Furthermore, the motor encoder feedback needs to be assigned an encoder channel. The user can set the *IDN* *P-0-1028 / 33796* to either of two values:

- 0: Channel 1
- 1: Channel 2

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-1028 / 33796	Motor Encoder Source Channel	Unsigned Integer (2 bytes)	N/A	0	0

Once the motor encoder type and channel has been configured, the position feedback type, resolution and control word need to be configured.

If using a rotary motor, set the *Motor Position Feedback Type* to rotary by setting bit 0 = 0. The resolution is then defined as the number of lines (quadrature counts divided by 4) of the encoder.

If using a linear motor, set the *Motor Position Feedback Type* to linear by setting bit 0 = 1. The resolution is then defined as the distance between lines (quadrature counts divided by 4) of the encoder.

The encoder control word can be set to either of two values depending on the incremental encoder properties:

- 0: Digital quadrature encoder
- 2: Analogue sin/cos encoder

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0116 / 116	Resolution of Motor Encoder	Unsigned Integer (4 bytes)	Lines per Rev	512	512
S-0-0277 / 277	Motor Position Feedback Type	Unsigned Integer (2 bytes)	Binary	0	0
P-0-0004 / 32772	Motor Encoder Control	Unsigned Integer (2 bytes)	None	0	2
P-0-0007 / 32775	Motor Encoder Linear Resolution	Unsigned Integer (4 bytes)	10^{-4} mm	0.02 mm	0.02 mm

5.2.2.6 Initial Current Loop Configuration

5.2.2.6.1 Description

The following calculation can be used to provide initial current loop tuning parameters. To optimise current loop performance, further tuning will be required, refer to [7.27 Tuning](#) for more information.

The motor resistance, R , and the motor inductance, L , is required for this calculation. This information should be available from the motor manufacturer's specification sheet.

$$T_{RL} = \frac{L}{R}, k_{RL} = \frac{2}{R}$$

$$k_i = \frac{T_{RL}}{4 \times 1.5T_s \times k_{RL}}$$

$$T_i = \min\left\{\frac{T_{RL}}{15T_s}\right\} \times 10^6$$

Where

L = Line to Line Motor Inductance [H]

R = Line to Line Motor Resistance [Ω]

T_s = Current Loop Sample Time [s]

k_i = Current Loop Gain [V/A]

T_i = Current Loop Integral Time [μ s]

On the AMD servo drive the current loop is updated at 62.5 μ s, therefore, $T_s = 62.5 \times 10^{-6}$.

Once calculated, set the Q & D axis proportional gains to k_i and the Q & D axis integral times to T_i .

5.2.2.6.2 Example

Motor Inductance (line to line) = 20mH

Motor Resistance (line to line) = 6 Ω

$$T_{RL} = \frac{20 \times 10^{-3}}{6}, k_{RL} = \frac{2}{6}$$

$$k_i = \frac{0.00333}{4 \times 1.5 \times 62.5 \times 10^{-6} \times \frac{1}{3}} = 26.6667$$

$$T_i = \min\left\{\frac{0.00333}{15 \times 62.5 \times 10^{-6}}\right\} \times 10^6 = 937.5$$

This results in the following parameters set on the drive:

Current Controller Q Axis Proportional Gain = 27

Current Controller D Axis Proportional Gain = 27

Current Controller Q Axis Integral Time = 938

Current Controller D Axis Integral Time = 938

5.2.2.6.3 Related IDNs

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0106 / 106	Current Controller Q Axis Proportional Gain	Unsigned Integer (2 bytes)	V/A	3	3
S-0-0107 / 107	Current Controller Q Axis Integral Time	Unsigned Integer (2 bytes)	μs	3750	3750
S-0-0119 / 119	Current Controller D Axis Proportional Gain	Unsigned Integer (2 bytes)	V/A	3	3
S-0-0120 / 120	Current Controller D Axis Integral Time	Unsigned Integer (2 bytes)	μs	3750	3750

5.2.2.7 Initial Velocity Loop Configuration

5.2.2.7.1 Description

The velocity tuning is dependent on the gearing and pitch of the machine. However, the following values can be used as a starting point. To optimise velocity loop performance, further tuning will be required, refer to [7.28 Tuning – Velocity Control](#) for more information.

$$\text{Velocity Controller Proportional Gain} = \frac{200}{\text{FeedConstant}} \times \frac{\text{GearInput}}{\text{GearOutput}}$$

Velocity Controller Integral Time = 100

Where

FeedConstant is the pitch in mm/rev

5.2.2.7.2 Example

Application has a 10:1 gearbox with a ball screw pitch of 5mm/rev:

$$\text{Velocity Controller Proportional Gain} = \frac{200}{5} \times \frac{10}{1} = 400$$

Velocity Controller Integral Time = 100

5.2.2.7.3 Related IDNs

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0106 / 106	Velocity Controller Proportional Gain	Unsigned Integer (2 bytes)	1/s	100	100
S-0-0107 / 107	Velocity Controller Integral Time	Unsigned Integer (2 bytes)	10 ⁻¹ μs	100	100



Note: If the motor begins vibrating upon 1st enabling or after a small amount of movement, a simple method to remove this is to halve the proportional gain and double the integral time until this is removed. Standard tuning methods to optimise performance then apply (refer to [7.28 Tuning – Velocity Control](#)).

5.2.2.8 I2T Overload Protection

5.2.2.8.1 Description

I2T Overload Protection is designed to detect a sustained unusually high motor current over a relatively short period. This is often an indication of excessive cutting forces in a machining application, or a mechanical crash or seizure, or deterioration of the motor commutation angle, which could be an indication of a fault in the encoder feedback.

5.2.2.8.2 General Operation

Figure 5-2 shows an example of a sudden and sustained increase in the motor current that will lead to Error 320.

Once the motor current $\left(\sqrt{i_d^2 + i_q^2}\right)$ exceeds the Motor Overload Threshold (P-0-1248 / 34016) then an accumulator begins to increment at the rate of $(i_d^2 + i_q^2) - (\text{Motor Overload Threshold})^2$ per second. Once this accumulator exceeds the Motor Load Limit (0x6410:03 [A² sec]), fault E320 will be triggered. If before the Load Limit is exceeded, the motor current decreases to a level below the Load Threshold, then the accumulator will decrement at the rate of $(\text{Motor Overload Threshold})^2 - (i_d^2 + i_q^2)$ per second until it reaches zero. The equivalent formula is:

$$I2T(k) = (i_d^2 + i_q^2) - (\text{Motor Overload Threshold})^2 + I2T(k - 1)$$

This behaviour allows the motor current to exceed the Load Threshold for short periods of time as required due to large acceleration demands and/or high disturbance loads.

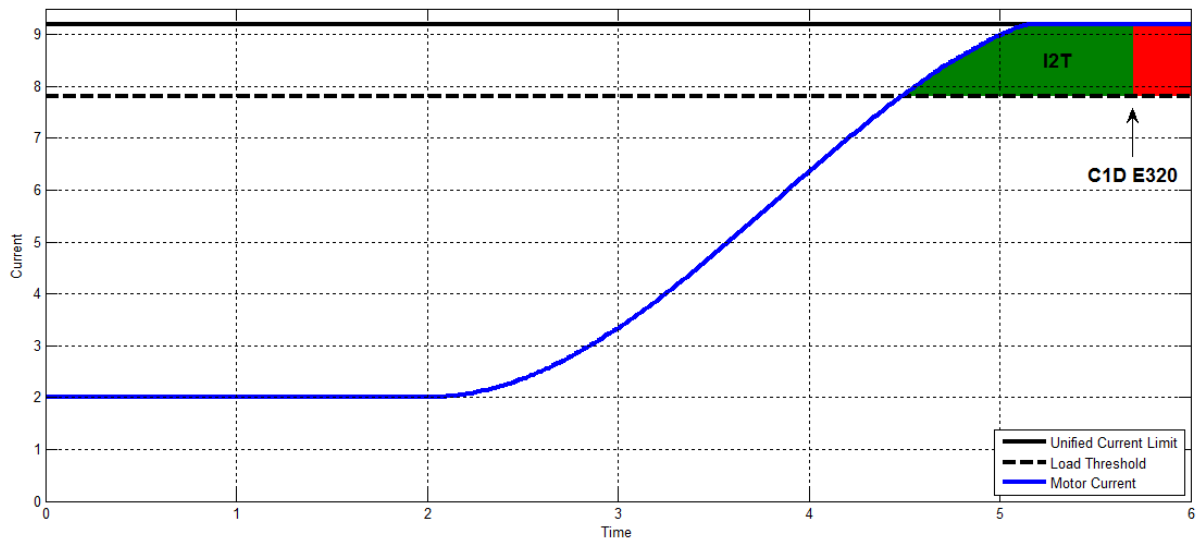


Figure 5-2 Motor I2T Overload Protection

5.2.2.8.3 Operation Example

A servo motor is operating a linear axis. It is desired that the I2T limit function be configured to detect the event that there is an obstruction in the axis. The axis never uses more than 3A under normal operating conditions and the motor has a peak current rating of 6A.

- Motor Overload Threshold – A threshold of 3.5A is chosen as it is a small amount above general operation current.
- Motor Load Limit – According to the machine design and motor capabilities, it is decided that the motor can operate at its peak current rating for 0.5 seconds before the error.

$$\text{Motor Load Limit} = \text{Time} \times (\text{Peak Motor Current}^2 - \text{Motor Overload Threshold}^2)$$

$$\text{Motor Load Limit} = 0.5 \times (6^2 - 3.5^2) = 11.875 \approx 12 \text{ A}^2\text{sec}$$

5.2.2.8.4 Related IDNs

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x

S-0-0109 / 109	Motor Peak Current Limit	Signed Integer (4 bytes)	Standard Current	0	0
P-0-1248 / 34016	Motor Overload Threshold	Signed Integer (4 bytes)	Standard Current	0.1	0.1
S-0-0114 / 114	Motor Load Limit	Unsigned Integer (2 bytes)	A ² s	10000	10000

5.3 Joint Direction

5.3.1 Description

The Invert Joint Direction parameter is used to configure the direction the motor turns for position, velocity or torque commands. Toggle the value between 1 and 0 to change the motor direction.

5.3.2 Related IDNs

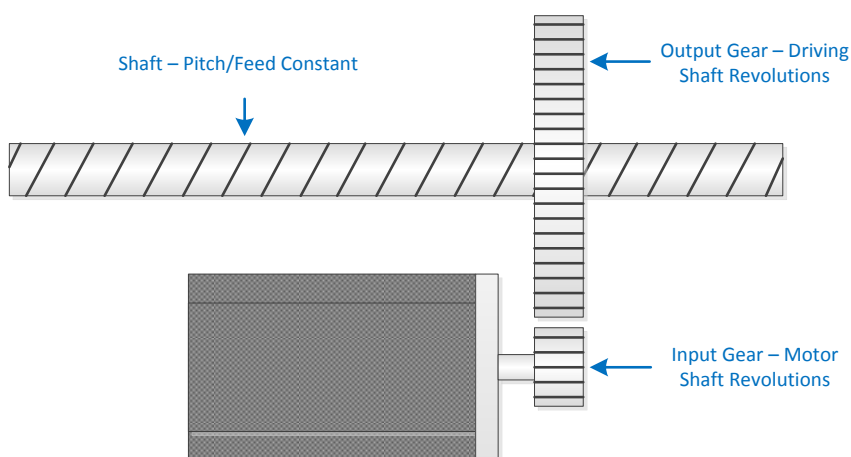
IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-0851 / 33619	Invert Joint Direction	Unsigned Integer (2 bytes)	Boolean	0	0

5.4 Units Selection

5.4.1 Motor to Joint Movement

5.4.1.1 Description

To configure how motor movement translates into axis movement, the gear ratios and feed constant can be configured via the parameters in the table below.



For a rotary axis, it is recommended that the feed constant be set to 1 rotation i.e. 360 degrees or 2π radians.

5.4.1.2 Related IDNs

IDN	Description	Data Type	Units	Default
-----	-------------	-----------	-------	---------

				AMD2000	AMD5x
S-0-0121 / 121	Motor Shaft Revolutions	Unsigned Integer (4 bytes)	N/A	1	1
S-0-0122 / 122	Driving Shaft Revolutions	Unsigned Integer (4 bytes)	N/A	1	1
S-0-0123 / 123	Feed Constant	Signed Integer (4 bytes)	Standard Position	5mm	5mm



Note: The velocity loop tuning is performed with respect to the load/axis. As a result, changing either the feed constant or gear ratio affects the velocity response of the motor. This can potentially lead to vibration or instability. Refer to [5.2.2.7 Initial Velocity Loop Configuration](#) for more information. These values will be effective as soon as motor moves.

5.4.2 Linear or Rotary Motion

5.4.2.1 Description

The units that are used to represent the axis over SoE can be configured to best suit the application. For most applications, simply configuring the axis as linear or modulo¹ rotary with the preferred/default units is sufficient.

Unit	Preferred Units	
	Linear	Rotary
Standard Position	10^{-7} m	10^{-4} deg
Standard Velocity	10^{-6} m/min	RPM
Standard Acceleration	10^{-6} m/s ²	10^{-3} rad/s ²

Setting the IDN in the table below will allow configuration between linear and rotary. The AMD servo drive defaults to linear preferred scaling.

Once modified, the new scaling is activated by running *Parameter Rescaling Procedure Command (P-0-0682 / 33450)*. Set the procedure command to 3 to run the command and it is considered complete when the procedure command acknowledge becomes 3.

If parameter/custom scaling is required, refer to [7.23 SoE Data Scaling](#) for more information on parameter scaling options.

¹ The feedback wraps back to the minimum value once the modulo value is reached. This is typically +/- 180 degrees.

5.4.2.2 Related IDNs

IDN	Description	Data Type	Units	Value for Preferred Units	
				Linear	Rotary
S-0-0076 / 76	Position Data Scaling Type	Unsigned Integer (2 bytes)	N/A	65	194
S-0-0044 / 44	Velocity Data Scaling Type	Unsigned Integer (2 bytes)	N/A	65	66
S-0-0160 / 160	Acceleration Data Scaling Type	Unsigned Integer (2 bytes)	N/A	65	66

5.5 Testing the Configuration

Once the basic motor configuration is complete, the motor operation should be tested to confirm motion is as expected.

5.5.1 MotionBench

If using MotionBench, the drive can be enabled by selecting 'Full Write' and then click 'Enable'. Navigating to the Motor Configuration Page will allow for a simple velocity test backwards and forwards.

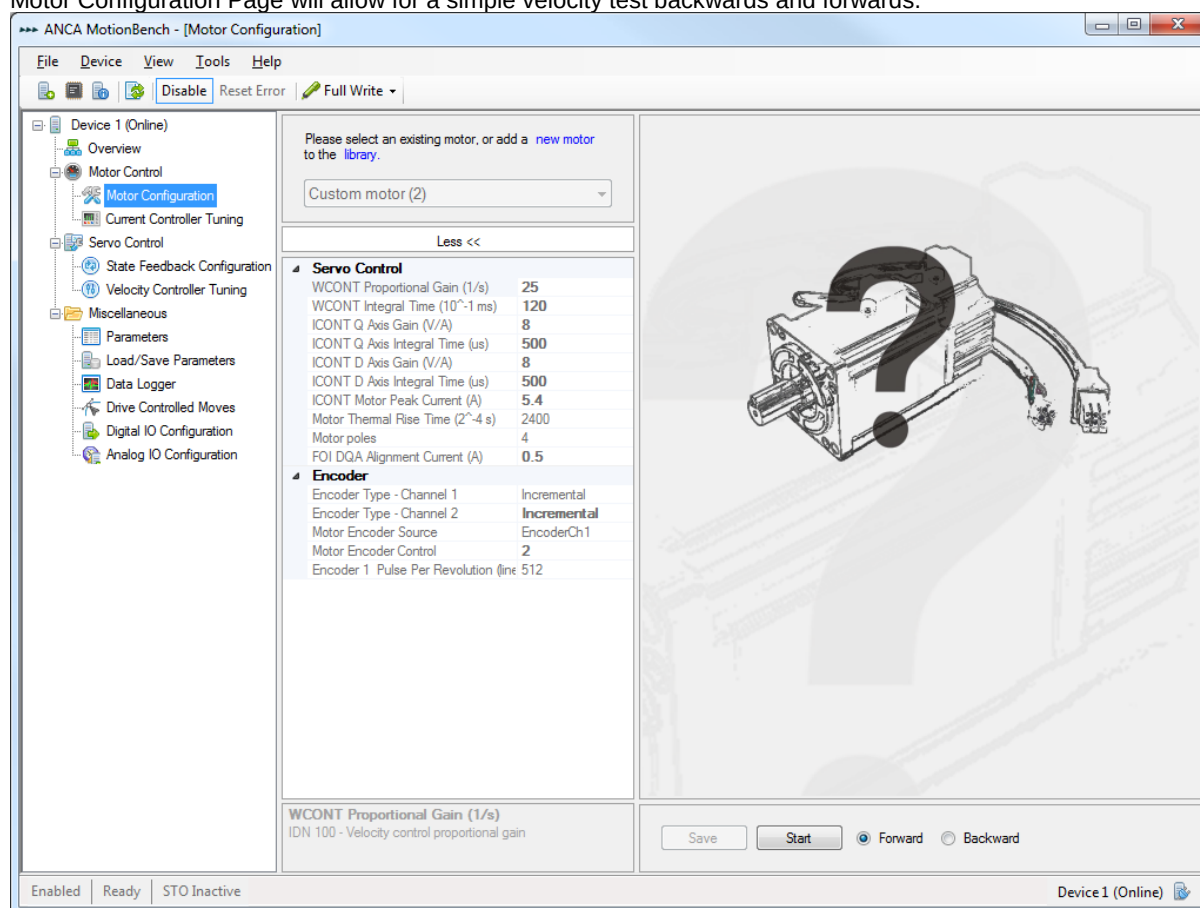


Figure 5-3 - MotionBench Motor Configuration page

For a more complex test, the Drive Controlled Moves page can be utilised to test the axis through a series of move profiles.

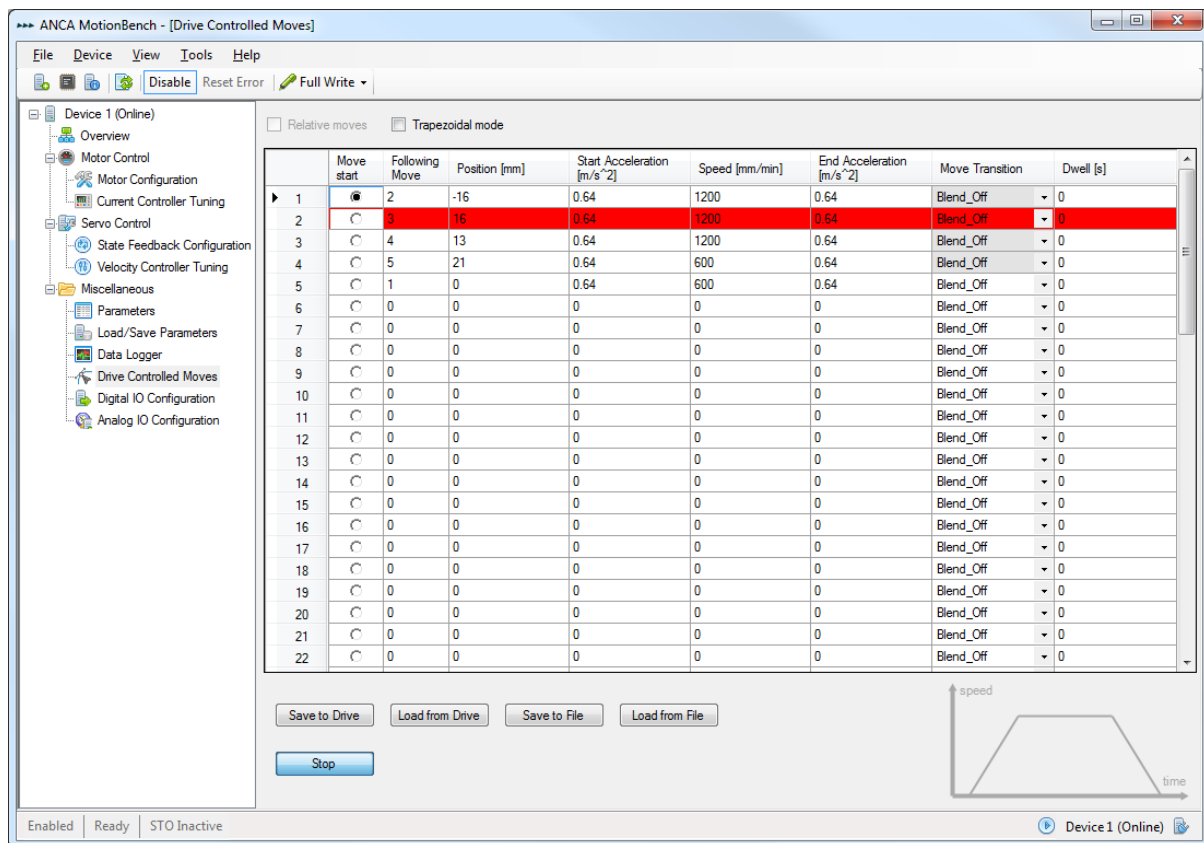


Figure 5-4 - MotionBench Drive Controlled Moves page

5.5.2 Using Operation Modes

If testing motor motion without MotionBench, a mode of operation will need to be used in conjunction with an EtherCAT master supporting SoE. For more information on the available modes of operation, refer to 6 [Operation Modes](#).

6 Operation Modes

6.1 What this Chapter Contains

This chapter describes the various motion control modes available.

6.2 Servo over EtherCAT (SoE)

6.2.1 Drive State Control

The AMD servo drive using SoE implements the drive state machine in accordance with the IEC 61800-7-204 standard. The drive state machine controls when power and torque is applied, how to stop the motor and how to handle a fault. The Master Control Word and Drive Status Word are used to interact with this state machine and are always mapped into cyclic/process data.

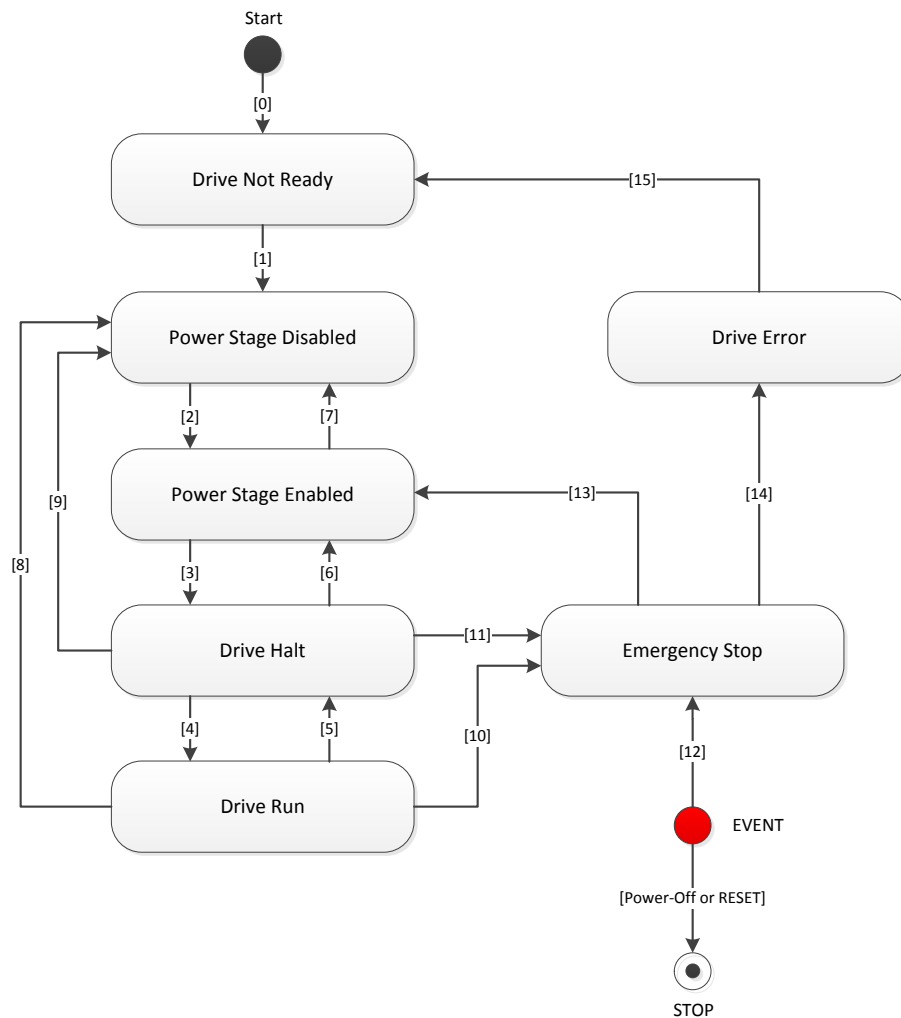


Figure 6-1 - Drive state machine operation

State	Description
Drive Not Ready	Drive not ready. Internal checks not yet concluded successfully.
Power Stage Disabled	Drive logic ready for main power (power stage) on.
Power Stage Enabled	Drive ready and main power applied. Motor is free of torque. Power stage pulses are blocked.
Drive Halt	Drive ready to operate. Power stage is active. Holding torque is applied to the motor. If required, a Halt Deceleration is applied to bring the motor to a halt.
Drive Run	Drive is enabled and able to perform motor control functions.

Emergency Stop	A transitory state to bring the motor to a stop. If required, an emergency stop deceleration is applied to bring the motor to a halt.
Drive Error	The drive has a Class 1 Diagnostic Error. Power stage pulses are blocked. The drive will remain in this state until the error is cleared. Refer to 9 Fault Tracing

Table 6-1 - Drive state description

Transition	Start State	End State	Comment
0	-	Drive Not Ready	Power On.
1	Drive Not Ready	Power Stage Disabled	The EtherCAT state must be in OP.
2	Power Stage Disabled	Power Stage Enabled	
3	Power Stage Enabled	Drive Halt	
4	Drive Halt	Drive Run	
5	Drive Run	Drive Halt	
6	Drive Halt	Power Stage Enabled	
7	Power Stage Enabled	Power Stage Disabled	
8	Drive Run	Power Stage Disabled	
9	Drive Halt	Power Stage Disabled	
10	Drive Run	Switch ON Disabled	
11	Drive Halt	Quick Stop Active	
12	ANY STATE	Emergency Stop	An error has been detected.
13	Emergency Stop	Power Stage Enabled	The motor has stopped. No error.
14	Emergency Stop	Drive Error	The motor has stopped due to an error.
15	Drive Error	Power Stage Disabled	Error has been cleared.
Power Off	ANY STATE	-	Stopped – no power to the system.
Reset	ANY STATE	-	Stopped – system still powered-up.

Table 6-2 - Drive state transition description



WARNING: The DC Bus of the AMD2000 is always charged when AC power is applied regardless of the drive state.

6.2.1.1 Master Control Word

The Master Control Word (MCW) is used to control the drive state, allowing for enabling, disabling and halting of the drive. Additionally, the MCW sets the current NC operation mode which can be changed during standard operation.

IDN	Description	Data Type	Units
S-0-0134 / 134	Master Control Word	Unsigned Integer (2 bytes)	Binary

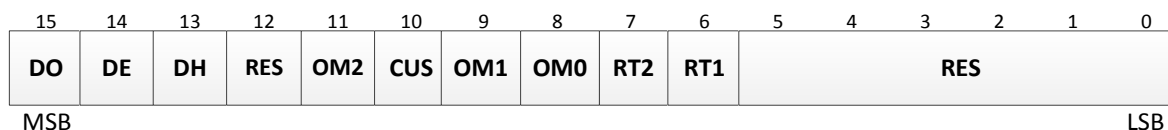


Figure 6-2 - Master Control Word definition

Bit	Key	Definition	Bit	Key	Definition
0	RES	Reserved	8	OM0	Operation Mode Bit 0
1	RES	Reserved	9	OM1	Operation Mode Bit 1
2	RES	Reserved	10	CUS	Control Unit Synchronization Bit
3	RES	Reserved	11	OM2	Operation Mode Bit 2
4	RES	Reserved	12	RES	Reserved
5	RES	Reserved	13	DH	Drive Halt
6	RT1	Real Time Control Bit 1	14	DE	Drive Enable
7	RT2	Real Time Control Bit 2	15	DO	Drive On

Table 6-3 - Master Control Word bit definitions

The Control Unit Synchronization Bit (IPOSYNC) is toggled with the control unit cycle time indicating the update of command values.

Operation Mode Bits 0-2 combine to encode an operation mode number with bit 0 the least significant bit as shown in [Table 6-4](#). Refer to [6.2.2.1 Pre-assigning Drive Modes of Operation](#).

The Real Time Control Bits can be used to control IDNs, refer to [7.18 Real Time Control Bits](#).

Operation Mode Bit			Selected Operation Mode
Bit 2	Bit 1	Bit 0	
0	0	0	Primary Operation Mode
0	0	1	Secondary Operation Mode 1
0	1	0	Secondary Operation Mode 2
0	1	1	Secondary Operation Mode 3
1	0	0	Secondary Operation Mode 4
1	0	1	Secondary Operation Mode 5
1	1	0	Secondary Operation Mode 6
1	1	1	Secondary Operation Mode 7

Table 6-4 - Operation Mode encoding in Master Control Word

Command	Bits of the Control Word			Transitions
	Bit 15	Bit 14	Bit 13	
Power Stage On	0	1	X	2
Enable Torque / Halt	1	1	0	2-3, 3, 5
Run	1	1	1	2-3-4, 3-4, 4
Disable Torque	0	1	X	10-13, 11-13
Power Stage Off	X	0	X	7, 8, 9
NOTE: Transitions 2-4 and 5-7 can occur automatically if the relevant command is given				

Table 6-5 - Command coding, 'X' refers to bits that are not examined. Refer to Figure 6-1 for transition definitions.

6.2.1.2 Drive Status Word

The status word reflects the current drive state. It can be used to check if the drive is enabled, disabled or in error. Additionally, the status word contains operation mode bits that indicate the current operation mode. Refer to [Table 6-1](#) for drive state definitions.

IDN	Description	Data Type	Units
S-0-0135 / 134	Drive Status Word	Unsigned Integer (2 bytes)	Binary

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RO1	RO0	C1D	RES	OM2	OM1	OM0	RT2	RT1	RES	SCV	RES	RES	RES	RES	RES
MSB								LSB							

Bit	Key	Definition	Bit	Key	Definition
0	RES	Reserved	8	OM0	Operation Mode Bit 0
1	RES	Reserved	9	OM1	Operation Mode Bit 1
2	RES	Reserved	10	OM2	Operation Mode Bit 2
3	SCV	Status Command Value Processing	11	RES	Reserved
4	RES	Reserved	12	RES	Reserved
5	RES	Reserved	13	C1D	Drive shutdown in Class 1 Diagnostic Error
6	RT1	Real Time Status Bit 1	14	RO0	Ready to Operate Bit 0
7	RT2	Real Time Status Bit 2	15	RO1	Ready to Operate Bit 1

Table 6-6 - Drive Status Word bit definitions

Status Command Value Processing is 1 when the drive will follow NC commands. This will be set to 0 when the drive is ignoring NC commands e.g. during Halt and drive controlled functions.

Operation Mode Bits 0-2 combine to encode an operation mode number with bit 0 the least significant bit in the same way as in the Master Control Word, see [Table 6-4](#). Refer to [6.2.2 Numerical Control](#) for information on using Operation Modes.

Drive Shutdown in Class 1 Diagnostic (C1D) Error will be 1 if the drive currently has a Class 1 Diagnostic Error. Refer to [9 Fault Tracing](#)

Ready to Operate Bits 0-1 combine to encode the current drive state, see [Table 6-7](#).

Drive State	Bits of the Status Word			
	Bit 15	Bit 14	Bit 13	Bit 3
Drive Not Ready	0	0	X	0
Power Stage Disabled	0	1	0	0
Power Stage Enabled	1	0	0	0
Drive Halt	1	1	0	0
Drive Run	1	1	0	X

Emergency Stop	X	0	X	0
Drive Error	0	0	1	0
Note: All other bits are all 'X'				

Table 6-7 - State Coding, 'X' refers to bits that are not examined

6.2.1.3 Halt Deceleration

The *Drive Halt Acceleration Bipolar* determines the rate at which deceleration to 0 occurs when the drive enters the halt state. The value is independent of the current direction of motor movement.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0372 / 372	Drive Halt Acceleration Bipolar	Signed Integer (4 bytes)	Standard Acceleration	6.4 m/s/s	6.4 m/s/s

6.2.1.4 Delay Times

Drive State Control contains a number of parameters to insert configurable time delays at selected state transitions.

The *Drive On Delay Time* is a time delay inserted in the transition from Power Stage Enabled to the Halt State (transition 3 as shown in [Figure 6-1](#)).

The *Drive Off Delay Time* is a time delay inserted in the transition from Emergency Stop to Power Stage Enabled state (transition 13 as shown in [Figure 6-1](#)).

The *Drive Off Maximum Delay Time* defines the maximum allowable time from when a Disable Torque command is given until the Power Stage Enabled state is reached. Once the Emergency Stop state is entered, the servo drive will enter the Power Stage Enabled state after the *Drive Off Maximum Delay Time* regardless of whether the axis has come to a halt or the *Drive Off Delay Time* has not yet been reached.

The *Drive Enable Delay Time* is a time delay inserted in the transition from Power Stage Disabled to Power Stage Enabled (transition 2 as shown in [Figure 6-1](#)).

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0206 / 206	Drive On Delay Time	Unsigned Integer (2 bytes)	4×10^{-3} s	0	0
S-0-0207 / 207	Drive Off Delay Time	Unsigned Integer (2 bytes)	4×10^{-3} s	0	0
S-0-0273 / 273	Drive Off Maximum Delay Time	Unsigned Integer (2 bytes)	4×10^{-3} s	0	0
S-0-0295 / 295	Drive Enable Delay Time	Unsigned Integer (2 bytes)	4×10^{-3} s	0	0

6.2.2 Numerical Control

Numerical Control is intended to operate with the trajectory generator in the control device. In a cyclic synchronous way, the command is updated by the control device which provides a control command value to the drive's control loops. A feedback value provides feedback of the motor operation for the control device. It is intended that these command and feedback IDNs are included in the cyclic data configuration. Refer to [8 Communication](#) for more information.

The cyclic synchronous nature of this mode relies on the use of distributed clocks to ensure the trajectory is communicated with the drive correctly. This is particularly important for position control modes.

The AMD servo drive supports the SoE profile for Numerical Control (NC) of a drive. The profile allows for the pre-setting of up to 8 operation modes for the drive. Each of these 8 modes or 'mode slots', can be configured for a particular drive mode of operation (e.g. position control).

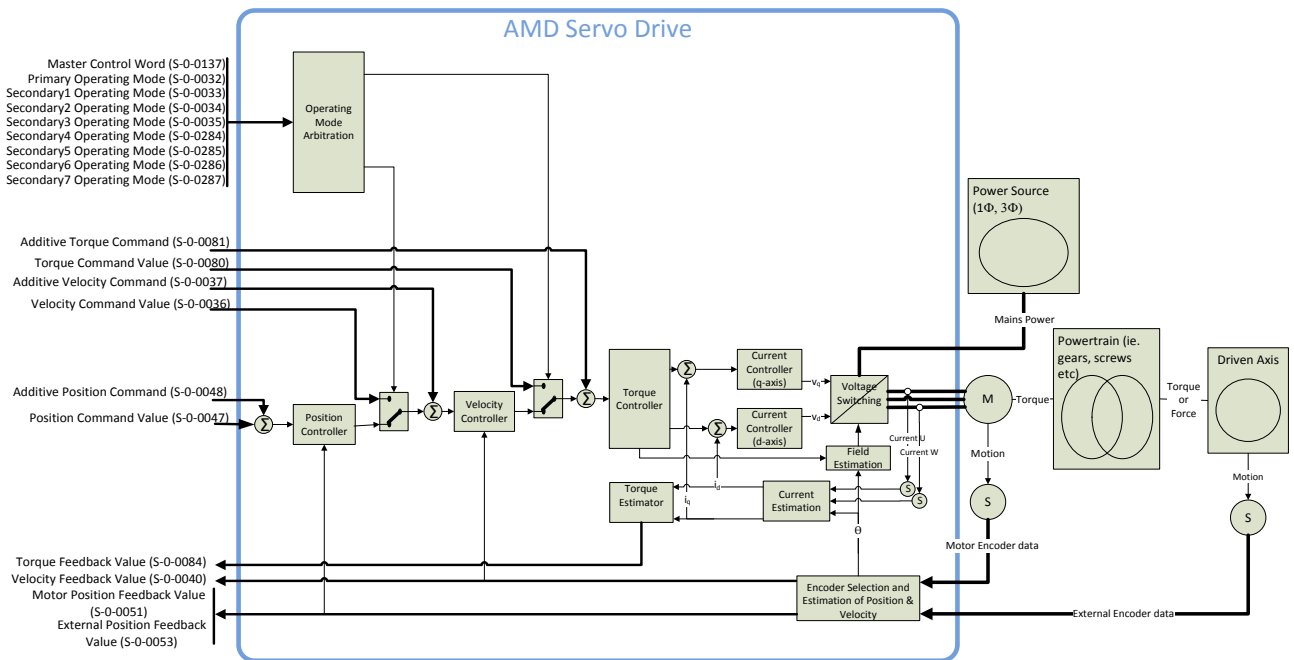


Figure 6-3 - AMD Servo Drive servo control architecture

The AMD servo drive supports up to 4 possible drive modes of operation to assign to any, or all, of these 8 available 'mode slots', namely:

- The position control drive mode, and this can be split into either motor or external encoder feedback modes,
- The velocity control drive mode,
- The torque control drive mode, and
- The mode of "no selected drive mode of operation."

The preceding 4 particular "drive modes of operation" can be pre-assigned by value to each 'mode slot' in the profile (see specific details below). When the drive is in operation, the user or external entity can then select into which mode they wish to place the drive by setting 3 bits in the **Master Control Word** to select from the 8 operation modes or 'slots'. These bits select which 'mode slot' the drive must query for determining its current "drive mode of operation," thus allowing it to quickly transition from one form of position control to another, or from velocity to torque control, etc. Similarly, 3 bits in the **Drive Status Word** reflect the currently active 'mode slot'. Note that this can be done "on the fly" while the drive is enabled and even moving, and usually within one scan at the default 4ms scan rate. However, mode transitions while in motion are not generally recommended.



WARNING: Switching between operation modes ('mode slots') while in motion should ONLY be done with care as unexpected motion can result.

In addition to the drive modes of operation being assigned, it is also possible to pre-assign the source of control set point data to be used for each 'mode slot'. Up to three different sources of data are available, including NC input, analogue input, or pulse train input.

6.2.2.1 Pre-assigning Drive Modes of Operation

6.2.2.1.1 Description

Each 'mode slot' of the drive can be assigned a value determining its associated drive mode of operation. The primary operation mode IDN (IDN S-0-0032 / 32) and 7 secondary operation mode IDN's (IDN's S-0-0033 / 33 → S-0-0035 / 35 and IDN's S-0-0284 / 284 → S-0-0287 / 287) can be filled with one of the following four values:

Value	Definition
0	No mode of operation
1	<i>Torque Control</i>
2	<i>Velocity Control</i>
3	<i>Position Control with Motor Encoder</i>
4	<i>Position Control with External Encoder</i>

Table 6-8 - Operation Mode Definition

6.2.2.1.2 Related IDNs

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0032 / 32	Primary Operation Mode	Unsigned Integer (2 bytes)	NC Op Mode	0	0
S-0-0033 / 33	Secondary Operation Mode 1	Unsigned Integer (2 bytes)	NC Op Mode	0	0
S-0-0034 / 34	Secondary Operation Mode 2	Unsigned Integer (2 bytes)	NC Op Mode	0	0
S-0-0035 / 35	Secondary Operation Mode 3	Unsigned Integer (2 bytes)	NC Op Mode	0	0
S-0-0284 / 284	Secondary Operation Mode 4	Unsigned Integer (2 bytes)	NC Op Mode	0	0
S-0-0285 / 285	Secondary Operation Mode 5	Unsigned Integer (2 bytes)	NC Op Mode	0	0
S-0-0286 / 286	Secondary Operation Mode 6	Unsigned Integer (2 bytes)	NC Op Mode	0	0
S-0-0287 / 287	Secondary Operation Mode 7	Unsigned Integer (2 bytes)	NC Op Mode	0	0

6.2.2.2 Pre-assigning Set Point Sources

6.2.2.2.1 Description

Each 'mode slot' of the drive can be assigned a value for its associated set point source, which is the data supplied to the controller for subsequent servo-control tracking. *NC Command Source* is an array of 8 elements, one for each of the 8 'mode slots.' The correspondence is in ascending order, so the first array element corresponds with the primary operation mode, the second element with the 1st of the 7 secondary operation modes, and so on. Each element can be filled with one of the following 3 values depending on which is the most appropriate source for set point information for the corresponding 'mode slot':

Value	Definition
0	Command via EtherCAT NC Command Set Point IDN for associated Operation Mode
1	Command via analogue voltage input. Only applies to the Velocity Control operation mode. Refer to 6.2.2.5 Analogue Input Velocity Control .

2	Command via Pulse / Stepper input. Only applies to the Position Control operation modes. Refer to 6.2.2.8 Pulse / Stepper Position Control .
---	--

Table 6-9 - NC Command Source Definition

6.2.2.2.2 Related IDNs

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-0531 / 33299	NC Command Source	Unsigned Integer (2 bytes) Array	NC Set Point Source	0,0,0,0,0,0,0	0,0,0,0,0,0,0

6.2.2.3 Torque Control

6.2.2.3.1 Description

To operate the AMD servo drive in Torque Control mode, set the active operation to 1. Commands are supplied from the control device to *NC Torque/Force Set Point Command (S-0-0080 / 80)*, while feedback is provided to the control device via *NC Torque/Force Feedback (S-0-0084 / 84)*.

Optionally, the *NC Additive Torque/Force Command (S-0-0081 / 81)* can be provided by the control device to offset the commanded set point. The polarity of these commands can also be reversed via *NC Torque/Force Polarity Configuration (S-0-0085 / 85)*.



WARNING: When in torque control mode, the motor can reach very high velocities without a proper limit value.

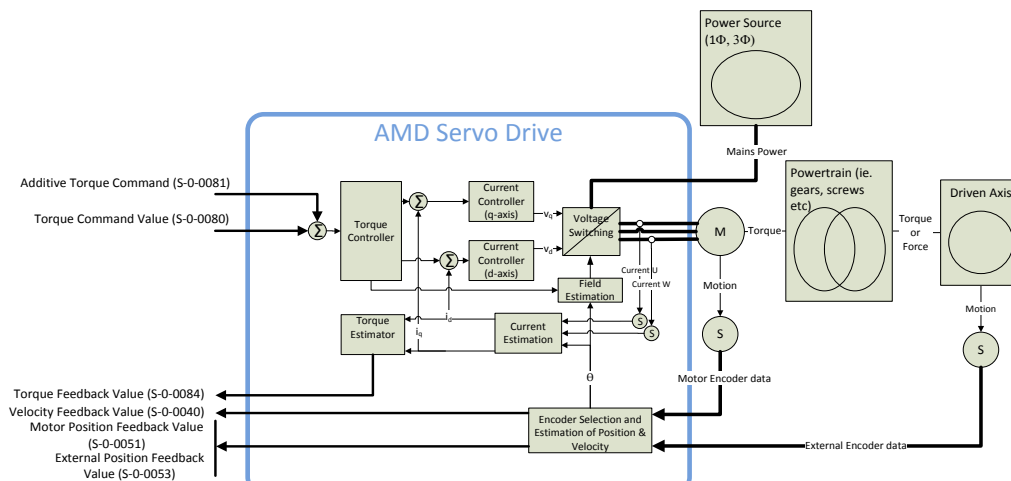


Figure 6-4 - Torque mode servo control

6.2.2.3.2 Related IDNs

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0080 / 80	NC Torque/Force Set Point Command	Signed Integer (2 bytes)	Standard Torque	0	0
S-0-0081 / 81	NC Additive Torque/Force Command	Signed Integer (2 bytes)	Standard Torque	0	0
S-0-0084 / 84	NC Torque/Force Feedback	Signed Integer (2 bytes)	Standard Torque	N/A	N/A
S-0-0085 / 85	NC Torque/Force Polarity Configuration	Unsigned Integer (2 bytes)	Binary	0	0

6.2.2.4 Velocity Control

6.2.2.4.1 Description

To operate the AMD servo drive in Velocity Control mode, set the active operation to 2. Commands are supplied from the control device to *NC Velocity Set Point Command (S-0-0036 / 36)*, while feedback is provided to the control device via *NC Velocity Feedback (S-0-0040 / 40)*.

Optionally, the *NC Additive Velocity Command (S-0-0037 / 37)* can be provided by the control device to offset the commanded set point. The polarity of these commands can also be reversed via *NC Velocity Polarity Configuration (S-0-0043 / 43)*.

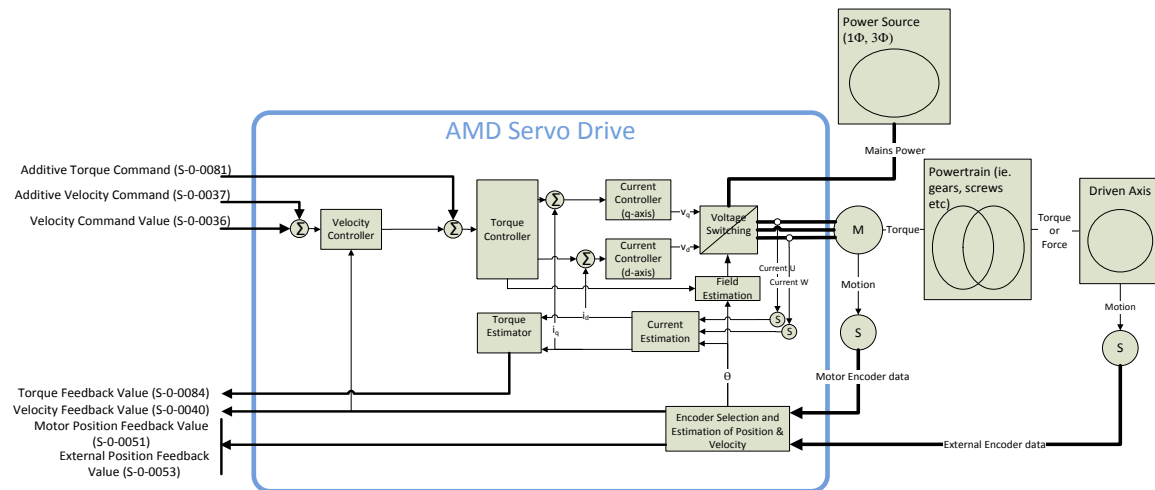


Figure 6-5 - Velocity mode servo control

6.2.2.4.2 Related IDNs

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0036 / 36	NC Velocity Set Point Command	Signed Integer (4 bytes)	Standard Velocity	0	0
S-0-0037 / 37	NC Additive Velocity Command	Signed Integer (4 bytes)	Standard Velocity	0	0
S-0-0040 / 40	NC Velocity Feedback	Signed Integer (4 bytes)	Standard Velocity	N/A	N/A
S-0-0043 / 43	NC Velocity Polarity Configuration	Unsigned Integer (2 bytes)	Binary	0	0

6.2.2.5 Analogue Input Velocity Control

6.2.2.5.1 Description



NOTE: This feature is only available on the AMD2000 series servo drives.

The AMD servo drive can be configured to operate in *Velocity Control* with a set point command sourced from an analogue input. This mode allows the AMD servo drive to be used with non-EtherCAT control devices / units or direct user velocity control via a control panel.

The desired velocity per volt gain can be configured to simplify axis configuration. [Global Acceleration Constraints](#) can also be applied if desired to ensure smooth servo motion. The [Encoder Pass-Through](#) feature is often used in conjunction with this operation mode to provide position feedback to the control device / unit providing the command.

Refer to the [AMD2000 Series D21xx Servo Drive - User Guide](#) for information on how to connect an analogue input.

6.2.2.5.2 General Operation

Analogue Input Velocity Control mode can be used in the following way:

1. Configure the [Analogue Inputs \(General Purpose\)](#) for the input to be used as required.
2. Configure *Analogue Set Point Source* for the input to be used.
3. Configure the *Analogue Input Velocity Command Gain*, the velocity per one volt, for the application.
4. Configure the set point source to analogue input. Refer to [6.2.2.2 Pre-assigning Set Point Sources](#).
5. Enable the axis with the active operation mode to Velocity Control. The axis will now respond to Analogue Input commands.

6.2.2.5.3 Related IDNs

IDN	Description	Data Type	Units	Default
				AMD2000
P-0-0527 / 33295	Analogue Set Point Source	Unsigned Integer (2 bytes)	Analogue Input	0
P-0-0528 / 33296	Analogue Input Velocity Command Gain	Signed Integer (4 bytes)	Standard Velocity	0

6.2.2.6 Position Control with Motor Encoder

6.2.2.6.1 Description

To operate the AMD servo drive in Position Control mode with Motor Encoder, set the active operation to 3. Commands are supplied from the control device to *NC Position Set Point Command (S-0-0047 / 47)*, while feedback is provided to the control device via *NC Motor Position Feedback (S-0-0051 / 51)*.

Optionally, the *NC Additive Position Command (S-0-0048 / 48)* can be provided by the control device to offset the commanded set point. The polarity of these commands can also be reversed via *NC Position Polarity Configuration (S-0-0055 / 55)*.

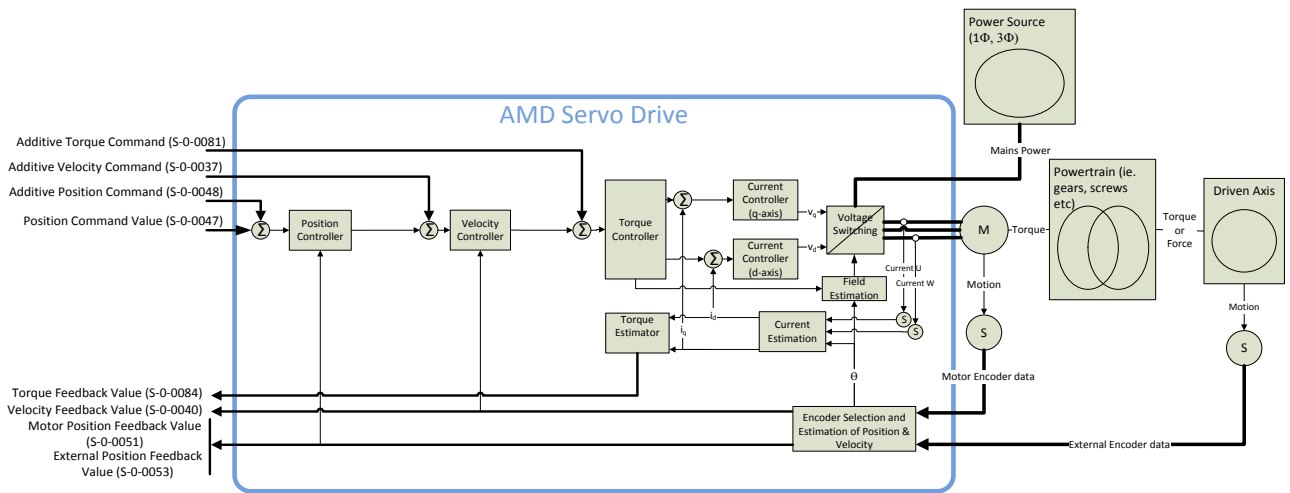


Figure 6-6 - Position mode servo control

6.2.2.6.2 Related IDNs

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0047 / 47	NC Position Set Point Command	Signed Integer (4 bytes)	Standard Position	0	0
S-0-0048 / 48	NC Additive Position Command	Signed Integer (4 bytes)	Standard Position	0	0
S-0-0051 / 51	NC Motor Position Feedback	Signed Integer (4 bytes)	Standard Position	N/A	N/A
S-0-0055 / 55	NC Position Polarity Configuration	Unsigned Integer (2 bytes)	Binary	0	0

6.2.2.7 Position Control with External Encoder

6.2.2.7.1 Description

To operate the AMD servo drive in Position Control mode with External Encoder, set the active operation to 4. Commands are supplied from the control device to *NC Position Set Point Command (S-0-0047 / 47)*, while feedback is provided to the control device via *NC External Position Feedback (S-0-0053 / 53)*.

Optionally, the *NC Additive Position Command (S-0-0048 / 48)* can be provided by the control device to offset the commanded set point. The polarity of these commands can also be reversed via *NC Position Polarity Configuration (S-0-0055 / 55)*.

6.2.2.7.2 Related IDNs

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0047 / 47	NC Position Set Point Command	Signed Integer (4 bytes)	Standard Position	0	0
S-0-0048 / 48	NC Additive Position Command	Signed Integer (4 bytes)	Standard Position	0	0
S-0-0051 / 53	NC External Position Feedback	Signed Integer (4 bytes)	Standard Position	N/A	N/A
S-0-0055 / 55	NC Position Polarity Configuration	Unsigned Integer (2 bytes)	Binary	0	0

6.2.2.8 Pulse / Stepper Position Control

6.2.2.8.1 Description



NOTE: This feature is only available on the AMD2000 series servo drives.

The AMD servo drive can be configured to operate in [Position Control with Motor Encoder](#) or [Position Control with External Encoder](#) with a set point command sourced from a pulse or step input. The input can be 'Quadrature' or 'Pulse + Direction' based. This mode allows the AMD servo drive to be used with non-EtherCAT control devices / units or slave the axis position to another device's encoder.

The desired distance per pulse can be configured to simplify axis configuration. [Global Acceleration and Velocity Constraints](#) can be applied to ensure smooth servo motion. The [Encoder Pass-Through](#) feature is often used in conjunction with this operation mode to provide position feedback to the control device / unit providing the command or fed as a pulse input into another AMD servo drive to slave the servo drives axis position.

Refer to the [AMD2000 Series D21xx Servo Drive - User Guide](#) for information on how to connect the Pulse / Stepper Position Control inputs.

6.2.2.8.2 Input Types

6.2.2.8.2.1 Quadrature

A quadrature input uses two input pulse trains, A & B, to determine both the distance to move and the direction of motion. [Figure 6-7](#) shows an example of a quadrature pulse sequence and the resulting count. When A leads B the count is up, and when B leads A the count is down. Reversing the direction of motion can be accomplished by inverting signal A or B using the [External Pulse Count Control Word](#).

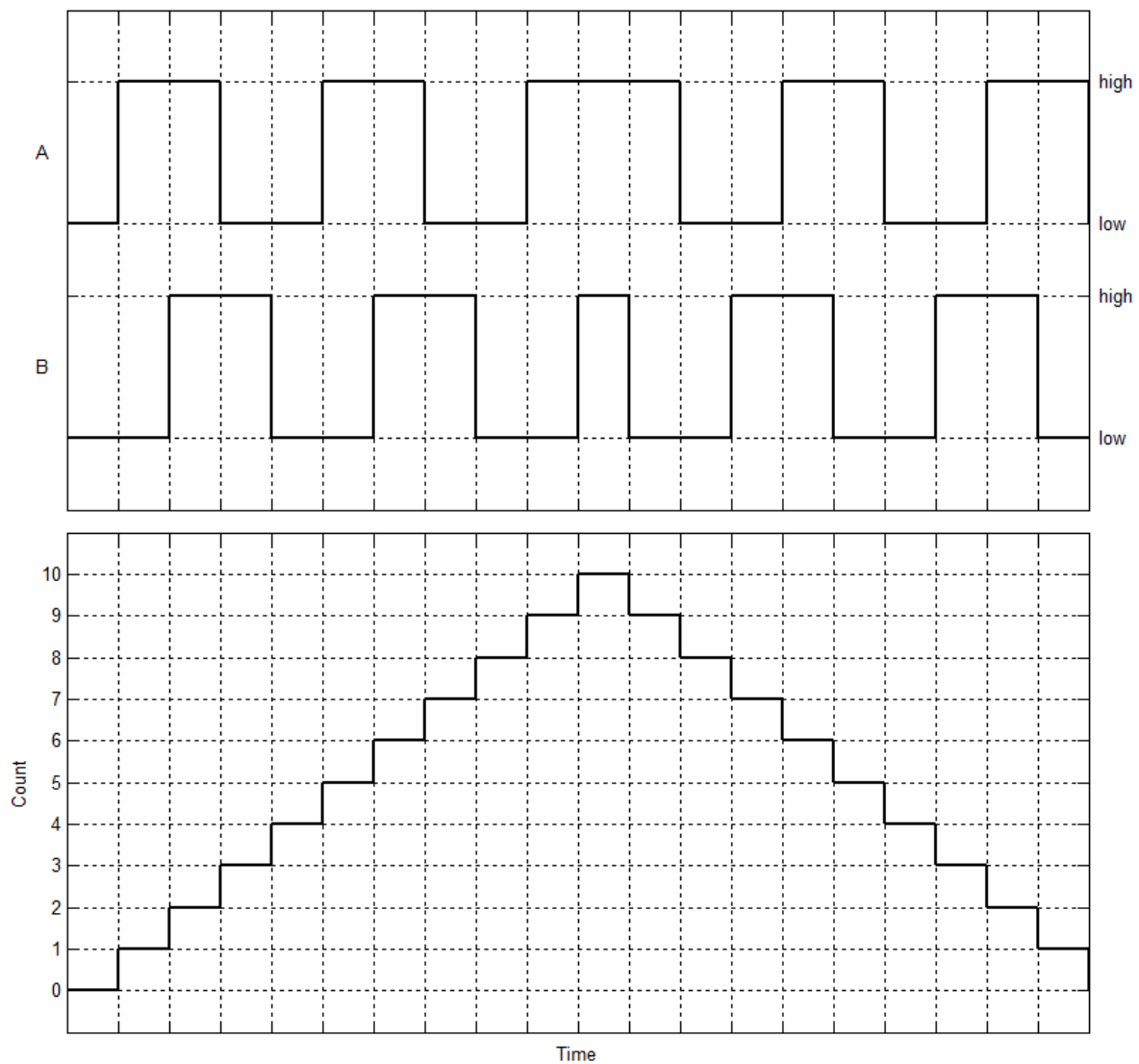


Figure 6-7 - Quadrature counting

6.2.2.8.2.2 Pulse + Direction

A pulse + direction input uses one input pulse train, A, to determine the distance to move and a dedicated direction input, B, to determine the direction of motion. [Figure 6-8](#) shows an example of a pulse + direction sequence and the resulting count. When the direction input is 1, the count will increment for every rising edge of A, and when the direction input is low the count will decrement for every rising edge of A. Reversing the direction of motion can be accomplished by inverting the direction input, B, using the *External Pulse Count Control Word*.

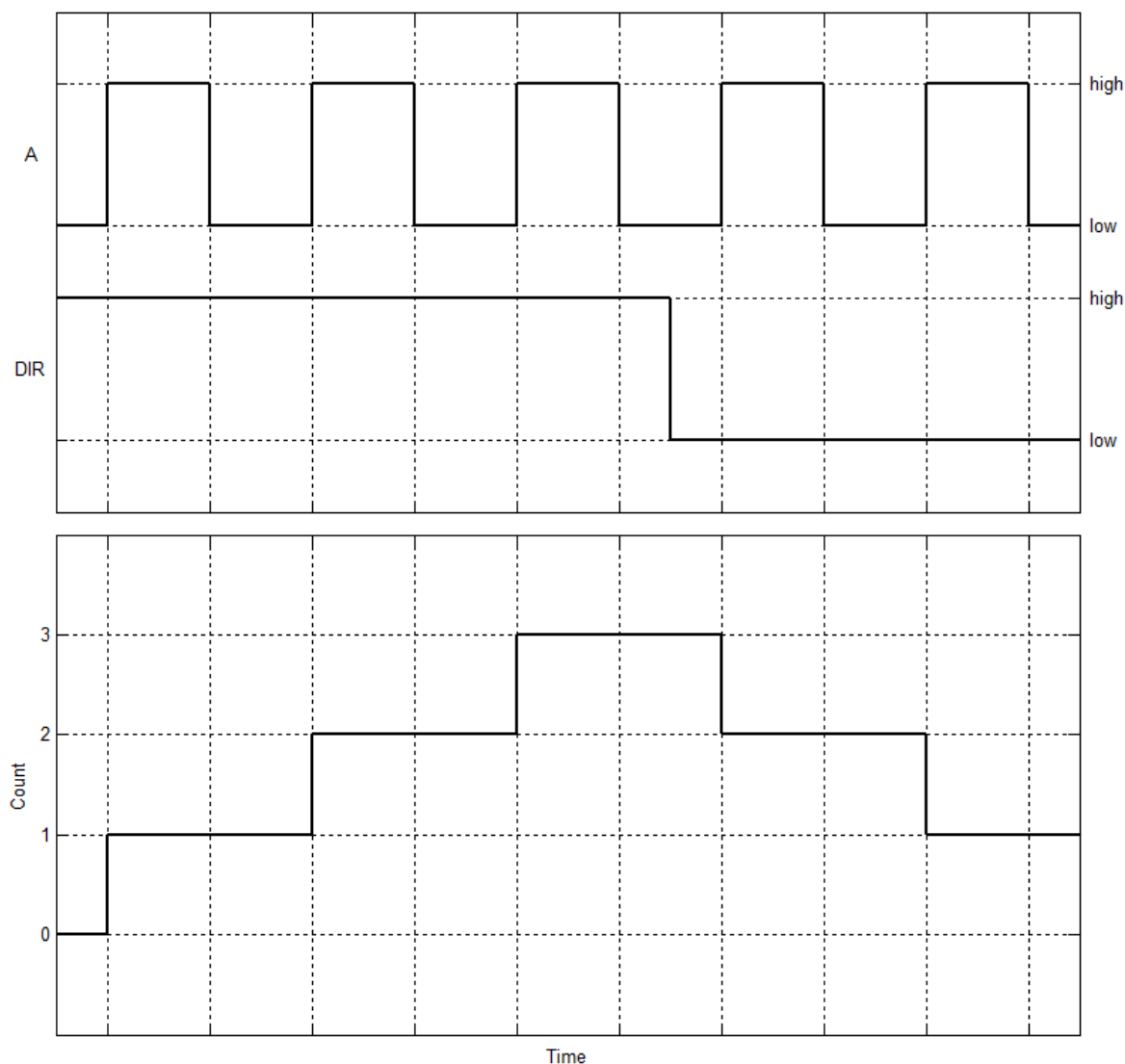


Figure 6-8 - Pulse + direction counting

6.2.2.8.3 General Operation



NOTE: The Pulse / Stepper Position Control feature cannot be used at the same time as the [Strobing](#) feature.

Pulse / Stepper Position Control mode can be used in the following way:

1. Configure the *External Pulse Count Control Word* for the desired input type and polarity.
2. Configure the *External Pulse Count Feed Rate*, the distance per count, for the application.
3. Configure the *External Pulse Count Relative Mode*, whether the position command generated from the pulse / stepper input is relative to the position the axis is enabled.
4. Configure the set point source to pulse commands. Refer to [6.2.2.2 Pre-assigning Set Point Sources](#).
5. Enable the axis with the active operation mode to Position Control with Motor Encoder or Position Control with External Encoder. The axis will now respond to Pulse / Stepper Position commands.



WARNING: Care must be taken when using absolute mode as unexpected jumps can occur upon enabling the motor depending on the starting position. The relative mode of operation is recommended for most applications.

6.2.2.8.4 External Pulse Count Control Word

The *External Pulse Count Control Word* is used to configure Pulse Stepper input.

Bit	Value	Definition
0-1	0	Input type – quadrature
	1	Input type – pulse + direction
	2-3	Reserved
1	0	Do not invert signal A
	1	Invert input signal A
2	0	Do not invert signal B / direction input
	1	Invert signal B / direction input

Table 6-10 - External Pulse Count Control Word Definition

6.2.2.8.5 Related IDNs

IDN	Description	Data Type	Units	Default
				AMD2000
P-0-0539 / 33307	External Pulse Count Relative Mode	Unsigned Integer (2 bytes)	Boolean	1
P-0-0569 / 33337	External Pulse Count Control Word	Unsigned Integer (2 bytes)	Binary	0000 0000
P-0-1452 / 34220	External Pulse Count Feed Rate	Signed Integer (4 bytes)	Standard Position	0.0005 mm/count
P-0-1453 / 34221	External Pulse Count	Signed Integer (4 bytes)	Count	N/A

6.2.3 Procedure Commands

6.2.3.1 Description

A control device / unit can write to Procedure Commands over SoE to initiate and control a procedure on the AMD servo drive. Procedure Commands can be used for a wide variety of functions in the drive, for example:

- Drive Controlled Homing
- Probing
- Saving to Non-Volatile Memory

Procedure Commands can be started, paused and stopped by the control device / unit. A list of all procedure commands can be read from the drive via *All Procedure Commands* (S-0-0025 / 25).

6.2.3.2 Procedure Command Control

The Procedure Command is controlled by writing command values to the IDN in the following format.

Bit	Value	Definition
0	0	Cancel Procedure Command
	1	Set Procedure Command
1	0	Pause Procedure Command execution
	1	Enable Procedure Command execution

Table 6-11 - Procedure Command control definition

6.2.3.3 Procedure Command Acknowledge

The status of the Procedure Command in the servo drive is represented by the Procedure Command Acknowledge in the following format.

Bit	Value	Definition
0	0	Procedure Command not set
	1	Procedure Command set
1	0	Procedure Command execution paused
	1	Procedure Command execution enabled
2	0	Procedure Command execution complete
	1	Procedure Command not yet executed
3	0	No Procedure Command error
	1	Procedure Command error. Execution halted and cannot continue.

Table 6-12 - Procedure Command Acknowledge definition

6.2.3.4 Procedure Command Execution

The Procedure Command execution process is presetned in [Figure 6-9](#) below; CC = Procedure Command Control and CA = Procedure Command Acknowledge.

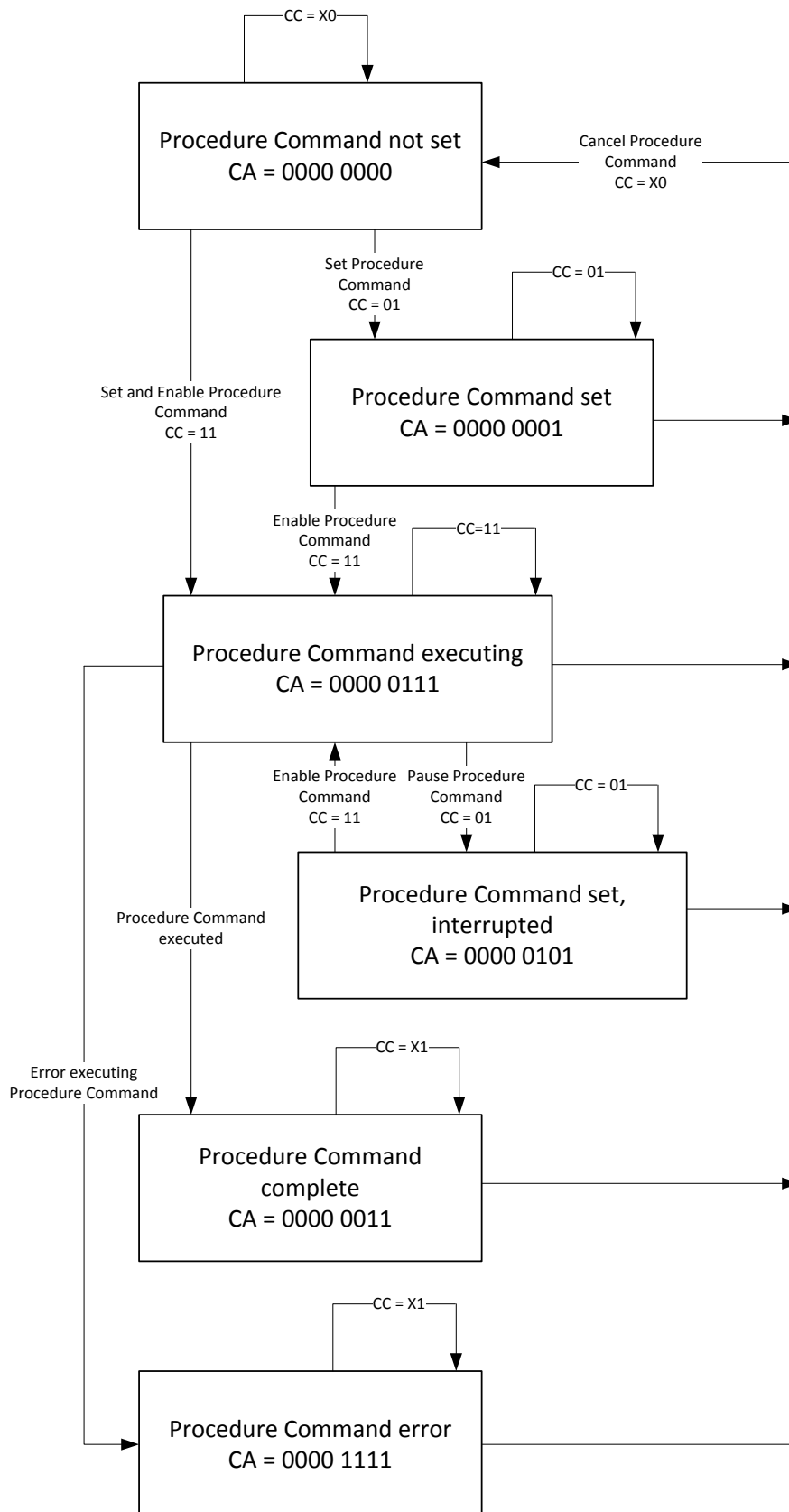


Figure 6-9 - Procedure Command Execution

6.2.3.5 Related IDNs

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0025 / 25	All Procedure Commands (Variable Length Array)	Unsigned Integer (2 bytes)	IDN	N/A	N/A

6.2.4 Homing

Homing is the process of referencing the incremental position feedback of the motor to a known location. Each time the servo drive is powered on, the position feedback will start at zero and count up or down as the motor moves depending on the direction moved. To ensure that a position on a machine is the same every time the servo is powered on, a homing process is performed to reference the servo drive to a known position or 'home' position. This is often referred to as 'zeroing' as the home position is often referenced as the zero position, however, this need not be the case and the location of the home position can be configured on the servo drive to any arbitrary position value.



WARNING: Care must be exercised so that all homing related IDN's are suitably configured whenever hardware (e.g. motor or encoder) changes are made, since unexpected behaviour may result.

6.2.4.1 Control Unit Controlled Homing

6.2.4.1.1 Description

Control Unit Controlled Homing (CUCH) is a method of homing where the control device/unit that would normally control axis motion via *Numerical Control* also controls the axis motion during the homing routine.

6.2.4.1.2 General Operation

Operation of Control Unit Controlled Homing is as follows:

1. The *CUCH Procedure Command (S-0-0146 / 146)* is set to 3 to begin the procedure command.
2. CUCH will commence when *Control Unit Controlled Homing Enable (S-0-0407 / 407)* is set to 1. This is generally done via *Real Time Control Bits* in the *Master Control Word*.
3. The control unit moves the axis to the home switch (if applicable). The home switch will be encountered and registered by the control unit or servo drive depending on the configuration.
4. The control unit then moves the axis to the marker / index pulse. The servo drive signals that the pulse has been found via *Control Unit Controlled Homing Reference Marker Pulse Registered* and latches the position. This is usually communicated via *Real Time Status Bits* in the *Drive Status Word*.
5. The *CUCH Procedure Command (S-0-0146 / 146)* is set to 0 to cancel the procedure command.
6. Calculation of the new servo drive position displacement is done by the control unit or by the servo drive depending on configuration. Refer to [6.2.4.2 Calculating Displacement](#).
7. Both the control unit and servo drive update their position frames of reference based on the calculated displacement. Refer to [6.2.4.3 Updating Position Reference Frame](#).

There are three configurable methods that the control unit may select from when using CUCH:

1. Control unit detection of home switch and initiation of drive marker / index pulse latching
2. Control unit detection of home switch via servo drive and initiation of servo drive marker / index pulse latching
3. Servo drive detection of home switch and initiation of servo drive marker / index pulse latching

6.2.4.1.2.1 Case 1

The control unit is connected to the home switch and determines when to look for drive marker / index pulse. The servo drive only latches the marker / index pulse location.

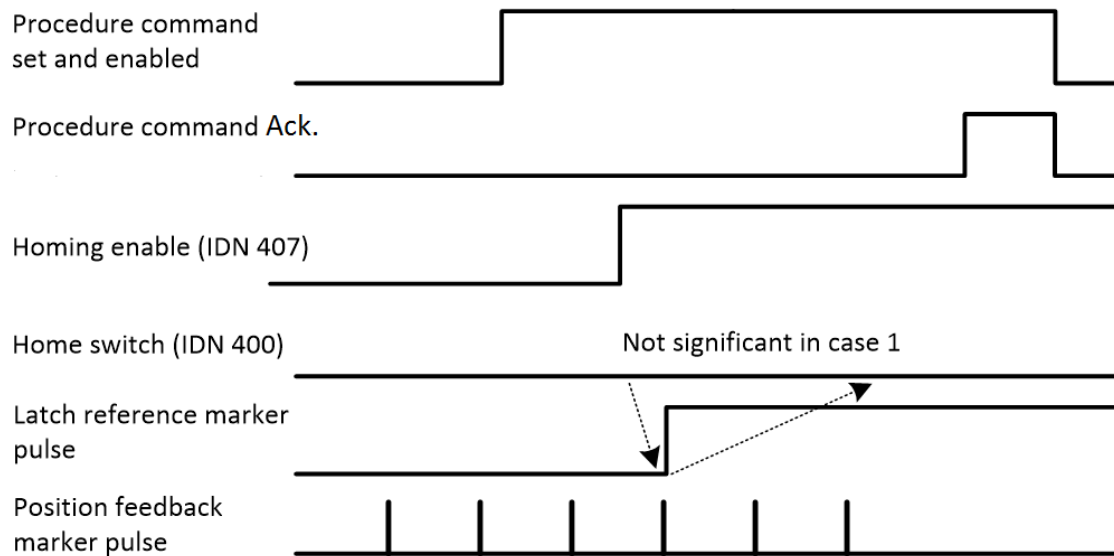


Figure 6-10: CUCH Case 1 timing diagram

6.2.4.1.2.2 Case 2

The servo drive is connected to the home switch and latches the marker / index pulse location. The control unit monitors the *Home Switch (S-0-0400 / 400)* via *Real Time Status Bits* from the drive and determines when to look for drive marker / index pulse.

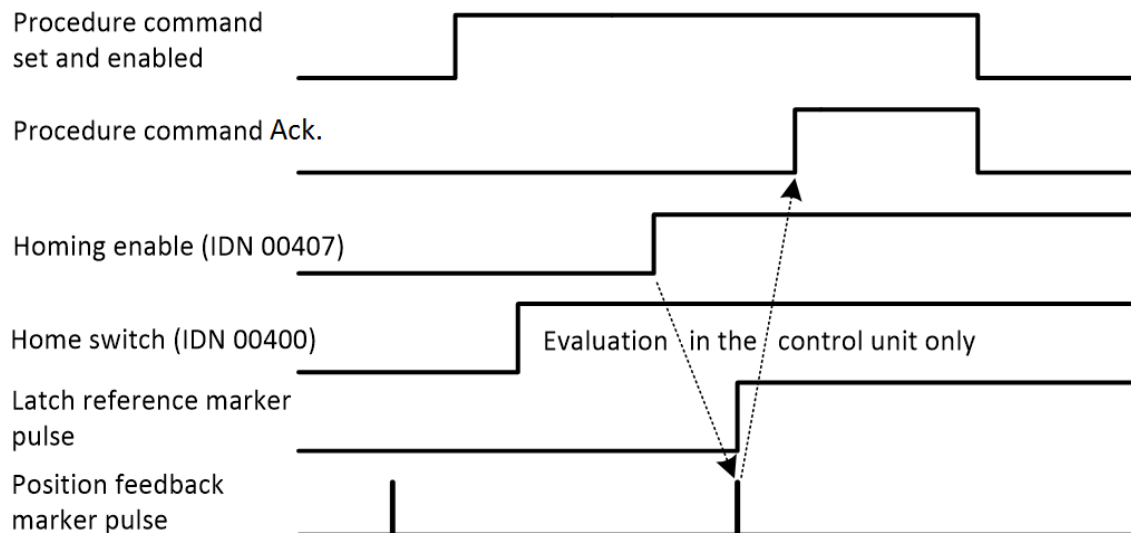


Figure 6-11: CUCH Case 2 timing diagram

6.2.4.1.2.3 Case 3

The servo drive monitors the home switch and latches the marker / index pulse location.

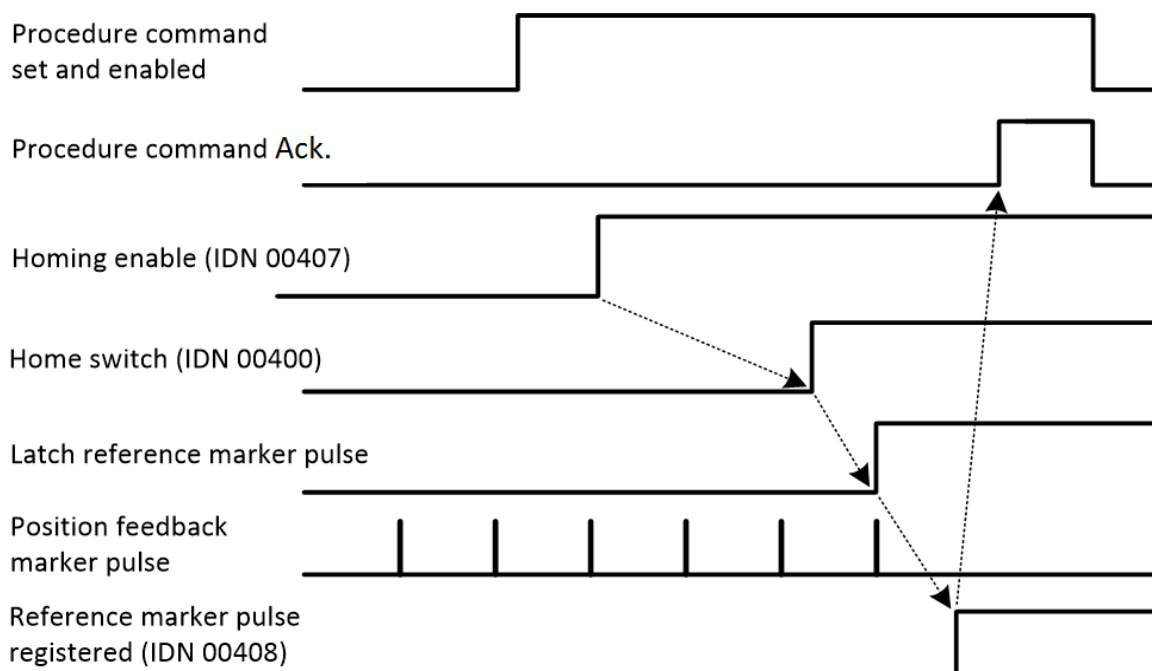


Figure 6-12: CUCH Case 3 timing diagram

Once the *Control Unit Controlled Homing Procedure Command* is complete, the control unit must calculate the new position reference frame displacement and update the reference frame. Refer to [6.2.4.2 Calculating Displacement](#) and [6.2.4.3 Updating Position Reference Frame](#) for more information.

Refer to [6.2.4.5 Homing Configuration](#) for more detail on how to configure CUCH behaviour.

6.2.4.1.3 Related IDNs

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0146 / 146	Control Unit Controlled Homing Procedure Command	Unsigned Integer (2 bytes)	Procedure Command	0	0
S-0-0147 / 147	Homing Parameter	Unsigned Integer (2 bytes)	Binary	0100 0100	0100 0100
S-0-0400 / 400	Home Switch	Unsigned Integer (2 bytes)	Boolean	N/A	N/A
S-0-0407 / 407	Control Unit Controlled Homing Enable	Unsigned Integer (2 bytes)	Boolean	0	0
S-0-0408 / 408	Control Unit Controlled Homing Reference Marker Pulse Registered	Unsigned Integer (2 bytes)	Boolean	N/A	N/A
P-0-0422 / 33190	Use a Digital Input in Place of an Index Pulse for Homing	Unsigned Integer (2 bytes)	Boolean	0	0
P-0-0432 / 33200	Extended Homing Parameter	Unsigned Integer(2 bytes)	Binary	0	0

6.2.4.2 Calculating Displacement

6.2.4.2.1 Description

Displacement between the initial drive zero point and the machine zero point needs to be calculated once the CUCH process is complete. This will then be applied to the drive to update it to the new position frame of reference. This calculation can be done by the servo drive or control unit.



NOTE: In cases where there are 2 versions of the same parameter e.g. Reference Offset 1 and Reference Offset 2, '1' refers to a motor encoder and '2' refers to an external encoder.

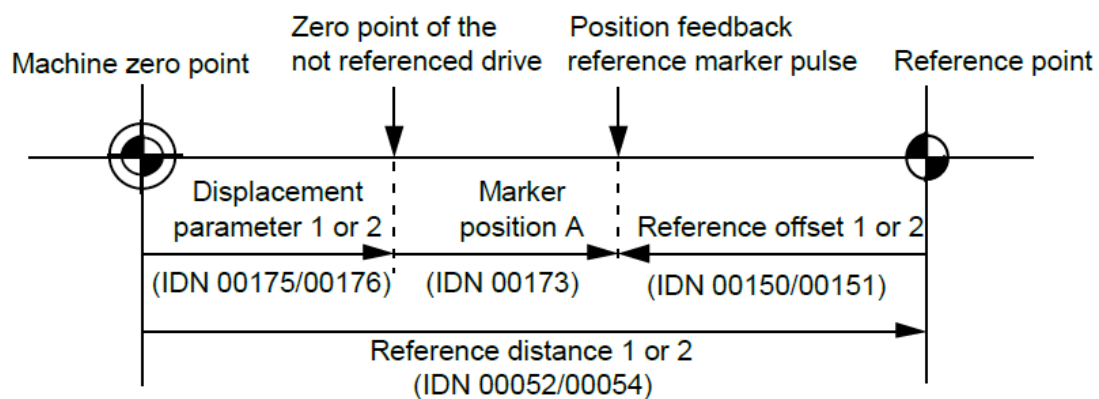


Figure 6-13 - Relationship between referenced and non-referenced positions

6.2.4.2.2 Drive Calculation

The servo drive will calculate the displacement upon the control unit running the *Calculate Displacement Procedure Command* (S-0-0171 / 171).

1. The control unit starts the procedure by setting Calculate Displacement Procedure Command (S-0-0171 / 171) to 3.
2. The servo drive calculates the Displacement Parameter using the configurable Reference Distance and Reference Offset with the CUCH latched Marker Position A in the following manner:
3. $\text{Displacement Parameter} = \text{Reference Distance} + \text{Reference Offset} - \text{Marker Position A}$
4. The drive reports the procedure command has completed successfully.
5. The control unit reads Displacement Parameter 1 (S-0-0175 / 175) or Displacement Parameter 2 (S-0-0176 / 176) to allow updating of the current command position value to the newly updated position reference frame.
6. The control unit cancels the procedure by setting Calculate Displacement Procedure Command (S-0-0171 / 171) to 0.

6.2.4.2.3 Control Unit Calculation

The control unit will calculate displacement by:

1. Marker Position A (S-0-0173 / 173) is read from the servo drive.
2. The control unit calculates the displacement utilising the control unit's configured distance of reference point to machine zero and with the CUCH latched Marker Position A in the following manner:

$$\text{Displacement Parameter} = \text{Reference Point to Machine Zero} - \text{Marker Position A}$$
3. The control unit writes to Displacement Parameter 1 (S-0-0175 / 175) or Displacement Parameter 2 (S-0-0176 / 176).

6.2.4.2.4 Related IDNs

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0052 / 52	Reference Distance 1	Signed Integer (4 bytes)	Standard Position	0	0
S-0-0054 / 54	Reference Distance 2	Signed Integer (4 bytes)	Standard Position	0	0
S-0-0150 / 150	Reference Offset 1	Signed Integer (4 bytes)	Standard Position	0	0
S-0-0151 / 151	Reference Offset 2	Signed Integer (4 bytes)	Standard Position	0	0
S-0-0171 / 171	Calculate Displacement Procedure Command	Unsigned Integer (2 bytes)	Procedure Command	0	0
S-0-0173 / 173	Marker Position A	Signed Integer (4 bytes)	Standard Position	N/A	N/A
S-0-0175 / 175	Displacement Parameter 1	Signed Integer (4 bytes)	Standard Position	0	0
S-0-0176 / 176	Displacement Parameter 2	Signed Integer (4 bytes)	Standard Position	0	0

6.2.4.3 Updating Position Reference Frame

6.2.4.3.1 Description

The position frame of reference in both the control unit supplied command and the servo drive supplied feedback must be updated in a controlled manner after homing to prevent undesired motor movement.

6.2.4.3.2 General Operation



NOTE: The reference feedback system will still change coordinate system even if the procedure command is cancelled before the position command has changed coordinate systems.

1. The process is started by setting *Set Reference Point Procedure Command* to 3.
2. In response, the servo drive will simultaneously update the position feedback data with *Displacement Parameter 1* or *Displacement Parameter 2* and set *Position Feedback Value Status* that the control unit can read via the [Real Time Status Bits](#).
3. Simultaneously, *Position Command Value Status* is set to 1 via [Real Time Control Bits](#) and the Position Command is updated with *Displacement Parameter 1* or *Displacement Parameter 2*. The drive will now process the position commands in the new position reference frame.
4. The servo drive will then signal the procedure command as successfully complete.
5. The control unit will set *Set Reference Point Procedure Command* to 0.

Use of *Reference Point Cancel Procedure Command* to remove the reference point offset is similar to setting it, however, it will not wait for the *Position Command Value Status* to be updated before signalling the procedure command has completed successfully.

6.2.4.3.3 Related IDNs

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0172 / 172	Set Reference Point Procedure Command	Unsigned Integer (2 bytes)	Procedure Command	0	0
S-0-0191 / 191	Reference Point Cancel Procedure Command	Unsigned Integer (2 bytes)	Procedure Command	0	0
S-0-0403 / 403	Position Feedback Value Status	Unsigned Integer (2 bytes)	Binary	N/A	N/A

S-0-0404 / 404	Position Command Value Status	Unsigned Integer (2 bytes)	Binary	0	0
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6.2.4.4 Drive Controlled Homing

6.2.4.4.1 Description

Drive Controlled Homing (DCH) is a method of homing where the servo drive controls axis motion. This can be utilized by control device/units to simplify their homing logic.

The DCH procedure command includes the *Calculating Displacement* function and the feedback component of *Updating Position Reference Frame*. Once complete, if a control device unit is utilized, it only needs to update the command component of *Updating Position Reference Frame*.

To execute DCH, the following conditions must be met:

- Position feedback is connected to the drive via one of the encoder inputs.
- The home switch (if used) is connected directly to the drive via one of its digital inputs.

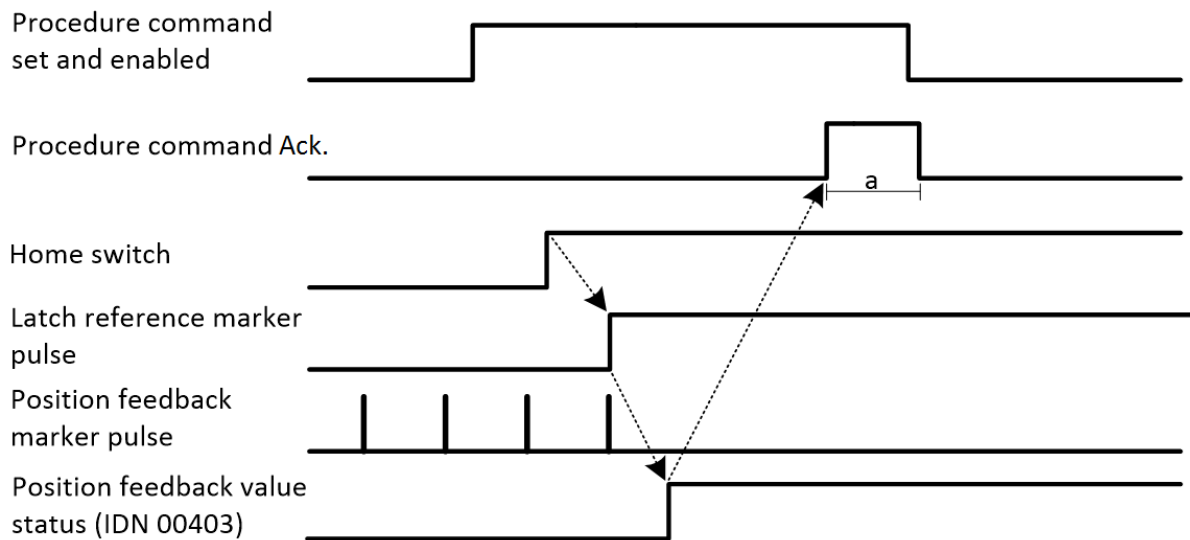


Figure 6-14 - Drive Controlled Homing Bit Sequence

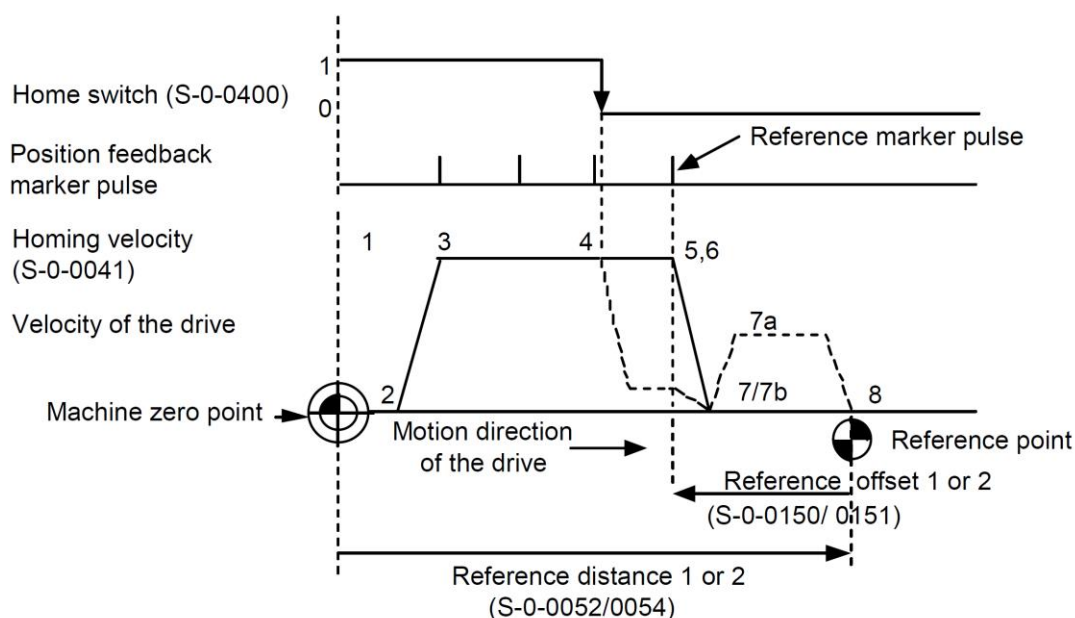


Figure 6-15 - Drive Controlled Homing

6.2.4.4.2 General Operation

Operation of Drive Controlled Homing is as follows:

1. The *Drive Controlled Homing Procedure Command* (S-0-0148 / 148) is set to 3 to begin the procedure command.
2. If the home switch is to be evaluated
 - a. If the drive detects it is already on the home switch, the axis will move opposite from the home switch direction at the configured *Maximum Homing Speed* and *Homing De-/Acceleration* for the configured *Pre-Home Switch Runoff Distance*. It is recommended that the distance be equal to the length of the home switch.
 - b. The drive will approach the home switch in the home switch direction at the configured *Home Switch Speed* and *Homing De-/Acceleration*.
 - c. Once the home switch is detected, the axis will move a *Post Home Switch Run Off Distance* at the configured *Maximum Homing Speed* and *Homing De-/Acceleration*. The direction of this move depends on the polarity of the *Post Home Switch Run Off Distance*.
3. If the marker / index pulse is to be evaluated
 - a. The axis will approach the marker pulse at the configured *Index Pulse Homing Speed* and *Homing De-/Acceleration*. The direction the marker / index pulse is approached from is determined by the polarity of the *Index Pulse Homing Speed*.
4. Once the final reference point has been evaluated, the position at this point is set using the *Reference Distance* and *Reference Offset* parameters as shown in Figure 6-15.
5. After successful completion of the DCH procedure command, the *Position Feedback Value Status* is set to 1 to indicate the position feedback reference frame has been updated.
6. If the control unit will control the servo drive in NC position mode, it will need to update its position command reference frame. Refer to 6.2.4.3 *Updating Position Reference Frame*.
7. The *Drive Controlled Homing Procedure Command* (S-0-0148 / 148) is set to 0 to cancel the procedure command.

Refer to 6.2.4.5 *Homing Configuration* for more detail on how to configure DCH behaviour.



WARNING: Care must be exercised so that all homing related IDN's are suitably configured whenever hardware (e.g. motor or encoder) changes are made, since unexpected behaviour may result.

6.2.4.4.3 Related IDNs

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0041 / 41	Homing Speed to Locate Home Switch	Signed Integer (4 bytes)	Standard Velocity	7.2 mm/min	7.2 mm/min
S-0-0042 / 42	Homing Max Acceleration	Signed Integer (4 bytes)	Standard Acceleration	0.64 m/s/s	0.64 m/s/s
S-0-0052 / 52	Reference Distance 1	Signed Integer (4 bytes)	Standard Position	0	0
S-0-0054 / 54	Reference Distance 2	Signed Integer (4 bytes)	Standard Position	0	0
S-0-0147 / 147	<i>Homing Parameter</i>	Unsigned Integer (2 bytes)	Binary	0100 0100	0100 0100
S-0-0148 / 148	Drive Controlled Homing Procedure Command	Unsigned Integer (2 bytes)	Procedure Command	0	0
S-0-0150 / 150	Reference Offset 1	Signed Integer (4 bytes)	Standard Position	0	0
S-0-0151 / 151	Reference Offset 2	Signed Integer (4 bytes)	Standard Position	0	0
S-0-0403 / 403	Position Feedback Value Status	Unsigned Integer (2 bytes)	Binary	N/A	N/A
S-0-0404 / 404	Position Command Value Status	Unsigned Integer (2 bytes)	Binary	0	0
P-0-0432 / 33200	<i>Extended Homing Parameter</i>	Unsigned Integer (2 bytes)	Binary	0	0
P-0-0433 / 33201	Home Switch from Master	Unsigned Integer (2 bytes)	Boolean	0	0
P-0-0437 / 33205	Homing Velocity to Locate Index Pulse	Signed Integer (4 bytes)	Standard Velocity	-7.2 mm/min	-7.2 mm/min
P-0-0438 / 33206	Pre Home Switch Run Off Distance	Signed Integer (4 bytes)	Standard Position	20 mm	20 mm
P-0-0443 / 33211	Post Home Switch Run Off Distance	Signed Integer (4 bytes)	Standard Position	0 mm	-1 mm
P-0-0444 / 33212	Home Switch Input	Unsigned Integer (2 bytes)	Boolean	0	0
P-0-0446 / 33214	Homing Velocity Max	Signed Integer (4 bytes)	Standard Velocity	120 mm/min	120 mm/min

6.2.4.5 Homing Configuration

6.2.4.5.1 Homing Reference Sources

6.2.4.5.1.1 Marker / Index Pulse

The majority of motor and external incremental encoders contain a marker or index pulse. On a rotary encoder this is generally a once per revolution signal to uniquely identify the position of the motor in the mechanical revolution. On a linear encoder there can be one or many marker / index pulses. In the case of many marker / index pulses, combination with a home switch or stall detection is required to uniquely identify a single home marker / index pulse.

As a marker / index pulse is built into an encoder, it is considered the most accurate reference point available for homing.

Refer to [6.2.4.5.2 Homing Parameter](#) for information on configuring homing with a marker / index pulse.

6.2.4.5.1.2 Home Switch

A home switch refers to a digital input that signifies a known axis location has been reached on the machine. On a linear axis this is generally placed at one end of the axis so that it always known in which direction to home.

The use of a digital input is not as accurate as a marker/index pulse so during homing, once a home switch has been found, the index is searched for at a slower velocity for an accurate position reference. The exception to this is when there may be slip between the encoder and the load where the home switch is located; in this case the index pulse of the encoder will not provide repeatable referencing.

The home switch can be connected to the servo drive or control unit. If the home switch is connected to the servo drive, the digital input can be mapped to the *Home Switch Input (P-0-0444 / 33212)* via [Digital Input to IDN Mapping](#). *Home Switch From Master (P-0-0433 / 33201)* is used in lieu of the drive connected home switch if a control unit connected home switch is selected in the [Homing Parameter](#).

Refer to [6.2.4.5.2 Homing Parameter](#) for information on configuring homing with a marker / index pulse.

6.2.4.5.1.3 Stall Detection

The AMD servo drive's homing procedures can be configured to detect an axis stall so that a proxy home switch is not required. This can be done by configuring the [Homing Parameter](#) to evaluate the home switch and set bit 0 in the [Extended Homing Parameter](#).

A stall is detected, resulting in 'home switch' detection, when the axis velocity is below the *stall detection velocity threshold* and the detected force (current with the default torque constant of 1) is greater than the *stall detection force threshold*. When using DCH, the *maximum homing force* can also be configured to prevent damage to the axis during stall detection homing.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-0440 / 33208	Homing Stall Detection Velocity Threshold	Signed Integer (4 bytes)	Standard Velocity	1.44 mm/min	1.44 mm/min
P-0-0441 / 33209	Homing Stall Detection Torque Threshold	Signed Integer (4 bytes)	Standard Torque 32bit	2 N	2 N
P-0-0445 / 33213	DCH Maximum Homing Torque	Signed Integer (4 bytes)	Standard Torque 32bit	4 N	4 N

6.2.4.5.2 Homing Parameter

The SoE standard Homing Parameter allows configuration of homing procedure behaviour. Some bits in the parameter may only affect DCH or CUCH behaviour.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0147 / 147	Homing Parameter	Unsigned Integer (2 bytes)	Binary	0100 0100	0100 0100

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RES									EMP	EHS	ID	HFS	HS	HSP	HD
MSB									LSB						

Figure 6-16 - Extended Homing Parameter definition

Bit	Key	Definition	Value	Description
0	HD	Homing Direction (DCH only)	0	Home in a positive direction
			1	Home in a negative direction

1	HSP	Home Switch Polarity	0	Home switch is active high
			1	Home switch is active low
2	HS	Home Switch (DCH only)	0	Connected to control unit
			1	Connected to drive
3	HFS	Homing Feedback Source	0	Use motor feedback
			1	Use external feedback
4	ID	Interpretation in the Drive	0	Home switch and homing enable
			1	Homing enable only
5	EHS	Evaluation of Home Switch	0	Home switch is evaluated
			1	Home switch is not evaluated
6	EMP	Evaluation of Marker Pulse	0	Marker pulse is evaluated
			1	Marker pulse is not evaluated
7-15	RES	Reserved	X	-

Table 6-13 - Homing Parameter bit definition

6.2.4.5.3 Extended Homing Parameter

The AMD specific Extended Homing Parameter allows for further homing procedure configuration beyond that defined in the SoE standard.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-0432 / 33200	Extended Homing Parameter	Unsigned Integer (2 bytes)	Binary	0	0



Figure 6-17 – Extended Homing Parameter definition

Bit	Key	Definition	Value	Description
0	SD	Stall Detection	0	Use digital input as home switch
			1	Use stall detection as home switch
1-15	RES	Reserved	X	-

Table 6-14 - Extended Homing Parameter bit definition

6.2.5 Drive Controlled Moves

6.2.5.1 Description

Drive Controlled Moves (DCM) allows for a series of up to 64 trapezoidal velocity profile moves to be pre-configured and the control device/unit to determine which move to run. Each move can be configured to automatically transition to another allowing for cyclic operation and more complex profiles. Alternatively, DCM allows for complex profiles when running in standalone mode, independent from a control unit.

6.2.5.2 General Operation

Operation of DCM occurs as described below.

1. DCM is initiated when the *DCM Procedure Command* (P-0-0256 / 33024) is set to 3. The move configured in *DCM First Move Index* will be started.
2. The axis will then follow the selected move's defined *acceleration*, *target velocity* and *deceleration* to its target position.
3. Behaviour at the termination of the move depends on if there is a *Next Move Index* defined for the move and the *Move Type* for the move.
 - a. *DCM Moves Procedure Command* is paused (= 1) a move is active: The axis will come to halt according to *DCM Halt Deceleration*.
 - b. *Next Move Index* = 0: The *DCM Procedure Command* is complete when the current move's target position is reached and the moves delay time has passed. The 'Target Reached' bit in the status word is set to 1. From here the procedure command can be resumed or cancelled.
 - c. *Next Move Index* is non-zero:
 - i. *Move Type* = 0: The current move will decelerate to a stop at its target position. The axis will wait the delay time set for the move before transitioning to the next move. Go to step 2.
 - ii. *Move Type* = 1: The current move will maintain the target velocity up to the target position. At this point it will transition to the next move. The delay time is disabled. Go to step 2.
 - iii. *Move Type* = 2: The current move will transition to the next move at the point that the move would begin to decelerate to the target position if the *Move Type* were set to 0. The delay time is disabled. Go to step 2.
4. The *DCM Procedure Command* is set to 0 to cancel the procedure command.



Note: If a drive controlled moves sequence forms a loop, the *DCM Cycle Counter* will increment every time the cycle is completed.

6.2.5.3 DCM Control Word

The *DCM Control Word* allows for additional configuration of DCM behaviour for all moves.

Bit	Value	Definition
0	0	Target positions are treated as absolute
	1	Target positions are treated as relative to where they start
1	0	Exact Mode: Deceleration may be adjusted during move to ensure a precise target position
	1	Trapezoidal Mode: Deceleration is constant

Table 6-15 - DCM Control Word definition



NOTE: Dwells are not supported when the Trapezoidal Mode option is selected.

6.2.5.4 DCM Range Detection

The Drive Controlled Moves function provides the ability to monitor if the axis is within an expected position range. Once *DCM Range Detection Limits* is configured with a minimum and maximum position, *DCM Range Detection State* can be used to monitor the state as shown in the table below.

Value	Definition
0	Axis not within range
1	Axis in range with a positive command velocity
2	Axis in range with a negative command velocity

Table 6-16 - DCM Range Detection State definition

6.2.5.5 Operation Example

The table below shows an example of programming five moves.

IDN	Description	Moves				
		1	2	3	4	5
P-0-0260 / 33028	DCM Target Position	100	200	300	-100	0
P-0-0261 / 33029	DCM Target Velocity	1000	2000	5000	10000	1000
P-0-0262 / 33030	DCM Acceleration	100	200	300	400	500
P-0-0263 / 33031	DCM Deceleration	500	400	300	200	100
P-0-0264 / 33032	DCM Delay Time	250	500	0	0	0
P-0-0265 / 33033	DCM Next Move	2	3	5	0	4

Table 6-17 - DCM Example

For this example the *DCM First Move Index* is set to 1. Setting the 'halt' bit in the control word to 0 results in the following motion:

1. Move to 100 mm at 1000 mm/min and pause for 1 second.
4. Move to 200 mm at 2000 mm/min and pause for 2 seconds.
5. Move to 300 mm at 5000 mm/min
6. Move to 0 mm at 1000 mm/min
7. Move to -100 mm at 10000mm/min
8. Stop

6.2.5.6 Related IDNs

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-0256 / 33024	DCM Procedure Command	Unsigned Integer (2 bytes)	Procedure Command	0	0

P-0-0260 / 33028	DCM Type (64 elements)	Unsigned Integer (2 bytes)	None	0,0...,0	0,0...,0
P-0-0260 / 33028	DCM Target Position (64 elements)	Signed Integer (4 bytes)	Standard Position	-16,16,13,21,0,0,...,0 mm	-16,16,13, 21,0,0,...,0 mm
P-0-0261 / 33029	DCM Maximum Absolute Velocity (64 elements)	Signed Integer (4 bytes)	Standard Velocity	1200,1200,1200,600,600,0,0,...,0 mm/min	1200,1200,1200,600,600,0,0,...,0 mm/min
P-0-0262 / 33030	DCM Acceleration (64 elements)	Signed Integer (4 bytes)	Standard Acceleration	0.64,0.64,0.64,0.64,0.64,0,...,0 m/s/s	0.64,0.64,0.64,0.64,0.64,0,0,...,0 m/s/s
P-0-0263 / 33031	DCM Deceleration (64 elements)	Signed Integer (4 bytes)	Standard Acceleration	0.64,0.64,0.64,0.64,0.64,0,0,...,0 m/s/s	0.64,0.64,0.64,0.64,0.64,0,0,...,0 m/s/s
P-0-0264 / 33032	DCM Delay Time (64 elements)	Unsigned Integer (2 bytes)	4x10 ⁻³ seconds	0.4,0.4,0.4,0.4,0.4,0,0,...,0 seconds	0.4,0.4,0.4,0.4,0.4,0,0,...,0 seconds
P-0-0265 / 33033	DCM Next Move Index (64 elements)	Unsigned Integer (2 bytes)	None	2,3,4,5,1,0,0,...,0	2,3,4,5,1,0,0,...,0
P-0-0266 / 33034	DCM First Move Index	Unsigned Integer (2 bytes)	None	1	1
P-0-0267 / 33035	DCM Halt Deceleration	Signed Integer (4 bytes)	Standard Acceleration	0.16 m/s/s	0.16 m/s/s
P-0-0268 / 33036	<i>DCM Control Word</i>	Unsigned Integer (2 bytes)	Binary	0000 0010	0000 0010
P-0-0269 / 33037	DCM Range Detection Limits (2 elements)	Signed Integer (4 bytes)	Standard Position	10, 20 mm	10, 20 mm
P-0-0270 / 33038	DCM Range Detection State	Unsigned Integer (2 bytes)	None	N/A	N/A
P-0-0271 / 33039	DCM Cycle Counter	Unsigned Integer (4 bytes)	None	N/A	N/A

7 Advanced Configuration

7.1 What this Chapter Contains

The following sections illustrate additional feature configuration for the advanced user.

7.2 Analogue Inputs (General Purpose)

7.2.1 Description



NOTE: This feature is only available on the AMD2000 series servo drives.

The AMD2000 features two general purpose analogue inputs, refer to the [AMD2000 Series D21xx Servo Drive - User Guide](#) for information on how these inputs can be connected to an external device. The analogue inputs also feature configuration options to allow for calibration and customization of the voltage for particular applications. Figure 7-1 illustrates the signal flow for general purpose analogue inputs in AMD2000.

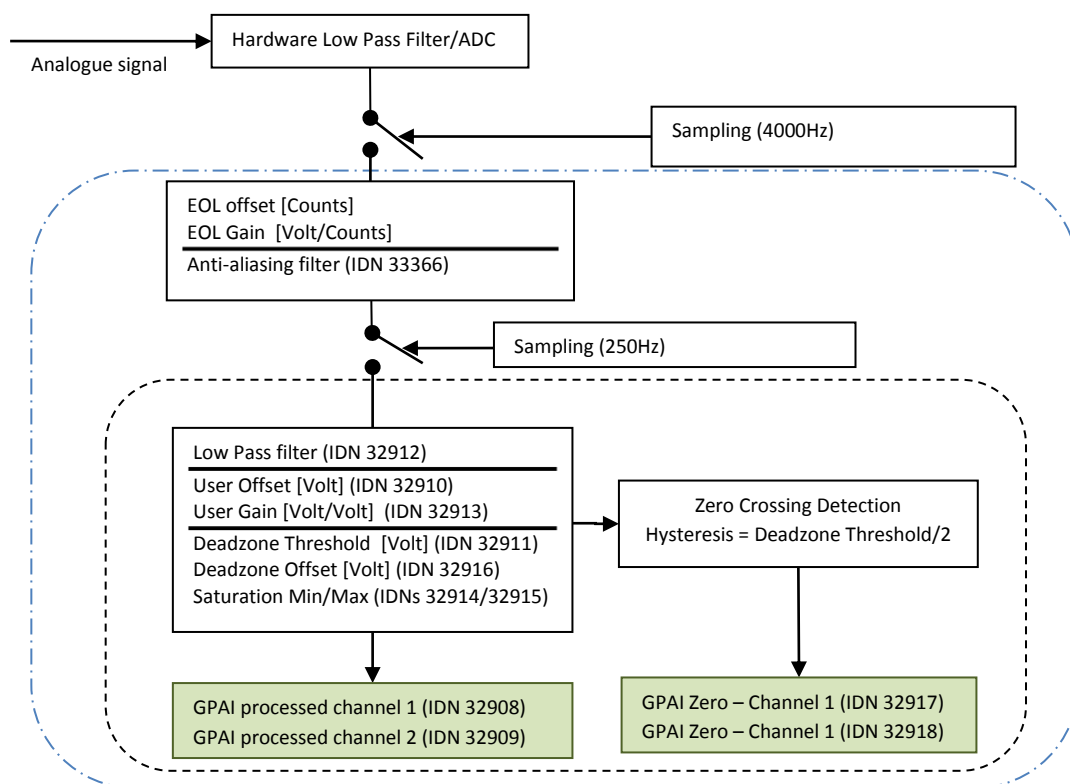


Figure 7-1 – General Purpose Analogue Inputs

First analogue signal is filtered and converted to digital value by hardware at sampling rate of 4000KHz. There are two other level of filtering at software level. One is applied for down sampling and the other one is a user defined filter. Filters implementation is simply a first order low pass filter as below

$$Y(k) = \text{filter coefficient} \times (U(k) - Y(k-1)) + Y(k-1)$$

End of production line calibration (EOL) is applied to generate signal value in volt. User can also define user defined offset and gain

$$Y_{\text{user-defined}} = (Y_{\text{EOL}} - \text{Analogue Input Offset}) \times \text{Analogue Input Gain}$$

Moreover, dead zone threshold, offset (refer to Figure 7-2) and saturation levels can be also applied to post process the signal.

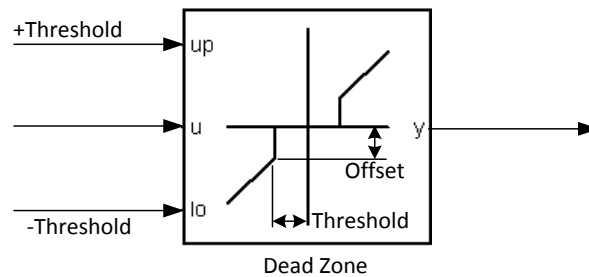


Figure 7-2 - Analogue Input Dead Zone

Related IDNs are given in the list below.

IDN	Description	Data Type	Units	Default
				AMD2000
P-0-0140 / 32908	Analogue Input 1	Signed Integer (4 bytes)	Standard Voltage	N/A
P-0-0141 / 32909	Analogue Input 2	Signed Integer (4 bytes)	Standard Voltage	N/A
P-0-0142 / 32910	Analogue Input Offset (2 elements)	Signed Integer (4 bytes) Array	Standard Voltage	0, 0
P-0-0143 / 32911	Analogue Input Dead Zone Threshold (2 elements)	Signed Integer (4 bytes) Array	Standard Voltage	0.5V, 0.5V
P-0-0144 / 32912	Analogue Input Filter Coefficient (2 elements)	Unsigned Integer (2 bytes) Array	%	25, 25
P-0-0145 / 32913	Analogue Input Gain (2 elements)	Signed Integer (4 bytes) Array	V/V	1, 1
P-0-0146 / 32914	Analogue Input Min (2 elements)	Signed Integer (4 bytes) Array	Standard Voltage	-15V, -15V
P-0-0147 / 32915	Analogue Input Max (2 elements)	Signed Integer (4 bytes) Array	Standard Voltage	15V, 15V
P-0-0148 / 32916	Analogue Input Dead Zone Offset (2 elements)	Signed Integer (4 bytes) Array	Standard Voltage	0.5V, 0.5V
P-0-0149 / 32917	Analogue Input Zero – Channel 1	Unsigned Integer (2 bytes)	Boolean	N/A
P-0-0150 / 32918	Analogue Input Zero – Channel 2	Unsigned Integer (2 bytes)	Boolean	N/A
P-0-0598 / 33366	Anti-aliasing Filter Coefficient (2 elements)	Unsigned Integer (2 bytes) Array	2^{-16}	25000, 25000

7.2.2 Definitions

7.2.2.1 Analogue Input 1

The measured voltage at drive analogue input 1 after the application of the offset, gain, filter, dead zone and saturation values. This value is intended to be fed in to an external device e.g. PLC.

7.2.2.2 Analogue Input 2

The measured voltage at drive analogue input 2 after the application of the offset, gain, filter, dead zone and saturation values. This value is intended to be fed in to an external device e.g. PLC.

7.2.2.3 Analogue Input Offset

Allows an offset to be applied to the measured voltage. This parameter, together with the *Analogue Input Gain*, allows for sensor commissioning. Each array index refers to an analogue input where the 1st item in the array applies to the 1st analogue input.

7.2.2.4 Analogue Input Dead Zone Threshold

Allows a dead zone to be applied to the measured voltage so that small voltage variations are still represented as 0V. Each array index refers to an analogue input where the 1st item in the array applies to the 1st analogue input.

7.2.2.5 Analogue Input Filter Coefficient

Alters the software filtering of the analogue input voltage. This defaults to the optimum value for the drive hardware and should therefore not require alteration for most applications. The lower this value the heavier the filtering. Each array index refers to an analogue input where the 1st item in the array applies to the 1st analogue input.

7.2.2.6 Analog Input Gain

Changes the scaling of *Analogue Input 1* and *Analogue Input 2* for a given voltage applied to the drive's inputs. This parameter, together with the *Analogue Input Offset*, allows for sensor commissioning. Each array index refers to an analogue input where the 1st item in the array applies to the 1st analogue input.

7.2.2.7 Analog Input Min

Allows a minimum saturation level to be applied to the measured voltage. Each array index refers to an analogue input where the 1st item in the array applies to the 1st analogue input.

7.2.2.8 Analog Input Max

Allows a maximum saturation level to be applied to the measured voltage. Each array index refers to an analogue input where the 1st item in the array applies to the 1st analogue input.

7.2.2.9 Analogue Input Dead Zone Offset

Allows an offset to be applied within dead zone threshold. The 'zero offset' dead zone (*Analogue Input Dead Zone Offset* = 0) modifies the analogue input reading to ensure it is continuous with no 'jump' once the dead zone threshold is exceeded. The 'offset' creates a discontinuity around the dead zone, however, there is no modification to the analogue input gradient.

7.2.2.10 Analogue Input Zero – Channel 1

A Boolean variable that is 1 when *Analogue Input 1* is equal to 0 i.e. in the dead zone.

7.2.2.11 Analogue Input Zero – Channel 2

A Boolean variable that is 1 when *Analogue Input 2* is equal to 0 i.e. in the dead zone.

7.2.2.12 Anti-aliasing Filter Coefficient

This defaults to the optimum value for the drive hardware and should therefore not require alteration for most applications. The lower this value the heavier the filtering. Each array index refers to an analogue input where the 1st item in the array applies to the 1st analogue input.

7.3 Analogue Outputs

7.3.1 Description



NOTE: This feature is only available on the AMD2000 series servo drives.

The AMD2000 features two general purpose analogue outputs, refer to [AMD2000 Series D21xx Servo Drive - User Guide](#) for information on how these outputs can be connected to an external device. The analogue outputs also feature configuration options to allow for calibration and customization of the voltage for particular applications. In all cases where there is an array to represent multiple analogue outputs in a single IDN, the first item in the array (index = 0) corresponds to analogue output 1.

End of production line calibration (EOL) is applied to generate output value in ADC counts. User can define user defined offset and gain to generate an output in milli-Volt.

$$Y_{user-defined} = \text{Analogue Output} \times \text{Analogue Output Gain} + \text{Analogue Output Offset}$$

The updated gain and offset will be effective as soon as analogue output is changed.

IDN	Description	Data Type	Units	Default
				AMD2000
P-0-151 / 32919	Analogue Output 1 Desired Voltage	Signed Integer (2 bytes)	mV	0
P-0-152 / 32920	Analogue Output 2 Desired Voltage	Signed Integer (2 bytes)	mV	0
P-0-153 / 32921	Analogue Output Voltage Offset (2 elements)	Signed Integer (2 bytes) Array	mV	0, 0
P-0-154 / 32922	Analogue Output Gain (2 elements)	Unsigned Integer (4 bytes) Array	2 ⁻¹⁴	16384, 16384
P-0-155 / 32923	Analogue Output Safe State Enable	Unsigned Integer (2 bytes)	Binary	0
P-0-156 / 32924	Analogue Output Safe State Value (2 elements)	Signed Integer (2 bytes)	mV	0, 0

7.3.2 Definitions

7.3.2.1 Analogue Output 1

The desired value of analogue output 1 that is to be applied to the connected external device.

7.3.2.2 Analogue Output 2

The desired value of analogue output 2 that is to be applied to the connected external device.

7.3.2.3 Analogue Output Voltage Offset

The offset applied to ensure the output signal matches the desired voltage. This parameter, together with the gain, allows for commissioning.

7.3.2.4 Analogue Output Gain

The gain to ensure the output signal matches the desired voltage. This parameter, together with the offset, allows for commissioning.

7.3.2.5 Analogue Output Safe State Enable

Each bit of *Analogue Output Safe State Enable* represents an analogue output with bit 0 (LSB) representing the first analogue output. If set to 1, when the servo drives EtherCAT state is not Operational the analogue output value will be equal to the voltage specified in *Analogue Output Safe State Value*. The drive will automatically enter EtherCAT state SAFEOP as a result of various abnormal conditions, for example, a loss of communication with the EtherCAT master, missing IPOSYNC, etc.

7.3.2.6 Analogue Output Safe State Value

For each analogue output, if the corresponding *Analogue Output Safe State Enable* bit is set to 1; when the servo drives EtherCAT state is not Operational the analogue output voltage will be equal to this value. The drive will automatically enter EtherCAT state SAFEOP as a result of various abnormal conditions, for example, a loss of communication with the EtherCAT master, missing IPOSYNC, etc.

7.4 Backlash Compensation

7.4.1 Description

The 'end effector' or 'working position' of a machine may have any number of backlash or tolerance stacks located in the drivetrain between itself and the driving torque of a motor (e.g. gaps in mating gear teeth, splines etc.). Consequently the position of the end effector is not always directly proportional to motor position. This is particularly true when the drive train is reversing direction and consequently has to traverse this region of backlash before all components re-engage and the motor can drive the end effector in a positive fashion. High performance machines generally minimise backlash, but cannot always eliminate it, especially with wear and differing temperatures between components.

The AMD servo drive's backlash compensation feature allows the servo drive to estimate the end effector position with only motor position feedback based on user commissioned values. The compensation algorithm monitors the controller's internal velocity demand to determine when a reversal in direction of motion is likely to occur. It is generally not necessary to use backlash compensation when the end effector position is measured directly via an external encoder.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-058 / 58	Backlash Compensation Distance	Signed Integer (4 bytes)	Standard Position	0	0
P-0-847 / 33615	Backlash Compensation Minimum Speed	Signed Integer (4 bytes)	Standard Velocity	0	0
P-0-849 / 33617	Backlash Compensation Slew Limit	Unsigned Integer (2 bytes)	%/ms	0	0

7.4.2 Definitions

7.4.2.1 Backlash Compensation Distance

The *Backlash Compensation Distance* is the fixed value of expected backlash by which the motor position is adjusted to correct the end effector position. A value of 0 will disable Backlash Compensation. It is possible for this value to be either positive or negative with respect to its reference value of 0.0.

The backlash distance can vary between gearboxes of the same design, therefore, for the best performance; this value should be individually commissioned on each machine. Commissioning involves a series of steps in one direction followed by a step in the opposite direction. The difference in the commanded distance and that measured at the end effector is the backlash distance. Measurements of this type are usually made with a dial gauge.

7.4.2.2 Backlash Compensation Minimum Speed

The *Backlash Compensation Minimum Speed* sets the lowest value of speed above which compensation will be applied.

7.4.2.3 Backlash Compensation Slew Limit

The *Backlash Compensation Slew Limit* determines the rate at which the *Backlash Compensation Distance* is applied to the position feedback. The units represent the percentage of the *Backlash Compensation Distance* that will be applied per millisecond. For example, setting this value to 10%/ms will result in the full *Backlash Compensation Distance* being applied linearly over a 10ms period.

The *Backlash Compensation Slew Limit* is typically application specific and may or may not need to be modified from the default value. Typical operation speeds of the application will impact the requirement on this parameter.

7.4.3 Backlash Compensation and Homing

The direction of homing is important for the successful application of this compensation technique, as it relates directly to the correct sign (+/-) of the applied clearance value and will vary depending on the particular machine setup. When the machine moves in the direction in which it was homed, effectively no clearance offset is applied, as positions generated when moving in this direction are taken as the reference for backlash. When moving in the opposite direction, however, the full clearance value will be applied after appropriate accrual due to slew limits. [Figure 7-3](#) and [Figure 7-4](#) illustrate the algorithm's operation.

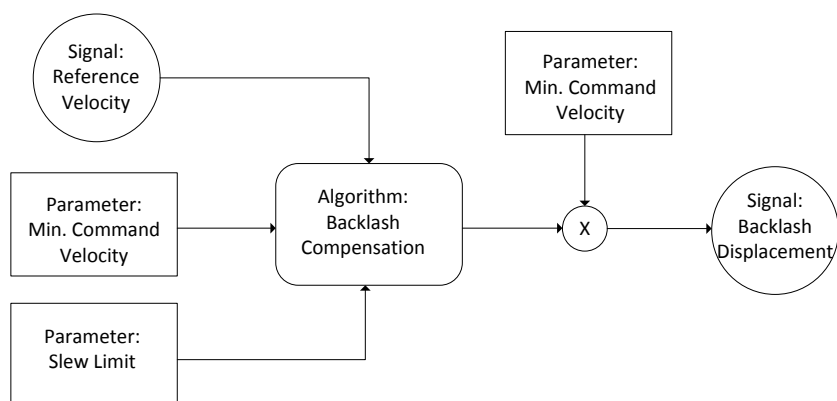


Figure 7-3 Backlash Compensation Data Flow

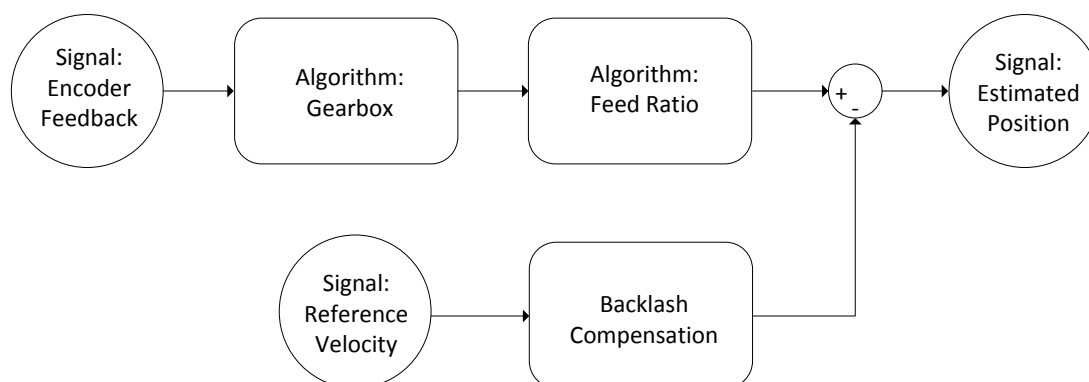


Figure 7-4 Illustrating where backlash compensation fits into the motor feedback path

7.5 Digital Input

7.5.1 Description



NOTE: This feature is only available on the AMD2000 series servo drives.

The AMD2000 comes with 8 general purpose digital inputs and 2 differential digital inputs. All 10 digital inputs are configured in the same way in the drive. Refer to [AMD2000 Series D21xx Servo Drive - User Guide](#) for information on how these inputs can be connected to an external device. Digital inputs can be directly read from the servo drive by the control device / unit or mapped into IDNs to command drive functions.

IDN	Description	Data Type	Units	Default
				AMD2000
P-0-0574/ 33342	Digital Input Polarity	Unsigned Integer (4 bytes)	Binary	0

P-0-0575 / 33343	Digital Input Data	Unsigned Integer (4 bytes)	Binary	N/A
P-0-0587 / 33355	Digital Input to IDN Mapping (32 elements)	Unsigned Integer (2 bytes) Array	IDN	0,0,0...,0
P-0-0588 / 33356	Digital Input Map Bitmask (32 elements)	Unsigned Integer (2 bytes) Array	Binary	65535,65535, ...,65535

7.5.2 Definitions

7.5.2.1 Digital Input Polarity

Digital Input Polarity can be used to invert each digital input. Each bit in the parameter corresponds to a digital input with the least significant bit representing the first digital input. The effective bits of the parameter are the number of digital inputs (i.e. maximum value is $2^{\text{number of digital inputs}} - 1$).

7.5.2.2 Digital Input Data

Each bit in *Digital Input Data* represents the state of a single digital input with the least significant bit representing the first digital input. For example, if the 3rd digital input was 'on', the *Digital Input Data* would read '0000 0100' which equal '4' in decimal. This data can then be used by the master as an input into another system e.g. PLC.

7.5.2.3 Digital Input to IDN Mapping



NOTE: Only IDNs of data type 'Unsigned Integer (2 bytes)' are supported by this feature.

This array can be used in conjunction with *Digital Input Map Bitmask* to allow a digital input to set an IDN in the servo drive. Each index in the array corresponds to a digital input with the 1st element (index = 0) in the array corresponding the first digital input. To map an IDN, set the contents of *Digital Input to IDN Mapping* at the index corresponding to the desired IDN. The feature is disabled if the IDN is set to 0.

7.5.2.4 Digital Input Map Bitmask

This array can be used in conjunction with *Digital Input to IDN Mapping* to use a digital input to set an IDN in the servo drive. Each index in the array corresponds to a digital input with the 1st element (index = 0) in the array corresponding the first digital input. The *Digital Input Map Bitmask* allows configuration of what bits in the mapped IDN are to be set by the digital input. For example, setting *Digital Input Map Bitmask* index 1 to 0000 0010 will cause the 2nd digital input to set the 2nd bit of the mapped IDN.

7.5.3 General Operation

Basic usage of this feature involves the control unit / device reading the Digital Input Data value and using this information as an input to another system such as a PLC.

For more advanced usage, the digital input can be used to set a parameter value in the drive itself without the control unit / devices involvement. To use this feature, enter the IDN to be set in the array index corresponding to the desired digital input and then set the value of *Digital Input Map Bitmask* in the same array index to the desired value to be set. For example, setting the 1st element of the *Digital Input to IDN Mapping* to 33024 and the 1st element of the *Digital Input Map Bitmask* to 0000 00011 would cause the 1st digital input to run the DCM procedure command and start Drive Controlled Moves.

7.6 Digital Output

7.6.1 Description



NOTE: This feature is only available on the AMD2000 series servo drives.

The AMD2000 comes with 6 digital outputs. These provide a powerful and flexible means of conveying information or making requests for actions external to the drive. They are highly configurable, being able to provide useful information during normal operation, as well as under those conditions where the servo drive, as an EtherCAT device, is no longer in its operating (OP) state. The user is able to manipulate up to 7 different IDN to configure the desired setting of the digital outputs, for both OP and Non-OP states of operation.

Refer to [AMD2000 Series D21xx Servo Drive - User Guide](#) for information on how these outputs can be connected to an external device.

IDN	Description	Data Type	Units	Default
				AMD2000
P-0-0576 / 33344	Digital Output Polarity	Unsigned Integer (4 bytes)	Binary	0
P-0-0577 / 33345	Digital Output Data	Unsigned Integer (4 bytes)	Binary	N/A
P-0-0582 / 33350	Digital Output Source IDN List (32 elements)	Unsigned Integer (2 bytes) Array	IDN	0,0,0,...,0
P-0-0583 / 33351	Digital Output Invert Mask (32 elements)	Unsigned Integer (2 bytes) Array	Binary	0,0,0,...,0
P-0-0584 / 33352	Digital Output Source Bitmask (32 elements)	Unsigned Integer (2 bytes) Array	Binary	65535,65535, ...,65535
P-0-0585 / 33353	Digital Output User Configurable Default Safe State – General Purpose (32 elements)	Unsigned Integer (2 bytes) Array	Binary	0,0,0,...,0
P-0-0586 / 33354	Digital Output User Configurable Default Safe State – Hardware Level	Unsigned Integer (4 bytes)	Binary	0

7.6.2 Definitions

7.6.2.1 Digital Output Polarity

The *Digital Output Polarity* can be used to invert each digital output. Each bit in the parameter corresponds to a digital output with the least significant bit representing the first digital output. The effective bits of the parameter are the number of digital outputs (i.e. maximum value is $2^{\text{number of digital outputs}} - 1$).

7.6.2.2 Digital Output Data

Each bit in *Digital Output Data* represents the state of a single digital output with the least significant bit representing the first digital output. For example, if *Digital Output Data* was set to '0000 1000' ('8' in decimal), digital output 4 would be 'on'. These outputs can be controlled by a control device / unit for tasks unrelated to the servo drive axis operations e.g. PLC logic.

7.6.2.3 Digital Output Source IDN List



NOTE: Only IDNs of data type 'Unsigned Integer (2 bytes)' are supported by this feature.

This array can be used in conjunction with *Digital Output Source Bitmask* and *Digital Output Invert Mask* to allow an IDN to set a digital output of the servo drive. Each index in the array corresponds to a digital output with the 1st element (index = 0) in the array corresponding the first digital output.

To map an IDN, set the contents of *Digital Output Source IDN List* at the index corresponding to the desired IDN. The feature is disabled if the IDN is set to 0.



NOTE: Any digital output with a non-zero IDN assigned to it via *Digital Output Source IDN List* will not be controllable by *Digital Output Data*.

7.6.2.4 Digital Output Invert Mask

This array can be used in conjunction with *Digital Output Source IDN List* and *Digital Output Source Bitmask* to allow an IDN to set a digital output of the servo drive. It also applies to *Digital Output User Configurable Default Safe State – General Purpose*. Each index in the array corresponds to a digital output with the 1st element (index = 0) in the array corresponding the first digital output.

The value of each entry into the *Digital Output Invert Mask* operates as a bit field with each bit inverting a bit in the value of the source IDN. For example, if a mapped IDN has the value of 7 (0...00000111), with a corresponding *Digital Output Invert Mask* value of 0...00000100, then the value that is passed to the bitmask function and then the digital output is 3 (0...00000011).

7.6.2.5 Digital Output Source Bitmask

This array can be used in conjunction with *Digital Output Source IDN List* and *Digital Output Invert Mask* to allow an IDN to set a digital output of the servo drive. It also applies to *Digital Output User Configurable Default Safe State – General Purpose*. Each index in the array corresponds to a digital output with the 1st element (index = 0) in the array corresponding the first digital output.

The value of each entry into the *Digital Output Source Bitmask* operates as a bit mask to isolate the bits of an IDN value that will set the digital output. As long as the output of this function is non-zero (at least one bit = 1) then the digital output will be set to on. For example, if a mapped IDN has the value of 7 (0...00000111), with a corresponding *Digital Output Source Bitmask* value of 0...00000011, then the final value is 3. This value is greater than 0 so the associated digital output will be set to on.

7.6.2.6 Digital Output User Configurable Default Safe State – General Purpose

This array will replace *Digital Output Source IDN List* when the servo drive is not in the EtherCAT OP state. The effects of *Digital Output Source Bitmask* and *Digital Output Invert Mask* still apply.

7.6.2.7 Digital Output User Configurable Default Safe State – Hardware Level

This IDN will replace *Digital Output Data* when the servo drive is not in the EtherCAT OP state. The effect of *Digital Output Polarity* still applies.

7.6.3 General Operation

The details of the process by which the digital outputs are set is described in this sections. The polarity of physical level of each digital output can be set by i^{th} bit in *Digital Output Polarity*. In addition, each digital output can be either configured as a **General Purpose** or **Hardware Level** digital output as explained below.

7.6.3.1 General Purpose Digital Output Configuration

First, i^{th} element of *Digital Output Source IDN List* is set to a desirable IDN as a source of information (S_i) in order to determine the state of the i^{th} digital output. It must be noted that if drive is not in EtherCAT OP state, the source of information is provided by i^{th} element of *Digital Output User Configurable Default Safe State – General Purpose*. Moreover, the state of i^{th} digital output can be inverted and masked by i^{th} element of *Digital Output Invert Mask* and *Digital Output Source Bitmask* respectively. The logic is presented in Table 7-1.

EtherCAT state	OP	Other than OP
S_i	i^{th} element in <i>Digital Output Source IDN List</i>	i^{th} element of <i>Digital Output User Configurable Default Safe State – General Purpose</i>
D_i	Boolean condition of $\{S_i \text{ XOR } \textit{Digital Output Invert Mask}[i^{\text{th}}] \text{ AND } \textit{Digital Output Source Bitmask}[i^{\text{th}}]\}$	
P_i	i^{th} bit in <i>Digital Output Polarity</i>	
Actual physical level of digital output	$D_i \text{ XOR } P_i$	

Table 7-1 - General Purpose Digital Output Logic

7.6.3.2 Hardware Level Digital Output Configuration

Hardware level configuration logic is presented in Table 7-2. As can be seen i^{th} bit in *Digital Output Data* can be directly set in order to determine the state of the i^{th} digital output and no source of information, i.e. S_i is required. Similarly, if drive is not in EtherCAT OP state, the digital output state is set by the i^{th} bit in *Digital Output User Configurable Default Safe State – Hardware Level*.

EtherCAT state	OP	Other than OP
D_i	i^{th} bit in <i>Digital Output Data</i>	i^{th} bit in <i>Digital Output User Configurable Default Safe State – Hardware Level</i>
P_i	i^{th} bit in <i>Digital Output Polarity</i>	
Actual physical level of digital output	$D_i \text{ XOR } P_i$	

Table 7-2 - Hardware Level Digital Output Logic

7.7 Drive Bypass Mode

7.7.1 Description

The AMD servo drive supports a feature to simulate the *Drive Status Word* as if the drive were enabled. This allows a drive to appear to be operational when in fact it is NOT. Uses for this feature include:

- The staging of drive power-up on a new machine (one drive at a time commissioning, where all drives may need to appear operational when testing the power-up sequences and evaluating their success)
- Troubleshooting a problematic drive
- The continued operation of a machine with a defective drive

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-0739 / 33507	Bypass Mode Enable	Unsigned Integer (2 bytes)	Boolean	0	0
P-0-0740 / 33508	Bypass Mode Status	Unsigned Integer (2 bytes)	Boolean	NA	NA

7.7.2 General Operation

When *Bypass Mode Enable* is set to '1':

- The motor is disabled regardless of the value of the *Master Control Word*.
- When the motor is commanded to enable via the *Master Control Word*, the *Drive Status Word* will indicate the motor is enabled.
- The *Drive Status Word* will reflect the commanded operation mode in the *Master Control Word*.
- All Class 1 Diagnostic Errors (C1D) are not reported by the *Drive Status Word*.
- The servo drives encoder based feedback still reflect the actual axis position which may affect monitoring by the control device / unit.

When *Bypass Mode Enable* is set to '0':

- The servo drive will operate as normal.
- If the servo drive is enabled via the *Master Control Word* and *Bypass Mode Enable* was set to '1', the servo drive must be commanded to disable and re-enable before the motor will enable.

Bypass Mode Status can be monitored to verify whether drive is in bypass mode.



NOTE: General Purpose Digital and Analogue IO is unaffected by the drive being in Bypass Mode.



WARNING: Since an axis which is in Bypass Mode will not move, there may be a risk of the machine joints crashing into each other.



WARNING: Vertical axes where the motor brake is **not** controlled by the drive may fall if the axis is placed in Bypass Mode (for example, when the brake is controlled through a PLC). When the motor brake **is** controlled by the drive, it will remain engaged even when in Bypass Mode.

7.8 Drive Data Logger

7.8.1 Description

The drive data logger can be used to synchronously sample data from the drives. Four IDNs can be logged simultaneously, each with up to 2048 data points. The log can be configured with a wide range of triggers and pre/post trigger sample components.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-1292 / 34060	Data Logger Procedure Command	Unsigned Integer (2 bytes)	Procedure Command	0	0
P-0-1293 / 34061	Data Logger Variable List (4 elements)	Unsigned Integer (2 bytes) Array	IDN	36,37,40,51	36,37,40,51
P-0-1294 / 34062	Data Logger Sample Period Factor	Unsigned Integer (2 bytes)	62.5µs	1	1
P-0-1295 / 34063	Data Logger Pre-Trigger Samples	Unsigned Integer (2 bytes)	Samples	0	0
P-0-1296 / 34064	Data Logger Trigger IDN	Unsigned Integer (2 bytes)	IDN	1	1
P-0-1297 / 34065	Data Logger Trigger Mask	Unsigned Integer (4 bytes)	Binary	0	0
P-0-1298 / 34066	Data Logger Trigger Value	Signed Integer (4 bytes)	None	0	0
P-0-1299 / 34067	Data Logger Control Word	Unsigned Integer (2 bytes)	Binary	0	0
P-0-1300 / 34068	Data Logger Variable List Indices (4 elements)	Unsigned Integer (2 bytes) Array	Array Index	0	0
P-0-1301 / 34069	Data Logger - Measured Signal - Channel 0 (2048 elements)	Unsigned Integer (4 bytes) Array	None	N/A	N/A
P-0-1302 / 34070	Data Logger - Measured Signal - Channel 1 (2048 elements)	Unsigned Integer (4 bytes) Array	None	N/A	N/A
P-0-1303 / 34071	Data Logger - Measured Signal - Channel 2 (2048 elements)	Unsigned Integer (4 bytes) Array	None	N/A	N/A
P-0-1304 / 34072	Data Logger - Measured Signal - Channel 3 (2048 elements)	Unsigned Integer (4 bytes) Array	None	N/A	N/A
P-0-1305 / 34073	Data Logger Error Word	Unsigned Integer (2 bytes)	Binary	N/A	N/A
P-0-1306 / 34074	Data Logger Main State	Unsigned Integer (2 bytes)	Binary	N/A	N/A

7.8.2 Definitions

7.8.2.1 Data Logger Variable List

The *Data Logger Variable List* specifies the IDNs of the variables to log. A complete listing is specified in the Digital Servo Drive SoE Parameter Reference. Up to 4 may be specified. An IDN of 0 disables the drive data logger channel.

- Specify the IDN for logging on channel 0 by placing the IDN of the variable of interest into the first element of the array contained by *Data Logger Variable List* (this element is considered to be element 0)
- Similarly specify IDN for channel 1: *Data Logger Variable List* (element 1)

- Similarly specify IDN for channel 2: *Data Logger Variable List* (element 2)
- Similarly specify IDN for channel 3: *Data Logger Variable List* (element 3)

7.8.2.2 Data Logger Sample Period Factor

The *Data Logger Sample Period Factor* specifies the period between samples and is defined as a multiple of 62.5µs. Specifically:

$$\text{Data Logger Sample Period Factor} = \frac{\text{Sample Period}}{62.5\mu\text{s}}$$

The supported sample rates are:

- Task0: *Data Logger Sample Period Factor* = 62.5µs / 62.5µs = 1 (current loop update period)
- Task1: *Data Logger Sample Period Factor* = 250µs / 62.5µs = 4 (position / velocity loop update period)
- Task2: *Data Logger Sample Period Factor* = 250µs / 62.5µs = 4 (optional communications update period)
- Task3: *Data Logger Sample Period Factor* = 4000µs / 62.5µs = 64 (default communications update period)

7.8.2.3 Data Logger Pre-trigger Samples

Configure a trigger to control under what conditions the data log will complete. The trigger mechanism is quite flexible; generally, any variable within the Profile can be selected as the trigger variable and the outcome of its comparison with a fixed value can be used to complete the data logging. The data logger is fixed at 2048 samples to be collected; however, the trigger mechanism allows the user to specify where within the 2048 samples the trigger level is to be detected. For example, the user may wish to commence sampling when a variable exceeds a certain threshold, they may wish to log both prior to and following a significant event, or they may wish to capture data prior to a certain point. All of these approaches utilise a single configuration in the AMD servo drive.

The user begins by specifying the number of pre-trigger samples to log. The Drive Data Logger uses a circular buffer to store data. *Data Logger Pre-Trigger Samples* informs the logger how many samples to keep prior to the trigger sample. In the following example, *Data Logger Pre-Trigger Samples* is set to 500. This means that the buffers returned on completion of the data log will include 500 samples taken immediately before the trigger event occurred and 1548 after (for a total of 2048 data points):

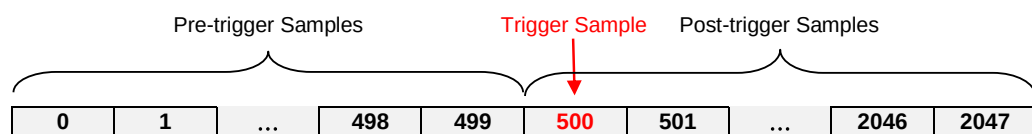


Figure 7-5 - Pre-trigger sample example

7.8.2.4 Data Logger Trigger IDN

The trigger IDN is specified in the *Data Logger Trigger IDN*. This value is compared against *Data Logger Trigger Value* to determine the trigger event. IDNs of many data types are supported including; signed 16-bit (S16), unsigned 16-bit (U16), signed 32-bit (S32) and unsigned 32-bit (U32).

7.8.2.5 Data Logger Trigger Mask

The *Data Logger Trigger Mask* can be applied to the data logger trigger value to isolate and compare individual bits (see table below). This is useful if the trigger variable *Data Logger Trigger IDN* is a bit field. If the *Data Logger Trigger Mask* is set to '0', the bit mask is disabled and the comparison reverts to a direct comparison.

Comparison	Pseudo Code	Notes
Equal To	(*34064 & 34065) == 34066	only U16 & U32
Not Equal To	(*34064 & 34065) != 34066	only U16 & U32
Greater Than	(*34064 & 34065) > 34066	only U16 & U32
Less Than	(*34064 & 34065) < 34066	only U16 & U32

Table 7-3 - Data Logger Trigger Mask

7.8.2.6 Data Logger Trigger Value

The trigger value is specified in the *Data Logger Trigger Value*. This value is compared against *Data Logger Trigger IDN* to determine the trigger event. IDNs of many data types are supported including; signed 16-bit (S16), unsigned 16-bit (U16), signed 32-bit (S32) and unsigned 32-bit (U32).

7.8.2.7 Data Logger Control Word

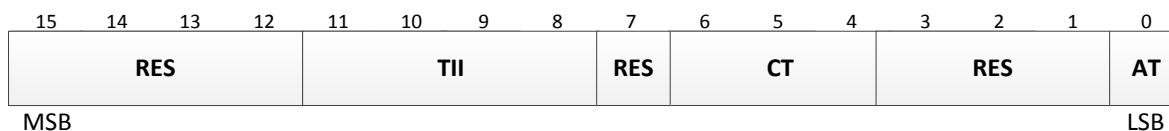


Figure 7-6 - Data Logger Control Word definition

Bit	Key	Definition	Description
0	AT	Automatic Trigger	If set to '1', the trigger IDN will be ignored and the data logger will automatically trigger.
1-3	RES	Reserved	-
4-6	CT	Comparison Type	Refer to Table 7-5 .
7	RES	Reserved	-
8-11	TII	Trigger IDN Index	Index of trigger IDN if it is an array. Set to '0' if IDN is not an array.
12-15	RES	Reserved	-

Table 7-4 - Data Logger Control Word bit definition

Comparison	Data Logger Control Word (bits 4-6)	Pseudo Code	Notes
Equal To	000	*IDN34064 == IDN34066	
Not Equal To	001	*IDN34064 != IDN34066	
Greater Than	010	*IDN34064 > IDN34066	
Less Than	011	*IDN34064 < IDN34066	
Absolute Greater Than	100	abs(*IDN34064) > IDN34066	only S16 & S32
Absolute Less Than	101	abs(*IDN34064) < IDN34066	only S16 & S32

Table 7-5 - Data Logger Control Word bits 4-6 definition

7.8.2.8 Data Logger Variable List Indices

If the IDN specified in the *Data Logger Variable List* refers to an array (rather than a single numeric variable), then the index of the element of interest in that array should be specified in *Data Logger Variable List Indices*. If the specified IDN is not an array then the associated element of *Data Logger Variable List Indices* should be set to 0.

- Specify the IDN index into the array for channel 0: *Data Logger Variable List Indices* (element 0)
- Specify the IDN index for channel 1: *Data Logger Variable List Indices* (element 1)
- Specify the IDN index for channel 2: *Data Logger Variable List Indices* (element 2)
- Specify the IDN index for channel 3: *Data Logger Variable List Indices* (element 3)

7.8.2.9 Data Logger - Measured Signal

There exists a *Data Logger – Measured Signal* for each data logger channel. Upon completion of the data log, each *Data Logger – Measured Signal* will contain 2048 samples of the value of the IDN as configured in the *Data Logger Variable List*.

- Data Logger - Measured Signal - Channel 0: 2048 samples of the IDN specified in *Data Logger Variable List* element 0
- Data Logger - Measured Signal - Channel 1: 2048 samples of the IDN specified in *Data Logger Variable List* element 1
- Data Logger - Measured Signal - Channel 2: 2048 samples of the IDN specified in *Data Logger Variable List* element 2
- Data Logger - Measured Signal - Channel 3: 2048 samples of the IDN specified in *Data Logger Variable List* element 3

7.8.2.10 Data Logger Error Word

The *Data Logger Error Word* is a bit field that can be used to determine the cause of the data logger procedure command returning an error. Each bit is '1' when the corresponding error occurred.

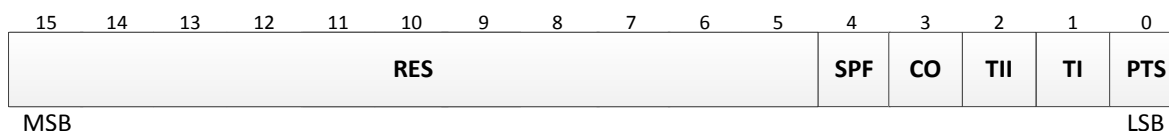


Figure 7-7 - Data Logger Control Word definition

Bit	Key	Definition	Description
0	PTS	Invalid Pre-Trigger Samples	The number of pre-trigger samples exceeds the buffer length (2048)
1	TI	Invalid Trigger IDN	The trigger IDN does not exist or is incompatible with the data logger
2	TII	Invalid Trigger IDN Index	The index does not exist for the selected trigger IDN
3	CO	Invalid Comparison Operator	The comparison operator configured does not exist or is not supported with the IDN's data type
4	SPF	Invalid Sample Period Factor	The sample period factor chosen is not supported

5-15	RES	Reserved	-
------	-----	----------	---

Table 7-6 - Data Logger Control Word bit definition

7.8.2.11 Data Logger Main State

The *Data Logger Main State* can be used by the control unit / device to monitor the status of the data logger, providing more information than the standard procedure command acknowledge.

Value	Description
0	Disabled
1	Collecting pre-trigger data
2	Armed and waiting for trigger
4	Collecting post-trigger data

Table 7-7 - Data Logger Main State definition

7.8.3 General Operation

The following instructions detail the configuration and execution of the drive data logger via IDN. Use of ANCA MotionBench simplifies the configuration process and is recommended for most scenarios. Refer to [7.8.5 MotionBench Interface](#).

1. Set the IDNs to be logged and other configuration options.
2. Arm the Drive Data Logger by setting *Data Logger Procedure Command* to 3.
3. Wait for the procedure command acknowledge to return successfully to indicate the data log is complete. Alternatively, the *Data Logger Main State* could be monitored.
4. Cancel the Drive Data Logger by setting the *Data Logger Procedure Command* to 0.
5. Read the logged data from *Data Logger – Measured Signal* for each channel used.

7.8.4 Example Usage

7.8.4.1 Trigger from Drive Stimulus Injection

This configuration is useful when collecting data for the velocity or current response.

Step 1: Specify the IDNs and IDN indexes to log, for example:

IDN 34061 [0] = 33006	Idq Current Command (IDN 33006)
IDN 34061 [1] = 33050	Idq Current Feedback (IDN 33050)
IDN 34061 [2] = 0	(not used)
IDN 34061 [3] = 0	(not used)
IDN 34068 [0] = 1	Element 1: Iq Current Command
IDN 34068 [1] = 1	Element 1: Iq Current Feedback
IDN 34068 [2] = 0	(not used)
IDN 34068 [3] = 0	(not used)

Step 2: Specify the sample period factor, for example:

IDN 34062 = 1	Log at Task0: 62.5μs / 62.5μs = 1
---------------	-----------------------------------

Step 3: Configure trigger, for example:

IDN 34064 = 34042	Use the Stimulus Status Word (IDN 34042) to trigger off
IDN 34065 = 3	Mask bits 0-1 of the Stimulus Status Word
IDN 34066 = 3	Compare to 3 (stimulus injection complete)
IDN 34067 = 0	Comparison type is "Equal To" (bits 4-6 = 000) and index into IDN 34042 is 0 (bits 8-11 = 0000)

Step 4: Specify the number of pre-trigger samples, for example:

IDN 34063 = 2047	Entire buffer contains the data before the trigger event occurred
------------------	---

Step 5: Enable the drive data logger, for example:

IDN 34060 = 3	Drive Data Logger Procedure Command
---------------	-------------------------------------

7.8.4.2 Trigger from Class 1 Diagnostic Fault

This configuration is useful when collecting data that coincides with a sporadic Class 1 Diagnostic (C1D) fault.

Step 1: Specify the IDNs and IDN indexes to log, for example:

IDN 34061 [0] = 32961	Position Command (IDN 32961)
IDN 34061 [1] = 33625	Position Feedback (IDN 33625)
IDN 34061 [2] = 32962	Velocity Command (IDN 32962)
IDN 34061 [3] = 33626	Velocity Feedback (IDN 33626)
IDN 34068 [0] = 0	Element 0 (not an array)
IDN 34068 [1] = 0	Element 0 (not an array)
IDN 34068 [2] = 0	Element 0 (not an array)
IDN 34068 [3] = 0	Element 0 (not an array)

Step 2: Specify the sample period factor, for example:

IDN 34062 = 4	Log at Task1: $250\mu\text{s} / 62.5\mu\text{s} = 4$
---------------	--

Step 3: Configure trigger, for example:

IDN 34064 = 33255	Class 1 Diagnostic (C1D) Error Status Word (IDN 33255)
IDN 34065 = 0	(disable mask)
IDN 34066 = 0	Compare to 0 (no C1D)
IDN 34067 = 16	Comparison type is "Not Equal To" (bits 4-6 = 001) and index into IDN 33255 is 0 (bits 8-11 = 0000)

Step 4: Specify the number of pre-trigger samples, for example:

IDN 34063 = 1024	First half of the data in the buffer is pre C1D Event and the second half is post C1D Event
------------------	---

Step 5: Enable the drive data logger, for example:

IDN 34060 = 3	Drive Data Logger Procedure Command
---------------	-------------------------------------

7.8.5 MotionBench Interface

MotionBench provides a convenient interface (see [Figure 7-8](#)) to configure the Drive Data Logger.

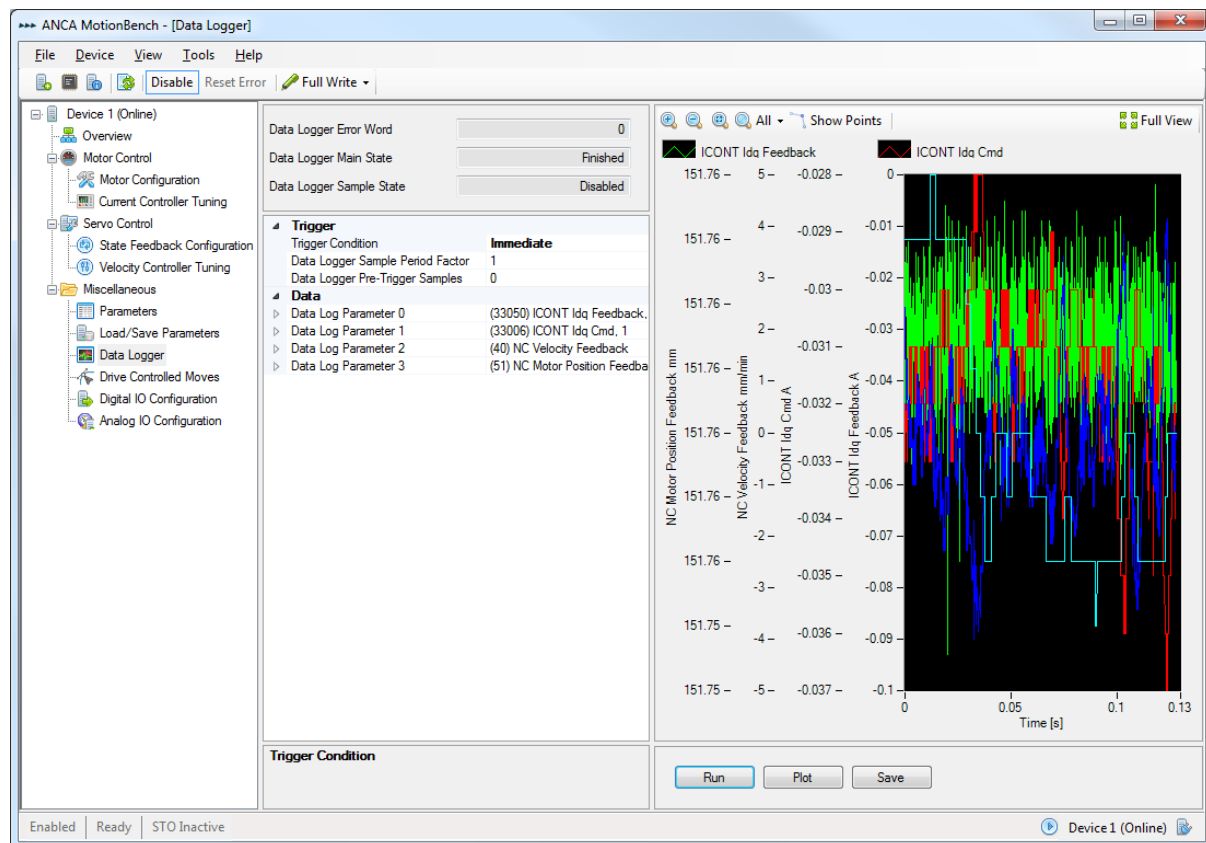


Figure 7-8 ANCA MotionBench Drive Data Logger Interface

7.9 Encoder Configuration

7.9.1 Description

Up to two incremental encoders may be connected to the AMD servo drive. Additionally, each encoder may be of digital quadrature or analogue Sin/Cos. Encoder support may vary depending on model. Refer to the relevant [AMD Servo Drive User Guide](#) for more information.

The encoders provide feedback information to the drive for closed-loop position or speed control and are also employed to some degree in current control. The encoder channels will be described using the terms “motor” and “external” encoder feedback in the discussion below, where the external feedback is often assigned to a linear scale or some other sensor that is closer to the tool tip / end effector of the machine under operation. The motor encoder feedback is usually directly connected to the motor itself. Either of these feedbacks can be connected to either encoder channel of the drive with suitable configuration described below. Each encoder channel needs to be configured separately.

7.9.2 Encoder Connection Setup

The drive needs to be configured to determine which channels are connected to an encoder. The user can set the following IDN's to either of the following two values:

- 0: No encoder connected
- 10: Incremental encoder connected

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-1432 / 34200	Encoder Type Connected - Channel 1	Unsigned Integer (2 bytes)	Encoder Type	10	10
P-0-1433 / 34201	Encoder Type Connected - Channel 2	Unsigned Integer (2 bytes)	Encoder Type	0	0

Furthermore, each channel needs to be specified as either “motor” encoder feedback, or “external” encoder feedback. The user can set the following IDN's to one of the following values:

- 0: Channel 1
- 1: Channel 2
- 4: None

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-1028 / 33796	Motor Encoder Source Channel	Unsigned Integer (2 bytes)	Encoder Channel	0	0
P-0-1029 / 33797	External Encoder Source Channel	Unsigned Integer (2 bytes)	Encoder Channel	4	4

Finally, the measured quadrature encoder counts can be monitored to ensure that fundamental wiring and setup is correct before further configuration is performed. It should be noted that encoder counts will overflow and ‘wrap’ when as if they were 16bit data types i.e. have a range of -32768 to 32767.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-1028 / 33796	Encoder Counter Sample - Ch1	Signed Integer (4 bytes)	Quadrature Counts	N/A	N/A
P-0-1029 / 33797	Encoder Counter Sample - Ch2	Signed Integer (4 bytes)	Quadrature Counts	N/A	N/A

7.9.3 Motor Encoder Feedback

If the *Motor Encoder Source Channel* has been associated with a channel which has an encoder connected (i.e. the channel's *Encoder Type Connected* has been set to ‘Incremental encoder connected’) then further motor encoder parameters must be configured.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0116 / 116	Resolution of Motor Encoder	Unsigned Integer (4 bytes)	Lines/rev	512	512
S-0-0121 / 121	Input Revolutions of Load Gear	Unsigned Integer (4 bytes)	Rev	1	1
S-0-0122 / 122	Output Revolutions of Load Gear	Unsigned Integer (4 bytes)	Rev	1	1
S-0-0123 / 123	Feed Constant	Signed Integer (4 bytes)	Standard Position	5 mm	5 mm
S-0-0277 / 277	Motor Encoder Feedback Type	Unsigned Integer (2 bytes)	Binary	0	0
P-0-0004 / 32772	Motor Encoder Control Word	Unsigned Integer (2 bytes)	Binary	0	1
P-0-0007 / 32775	Motor Encoder Linear Resolution	Signed Integer (4 bytes)	10 ⁻⁴ mm	0.02 mm	0.02 mm

P-0-1052 / 33820	Motor Velocity Feedback Low Pass Filter Coefficient	Unsigned Integer (2 bytes)	%	100	100
------------------	---	----------------------------	---	-----	-----



Note: The maximum admissible 'Output Revolutions' to 'Input Revolutions' of load gear ratio is 100.

7.9.3.1 Motor Encoder Feedback Type

The *Motor Encoder Feedback Type* defines different characteristics of the motor encoder feedback.

Bit	Value	Definition
0	0	Rotary external encoder feedback. Reference <i>Resolution of Motor Encoder</i> for encoder resolution configuration.
	1	Linear external encoder feedback. Reference <i>Motor Encoder Linear Resolution</i> for encoder resolution configuration.
3	0	Natural motor encoder direction.
	1	Reverse motor encoder direction.

Table 7-8 - Motor Encoder Feedback Type bit definition

7.9.3.2 Motor Encoder Control Word

The *Motor Encoder Control Word* is used to configure the motor encoder type. The encoder control word can be set to either of two values:

- 0: Digital quadrature encoder
- 2: Analogue sin/cos encoder

7.9.3.3 Motor Encoder to Axis Position Scaling

The AMD servo drive provides the ability to configure the significance of motor encoder counts to the axis position/velocity.

When using a linear motor, this is done via the *Motor Encoder Linear Resolution* parameter.

When using a rotary motor, this is done via the use of the *Input Revolutions of Load Gear*, *Output Revolutions of Load Gear* and *Feed Constant*.

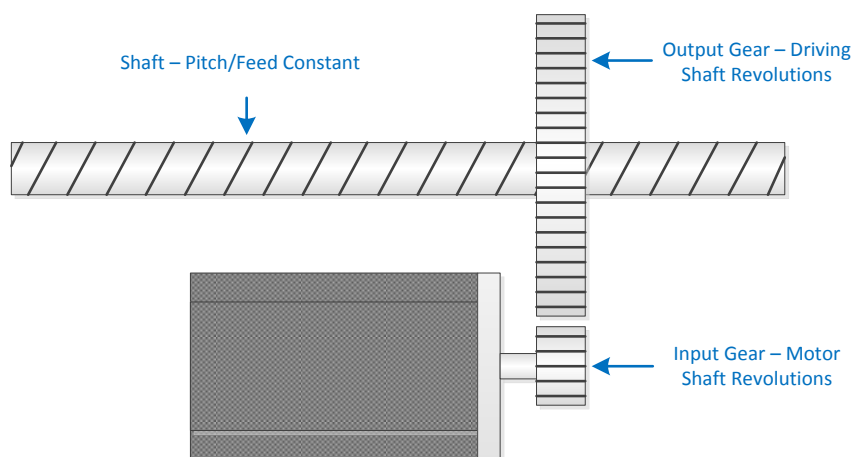


Figure 7-9 - Motor Encoder to Axis Position Scaling

These parameters combine such that:

$$\text{Axis Position} = \text{Feed Constant} \times \frac{\text{Output Revolutions of Load Gear}}{\text{Input Revolutions of Load Gear}} \times \frac{\text{Motor Encoder Count}}{\text{Resolution of Motor Encoder}}$$



NOTE: For a rotary axis, it is recommended that the feed constant be set to 1 rotation i.e. 360 degrees or 2π radians.

7.9.3.4 Motor Velocity Feedback Low Pass Filter Coefficient

The *Motor Velocity Feedback Low Pass Filter Coefficient* allows configuration of the low pass filter applied to measured velocity from the encoder feedback. The filter is intended for use with coarse pitch encoders to smooth the velocity feedback signal. This smoothing will add delay to the signal, which can lead to instability so it is a trade-off. The filter equation is shown below with the filter coefficient represented by α .

$$\text{Motor Velocity Measurement} = \alpha \times \text{Unfiltered Velocity} + (\alpha - 1) \times \text{Motor Velocity Measurement Last Sample}$$

It can be seen that when the filter coefficient =1, the filter is disabled and when the filter coefficient is 0, the measured velocity will never update.

7.9.3.5 Analogue Incremental Encoders Only

Analogue (sin/cos) encoders can be commissioned to increase position feedback accuracy. Both a gain and offset can be calibrated for both the sine and cosine components of the signal. The two IDNs listed in the following table are each an array containing two elements. The first element in each array relates to the cosine information from the sensor while the second element relates to the sine information (hence 'CosSin' in the description). For maximum position accuracy these parameters can be commissioned on each motor to account for manufacturing tolerances.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-1035 / 33803	Motor Encoder CosSin Gain	Signed Integer (2 bytes)	2^{-14}	16384	16384
P-0-1036 / 33804	Motor Encoder CosSin Offset	Signed Integer (2 bytes)	ADC	1368	0

Refer to [7.9.5 Analogue Encoder Compensation](#) for details on how to commission the gain and offset parameters.

7.9.4 External Encoder Feedback

If the *External Encoder Source Channel* has been associated with a channel which has an encoder connected (i.e. the channel's *Encoder Type Connected* has been set to 'Incremental encoder connected') then further external encoder parameters must be configured.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0115 / 115	External Encoder Feedback Type	Unsigned Integer (2 bytes)	Binary	0	0
S-0-0117 / 117	Resolution of Rotary External Encoder	Unsigned Integer (4 bytes)	Lines/rev	512	512
S-0-0118 / 118	Resolution of Linear External Encoder	Unsigned Integer (4 bytes)	10^{-4} mm	0.8 mm	0.8 mm
P-0-0005 / 32773	External Encoder Control Word	Unsigned Integer (2 bytes)	Binary	0000 0000	0000 0010

7.9.4.1 External Encoder Feedback Type

The *External Encoder Feedback Type* defines different characteristics of the external encoder feedback.

Bit	Value	Definition
0	0	Rotary external encoder feedback. Reference <i>Resolution of Rotary External Encoder</i> for encoder resolution configuration.
	1	Linear external encoder feedback. Reference <i>Resolution of Linear External Encoder</i> for encoder resolution configuration.
3	0	Natural external encoder direction.
	1	Reverse external encoder direction.

Table 7-9 - External Encoder Feedback Type bit definition



WARNING: Opposite polarities of motor and external encoders can cause incorrect motor movement. Ensure both motor and external encoder positions increment in the same direction before enabling the axis.

7.9.4.2 External Encoder Control Word

The *External Encoder Control Word* is used to configure the external encoder type. The encoder control word can be set to either of two values:

- 0: Digital quadrature encoder
- 2: Analogue sin/cos encoder

7.9.4.3 Analogue Incremental Encoders Only

Analogue (sin/cos) encoders can be commissioned to increase position feedback accuracy. Both a gain and offset can be calibrated for both the sine and cosine components of the signal. The two IDNs listed in the following table are each an array containing two elements. The first element in each array relates to the cosine information from the sensor while the second element relates to the sine information (hence 'CosSin' in the description). For maximum position accuracy these parameters can be commissioned on each external encoder to account for manufacturing tolerances.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-1075 / 33843	External Encoder CosSin Gain	Signed Integer (2 bytes)	2^{-14}	16384	16384
P-0-1076 / 33844	External Encoder CosSin Offset	Signed Integer (2 bytes)	ADC	-2232	0

Refer to [7.9.5 Analogue Encoder Compensation](#) for details on how to commission the gain and offset parameters.



NOTE: Changing analog encoder or encoder cable while drive is powered results incorrect analog position. Power-cycle is required for phase initialization.

7.9.5 Analogue Encoder Compensation

7.9.5.1 Description



NOTE: This feature is only applicable when using a motor with an analogue SINCOS encoder.

Analogue encoder compensation provides the user with the ability to adjust the raw SINCOS encoder signals from the motor in order to compensate for static zero level offsets or linear scaling imperfections in the encoder feedback. These imperfections can be seen as a ripple in velocity feedback when the axis is known to be moving at a constant speed. Consider using encoder compensation when a plot of the raw SIN vs COS encoder feedback signals is not centred on zero or is not a perfect circle. Such a plot can be generated using the ANCA MotionBench software.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-1014 / 33782	Adjusted Cos/Sin Signals – Motor Encoder (2 elements)	Signed Integer (2 bytes) Array	ADC	N/A	N/A
P-0-1015 / 33783	Adjusted Cos/Sin Signals – External Encoder (2 elements)	Signed Integer (2 bytes) Array	ADC	N/A	N/A
P-0-1016 / 33784	Unadjusted Cos/Sin Signals – External Encoder (2 elements)	Signed Integer (2 bytes) Array	ADC	N/A	N/A
P-0-1041 / 33809	Unadjusted Cos/Sin Signals – Motor Encoder (2 elements)	Signed Integer (2 bytes) Array	ADC	N/A	N/A

7.9.5.2 Procedure for Testing and Setting Up Encoder Compensation

Step 1: Using the ANCA MotionBench to capture and plot SINCOS encoder signals

Make sure that the SINCOS encoder is connected per the applicable User Guide and configured (refer to [7.9 Encoder Configuration](#)) and your personal computer is connected to the drive (refer to [4.6 Connecting the AMD Servo Drive to a PC](#)). Start-up ANCA MotionBench (refer to [4.8 Configuring the AMD Servo Drive](#)), and navigate to the Circle Graph interface. Click *Start* to begin measuring the encoder feedback. If possible, rotate the motor shaft by hand, or alternatively, use the *Start Motor* button on the page. You should then see data points begin to appear on the page in an approximately circular shape.

[Figure 7-10](#) and [Figure 7-11](#) show MotionBench when running the associated Circle Graph for a typical SINCOS encoder setup. The horizontal axis, or abscissa, represents the scale for the cosine signal from the encoder, whereas the vertical axis, or ordinate, is the sine signal scale. Units for the two axes are ADC (analogue to digital conversion) counts, and both the maximum and minimum allowable levels of ADC are plotted as circles on the graph. The raw encoder signals should fall within these two bounds. Both the raw encoder output and the user adjusted (“Adj.”) outputs will be shown plotted on the same Circle Graph. The user determines the shape of the adjusted output by [Figure 7-10](#). The user receives immediate visual confirmation of the effect of their changes by inspecting the Circle Graph “Adj.” outputs.

If the user prefers to see all the encoder signals rendered individually in the time domain (rather than sine versus cosine), then they can choose instead to examine the Tab labelled “Time Domain” rather than the default Tab “Circle Graph.” The same operations can be performed when viewing the time domain data (see [Figure 7-11](#)).

Step 2: Measure the minimum and maximum values of the raw sin and cos signals

For convenience the following symbols are defined:²

- S_{min} – minimum value of raw unadjusted sin signal
- S_{max} – maximum value of raw unadjusted sin signal
- C_{min} – minimum value of raw unadjusted cos signal
- C_{max} – maximum value of raw unadjusted cos signal

² These symbols do not refer to the minimum and maximum allowable encoder values shown in the plots, but are instead the minimum and maximum values of the raw encoder outputs.

It is recommended that the user select from the appropriate Zoom buttons on either Tab to obtain a high resolution plot of the region close to the appropriate minimum and maximum data that they wish to collect. This will maximise the resolution of ADC values for the purposes of calculation in the next step.

Step 3: Calculate the offsets and gains for correcting the sin and cos signals

Step 3.1 Calculate the gains for correcting the SINCOS signals.

$$\text{Gain}_{\text{sine}} = \frac{1}{2} + \frac{1}{2} \left(\frac{C_{\text{max}} - C_{\text{min}}}{S_{\text{max}} - S_{\text{min}}} \right), \text{Gain}_{\text{cosine}} = \frac{1}{2} + \frac{1}{2} \left(\frac{S_{\text{max}} - S_{\text{min}}}{C_{\text{max}} - C_{\text{min}}} \right).$$

Step 3.2 Calculate the offsets for correcting the SINCOS signals.

$$\text{Offset}_{\text{sine}} = -\frac{S_{\text{max}} + S_{\text{min}}}{2}, \text{Offset}_{\text{cosine}} = -\frac{C_{\text{max}} + C_{\text{min}}}{2}.$$

Step 4: Visually verify the SINCOS signals have been properly compensated

Check the updated Circle Graph for whether the adjusted SINCOS signals are more closely reflecting a perfect circle than the raw unadjusted data. This can be done while taking measurements so the adjustments are seen live on axis.

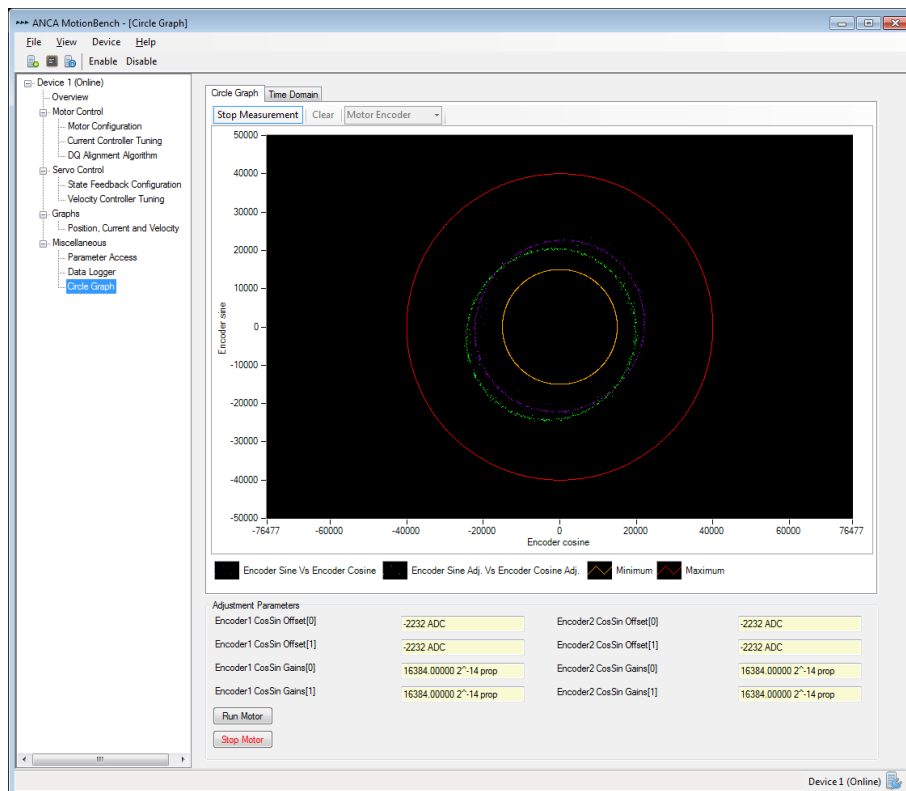


Figure 7-10 Circle Graph Tab

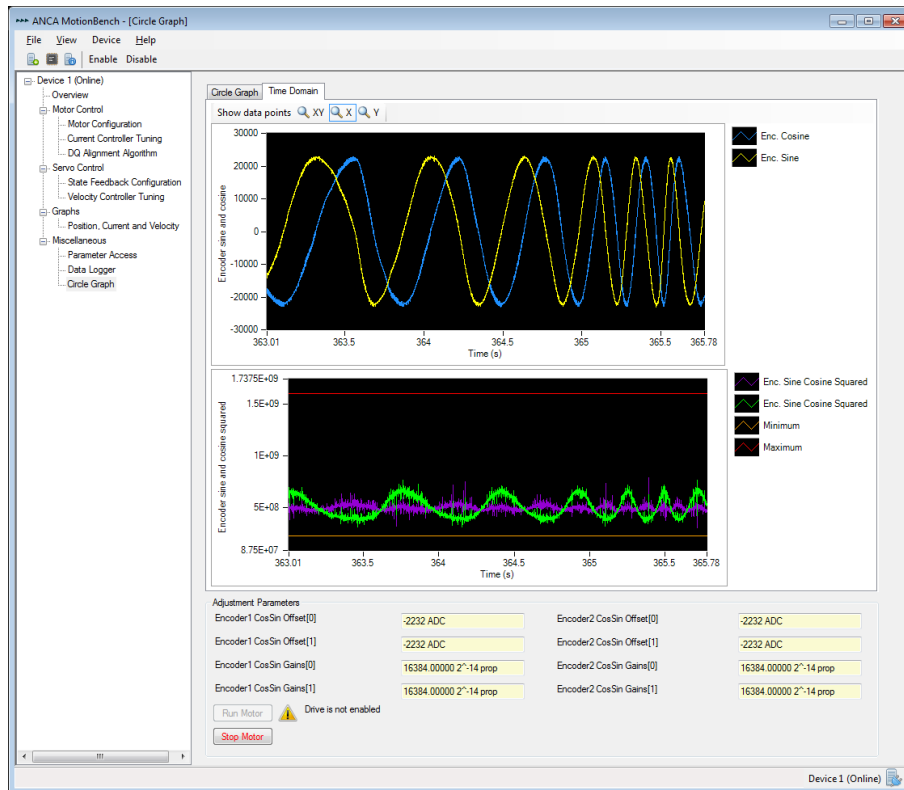


Figure 7-11 Time Domain Tab

7.9.6 Analogue Encoder Signal Monitoring

7.9.6.1 Description



NOTE: This feature is only applicable when using a motor with an analogue SINCOS encoder.

When using an analogue incremental encoder, the AMD servo drive will monitor the encoder signal integrity to ensure it is within optimal operating range. The operating range may vary between encoder types; therefore, the allowable signal strength is configurable via minimum and maximum encoder signal values.

When monitoring the analogue encoder signal, the measured cos and sin encoder signals are compared to the minimum/maximum Cos/Sin values as shown in the equation below. If the signal exceeds the minimum or maximum limit, a C1D error is asserted.

$$\text{Min}^2 \leq \text{Cos}^2 + \text{Sin}^2 \leq \text{Max}^2$$

Detection of an encoder fault will cause the drive will to perform [Field Orientation Initialisation](#) again. It is also recommended to re-home the axis.

When using an analogue motor encoder, analogue encoder signal monitoring is automatically enabled when the encoder is configured to use analogue encoder interpolation via the [Motor Encoder Control Word](#). When using an analogue external encoder, analogue encoder signal monitoring is enabled via the [External Encoder Control Word](#) and [Cos/Sin Check Enable – External Encoder](#) parameter.

Separate limits and C1D errors are used for the adjusted and unadjusted encoder signal. Refer to [7.9.5 Analogue Encoder Compensation](#) for more information on adjusted encoder signals.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-1012 / 33780	Adjusted ADC Squared – Motor Encoder	Signed Integer (4 bytes)	ADC ²	N/A	N/A
P-0-1013 / 33781	Adjusted ADC Squared – External Encoder	Signed Integer (4 bytes)	ADC ²	N/A	N/A
P-0-1022 / 33790	Cos/Sin Check Enable – External Encoder	Unsigned Integer (2 bytes)	Boolean	0	0
P-0-1023 / 33791	Minimum Cos/Sin Value – External Encoder	Unsigned Integer (2 bytes)	ADC	15000	15000
P-0-1024 / 33792	Maximum Cos/Sin Value – External Encoder	Unsigned Integer (2 bytes)	ADC	40000	40000
P-0-1025 / 33793	Unadjusted ADC Squared – External Encoder	Signed Integer (4 bytes)	ADC ²	N/A	N/A
P-0-1032 / 33800	Unadjusted ADC Squared – Motor Encoder	Signed Integer (4 bytes)	ADC ²	N/A	N/A
P-0-1033 / 33801	Minimum Cos/Sin Value – Motor Encoder	Unsigned Integer (2 bytes)	ADC	15000	15000
P-0-1034 / 33802	Maximum Cos/Sin Value – Motor Encoder	Unsigned Integer (2 bytes)	ADC	40000	40000
P-0-1041 / 33809	Unadjusted Cos/Sin Signals – Motor Encoder (2 elements)	Signed Integer (2 bytes) Array	ADC	N/A	N/A

7.9.6.2 Definitions

7.9.6.2.1 Adjusted ADC Squared – Motor Encoder

The *Adjusted ADC Squared – Motor Encoder* is the current measurement of the motor encoder's adjusted sin signal² + cos signal². This is compared to *Minimum Cos/Sin Value – Motor Encoder* and *Maximum Cos/Sin Value – Motor Encoder* to determine if the encoder signal is in range. If this value is determined to be out of range, C1D E0215 or E0216 will be asserted.

7.9.6.2.2 Adjusted ADC Squared – External Encoder

The *Adjusted ADC Squared – External Encoder* is the current measurement of the external encoder's adjusted sin signal² + cos signal². This is compared to *Minimum Cos/Sin Value – External Encoder* and *Maximum Cos/Sin Value – External Encoder* to determine if the encoder signal is in range. If this value is determined to be out of range, C1D E0227 or E0228 will be asserted.

7.9.6.2.3 Cos/Sin Check Enable – External Encoder

Set *Cos/Sin Check Enable – External Encoder* to 1 enable analogue encoder signal monitoring on an external encoder.

7.9.6.2.4 Minimum Cos/Sin Value – External Encoder

The *Minimum Cos/Sin Value – External Encoder* defines the minimum allowable value of the cos/sin signal this applies to both the adjusted and unadjusted external encoder signal.

7.9.6.2.5 Maximum Cos/Sin Value – External Encoder

The *Maximum Cos/Sin Value – External Encoder* defines the minimum allowable value of the cos/sin signal this applies to both the adjusted and unadjusted external encoder signal.

7.9.6.2.6 Unadjusted ADC Squared – External Encoder

The *Unadjusted ADC Squared – External Encoder* is the current measurement of the external encoder's unadjusted sin signal² + cos signal². This is compared to *Minimum Cos/Sin Value – External Encoder* and *Maximum Cos/Sin Value – External Encoder* to determine if the encoder signal is in range. If this value is determined to be out of range, C1D E0027 or E0028 will be asserted.

7.9.6.2.7 Unadjusted ADC Squared – Motor Encoder

The *Unadjusted ADC Squared – Motor Encoder* is the current measurement of the external encoder's unadjusted sin signal² + cos signal². This is compared to *Minimum Cos/Sin Value – Motor Encoder* and *Maximum Cos/Sin Value – Motor Encoder* to determine if the encoder signal is in range. If this value is determined to be out of range, C1D E0015 or E0016 will be asserted.

7.9.6.2.8 Minimum Cos/Sin Value – Motor Encoder

The *Minimum Cos/Sin Value – Motor Encoder* defines the minimum allowable value of the cos/sin signal this applies to both the adjusted and unadjusted motor encoder signal.

7.9.6.2.9 Maximum Cos/Sin Value – Motor Encoder

The *Maximum Cos/Sin Value – Motor Encoder* defines the minimum allowable value of the cos/sin signal this applies to both the adjusted and unadjusted motor encoder signal.

7.9.7 Missing Encoder Count Detection



NOTE: Lost incremental encoder counts can result from a faulty encoder, a poor connection between encoder and servo drive or exceeding the specifications of the encoder and/or drive.

An incremental encoder relies on detecting every change in signal level and counting these level changes. If a signal change is not detected or a false signal is detected then the resulting position feedback will be incorrect. The incorrect position will also affect field orientation reducing the efficiency of motor operation.

The AMD servo drive contains the ability to detect lost encoder counts based on the number of counts measured each time the motor reference/index pulse is encountered. If the servo drive determines a loss of encoder counts has occurred, a C1D error (E0407) will be asserted and the drive will need to perform *Field Orientation Initialisation* again. In the event of any encoder fault, it is also recommended to re-home the axis.

To enable this feature, the *Missing Encoder Counts Threshold* should be set to the allowable change in electrical angle that can occur at the motor index pulse before an error is asserted. For diagnostic purposes, the measured change in electrical angle is also available from *Missing Encoder Counts*.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-0326 / 33094	Missing Encoder Count Electrical Angle Threshold	Signed Integer (4 bytes)	10 ⁻⁴ elec rev	0	0
P-0-0327 / 33095	Change in Electrical Angle at Index Pulse	Signed Integer (4 bytes)	10 ⁻⁴ elec rev	N/A	N/A

7.10 Encoder Pass-Through

7.10.1 Description



NOTE: This feature is only available on the AMD2000 series servo drives.

The AMD2000 series servo drive provides a quadrature digital encoder output to provide feedback to non-EtherCAT control devices/units. It can also be used together with *Pulse / Stepper Position Control* to allow servo drives to slave to an encoder signal. This output can be configured to pass through either encoder input.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-0605 / 33373	Encoder Pass-Through Channel Selection	Unsigned Integer (2 bytes)	None	0	0

7.10.2 Definitions

7.10.2.1 Encoder Pass-Through Channel Selection

The *Encoder Pass-Through Channel Selection* allows configuration of the Encoder Pass-Through feature.

Value	Description
0	Off
1	Channel 1
2	Channel 2

Table 7-10 - Encoder Pass-Through Channel Selection definition

7.11 Field Orientation Initialisation

7.11.1 Overview

7.11.1.1 Description

A motor rotates due to the forces of attraction/repulsion between magnetic fields situated on the rotor and the stator where these fields can be generated in a number of different ways. The torque applied to the rotor is a resolved component of these forces acting around the motor shaft. It is proportional to the flux density of the magnetic fields, as well as a number of other parameters. Where electrical windings are used to generate these magnetic fields, the field's flux density is proportional to the current flowing through the winding.

One mechanical revolution of the rotor is usually more than one complete cycle traversing all the winding phases. That is to say, the sequence of electrical windings for each phase of electricity driving the motor is usually repeated more than once around the circumference of the motor. Hence, if a single vector is used to collectively represent the current for all these electrical phases, it must traverse a full 360 'electrical' degrees a number of times (depending on number of times the phase winding repeat) before completing one mechanical revolution. It is possible to represent such a single current vector as two component parts; one "quadrature" current, and a "direct" current. The "quadrature" component of the current vector is most closely associated with the magnetic forces that act around the motor shaft, and reaches its highest value when the electrical angle between the stator and rotor magnetic fields is near 90 degrees. For optimum torque delivery and motor efficiency, it is essential to keep this "field angle" at 90 degrees. The algorithm for doing this task is called commutation.

For Permanent Magnet AC (PMAC) motors, successful commutation requires correct initialisation wherein the rotor field angle is determined relative to a reference position on the stator. This initialisation has many names, for example: commutation initialisation, field orientation initialisation, phase initialisation. In the AMD servo drives it is known as Field Orientation Initialisation (FOI).

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0123 / 123	Feed Constant	Signed Integer (4 bytes)	Standard Position	5 mm	5 mm
P-0-0006 / 32774	Motor Poles	Unsigned Integer (2 bytes)	Poles / rev	4	4
P-0-0283 / 33051	Reverse Phase Sequence	Unsigned Integer (2 bytes)	Boolean	0	0
P-0-0292 / 33060	FOI Control	Unsigned Integer (2 bytes)	None	1	1
P-0-0297 / 33065	FOI Type	Unsigned Integer (2 bytes)	None	1	1
P-0-0940 / 33708	Release Brake at FOI	Unsigned Integer (2 bytes)	Boolean	0	0

7.11.1.2 Definitions

7.11.1.2.1 Feed Constant

If using a linear motor, the *Feed Constant* must be populated with the pole pitch (N-N distance) of the motor. This information should be available from the motor manufacturer's specification sheet.

7.11.1.2.2 Motor Poles

If using a rotary motor, the *Motor Poles* parameter must be populated with the number of motor poles per revolution (2 x the number of pole pairs) of the permanent magnet motor. This information should be available from the motor manufacturer's specification sheet.

7.11.1.2.3 Reverse Phase Sequence

If the *Reverse Phase Sequence* is set to 1, the U and W phases are reversed in the commutation sequence. This is equivalent to physically swapping U and W phase wires on the motor armature/power cable but only with regard to the field orientation. U and W current readings etc. will operate the same regardless of this parameter's value.

7.11.1.2.4 FOI Control

The *FOI Control* allows the user to configure when to perform FOI.

Value	Description
0	Do not perform FOI (advanced users only)
1	Only perform FOI when enabling the motor for the first time
2	Always perform FOI when enabling the motor

Table 7-11 - FOI Control definition



NOTE: Setting FOI Control = '2' allows for simpler commissioning of FOI, allowing for repeated tests to be run without removing power from the servo drive.

7.11.1.2.5 FOI Type

The *FOI Type* allows the user to configure the FOI type to be used with the motor.

Value	Description
1	DQ Alignment
2	Pre-set Offset (advanced users only)
3	Absolute (Commutation Track or UVW Wire-Saving)
4	Acceleration Observer
5	Braked Compliance

Table 7-12 - FOI Control definition

7.11.1.2.6 Release Brake at FOI

When a braked motor is used, setting *Release Brake at FOI* to 1 will release the brake while FOI is being performed. This allows DQ Alignment or the Acceleration Observer to be used. This is only appropriate for axes that will not move of its own accord when the brake is released i.e. horizontal or counterbalanced vertical axes.

7.11.1.3 General Operation

Regardless of the FOI mode, the FOI will operate in the following manner:

1. Upon first drive power up, no FOI has been completed.
2. When the servo drive is commanded to enable via the *Master Control Word*, the servo drive will perform FOI depending on the value of *FOI Control* once voltage has been applied to the motor. The drive reports that it is not yet ready to follow NC commands via the *Drive Status Word*.
3. Once the FOI is complete, the drive reports it is ready to follow NC commands via the *Drive Status Word* and enters the selected operating mode. The drive records that FOI is complete.
4. Subsequent disabling and re-enabling of the servo drive will only trigger FOI if *FOI Control* is set to '2'.
5. Encoder related Class 1 Diagnostic Errors that occur during operation will trigger FOI to be performed again upon next servo drive enable if *FOI Control* is set to '1' or '2'.

7.11.1.4 Supported FOI Types

AMD servo drives support several FOI techniques including:

- *DQ Alignment*
- *Commutation Track*
- *Acceleration Observer*
- *UVW Wire-Saving*
- *Braked Compliance*

For the above FOI techniques:

- *DQ Alignment* and *Acceleration Observer* can be used for any un-braked motor with an incremental encoder.
- *Braked Compliance* can be used for any braked motor with an incremental encoder.
- *Commutation Track* requires a servo motor with a supported encoder.
- *UVW Wire-Saving* is recommended for ANCA Motion Alpha series motors.

Table 7-13 below lists the supported FOI options for different drives and encoders. It should be noted that to further improve FOI accuracy for some techniques, post processing (*Alignment Off Index Pulse*) can be conducted for incremental encoders that possess such an index or reference pulse.

Encoder Type	FOI Techniques					Post Processing
	<i>DQ Alignment</i>	<i>Commutation Track</i>	<i>Acceleration Observer</i>	<i>UVW Wire-Saving</i>	<i>Braked Compliance</i>	<i>Alignment Off Index Pulse</i>
Incremental	✓	✓	✓			✓
Incremental with motor brake		✓		✓	✓	✓ (highly recommended)
AMD2000	✓		✓	✓	✓	
AMD5x	✓	✓	✓		✓	

Table 7-13 Drive and Encoder Types and Supported FOI Algorithms

7.11.2 DQ Alignment

7.11.2.1 Description

DQ Alignment injects magnetising current into the motor windings, resulting in the rotor moving to align with the stator magnetic field. This resulting position is latched and used to determine the correct offset. To verify the alignment, the motor commutation angle is commanded to move to seven different configurable positions in order to verify the expected rotor movements. By default, a PMAC motor will move $360/(\text{motor poles}/2)$ mechanical degrees during DQ Alignment FOI. For a motor with 8 poles, this is 90 mechanical degrees.



WARNING: Care should be taken to ensure that the resulting test move will not result in a mechanical collision.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-0300 / 33068	FOI DQA Alignment Time	Unsigned Integer (2 bytes)	250 μ s	0.3 s	0.75 s
P-0-0301 / 33069	FOI Alignment Current	Signed Integer (4 bytes)	Standard Current	4.0 A	3.0 A
P-0-0302 / 33070	FOI Alignment Current Slew	Signed Integer (4 bytes)	Standard Current	0.01 A	0.01 A
P-0-0303 / 33071	FOI Alignment Current Tolerance	Signed Integer (4 bytes)	Standard Current	0.5 A	1.0 A
P-0-0304 / 33072	FOI DQA Electrical Angle Slew Limit	Signed Integer (4 bytes)	10^{-4} electrical rev/s	0.4	0.4
P-0-0305 / 33073	FOI DQA Test Angles	Signed Integer (4 bytes) Array	10^{-4} electrical rev	0,0.3,0.6,1.0,0.7,0.4,0 elec rev	
P-0-0306 / 33704	FOI DQA Test Angle Offset	Signed Integer (4 bytes) Array	10^{-4} electrical rev	0.17,-0.83,1.17,0 elec rev	

7.11.2.2 Definitions

7.11.2.2.1 FOI DQA Alignment Time

The *FOI DQA Alignment Time* specifies the time for holding the electrical angle in each of the seven tests that are conducted in DQA FOI. This may need to be increased for axes that do not stop moving before progressing to the next angle.

7.11.2.2.2 FOI Alignment Current

The *FOI Alignment Current* specifies the current used during DQA movements. This value is axis and motor specific and may need to be adjusted depending on the application. The current needs to be sufficient to move the motor to the test points and within the current rating of the motor.

7.11.2.2.3 FOI Alignment Current Slew

FOI Alignment Current Slew specifies the initial rate of current increase to *FOI Alignment Current*.

7.11.2.2.4 FOI Alignment Current Tolerance

FOI Alignment Current Tolerance specifies the allowed difference between the commanded and actual current. Exceeding this value results in a Class 1 Diagnostic Error (E405). This parameter should not require modification in general operation, exceeding this value is a result of poor current control which can indicate poor current loop tuning.

7.11.2.2.5 FOI DQA Electrical Angle Slew Limit

FOI DQA Electrical Angle Slew Limit specifies the rate of change of electrical angle between the DQA test points. Reducing this value may be required for heavier and stiffer axes.

7.11.2.2.6 FOI DQA Test Angles

FOI DQA Test Angles defines the test angles used in the DQA algorithm. It is not recommended to change these values as they encompass a full electrical revolution of the motor to ensure correct operation.

7.11.2.2.7 FOI DQA Test Angle Offset

FOI DQA Test Angle Offset defines an offset that is applied to *FOI DQA Test Angles*. In operation, the 1st element of this array is applied to *FOI DQA Test Angles*. If the tests are determined to be successful then FOI is complete, however, if not, then the 2nd element of *FOI DQA Test Angle Offset* is applied to *FOI DQA Test Angles* and the DQA algorithm is run again. If the DQA tests fail a 2nd time then a Class 1 Diagnostic Error is reported (E402).

7.11.2.3 General Operation

To use the DQ Alignment FOI:

1. Ensure the number of motor poles and encoder line count is correct.
2. Set *FOI Type* = '1' to select DQ Alignment as the FOI mode.
3. Configure DQA parameters as required.
4. Once FOI is initiated via the *Master Control Word*, the rotor will move to align with the stator magnetic field.
5. The rotor will then be rotated through the test electrical angles.
6. If the test is not passed, steps 3 and 4 will be repeated with a different offset applied.
7. If DQA is successful, the servo drive will enter the requested operation mode and follow NC commands.

7.11.2.4 Troubleshooting

The following common errors may occur during DQ Alignment FOI.

E403: DQ Alignment Invalid Movement Detected

Motor movement error (the difference between the commutation angle and the measured electrical angle) exceeded specified tolerance (0.15 elec rev or 54 elec deg). This error is usually triggered by the motor not moving, moving too far or not enough, or moving in the wrong direction.

E405: DQ Alignment Current Control Error

Alignment current error (the difference between the command and measured motor current) exceeded specified tolerance (IDN 33071). This error may be caused by sensor failure, current loop poorly tuned, Safe Torque Off (STO) or motor/cable fault.

7.11.2.5 Testing and Diagnosing Faults

1. Using the DQ Alignment Algorithm page in MotionBench, monitor the motor currents and the electrical angle, while FOI is executing.
2. If E403 is triggered, check the relative movements between Motor Electrical Angle and Motor Commutation Command.
 - *Figure 7-12* shows an example of a good alignment. Notice how the offset between the Motor Commutation Command and the Motor Electrical Angle remains approximately constant.
 - *Figure 7-13* shows an example of the Motor Commutation Command and the Motor Electrical Angle diverging. This will lead to E403. The likely cause is: incorrect number of motor poles (IDN 32774) and/or incorrect encoder resolution (IDN 116).
 - *Figure 7-14* shows an example of the Motor Commutation Command and the Motor Electrical Angle moving in opposite directions. This will lead to E403. The likely cause is: incorrect motor phase sequence or inverted encoder feedback (IDN 277).

3. If E405 is triggered, check the Alignment Current Command versus Alignment Current Feedback.
 - [Figure 7-12](#) shows an example of a good alignment. Notice how the Alignment Current Feedback follows Alignment Current Command correctly.
 - [Figure 7-14](#) shows an example of the Alignment Current Feedback not following the Alignment Current Command correctly. Likely cause: DC bus not powered, Safe Torque OFF (STO) active, or current loop properly tuned.

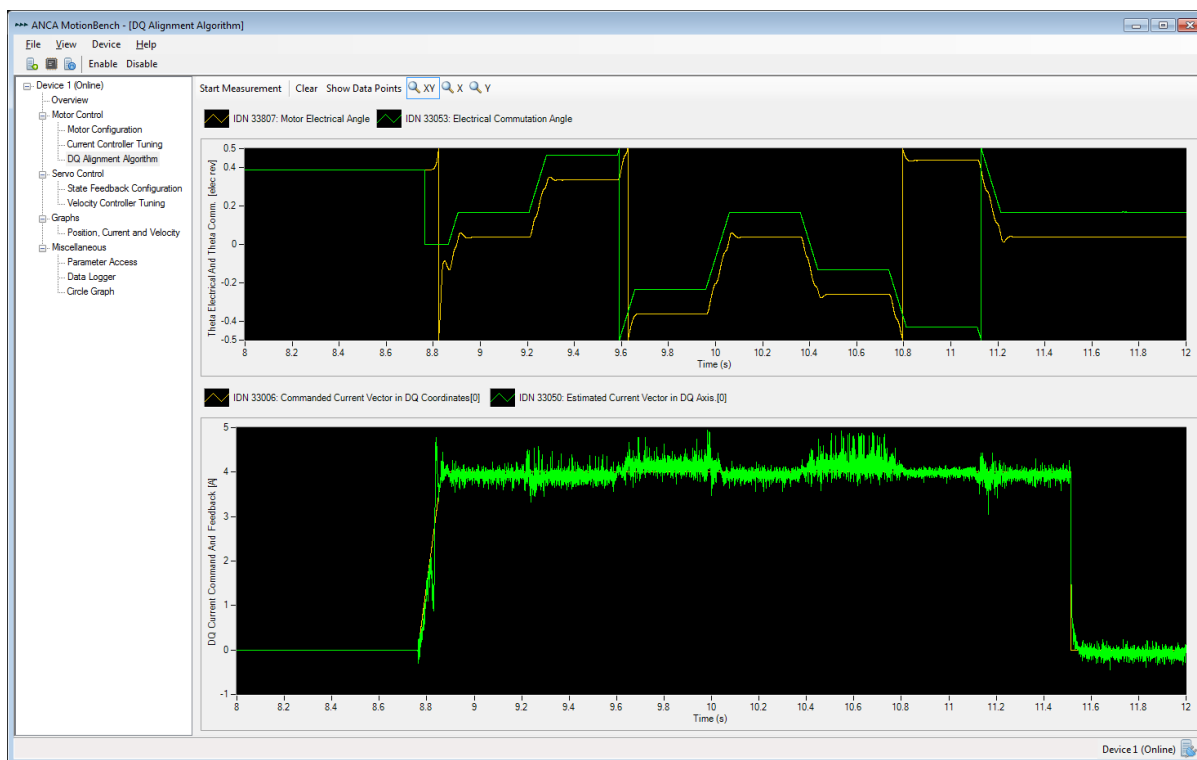


Figure 7-12 - Motor Electrical Angle follows Motor Commutation Command correctly. Alignment Current Feedback follows Alignment Current Command correctly.

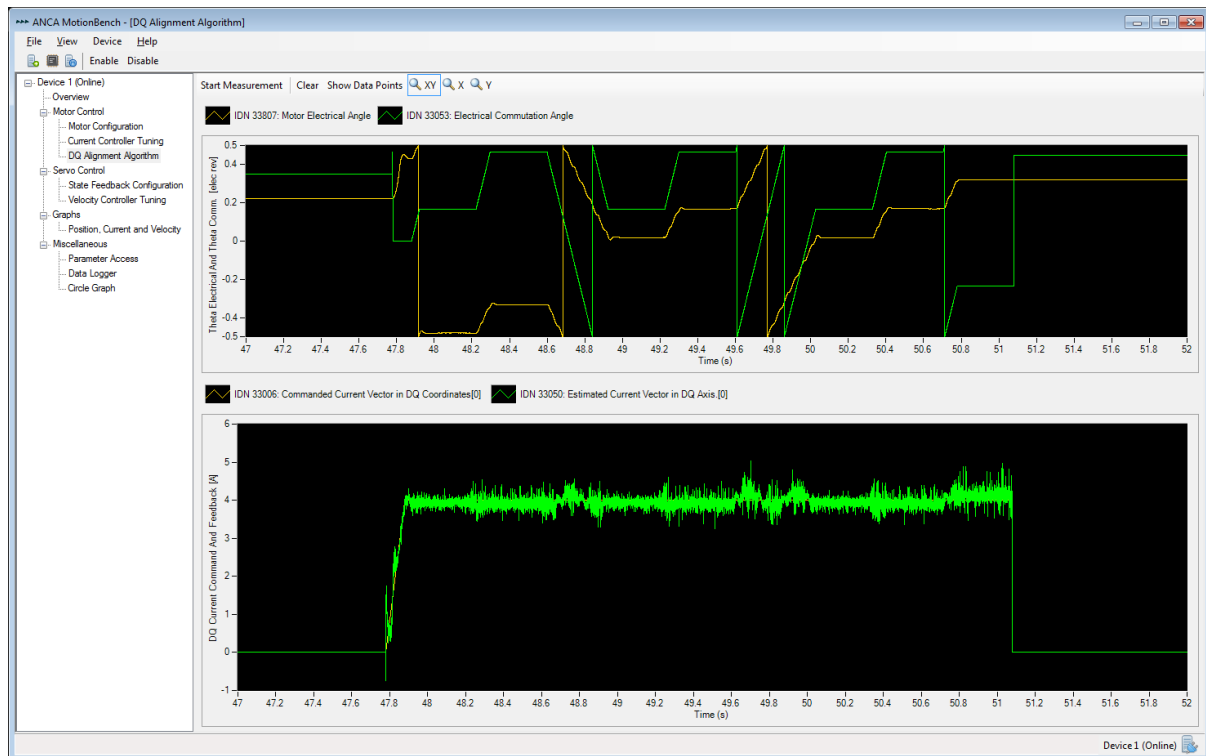


Figure 7-13 - Motor Electrical Angle diverges from Motor Commutation Command. Likely cause: incorrect number of motor poles and/or incorrect encoder resolution.

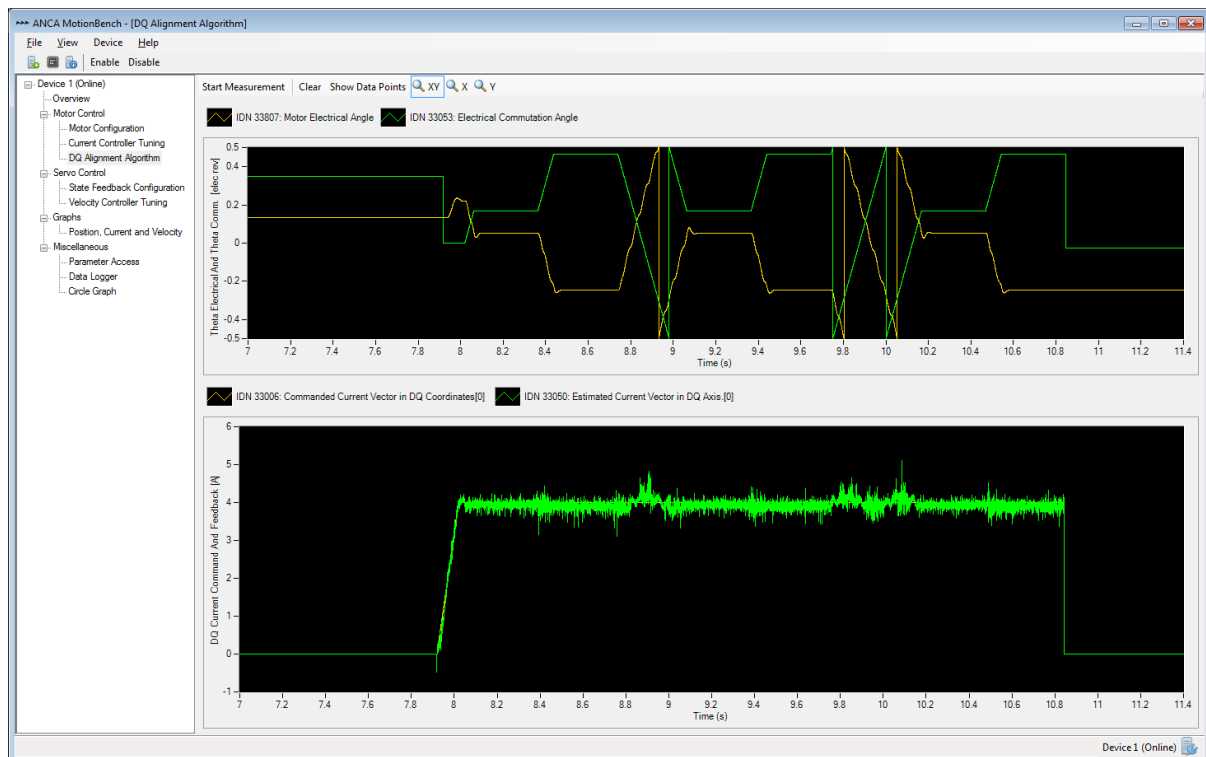


Figure 7-14 - Motor Electrical Angle moves in the opposite direction to the Motor Commutation Command. Likely cause: incorrect motor phase sequence or inverted encoder feedback.

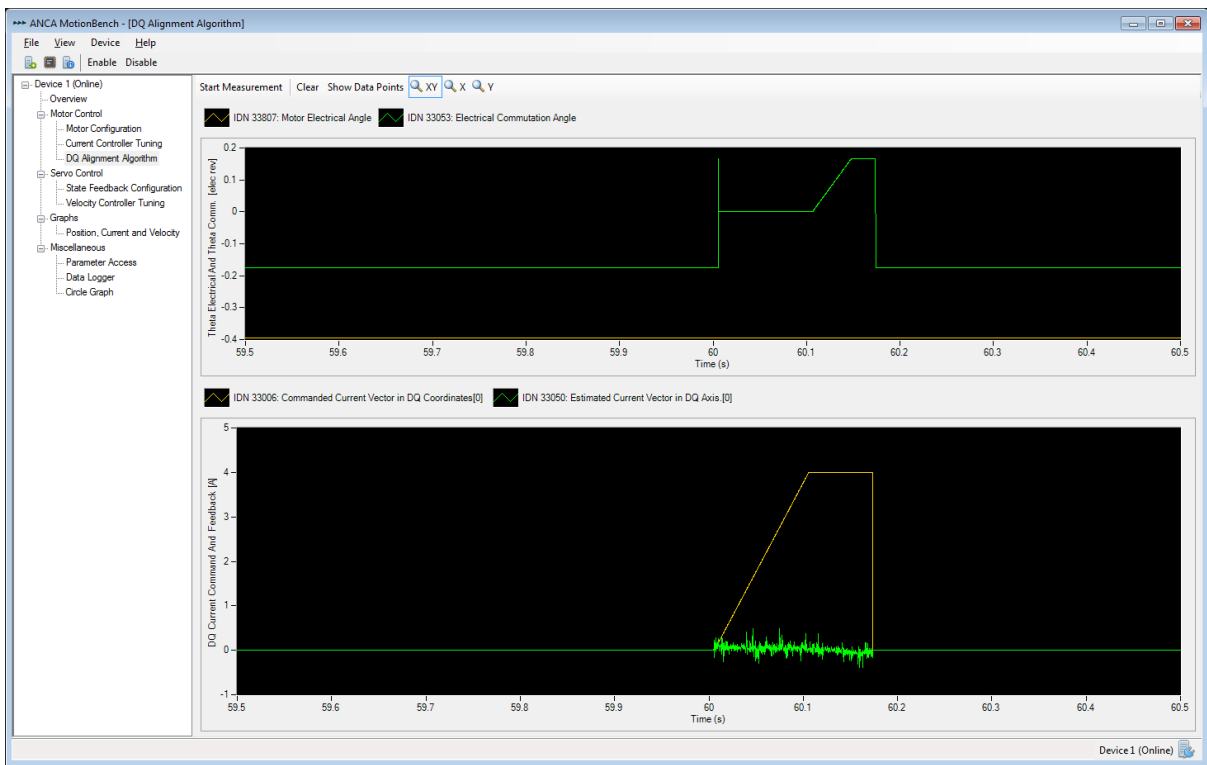


Figure 7-15 - Alignment Current Feedback does not follow Alignment Current Command. Likely cause: DC bus not powered, or current loop poorly tuned.

7.11.3 Commutation Track

7.11.3.1 Description



NOTE: This feature is only available on the AMD5x series servo drives.

Analogue commutation tracks (sinCOM / cosCOM) of an analogue (sin / cos) incremental encoder are used to provide once per electrical revolution feedback. They are typically aligned to the index pulse i.e. the index pulse is located at the zero degree location of the commutation track (sin = 0 & cos = 0). If the offset angle between index pulse and the stator magnetic field is known, this feedback can be used to perform FOI. Typically this offset angle is zero, if this is not the case, then an offset can be added using the *Electrical Angle Pre-Set Offset*.

IDN	Description	Data Type	Units	Default
				AMD5x
P-0-0298 / 33066	Electrical Angle Pre-Set Offset	Signed Integer (4 bytes)	10^{-4} Electrical Rev	0
P-0-0312 / 33080	Absolute Feedback Type	Unsigned Integer (2 bytes)	None	0
P-0-0318 / 33086	Commutation Track CosSin Offset	Unsigned Integer (2 bytes) Array	ADC	32768, 32768
P-0-0319 / 33087	Commutation Track CosSin Scale Factor	Signed Integer (2 bytes)	None	-1

7.11.3.2 Definitions

7.11.3.2.1 Electrical Angle Pre-Set Offset

The *Electrical Angle Pre-Set Offset* defines the angle between the zero degree location of the commutation track ($\sin(0) = 0$ & $\cos(0) = 1$) and the direction of the magnetic field generated by U phase current.

7.11.3.2.2 Absolute Feedback Type

The *Absolute Feedback Type* is used to determine the method of absolute feedback when the *FOI Type* is set to 'absolute'. Set this to '2' to use fg Wire-Saving FOI.

Value	Description
0	None (advanced users only)
1	Commutation Track
2	UVW Wire-Saving

Table 7-14 - Absolute Feedback Type definition

7.11.3.3 General Operation

To use Commutation Track FOI:

1. Ensure the number of motor poles and encoder line count is correct.
2. Set *FOI Type* = '3' to select Absolute Feedback as the FOI mode.
3. Set *Absolute Feedback Type* = '1'.
4. Once FOI is initiated via the *Master Control Word*, power cycle the encoder and latch the commutation track feedback.
5. If successful, the servo drive will enter the requested operation mode and follow NC commands.

7.11.3.4 Troubleshooting

E406: Absolute Encoder Alignment Error.

Absolute encoder used for field orientation initialisation has failed to latch an alignment angle.

Possible causes for this error are:

1. Encoder is faulty. Power cycling the drive/encoder may resolve the issue.
2. Drive is faulty. Please contact ANCA Motion for support.

E013: Encoder Amplitude Low - Motor Commutation Track

The magnitude of the signals coming from the motor encoder analogue commutation track is too low. In particular, the amplitude of the encoder commutation track signals ($\cos^2 + \sin^2$) is below its minimum limit (10000, hard coded). That is, the following condition is true

$$(\text{ID33364}[0] - 32768)^2 + (\text{ID33364}[1] - 32768)^2 < 10000^2$$

Possible causes for this error are:

1. Encoder cable is disconnected.
2. Encoder cable is wired incorrectly.
3. Encoder does not support commutation track.
4. Encoder is not outputting the correct voltage.
5. Encoder is faulty.

6. Drive is faulty. Please contact ANCA Motion for support.

E014: Commutation Track Amplitude High

The magnitude of the signals coming from the motor encoder analogue commutation track is too high. In particular, the amplitude of the encoder commutation track signals ($\cos^2 + \sin^2$) is larger than its maximum limit (32700, hard coded). That is, the following condition is true

$$(\text{ID33364}[0] - 32768)^2 + (\text{ID33364}[1] - 32768)^2 > 32700^2$$

Possible causes for this error are:

1. Encoder cable is wired incorrectly.
2. Encoder does not support commutation track.
3. Encoder is not outputting the correct voltage.
4. Encoder is faulty.
5. Drive is faulty. Please contact ANCA Motion for support.

7.11.4 Wire-Saving UVW

7.11.4.1 Description



NOTE: This feature is only available on the AMD2000 series servo drives using the 2nd encoder channel.



NOTE: This feature is only available with ANCA Motion Alpha series motors.

The UVW Wire-Saving FOI technique utilises a feature of the ANCA Motion Alpha servo motors to obtain hall sensor feedback from the encoder for a short period of time upon first powering up the encoder. This allows an approximate electrical angle offset to be calculated to perform commutation with.

The result of this method of FOI is only accurate to within +/-30 degrees and should therefore be combined with [Alignment Off Index Pulse](#) to maximise the output torque and efficiency of the motor.

IDN	Description	Data Type	Units	Default
				AMD2000
P-0-0312 / 33080	Absolute Feedback Type	Unsigned Integer (2 bytes)	None	0
P-0-0315 / 33083	Hexant Input Map	Signed Integer (2 bytes)		[-1, 5, 1, 0, 3, 4, 2, -1]
P-0-0316 / 33084	Hexant Edge Angles	Signed Integer (4 bytes)	Electrical Rev	1/6 * [0, 1, 2, 3, -2, -1]

7.11.4.2 Definitions

7.11.4.2.1 Absolute Feedback Type

The *Absolute Feedback Type* is used to determine the method of absolute feedback when the *FOI Type* is set to 'absolute'. Set this to '2' to use UVW Wire-Saving FOI.

Value	Description
0	None (advanced users only)
1	Commutation Track

2	UVW Wire-Saving
---	-----------------

Table 7-15 - Absolute Feedback Type definition

7.11.4.3 Hexant Input Map and Hexant Edge Angle

The *Hexant Input Map* and *Hexant Edge Angle* are used as a look-up-table to map the detected hexant to the electrical angle. This mapping is encoder specific and default values are for Tomagawa wire-saving digital encoder, standard ANCA Motion alpha series motor encoder. '-1' indicates an invalid state.

7.11.4.4 General Operation

To use UVW Wire-Saving FOI:

1. Ensure the number of motor poles and encoder line count is correct.
2. Set *FOI Type* = '3' to select Absolute Feedback as the FOI mode.
3. Set *Absolute Feedback Type* = '2'.
4. Configure *Alignment Off Index Pulse*.
5. Once FOI is initiated via the *Master Control Word*, power cycle the encoder and latch the hall sensor feedback.
6. If successful, the servo drive will enter the requested operation mode and follow NC commands.
7. *Alignment Off Index Pulse* will be triggered when the next index pulse is encountered during normal operation.



WARNING: While this FOI procedure is being performed the axis should not move or be moved by external forces. If this is likely in a given application, it is recommended that a braked motor or other FOI method is used.

7.11.4.5 Troubleshooting

The following common errors may occur during the UVW Wire-Saving FOI.

E406: Absolute Encoder Alignment Error.

Absolute encoder used for field orientation initialisation has failed to latch an alignment angle.

Possible causes for this error are:

1. Encoder is faulty. Power cycling the drive/encoder may resolve the issue.
2. Motor phases are not in the correct order.
3. Drive is faulty. Please contact ANCA Motion for support.

7.11.5 Acceleration Observer

7.11.5.1 Description

The *Acceleration Observer* technique is a minimum displacement FOI method. This means that it is suited to applications where excessive moment on the motor during initialisation is not acceptable.

The general idea behind this technique is to inject *d*-axis current and repeatedly sweep the commutation angle through 360 degrees. The frequency of the commutation angle must be fast enough so that the rotor does not begin tracking the rotating magnetic field, but slow enough to ensure acceptable convergence to a result. Given this, the algorithm consists of the following major components:

1. **Stimulus generation** - generate *d*-axis current with defined amplitude and frequency.

2. **Data acquisition** - measure the motor reaction (electrical displacement) as a result of the stimulus.
3. **Torque estimation** - estimate electrical torque as a function of electrical angle.
4. **Alignment angle offset estimation** - estimate the electrical angle offset for optimum torque generation and efficiency.
5. **Alignment angle offset validation** - initiate motion using the estimated alignment angle offset to confirm motor movement is as expected.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-1132 / 33900	FOI AO Stimulus Amplitude	Signed Integer (4 bytes)	Standard Current	3 A	3 A
P-0-1133 / 33901	FOI AO Stimulus Frequency	Signed Integer (4 bytes)	10^{-4} Electrical Rev/s	80 elec rev/s	80 elec rev/s
P-0-1135 / 33903	FOI AO Repeat Count	Unsigned Integer (2 bytes)	None	0	0
P-0-1136 / 33904	FOI AO Stimulus Time	Unsigned Integer (2 bytes)	250 μ s	2 s	2 s
P-0-1137 / 33905	FOI AO Max Deviation	Signed Integer (4 bytes)	10^{-4} Electrical Rev	0.05 elec rev	0.05 elec rev
P-0-1145 / 33913	FOI AO Model Inertia	Signed Integer (4 bytes)	Kg mm ²	99.4718	99.4718
P-0-1148 / 33916	FOI AO LPF1 Coefficient	Unsigned Integer (2 bytes)	2^{-15}	13116	13116
P-0-1156 / 33924	FOI AO Alignment Current Tolerance	Signed Integer (4 bytes)	Standard Current	0.5 A	0.5 A
P-0-1160 / 33928	FOI AO Validation Velocity Threshold	Signed Integer (4 bytes)	10^{-4} Mechanical Rev/s	0.2 mech rev/s	0.2 mech rev/s

7.11.5.2 Definitions

7.11.5.2.1 FOI AO Stimulus Amplitude

The *FOI AO Stimulus Amplitude* determines the magnitude of current used during Analogue observer operation. This applies to both the estimation and validation phases.

7.11.5.2.2 FOI AO Stimulus Frequency

The *FOI AO Stimulus Frequency* determines the frequency at which the commutation angle is swept.

7.11.5.2.3 FOI AO Repeat Count

The *FOI AO Repeat Count* determines the number of times the commutation angle is swept for a duration of *FOI AO Stimulus Time*. Set this value to 0 to only perform the sweep once.

7.11.5.2.4 FOI AO Stimulus Time

The *FOI AO Stimulus Time* determines the duration for which the commutation angle is swept.

7.11.5.2.5 FOI AO Max Deviation

The *FOI AO Max Deviation* determines the maximum allowable movement during operation of the Analogue Observer.

7.11.5.2.6 FOI AO Model Inertia

The *FOI AO Model Inertia* is the approximate inertia of the axis used in the inertia torque estimator model.

7.11.5.2.7 FOI AO LPF1 Coefficient

The *FOI AO LPF1 Coefficient* determines the cut-off frequency of the low pass filter used to separate the estimated torque from high frequency noise. Refer to the configuration section below for information on how to determine this value.

7.11.5.2.8 FOI AO Alignment Current Tolerance

FOI AO Alignment Current Tolerance specifies the allowed difference between the commanded and actual current. Exceeding this value results in a Class 1 Diagnostic Error (E412). This parameter should not require modification in general operation, exceeding this value is a result of poor current control which can indicate poor current loop tuning.

7.11.5.2.9 FOI AO Validation Velocity Threshold

During the validation phase, if the measured velocity is above the *FOI AO Validation Velocity Threshold* the FOI is determined to be successful. It is important that this threshold is higher than any encoder feedback noise and is greater than 3 encoder counts when using a digital encoder. A lower value will result in reduced motion during the validation operation.

7.11.5.3 Configuration

The following guidelines can be used for configuring the Acceleration Observer on different axis types.

- Model configuration:
 - Axis inertia (*FOI AO Model Inertia*) does not need to be exact; typically a larger value will ensure that a suitable torque estimate as a function of stimulus electrical angle is achieved.
- Stimulus configuration:
 - *FOI AO Stimulus Amplitude* should be selected through experimentation. Start with a small value and then gradually increased in small steps. A large amplitude can result in excessive vibration and mechanical damage.
 - *FOI AO Stimulus Frequency* should be selected through experimentation. Start with a large value and then gradually reduce in small steps. A small value may result in the motor tracking the rotating magnet field; resulting in excessive displacement. In short, the stimulus frequency must be well above the axis position-loop bandwidth.
 - The total time taken by FOI AO = *FOI AO Stimulus Time* x *FOI AO Repeat*. For example, if *FOI AO Stimulus Time* = 0.1875 [sec] and *FOI AO Repeat* = 10, then the test time = 1.875 [sec].
- Alignment angle offset estimation:
 - *FOI AO LPF1 Coefficient* should be set based on the *FOI AO Stimulus Frequency* according to the following formula:

$$\frac{7.85 \times 10^{-7} \times FOI\ AO\ Stimulus\ Frequency}{1 + 7.85 \times 10^{-7} \times FOI\ AO\ Stimulus\ Frequency} \times 2^{15}$$

7.11.5.4 General Operation

To use the Acceleration Observer FOI:

1. Ensure the number of motor poles and encoder line count is correct.
2. Set *FOI Type* = '4' to select Acceleration Observer as the FOI mode.
3. Configure Acceleration Observer parameters as required.
4. Once FOI is initiated via the *Master Control Word*, the rotor will vibrate at the configured frequency and amplitude.
5. The axis will then rotate a short distance to validate the result.
6. If FOI is successful, the servo drive will enter the requested operation mode and follow NC commands.

7.11.5.5 Troubleshooting

The following common errors may occur during the Acceleration Observer FOI.

E414: Acceleration Observer Torque Response Amplitude Low

The fundamental frequency component of the torque response is below the minimum amplitude threshold (0.1 Nm). The system may not have been stimulated correctly and hence the alignment estimate may not be accurate.

E415: Acceleration Observer Torque Response Amplitude High

The fundamental frequency component of the torque response is above the maximum amplitude threshold (2 Nm). The system may not have been stimulated correctly and hence the alignment estimate may not be accurate.

E416: Acceleration Observer Torque Response Mean Squared Error

The difference between the predicted torque and the estimated (measured) torque is greater than the threshold (1 Nm).

7.11.6 Braked Compliance

7.11.6.1 Description



NOTE: This feature is recommended for use with analogue motor encoders only.

The *Braked Compliance* FOI algorithm is designed for axes locked by brakes that cannot be disengaged before the drives are enabled. Typical applications include vertical axes (z-axis) without com tracks and x- and y-axis using linear motors.

This technique injects *d*-axis current and sweeps the commutation angle through 360 electrical degrees. During the execution of the algorithm the commutation angle is stepped through a configurable number of steps. By default, 512 steps are used, which leads to a resolution of $360/512=0.703$ electrical degrees. The commutation angle is held for 4ms at each step, allowing to detect valid axis relative displacement to the brake.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-0301 / 33069	FOI Alignment Current	Signed Integer (4 bytes)	Standard Current	4.0 A	3.0 A
P-0-0302 / 33070	FOI Alignment Current Slew	Signed Integer (4 bytes)	Standard Current	0.01 A	0.01 A
P-0-0303 / 33071	FOI Alignment Current Tolerance	Signed Integer (4 bytes)	Standard Current	0.5 A	1.0 A
P-0-1116 / 33934	Braked Compliance - Test Angle Step Number	Unsigned Integer (2 bytes)	Steps	512	512
P-0-1117 / 33935	Braked Compliance - Minimum Positive Deflection	Signed Integer (4 bytes)	10^{-6} electrical rev	0.0003 elec rev	0.0003 elec rev
P-0-1118 / 33936	Braked Compliance - Maximum Absolute Deflection	Signed Integer (4 bytes)	10^{-6} electrical rev	0.005 elec rev	0.005 elec rev

7.11.6.2 Definitions

7.11.6.2.1 FOI Alignment Current

The *FOI Alignment Current* specifies the current used during Braked Compliance movements. This value is axis and motor specific and may need to be adjusted depending on the application. The current needs to be sufficient to move the motor to the test points and within the current rating of the motor.

7.11.6.2.2 FOI Alignment Current Slew

FOI Alignment Current Slew specifies the initial rate of current increase to *FOI Alignment Current*.

7.11.6.2.3 FOI Alignment Current Tolerance

FOI Alignment Current Tolerance specifies the allowed difference between the commanded and actual current. Exceeding this value results in a Class 1 Diagnostic Error (E405). This parameter should not require modification in general operation, exceeding this value is a result of poor current control which can indicate poor current loop tuning.

7.11.6.2.4 Braked Compliance - Test Angle Step Number

Braked Compliance - Test Angle Step Number defines the number of steps by which the field is swept through in one electrical revolution. The minimum possible value is 64 and the value must be a power of two (if this condition is not met, this will be rounded to by the drive).

7.11.6.2.5 Braked Compliance - Minimum Positive Deflection

Braked Compliance - Minimum Positive Deflection specifies the minimum allowed axis positive deflection for the final Braked Compliance result. It is recommended that this parameter is set to the equivalent of 1 to 2 encoder quadrature counts (electrical revolution = mechanical revolution / motor pole pairs). If movement is measured below this value at any step then a Class 1 Diagnostic Error (E408) will be triggered.

7.11.6.2.6 Braked Compliance - Maximum Absolute Deflection

Braked Compliance - Maximum Absolute Deflection defines the maximum allowed axis deflection for each step. If movement is measured above this value at any step then a Class 1 Diagnostic Error (E410) will be triggered. This threshold can be used to detect unexpected scenarios such as the brake being disabled.

7.11.6.3 General Operation

To use the Braked Compliance FOI:

1. Ensure the number of motor poles and encoder line count is correct.
2. Set *FOI Type* = '5' to select Braked Compliance as the FOI mode.
3. Configure Braked Compliance parameters as required.
4. Once FOI is initiated via the Master Control Word, the rotor will make a series of movements as the algorithm steps through 360 electrical degrees. This may not be visible to the naked eye due to the motor brake action but should be visible through encoder feedback.
5. If FOI is successful, the servo drive will enter the requested operation mode and follow NC commands.

7.11.6.4 Troubleshooting

The following common errors may occur during the Braked Compliance FOI.

E405: FOI Current Control Error

Alignment current error, the difference between the commanded alignment current and the measured motor current, exceeds a specified tolerance (IDN 33071).

This error may be caused by:

1. Incorrect DC bus voltage
2. Current loop poorly tuned
3. Motor/cable fault
4. Sensor failure

E408: Excessive Movement Detected

E408 is triggered if the axis movement (deflection) is larger than a value defined by IDN 33936 in [elec rev].

This error may be caused by:

1. A faulty brake
2. The brake mistakenly disengaged
3. High testing current (IDN 33069)

E410: Insufficient Positive Movement Detected

E410 is triggered if the axis maximum movement (deflection) in the positive direction is less than a value defined by IDN 33935 in [elec rev].

This error may be caused by:

1. Low testing current (IDN 33069)

7.11.7 Alignment Off Index Pulse

7.11.7.1 Description

For motors with an incremental encoder, it is possible to align the electrical angle off the index pulse in order to improve the accuracy of electrical angle. To this end, the index pulse offset (namely the distance between the index pulse and magnetic field generated by U phase current) needs to be commissioned for each individual motor and encoder assembly. Failure to properly configure this feature can potentially result in inefficient motor operation and potential loss of motor control.

When using rotary motors, an index pulse occurs once per mechanical revolution and always the same electrical angle every revolution (the number of electrical revolutions per mechanical revolution = number of pole pairs). This means that as soon as an index pulse is seen, the alignment off index pulse can be run. When using a linear motor with a linear encoder with a repeated index pulse, it is unlikely that each index pulse is repeated at a multiple of the pole pitch. Therefore, only a single index pulse along the entire stroke must be used for Alignment Off the Index Pulse. In this case, Alignment Off the Index Pulse can be configured to run when the index pulse is found during homing.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-0288 / 33056	FOI AIP Control Word	Unsigned Integer (2 bytes)	Binary	0	0
P-0-0289 / 33057	FOI AIP Complete	Unsigned Integer (2 bytes)	Boolean	N/A	N/A
P-0-0290 / 33058	FOI AIP Difference Threshold	Signed Integer (4 bytes)	10^{-4} electrical rev	0.17 elec rev	0.17 elec rev
P-0-0294 / 33062	FOI AIP Control	Unsigned Integer (2 bytes)	None	0	0
P-0-0295 / 33063	FOI AIP Measured Electrical Angle at Index Pulse	Signed Integer (4 bytes)	10^{-4} electrical rev	N/A	N/A
P-0-0296 / 33064	FOI AIP Commissioned Electrical Angle at Index Pulse	Signed Integer (4 bytes)	10^{-4} electrical rev	0	0

7.11.7.2 Definitions

7.11.7.2.1 FOI AIP Control Word

The *FOI AIP Control Word* allows configuration of Alignment Off Index Pulse behaviour.

Value	Description
0	Use any index pulse
1	Use index pulse found during homing

Table 7-16 - FOI AIP Control Word definition

7.11.7.2.2 FOI AIP Complete

FOI AIP Complete is a Boolean value that is '1' when the Alignment Off Index Pulse process has been completed.

7.11.7.2.3 FOI AIP Difference Threshold

The *FOI AIP Difference Threshold* is the maximum allowed difference between the FOI result before the Alignment Off Index Pulse was run and afterwards. Essentially, if the difference between *FOI AIP Measured Electrical Angle at Index Pulse* and *FOI AIP Commissioned Electrical Angle at Index Pulse* is greater than *FOI AIP Difference Threshold*, a Class 1 Diagnostic Error (E403) will be triggered.

7.11.7.2.4 FOI AIP Control

The *FOI AIP Control* is used to enable and disable the Alignment Off Index Pulse Function.

Value	Description
0	Disable
1	Commissioning
2	Enable

Table 7-17 - FOI AIP Control definition

7.11.7.2.5 FOI AIP Measured Electrical Angle at Index Pulse

When *FOI AIP Control* is not set to disable, *FOI AIP Measured Electrical Angle at Index Pulse* will be populated with the electrical angle at the index pulse based on the electrical angle offset calculated in the initial FOI. This can be used for commissioning when combined with an accurate initial FOI result e.g. a well configured DQ Alignment.

7.11.7.2.6 FOI AIP Commissioned Electrical Angle at Index Pulse

The *FOI AIP Commissioned Electrical Angle at Index Pulse* is the value that is configured by the user to be applied by Alignment Off Index Pulse.

7.11.7.3 Commissioning and Configuration

If a motor model has a fixed relationship between its stator and index pulse location then commissioning only needs to be performed when a new motor model is employed. Otherwise, the following commissioning procedure will need to be performed for every single motor individually.

Step1: Determine the angle between the index pulse and the *d*-axis, that is, the index pulse offset.

1. Ensure that the drive is disabled.
2. Set *Alignment Off Index Pulse* to *Commissioning Mode* (*FOI AIP Control* = 1).
3. Set *FOI Control* to *Always* (= 2).
4. Set *FOI Type* to *DQ Alignment* (= 1).
5. Monitor *FOI AIP Complete* and *FOI AIP Measured Electrical Angle at Index Pulse*.
6. Enable the drive and allow the *DQ Alignment* algorithm to complete. At this stage *FOI AIP Complete* should be zero. Slowly move the motor until the index pulse is found (*FOI AIP Complete*). Record the value of *FOI AIP Measured Electrical Angle at Index Pulse*. Disable the drive and repeat this step a number of times to ensure that the calculated value in *FOI AIP Measured Electrical Angle at Index Pulse* is repeatable.

Step2: Enable alignment off index pulse.

1. Ensure that the drive is disabled.
2. Set *FOI AIP Commissioned Electrical Angle at Index Pulse* to the average value of *FOI AIP Measured Electrical Angle at Index Pulse* that was recorded in the previous step.
3. Set *Alignment Off Index Pulse* to *Enable* (= 2).
4. Set *FOI Type* as required for the application.
5. Set *FOI Control* to *Once* (IDN 33060 = 1).
6. Save these parameters to the drive's non-volatile memory or control device/unit.

Once this procedure is complete, on subsequent power cycles of the drive the alignment angle will automatically be corrected once the index pulse is located.

7.11.7.4 General operation

To use the Alignment Off Index Pulse:

1. Complete the procedure outlined in [7.11.7.3 Commissioning and Configuration](#) to setup the required parameters.
2. The selected FOI method will be run when the servo drive is enabled for the first time with the *Master Control Word*.
3. Once FOI is successful, the servo drive will enter the requested operation mode and follow NC commands.
4. The Alignment Off Index Pulse will occur depending on the value of the *FOI AIP Control Word*:
 - a) If *FOI AIP Control Word* = 0: As soon as the index pulse of the motor is encountered through normal operation, Alignment Off Index Pulse will occur.
 - b) If *FOI AIP Control Word* = 1: Alignment Off Index Pulse will only be triggered during the homing procedure when the index pulse is found.

7.11.7.5 Troubleshooting

The following common errors may occur during the Alignment Off the Index Pulse.

E403: Alignment Off Index Pulse Error

The commissioned index pulse offset value (IDN 33064) is determined by a machine commissioner. Upon machine power up the motor will execute Field Orientation Initialisation to achieve field alignment. Subsequent to this, and once the index pulse has been located, a new estimate of the index pulse offset (IDN 33063) is calculated. This error is produced when the difference between this estimate (IDN 33063) and the commissioned value (IDN 33064) exceeds the tolerance defined by IDN 33058, where the default value is 0.17 [elec rev]. Possible causes of this error are:

1. Incorrect encoder configuration by the commissioner (i.e. UVW hexant binary)
5. A change in the stored commissioned value
6. A change in the hardware configuration (relative movement between the motor and encoder)

7.12 LED Display User Menu

7.12.1 Description



NOTE: This feature is only available on the AMD2000 series servo drives.

The AMD2000 series servo drive provides a facility to view and edit IDNs via the LED display and push buttons located on the front of the device. The available menu items are based on parameters saved in [Non-Volatile Parameters](#) and additional read only parameters.

IDN	Description	Data Type	Units	Default
				AMD2000
P-0-0052 / 32820	Menu Display List (8 elements)	Unsigned Integer (2 bytes) Array	IDN	51,40,0,0,...,0
P-0-0354 / 33122	Selected NV Parameter(s) List (50 elements)	Unsigned Integer (2 bytes)	IDN	0,0,...,0

7.12.2 Definitions

7.12.2.1 Menu Display List

The *Menu Display List* contains IDNs for read-only parameters to be displayed in the LED Display User Menu. This can include items such as position and velocity feedback.

7.12.2.2 Selected NV Parameter(s) List

The *Selected NV Parameter(s) List* defines the IDNs to be saved in the servo drive's Non-Volatile Memory and the editable parameters available in the LED Display User Menu. Refer to [7.16 Non-Volatile Parameters](#).

7.12.3 General Operation

Perform the following steps to modify a parameter listed in the *Selected NV Parameter(s) List* via the LED user menu.

1. Enter the parameter selection screen by pressing the third button from the left as shown below.

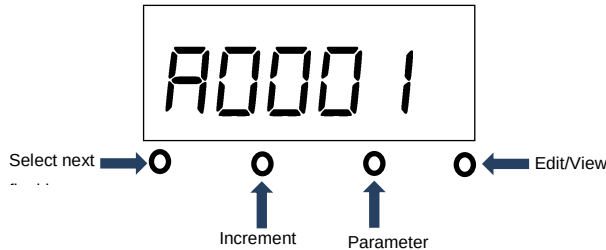


2. In the parameter selection screen, cycle through available IDNs and select the item to be modified. Note that IDNs listed in *Menu Display List* cannot be modified. Press the select button to enter the array index (if the IDN is variable length) or parameter edit screen.

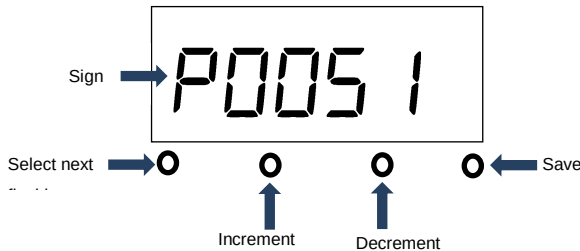


3. If a variable length IDN is selected the menu will show the array index screen. This allows the user to select the array element that will appear in the parameter edit/view screen. To input the index use the 'Select next' button to choose the appropriate digit (as indicated by the flash) and the 'increment' button

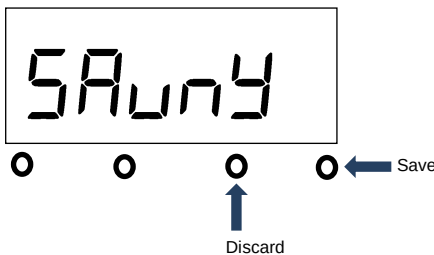
to the set the desired value. Note that the selected digit will wrap if incremented higher than the maximum number of elements in the array. To proceed to the parameter edit/view screen press the edit/view button.



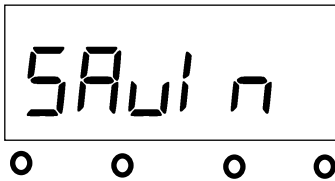
- The parameter edit/view allows the user to modify operation data of IDNs listed in the *Selected NV Parameter(s) List* and view the operation data of IDNs listed in the *Menu Display List*. To modify the value use the 'Select next' button to choose the appropriate digit (as indicated by the flash) and the 'increment' button to the set the desired value. It is also possible to change the value from negative to positive or vice versa by selecting the 'P' (positive) on n (negative) character and using the increment or decrement button. Press the save button to move to the save screen, note that if the parameter is read only the parameter select screen will be shown instead.



- The next screen asks for confirmation before saving to the non-volatile memory. If the user selects 'n' the changes will be discarded and the parameter select screen will be displayed instead. If the save button is pressed the saving screen will be displayed.



- The saving screen is displayed while the drive is writing the modified value to non-volatile memory. Once completed the parameter selection screen will be displayed.



7.12.4 Error Codes

The following error conditions may occur when editing parameters with the LED menu. All error codes will be prefixed with the 'E-' characters. Press any button to continue.

Code	Description
1	Array index exceeds the maximum length of the array.
2	Invalid data type.
4	Input value is greater than the maximum allowed for this IDN.
5	Input value is lesser than the minimum allowed for this IDN.
6	IDN has an invalid conversion factor.
7	IDN does not exist.
8	Error when triggering non-volatile memory procedure command.
9	IDN is read only.
10	Error when saving value to non-volatile memory.
11	IDN has an invalid display type.

Table 7-18 - LED Display User Menu error codes

7.13 Motor Brake Control

7.13.1 Description

The AMD servo drive supports braked motors, automatically engaging and disengaging the brake when the motor is disabled and enabled to prevent any uncontrolled movement. The brake can engage while the drive is enabled and halted for a configurable period of time to compensate for any mechanical delay in the brake engaging.

The motor brake is controlled via a dedicated brake output or an assigned general purpose digital output depending on the servo drive model. Refer to the [AMD Servo Drive User Guide](#) for more information.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-0308 / 33076	Brake Engage Hold Time	Unsigned Integer (2 bytes)	4×10^{-3} s	0	0
P-0-0309 / 33077	Brake Dis-Engage Hold Time	Unsigned Integer (2 bytes)	4×10^{-3} s	0.3 s	0.3 s
P-0-0578 / 33346	Motor Brake Release	Unsigned Integer (2 bytes)	Boolean	N/A	N/A

7.13.2 Definitions

7.13.2.1 Brake Engage Hold Time

The *Brake Engage Hold Time* is a configurable period of time after the brake is commanded to engage (power removed) before the motor is de-energised. This allows the servo drive to compensate for delay in engaging the brake.

7.13.2.2 Brake Dis-Engage Hold Time

The *Brake Dis-Engage Hold Time* is a configurable period of time before the brake is commanded to dis-engage (power applied) after the motor is energised. This allows the servo drive to energise the motor with the brake still engaged to prevent any uncontrolled movement upon enable.

7.13.2.3 Motor Brake Release

The *Motor Brake Release* signal reflects the status of the brake release digital output. When this signal is 1 the brake output is energised to release the brake. To assign a general digital output to the brake function, map this signal to the selected digital output. Refer to [7.6 Digital Output](#) for more information.

7.14 Motion Constraints and Limits

7.14.1 Description

In practice most machines, and thereby machine axes, cannot move through an unlimited range of motion. People or machine parts can impede the motion. There may also be fundamental limitations governing rates and accelerations to do with the machining process itself. The AMD servo drives provide a number of useful configuration parameters for restraining its controlled axis' movements.

These restraints can be applied to both the **command** and **feedback** quantities for position, velocity, acceleration, and force/torque.

- **Constraints** are applied to **commands** (i.e. control demands) so that the axis can never exceed certain settings. If a particular demand exceeds a constraint value, it is merely held at the constraint value, but otherwise no errors or warnings are issued. The AMD servo drive contains multiple constraint sources. The smallest of that which is enabled will be applied:
 - Global
 - NC
 - Safety related
- **Limits** or trip level are applied to **feedback** variables and used for monitoring axis behaviour. They result in Class 1 Diagnostic (C1D) errors if exceeded.

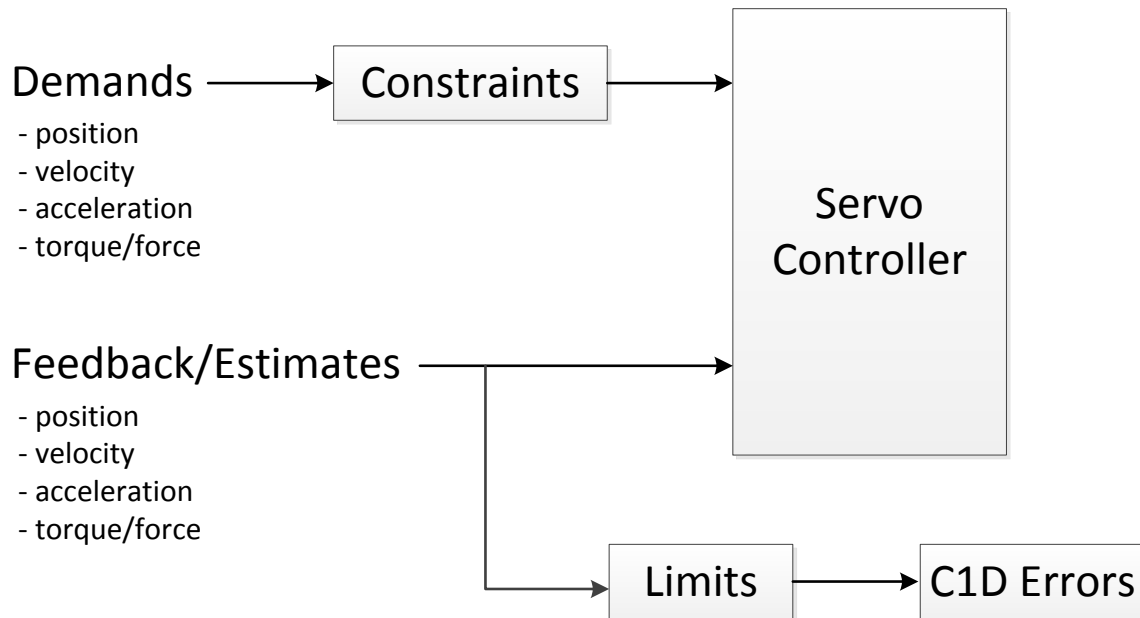


Figure 7-16 - Overview of Motion Constraints and Limits

7.14.2 Global Constraints

7.14.2.1 Description

Global constraints specify the minimum and maximum values associated with demands issued to the servo control loops. In the case where a particular constraint is not enabled, its global constraint is set to the maximum internally representable value. The full set of adjustable constraints is listed below, and each of these constraints is enabled by setting its associated bit to 1 (ON) in the *Global Constraints Enable Flag*. The values residing in the following list of IDN's, if so enabled, will be applied to constrain their associated demands.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-0099 / 32867	Global Constraints Enable Flag	Unsigned Integer (2 bytes)	Binary	0	0
P-0-0100 / 32868	Global Maximum Position Constraint	Signed Integer (4 bytes)	Standard Position	100 mm	100 mm
P-0-0101 / 32869	Global Minimum Position Constraint	Signed Integer (4 bytes)	Standard Position	-100 mm	-100 mm
P-0-0102 / 32870	Global Maximum Velocity Constraint	Signed Integer (4 bytes)	Standard Velocity	312 mm/min	312 mm/min
P-0-0103 / 32871	Global Minimum Velocity Constraint	Signed Integer (4 bytes)	Standard Velocity	-312 mm/min	-312 mm/min
P-0-0104 / 32872	Global Acceleration Constraint – Positive Velocity Region	Signed Integer (4 bytes)	Standard Acceleration	0.0064 m/s/s	0.0064 m/s/s
P-0-0105 / 32873	Global Acceleration Constraint – Negative Velocity Region	Signed Integer (4 bytes)	Standard Acceleration	-0.0064 m/s/s	-0.0064 m/s/s
P-0-0106 / 32874	Global Deceleration Constraint – Positive Velocity Region	Signed Integer (4 bytes)	Standard Acceleration	-0.0064 m/s/s	-0.0064 m/s/s
P-0-0107 / 32875	Global Deceleration Constraint – Negative Velocity Region	Signed Integer (4 bytes)	Standard Acceleration	0.0064 m/s/s	0.0064 m/s/s
P-0-0108 / 32876	Global Maximum Force/Torque Constraint	Signed Integer (4 bytes)	Standard Torque	4 Nm	4 Nm
P-0-0109 / 32877	Global Minimum Force/Torque Constraint	Signed Integer (4 bytes)	Standard Torque	-4 Nm	-4 Nm

7.14.2.2 Definitions

7.14.2.2.1 Global Constraints Enable Flag

The *Global Constraints Enable Flag* allows for selective enabling of the different global constraints.

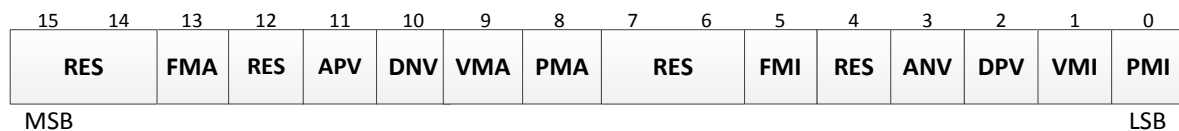


Figure 7-17 - Global Constraints Enable Flag definition

Bit	Key	Definition	Bit	Key	Definition
0	PMI	Position Minimum	8	PMA	Position Maximum
1	VMI	Velocity Minimum	9	VMA	Velocity Maximum
2	DPV	Deceleration – Positive Velocity	10	DNV	Deceleration – Negative Velocity
3	ANV	Acceleration – Negative Velocity	11	APV	Acceleration – Positive Velocity
4	RES	Reserved	12	RES	Reserved
5	FMI	Force/Torque Minimum	13	FMA	Force/Torque Maximum
6	RES	Reserved	14	RES	Reserved
7	RES	Reserved	15	RES	Reserved

Table 7-19 - Global Constraints Enable Flag bit definitions

7.14.2.2.2 Global Maximum Position Constraint

The maximum allowable position when enabled via the *Global Constraints Enable Flag*.

7.14.2.2.3 Global Minimum Position Constraint

The minimum allowable position when enabled via the *Global Constraints Enable Flag*.

7.14.2.2.4 Global Maximum Velocity Constraint

The maximum allowable velocity when enabled via the *Global Constraints Enable Flag*. In most applications this should be a positive value.

7.14.2.2.5 Global Minimum Velocity Constraint

The minimum allowable velocity when enabled via the *Global Constraints Enable Flag*. In most applications this should be a negative value.

7.14.2.2.6 Global Acceleration Constraint – Positive Velocity Region

The maximum allowable acceleration when moving in the positive velocity direction. In most applications this should be a positive value and must be enabled via the *Global Constraints Enable Flag*.

7.14.2.2.7 Global Acceleration Constraint – Negative Velocity Region

The minimum allowable acceleration when moving in the negative velocity direction. In most applications this should be a negative value and must be enabled via the *Global Constraints Enable Flag*.

7.14.2.2.8 Global Deceleration Constraint – Positive Velocity Region

The minimum allowable deceleration when moving in the positive velocity direction. In most applications this should be a negative value and must be enabled via the *Global Constraints Enable Flag*.

7.14.2.2.9 Global Deceleration Constraint – Negative Velocity Region

The maximum allowable acceleration when moving in the negative velocity direction. In most applications this should be a positive value and must be enabled via the *Global Constraints Enable Flag*.

7.14.2.2.10 Global Maximum Force/Torque Constraint

The maximum allowable force/torque when enabled via the *Global Constraints Enable Flag*. In most applications this should be a positive value.

7.14.2.2.11 Global Minimum Force/Torque Constraint

The minimum allowable force/torque when enabled via the *Global Constraints Enable Flag*. In most applications this should be a negative value.

7.14.2.3 Operation Example

Figure 7-18 shows an example of configuring the global constraints. The grey region “inside the onion” shows the allowable region of operation.

The parameters settings are for position, velocity and acceleration, BUT not force, so only the 4 LSB and the mid-bits need to be activated. The following is applied:

IDN	Description	Value
P-0-0099 / 32867	Global Constraints Enable Flag	0000 1111 0000 1111
P-0-0100 / 32868	Global Maximum Position Constraint	0.4 m
P-0-0101 / 32869	Global Minimum Position Constraint	-0.1 m
P-0-0102 / 32870	Global Maximum Velocity Constraint	1200 mm/min
P-0-0103 / 32871	Global Minimum Velocity Constraint	-900 mm/min
P-0-0104 / 32872	Global Acceleration Constraint – Positive Velocity Region	0.003 m/s/s
P-0-0105 / 32873	Global Acceleration Constraint – Negative Velocity Region	-0.003 m/s/s
P-0-0106 / 32874	Global Deceleration Constraint – Positive Velocity Region	-0.001 m/s/s
P-0-0107 / 32875	Global Deceleration Constraint – Negative Velocity Region	0.001 m/s/s

Table 7-20 - Global Constraints Example

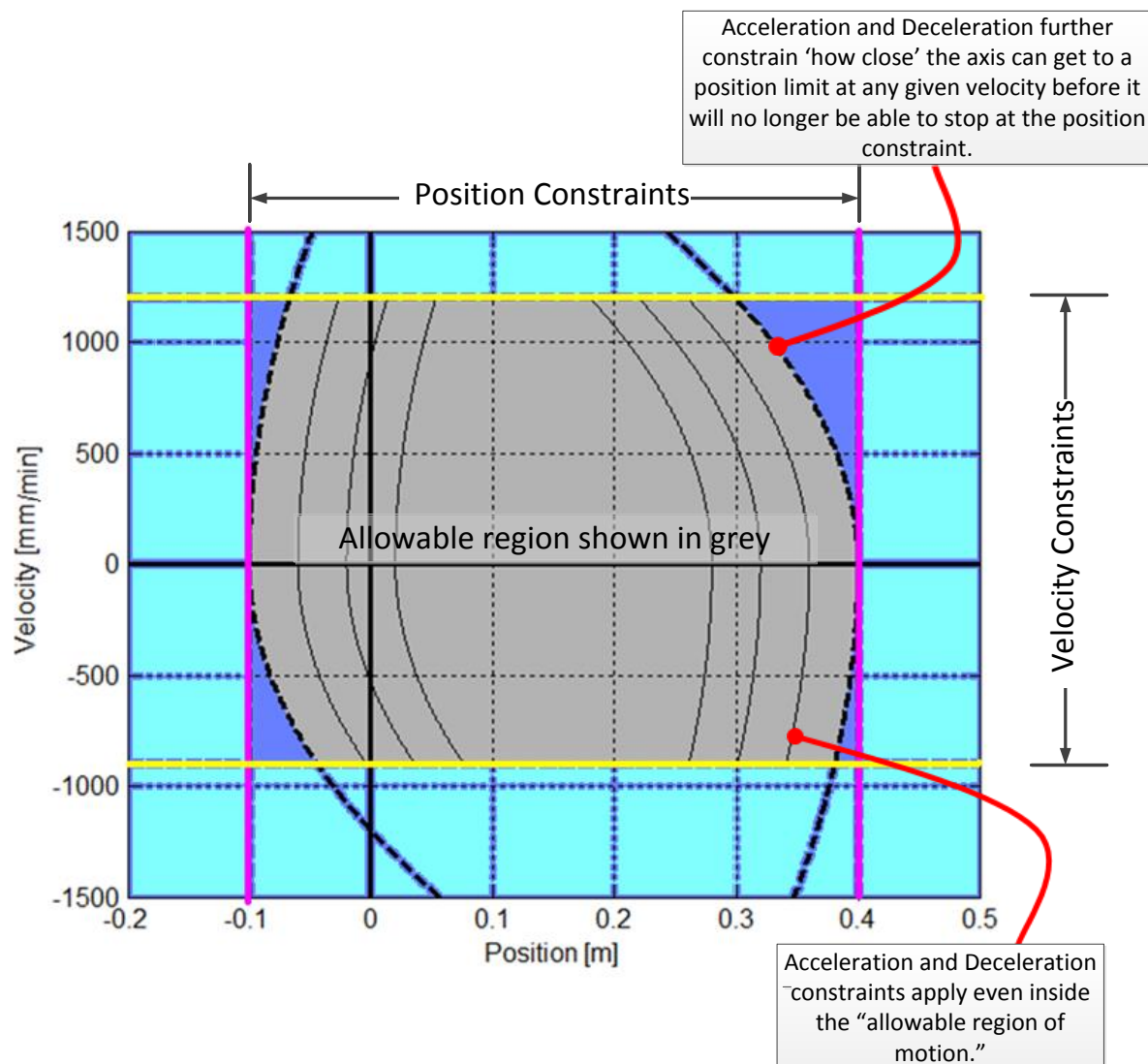


Figure 7-18 Global Constraints Example



NOTE: The acceleration/deceleration constraints are enforced anywhere within the working envelope (position vs. velocity space), not just those points dictated by the position constraints.

7.14.3 NC Constraints

7.14.3.1 Description

NC constraints specify the minimum and maximum values associated with demands issued to the servo control loops when operating in the *Numerical Control* operation mode. In the case where a particular constraint is not enabled, its constraint is set to the maximum internally representable value. The full set of adjustable constraints is listed below with each of these constraints is enabled by setting its associated bit to 1 (ON) in the *NC Constraints Enable Flag*. The values residing in the following list of IDN's, if so enabled, will be applied to constrain their associated demands.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x

P-0-0802 / 33570	NC Constraints Enable Flag	Unsigned Integer (2 bytes)	Binary	0	0
P-0-0803 / 33571	NC Maximum Position Constraint	Signed Integer (4 bytes)	Standard Position	0	0
P-0-0804 / 33572	NC Minimum Position Constraint	Signed Integer (4 bytes)	Standard Position	0	0
P-0-0805 / 33573	NC Maximum Velocity Constraint	Signed Integer (4 bytes)	Standard Velocity	0	0
P-0-0806 / 33574	NC Minimum Velocity Constraint	Signed Integer (4 bytes)	Standard Velocity	0	0
P-0-0807 / 33575	NC Acceleration Constraint – Positive Velocity Region	Signed Integer (4 bytes)	Standard Acceleration	0	0
P-0-0808 / 33576	NC Acceleration Constraint – Negative Velocity Region	Signed Integer (4 bytes)	Standard Acceleration	0	0
P-0-0809 / 33577	NC Deceleration Constraint – Positive Velocity Region	Signed Integer (4 bytes)	Standard Acceleration	0	0
P-0-0810 / 33578	NC Deceleration Constraint – Negative Velocity Region	Signed Integer (4 bytes)	Standard Acceleration	0	0
P-0-0811 / 33579	NC Maximum Force/Torque Constraint	Signed Integer (4 bytes)	Standard Torque	0	0
P-0-0812 / 33580	NC Minimum Force/Torque Constraint	Signed Integer (4 bytes)	Standard Torque	0	0

7.14.3.2 Definitions

7.14.3.2.1 NC Constraints Enable Flag

The *NC Constraints Enable Flag* allows for selective enabling of the different global constraints.

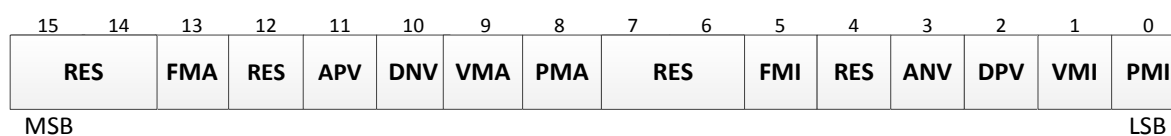


Figure 7-19 - NC Constraints Enable Flag definition

Bit	Key	Definition	Bit	Key	Definition
0	PMI	Position Minimum	8	PMA	Position Maximum
1	VMI	Velocity Minimum	9	VMA	Velocity Maximum
2	DPV	Deceleration – Positive Velocity	10	DNV	Deceleration – Negative Velocity
3	ANV	Acceleration – Negative Velocity	11	APV	Acceleration – Positive Velocity
4	RES	Reserved	12	RES	Reserved

5	FMI	Force/Torque Minimum	13	FMA	Force/Torque Maximum
6	RES	Reserved	14	RES	Reserved
7	RES	Reserved	15	RES	Reserved

Table 7-21 - NC Constraints Enable Flag bit definitions

7.14.3.2.2 NC Maximum Position Constraint

The maximum allowable position when enabled via the *NC Constraints Enable Flag*.

7.14.3.2.3 NC Minimum Position Constraint

The minimum allowable position when enabled via the *NC Constraints Enable Flag*.

7.14.3.2.4 NC Maximum Velocity Constraint

The maximum allowable velocity when enabled via the *NC Constraints Enable Flag*. In most applications this should be a positive value.

7.14.3.2.5 NC Minimum Velocity Constraint

The minimum allowable velocity when enabled via the *NC Constraints Enable Flag*. In most applications this should be a negative value.

7.14.3.2.6 NC Acceleration Constraint – Positive Velocity Region

The maximum allowable acceleration when moving in the positive velocity direction. In most applications this should be a positive value and must be enabled via the *NC Constraints Enable Flag*.

7.14.3.2.7 NC Acceleration Constraint – Negative Velocity Region

The minimum allowable acceleration when moving in the negative velocity direction. In most applications this should be a negative value and must be enabled via the *NC Constraints Enable Flag*.

7.14.3.2.8 NC Deceleration Constraint – Positive Velocity Region

The minimum allowable deceleration when moving in the positive velocity direction. In most applications this should be a negative value and must be enabled via the *NC Constraints Enable Flag*.

7.14.3.2.9 NC Deceleration Constraint – Negative Velocity Region

The maximum allowable acceleration when moving in the negative velocity direction. In most applications this should be a positive value and must be enabled via the *NC Constraints Enable Flag*.

7.14.3.2.10 NC Maximum Force/Torque Constraint

The maximum allowable force/torque when enabled via the *NC Constraints Enable Flag*. In most applications this should be a positive value.

7.14.3.2.11 NC Minimum Force/Torque Constraint

The minimum allowable force/torque when enabled via the *NC Constraints Enable Flag*. In most applications this should be a negative value.

7.14.4 Non-Functional Safety Constraints**7.14.4.1 Description**

WARNING: These constraints are NOT CE Safety certified safety functions, they provide safety related capability for the AMD servo drive without the associated claims for reliability.

Safety constraints are intended to be dynamically enabled and disabled while the drive is operating. Setting the *Safety Constraints Master Enable* to '1' will result in the safety constraints being applied. A separate *Safety Constraints Enable Flag* is used to individually specify which constraints are to be enabled, similar to the global constraints described above. An example application of this feature may be for the control device / unit to dynamically implement a conservative set of constraints under potentially dangerous situations, such as when the axis is not homed or the machine door is open.

The full set of adjustable constraints is listed below, and each of these constraints is enabled by setting its associated bit to '1' in the *Safety Constraints Enable Flag*. The values residing in the following list of IDNs then provide the relevant constraint levels.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-0114 / 32882	Safety Constraints Master Enable	Unsigned Integer (2 bytes)	Boolean	1	1
P-0-115 / 32883	Safety Constraints Enable Flag	Unsigned Integer (2 bytes)	Binary	0	0
P-0-0116 / 32884	Safety Maximum Position Constraint	Signed Integer (4 bytes)	Standard Position	0	0
P-0-0117 / 32885	Safety Minimum Position Constraint	Signed Integer (4 bytes)	Standard Position	0	0
P-0-0118 / 32886	Safety Maximum Velocity Constraint	Signed Integer (4 bytes)	Standard Velocity	0	0
P-0-0119 / 32887	Safety Minimum Velocity Constraint	Signed Integer (4 bytes)	Standard Velocity	0	0
P-0-0120 / 32888	Safety Acceleration Constraint – Positive Velocity Region	Signed Integer (4 bytes)	Standard Acceleration	0	0
P-0-0121 / 32889	Safety Acceleration Constraint – Negative Velocity Region	Signed Integer (4 bytes)	Standard Acceleration	0	0
P-0-0122 / 32890	Safety Deceleration Constraint – Positive Velocity Region	Signed Integer (4 bytes)	Standard Acceleration	0	0
P-0-0123 / 32891	Safety Deceleration Constraint – Negative Velocity Region	Signed Integer (4 bytes)	Standard Acceleration	0	0
P-0-0124 / 32892	Safety Maximum Force/Torque Constraint	Signed Integer (4 bytes)	Standard Torque	0	0
P-0-0125 / 32893	Safety Minimum Force/Torque Constraint	Signed Integer (4 bytes)	Standard Torque	0	0

7.14.4.2 Definitions

7.14.4.2.1 Safety Constraints Master Enable

The *Safety Constraints Master Enable* allows a single point of control to control the use of the Non-Functional Safety Constraints. Set this parameter to '1' to enable.

7.14.4.2.2 Safety Constraints Enable Flag

The *Safety Constraints Enable Flag* allows for selective enabling of the different safety constraints.

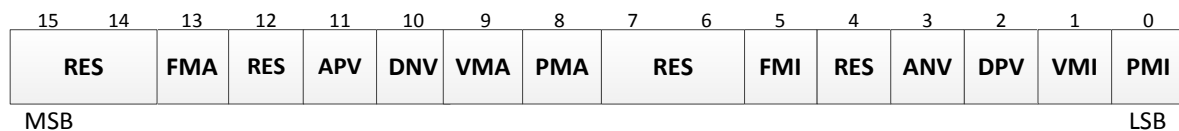


Figure 7-20 - Safety Constraints Enable Flag definition

Bit	Key	Definition	Bit	Key	Definition
0	PMI	Position Minimum	8	PMA	Position Maximum
1	VMI	Velocity Minimum	9	VMA	Velocity Maximum
2	DPV	Deceleration – Positive Velocity	10	DNV	Deceleration – Negative Velocity
3	ANV	Acceleration – Negative Velocity	11	APV	Acceleration – Positive Velocity
4	RES	Reserved	12	RES	Reserved
5	FMI	Force/Torque Minimum	13	FMA	Force/Torque Maximum
6	RES	Reserved	14	RES	Reserved
7	RES	Reserved	15	RES	Reserved

Table 7-22 - Safety Constraints Enable Flag bit definitions

7.14.4.2.3 Safety Maximum Position Constraint

The maximum allowable position when enabled via the *Safety Constraints Enable Flag*.

7.14.4.2.4 Safety Minimum Position Constraint

The minimum allowable position when enabled via the *Safety Constraints Enable Flag*.

7.14.4.2.5 Safety Maximum Velocity Constraint

The maximum allowable velocity when enabled via the *Safety Constraints Enable Flag*. In most applications this should be a positive value.

7.14.4.2.6 Safety Minimum Velocity Constraint

The minimum allowable velocity when enabled via the *Safety Constraints Enable Flag*. In most applications this should be a negative value.

7.14.4.2.7 Safety Acceleration Constraint – Positive Velocity Region

The maximum allowable acceleration when moving in the positive velocity direction. In most applications this should be a positive value and must be enabled via the *Safety Constraints Enable Flag*.

7.14.4.2.8 Safety Acceleration Constraint – Negative Velocity Region

The minimum allowable acceleration when moving in the negative velocity direction. In most applications this should be a negative value and must be enabled via the *Safety Constraints Enable Flag*.

7.14.4.2.9 Safety Deceleration Constraint – Positive Velocity Region

The minimum allowable deceleration when moving in the positive velocity direction. In most applications this should be a negative value and must be enabled via the *Safety Constraints Enable Flag*.

7.14.4.2.10 Safety Deceleration Constraint – Negative Velocity Region

The maximum allowable acceleration when moving in the negative velocity direction. In most applications this should be a positive value and must be enabled via the *Safety Constraints Enable Flag*.

7.14.4.2.11 Safety Maximum Force/Torque Constraint

The maximum allowable force/torque when enabled via the *Safety Constraints Enable Flag*. In most applications this should be a positive value.

7.14.4.2.12 Safety Minimum Force/Torque Constraint

The minimum allowable force/torque when enabled via the *Safety Constraints Enable Flag*. In most applications this should be a negative value.

7.14.5 Error Limits

7.14.5.1 Description

Error limits are applied similarly to the above constraints but they influence the performance in an entirely different fashion. Error limits are applied to monitored feedback variables and, therefore, do NOT limit the demands going to the servo controller. They operate via a combination of applied hard and soft limits. When error limits are exceeded then Class 1 Diagnostic (C1D) errors are asserted to stop the drive from further motor movement. Error limits can be used to define the boundary between safe and hazardous operation, such as the ends of a linear operating boundary, or the maximum safe velocity.

If the position is within the soft limit error region (refer to [Figure 7-21](#)), the drive will be unable to command decelerations greater than the drive maximum deceleration to stop the joint at the soft position limit specified, the drive will then trigger C1D E304 or E305 and shut down.

The hard error limits are a simple position and/or velocity parameter value comparison to the position/velocity feedback (refer to [Figure 7-22](#)). If estimated position exceeds the position hard limit then error C1D E330 or E331 is asserted.

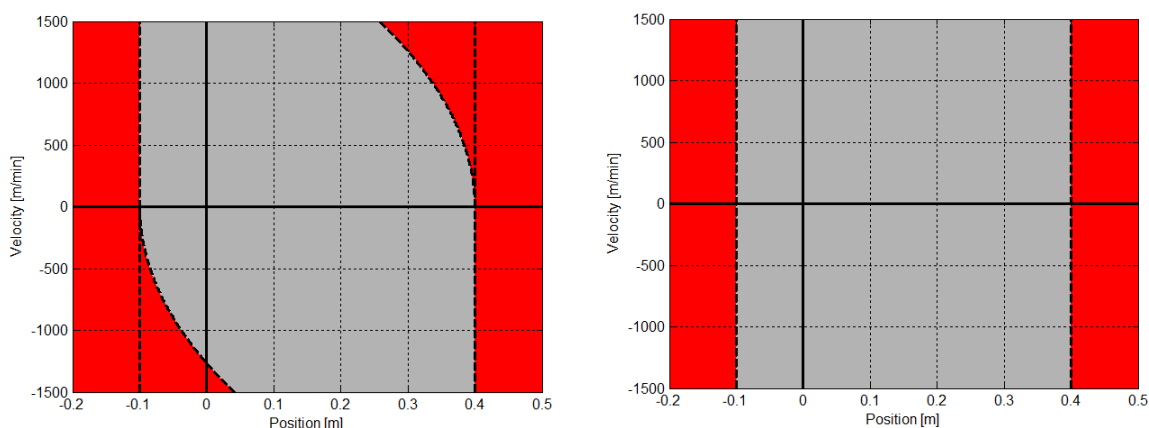


Figure 7-21 - Position Soft (left) and Hard (right) Error Limits = [-0.1, 0.4] metres

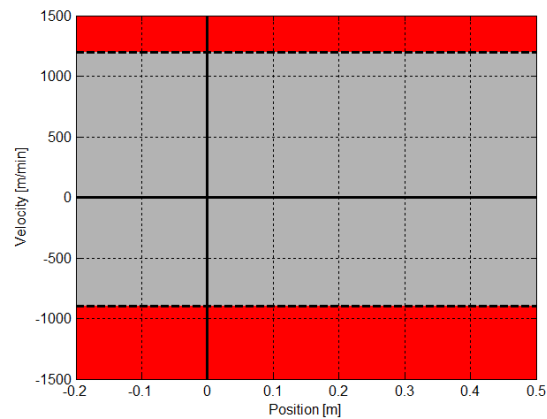


Figure 7-22 - Velocity Hard Error Limit = [-900, 1200] mm/min

For rotary axes with limited stroke, where the motion is constrained to within less than one full revolution, there is a special requirement for configuration of Error Limits. The feature must be enabled via *Rotary Joint with Limited Stroke Enable* and position end stop limits defined via *Location of End Stops on a Rotary Joint*.

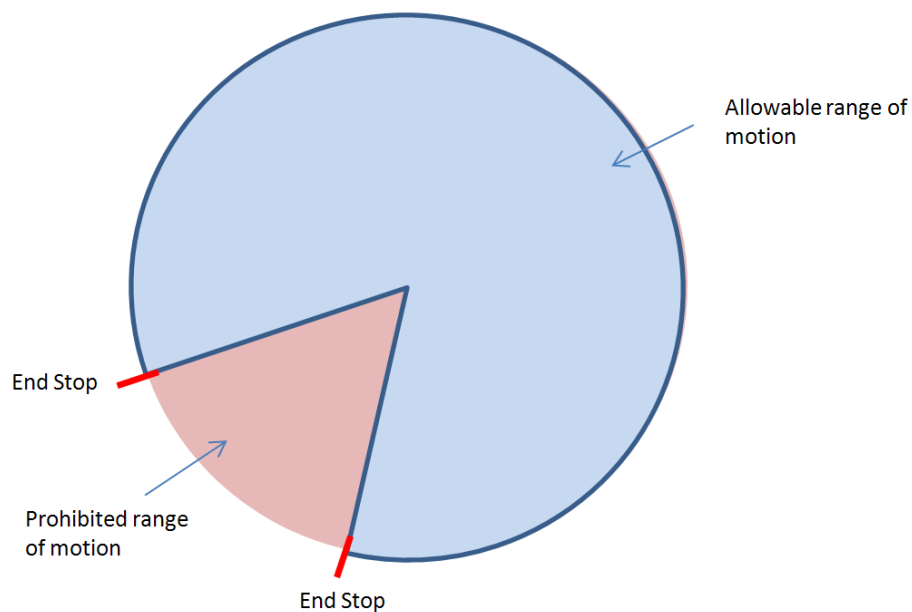


Figure 7-23 - Modulo Rotary Axis with Limited Stroke

To enable error limits, the *Error Limits Master Enable* must be set to '1' and individual bits corresponding to desired limits in the *Error Limits Enable Flags* must be configured. The values residing in the following list of IDN's are relevant to setting Error Limits.



NOTE: Axis shall be homed for position soft and hard limits to function.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-0126 / 32894	Error Limits Master Enable	Unsigned Integer (2 bytes)	Boolean	1	1

P-0-0127 / 32895	Error Limits Enable Flags	Unsigned Integer (2 bytes)	Binary	0	0
P-0-0128 / 32896	Position Hard Min and Max (2 elements)	Signed Integer (4 bytes) Array	Standard Position	0, 150 mm	0, 150 mm
P-0-0129 / 32897	Position Soft Min and Max (2 elements)	Signed Integer (4 bytes) Array	Standard Position	0, 150 mm	0, 150 mm
P-0-0131 / 32899	Rotary Joint with Limited Stroke Enable	Unsigned Integer (2 bytes)	Boolean	0	0
P-0-0132 / 32900	Location of End Stops on a Rotary Joint (2 elements)	Signed Integer (4 bytes)	Standard Position	-360 degrees	360 degrees
P-0-0133 / 32901	Velocity Hard Min and Max (2 elements)	Signed Integer (4 bytes)	Standard Velocity	-720,000, 720,000 mm/min	-720,000, 720,000 mm/min
P-0-0134 / 32902	Soft Limit Max Deceleration (2 elements)	Signed Integer (4 bytes) Array	Standard Acceleration	0	0
P-0-0135 / 32903	Positive Limit Switch	Unsigned Integer (2 bytes)	Boolean	0	0
P-0-0136 / 32904	Negative Limit Switch	Unsigned Integer (2 bytes)	Boolean	0	0

7.14.5.2 Definitions

7.14.5.2.1 Error Limits Master Enable

The *Error Limits Master Enable* allows a single point of control to control the use of the Error Limits. Set this parameter to '1' to enable.

7.14.5.2.2 Error Limits Enable Flag

The *Error Limits Enable Flag* allows for selective enabling of the different error limits.

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RES	VHMA	RES	PDMA	PHMA	PSMA	RES	VHMI	RES	PDMI	PHMI	PSMI				
MSB								LSB							

Figure 7-24 – Error Limits Enable Flag definition

Bit	Key	Definition	Bit	Key	Definition
0	PSMI	Position Soft Minimum	8	PSMA	Position Soft Maximum
1	PHMI	Position Hard Minimum	9	PHMA	Position Hard Maximum
2	PDMI	Position Deadstop Minimum	10	PDMA	Position Deadstop Maximum
3	RES	Reserved	11	RES	Reserved
4	RES	Reserved	12	RES	Reserved
5	VHMI	Velocity Hard Minimum	13	VHMA	Velocity Hard Maximum
6	RES	Reserved	14	RES	Reserved

7	RES	Reserved	15	RES	Reserved
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Table 7-23 - Error Limits Enable Flag bit definitions

7.14.5.2.3 Position Hard Min and Max

The *Position Hard Min and Max* is a two element array where the 1st element (index = 0) is the minimum allowed position hard error limit and the 2nd element (index = 1) is the maximum position hard error limit. If the axis position feedback exceeds these limit C1D errors E330 and E331 are asserted.

7.14.5.2.4 Position Soft Min and Max

The *Position Soft Min and Max* is a two element array where the 1st element (index = 0) is the minimum allowed position hard error limit and the 2nd element (index = 1) is the maximum position hard error limit. If the axis is predicted to not be able to stop using the *Soft Limit Max Deceleration* before these limits are exceeded, then C1D errors E304 and E305 are asserted.

7.14.5.2.5 Rotary Joint with Limited Stroke Enable

When utilising a [Modulo](#) rotary axis with position error limits, the *Rotary Joint with Limited Stroke Enable* must be set to '1' and the *Location of End Stops on a Rotary Joint* must be configured.

7.14.5.2.6 Location of End Stops on a Rotary Joint

The *Location of End Stops on a Rotary Joint* is used when *Rotary Joint with Limit Stroke Enable* is set to '1' and the axis is configured as [Modulo](#) rotary to define the axis position end stops (refer to [Figure 7-23](#))

7.14.5.2.7 Velocity Hard Min and Max

The *Velocity Hard Min and Max* is a two element array where the 1st element (index = 0) is the minimum allowed velocity hard error limit and the 2nd element (index = 1) is the maximum velocity hard error limit. In most applications, the minimum velocity hard limit will be set to a negative value to indicate the maximum speed in the negative direction. If the axis velocity feedback exceeds these limit C1D errors E330 and E331 are asserted.

7.14.5.2.8 Soft Limit Max Deceleration

The Soft Limit Max Deceleration is a two element array where the 1st element (index = 0) is the maximum allowed deceleration to the position soft limit minimum error limit and the 2nd element (index = 1) is the maximum allowed deceleration to the position soft limit maximum error limit. If an element in this IDN is set to 0 then the smaller of the [Global Constraints](#) and [Non-Functional Safety Constraints](#) acceleration constraints will be used.

7.14.5.2.9 Positive Limit Switch

If *Positive Limit Switch* is set to '1', the C1D E306 will be asserted. This parameter is intended to have a limit switch signal mapped to it either from the control unit / device or a servo drive [Digital Input](#).

7.14.5.2.10 Negative Limit Switch

If *Negative Limit Switch* is set to '0', the C1D E307 will be asserted. This parameter is intended to have a limit switch signal mapped to it either from the control unit / device or a servo drive [Digital Input](#).

7.15 Motor Control

7.15.1 Overview

The AMD servo drive control loops utilise the architecture for position, velocity and torque/current control displayed by [Figure 7-25](#). On the AMD servo drive, the term Motor Control refers to the current control loop, the torque gain scheduler and all of the related switches to toggle between a variety of magnetic field alignment methods. It can be seen in diagram that the accuracy of the position and velocity controllers must, in part, be determined by the Motor Control.³

³ Details for the motivation in having two current controllers, one for "quadrature" and one for "direct" current (so-called q and d currents), is given elsewhere in this configuration manual under section [7.11 Field Orientation Initialisation](#).

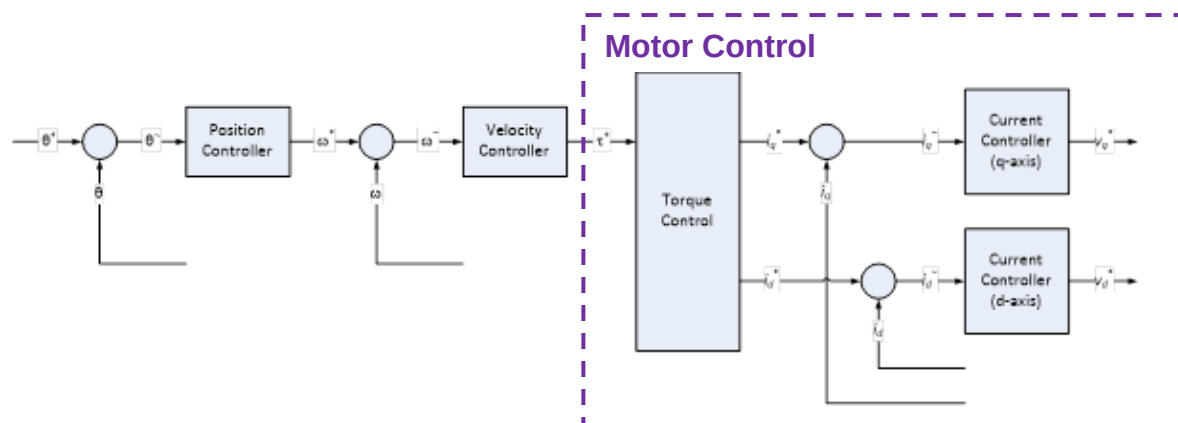


Figure 7-25 - Cascaded control architecture found in IEC 61491

As the name is intended to imply, the Motor Control portion of the control loops is intimately related to the type of motor under control. The AMD servo drive supports control for two dominant types of motor:

1. Permanent Magnet Synchronous Motors
2. Induction Motors.

The AMD servo drive's Motor Control is broken down into a number of sub-systems that must all be configured correctly if they are to be used effectively in controlling either of these two motor types. The primary sub-systems of concern to the user are:

1. Torque controller
2. Current controller

Torque and current are intimately related to one another, where the relationship between torque, τ , and q-axis motor current, i_q , is described using a units scaling (or proportional gain) K_t , as follows:

$$\tau = K_t i_q$$

The AMD servo drive makes the assumption that $K_t = 1$ under all conditions, except for *Field Weakening*, in order to simplify this relationship further. Under this condition the torque and current control commands are the same value.

However, as far as the servo control architecture is concerned, the two physical quantities remain distinct. As a consequence, it is necessary for the user to select the appropriate motor type for both the torque and current controller subsystems separately.



WARNING: Care must be exercised so that configuration changes to the type of motor under control are made to both the Torque Controller AND the Current Controller subsystems in the Motor Control. Failure to do this can cause unexpected behaviour.

In addition to the flexible configuration of both the Torque and Current Controllers, the following section concerning Motor Control will describe details including:

- Current loop integral gain scheduler
- Current/Torque limits

7.15.2 Configuring Motor Type

7.15.2.1 Description

The AMD servo drive can be configured to control a Permanent Magnet Synchronous Motor (PMSM) or Induction Motor via Velocity over Frequency. Configuration IDNs are broken up into the various subsystems of the servo drive; however, they can generally be configured as a set of parameters depending on the motor type.

Description	IDN	Motor Type	
		PMSM	IM V/F
Primary mode of current control	P-0-0503 / 33271	10	15
Primary mode of torque control	P-0-0222 / 32990	10	15
Technique of motor commutation	P-0-0506 / 33274	10	15
Technique of voltage control	P-0-0041 / 32809	0	1

Table 7-24 - Parameter set configuration for each motor type

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-0041 / 32809	Technique of Voltage Control	Unsigned Integer (2 bytes)	None	0	0
P-0-0222 / 32990	Primary Mode of Torque Control	Unsigned Integer (2 bytes)	None	10	10
P-0-0503 / 33271	Primary Mode of Current Control	Unsigned Integer (2 bytes)	None	10	10
P-0-0506 / 33274	Technique of Motor Commutation	Unsigned Integer (2 bytes)	None	10	10

7.15.2.2 Definitions

7.15.2.2.1 Technique of Voltage Control

The *Technique of Voltage Control* determines the voltage control type for the connected motor.

Value	Description
0	PMSM – D and Q axis current
1	IM V/F – Total current feedback

Table 7-25 - Technique of Voltage Control definition

7.15.2.2.2 Primary Mode of Torque Control

The *Primary Mode of Torque Control* determines the torque control type for the connected motor.

Value	Description
10	Permanent Magnet Synchronous Motor
15	Induction Motor via Velocity over Frequency

Table 7-26 - Primary Mode of Torque Control definition

7.15.2.2.3 Primary Mode of Current Control

The *Primary Mode of Current Control* determines the current control type for the connected motor.

Value	Description
-------	-------------

10	Permanent Magnet Synchronous Motor
15	Induction Motor via Velocity over Frequency

Table 7-27 - Primary Mode of Current Control definition

7.15.2.2.4 Technique of Motor Commutation

The *Technique of Motor Commutation* determines the current control type for the connected motor.

Value	Description
10	Permanent Magnet Synchronous Motor
15	Induction Motor via Velocity over Frequency

Table 7-28 - Technique of Motor Commutation definition

7.15.3 PMSM Control

7.15.3.1 Description

PMSM torque control shall be used when controlling a Permanent Magnet Synchronous Motor where torque command is directly transformed to q -axis current command for the current control loop via the motor torque constant relationship.

7.15.3.2 Field Weakening

7.15.3.2.1 Description



NOTE: The field weakening curve is a property of the motor. Consult the motor manufacturer for more information.



NOTE: The feield weakening is only supported in AMD5x series drives.

The field weakening technique may be utilised to command higher velocity at the trade-off of lower torque when the motor has reached its maximum power output. The general approach is to reduce the torque producing current (q -axis) limit and increase the field weakening current (d -axis) command. The AMD servo drive achieves this by reducing the q -axis limit such that the magnitude of the total current is always below the *Commanded Current Limit*.

IDN	Description	Data Type	Units	Default
				AMD5x
P-0-0927	Field Weakening Enable	Unsigned Integer (2 bytes)	Boolean	0
P-0-0929 / 33697	Field Weakening Lookup Table - Number of Break Points	Unsigned Integer (2 bytes)	Array Length	11
P-0-0930 / 33698	Field Weakening Lookup Table - Velocity Break Points (12 elements)	Signed Integer (4 bytes)	Standard Velocity	125,131,138,144,150,156,162,169,175,181,188,0 m/min
P-0-0931 / 33699	Field Weakening Lookup Table - Field Weakening (d -axis) Current Command Break Points	Signed Integer (4 bytes)	Standard Current	0,0.4,0.8,1.2,1.6,2,2.4,2.8,3.2,3.6,4,0 A

7.15.3.2.2 Definitions

7.15.3.2.2.1 Field Weakening Lookup Table – Number of Break Points

This parameter determines the number of points in the Field Weakening Lookup Table that are used to generate the velocity / D axis current curve.

7.15.3.2.2.2 Field Weakening Lookup Table – Velocity Break Points

This parameter contains the velocity points in the lookup table. Each velocity will have a corresponding current command.

7.15.3.2.2.3 Field Weakening Lookup Table – Field Weakening (d-axis) Current Command Break Points

This parameter contains the d-axis current command points in the lookup table. Each current will have a corresponding velocity.

7.15.3.2.3 Operation Example

Below is an example of a typical field weakening curve. Here, the *ICONT Iq Limit* is 50A. At 5000RPM, field weakening begins. The user needs to define the Field Weakening Current Command profile via the *Field Weakening Lookup Table* (P-0-0929 / 33697, P-0-0930 / 33698, P-0-0931 / 33699). The drive will then calculate the torque producing current (q-axis) limit so as not to violate the overall *Commanded Current Limit*.

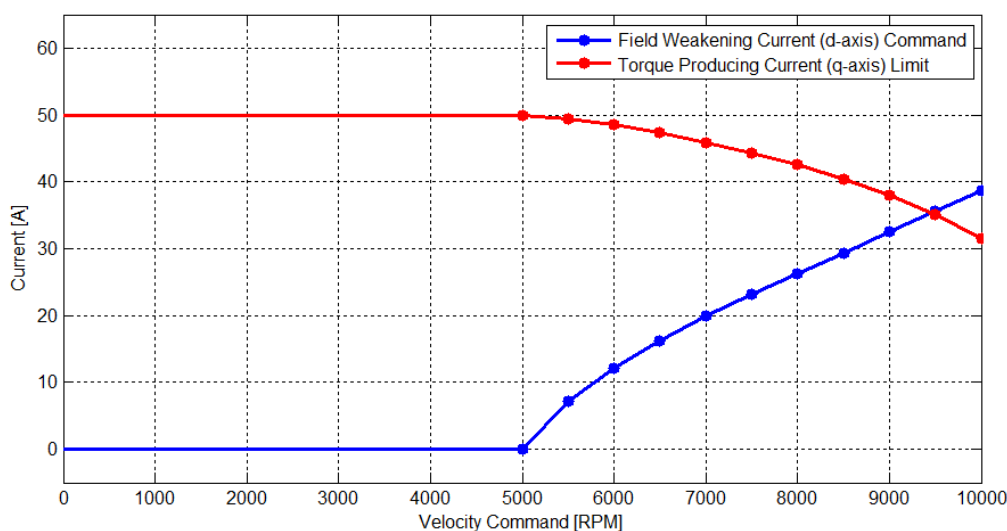


Figure 7-26 Example of Field Weakening

Parameter	Value
Number of Break Points	11
Velocity Break Points	5000, 5500, 6000, 6500, 7000, 7500, 8000, 8500, 9000, 9500, 10000 RPM
Current Command Break Points	0, 7125, 12068, 16159, 20000, 23125, 26250, 29375, 32500, 35625, 38750 mA

Table 7-29 - Example of Field Weakening Lookup Table

7.15.4 Induction Motor V/F Control

7.15.4.1 Description



WARNING: It is NOT recommended practice for Induction Motor V/F Control to be driven from the position control loop, as difficulties with low speed control using this technique may cause undesirable or unexpected motions.



NOTE: The feield weakening is only supported in AMD5x series drives.

The AMD servo drive gives the user the ability to control the speed of induction motors using Voltage over Frequency (V/F). V/F control is active when in position or velocity control modes and the velocity command from either the position loop output or control unit exceeds the *VF Control - Minimum Velocity Command*. While active, commanded current is set by *VF Control - Max Current*. If the velocity drops below the *VF Control - Minimum Velocity Command*, braking is executed with the *VF Control - Stop Current* and *VF Control - Stop Voltage* for the duration of *VF Control - Stop Time*. After this period of time, power is removed from the motor.

The V/F motor control is configured via a lookup table based VF Curve along with the number of motor poles. The lookup table is composed of a configurable number of velocity command and voltage pairs. The VF Curve and number of motor poles is a characteristic of the motor and should be available from the motor manufacturer.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-0006 / 32774	Motor Poles	Unsigned Integer (2 bytes)	Poles	4	4
P-0-212 / 32980	VF Curve - Number of Break Points	Unsigned Integer (2 bytes)	None	2	2
P-0-213 / 32981	VF Curve - Velocity Break Points	Signed Integer (4 bytes)	Standard Velocity	28,384,0,0,0 RPM	28,384,0,0,0 RPM
P-0-214 / 32982	VF Curve - Voltage Break Points	Signed Integer (2 bytes)	Standard Voltage	45,600,0,0,0 V	45,600,0,0,0 V
P-0-0215 / 32983	VF Control - Max Current	Signed Integer (4 bytes)	Standard Current	35 A	35 A
P-0-0216 / 32984	VF Control - Stop Current	Signed Integer (4 bytes)	Standard Current	5.66 A	5.66 A
P-0-0217 / 32985	VF Control - Stop Voltage	Signed Integer (2 bytes)	Standard Voltage	600 V	600 V
P-0-0218 / 32986	VF Control - Stop Time	Unsigned Integer (2 bytes)	250×10^{-6} sec	3 s	3 s
P-0-0219 / 32987	VF Control - Minimum Velocity Command	Signed Integer (4 bytes)	Standard Velocity	6 RPM	6 RPM
P-0-0220 / 32988	VF Control - Velocity Command Scale Factor	Signed Integer (2 bytes)	2^{-10}	1024	1024

7.15.4.2 Definitions

7.15.4.2.1 Motor Poles

Motor Poles defines the number of motor poles (2 x the number of pole pairs) of the induction or permanent magnet motor.

7.15.4.2.2 VF Curve - Number of Break Points

This parameter determines the number of points in the V/F Lookup Table that are used to generate the voltage / frequency curve.

7.15.4.2.3 VF Curve - Velocity Break Points

This parameter contains the velocity command break points in the V/F lookup table. Each velocity will have a corresponding voltage.

7.15.4.2.4 VF Curve - Voltage Break Points

This parameter contains the voltage break points in the V/F lookup table. Each velocity will have a corresponding velocity.

7.15.4.2.5 VF Control - Max Current

The *VF Control - Max Current* defines the commanded current when the motor commanded velocity is above the *VF Control - Minimum Velocity Command*.

7.15.4.2.6 VF Control - Stop Current

The *VF Control - Stop Current* defines the commanded current used to stop the motor when the commanded velocity is below the *VF Control - Minimum Velocity Command*.

7.15.4.2.7 VF Control - Stop Voltage

The *VF Control - Stop Voltage* defines the commanded voltage used to stop the motor when the commanded velocity is below the *VF Control - Minimum Velocity Command*.

7.15.4.2.8 VF Control - Stop Time

VF Control - Stop Time defines the time that *VF Control - Stop Voltage* and *VF Control - Stop Current* is applied when the commanded velocity is below the *VF Control - Minimum Velocity Command* to stop the motor. After this period of time, power is removed from the motor.

7.15.4.2.9 VF Control - Minimum Velocity Command

This parameter defines the minimum controllable velocity to for V/F control, below this value the velocity command we be treated as zero.

7.15.4.2.10 VF Control - Velocity Command Scale Factor

The *VF Control - Velocity Command Scale Factor* defines a multiplication factor that can be applied to the velocity command to increase or decrease the motor speed. This scale factor does not have an impact on the input to the VF curve or minimum velocity command threshold.

7.15.4.3 Operation Example

The figure below demonstrates a typical Voltage / Frequency Curve for an induction motor.

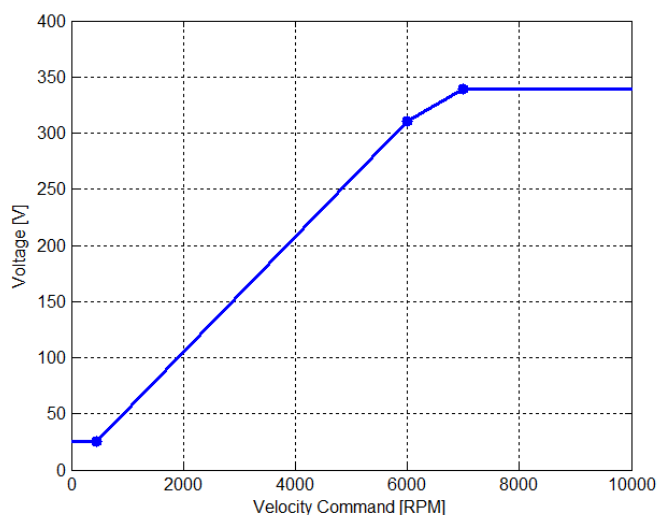


Figure 7-27 Voltage / Frequency Curve Example

Parameter	Value
Number of Break Points	3
Velocity Break Points	450, 6000, 7000 RPM
Voltage Break Points	25, 294, 339 V

Table 7-30 - Example of Voltage Frequency Curve Lookup Table

7.15.5 Current Control Loop

7.15.5.1 Overview

The AMD servo drive current control is performed by parallel Proportional-Integral components for Q and D axis current. These controllers contain gain parameters that allow the user to 'tune' the response of the system, limits to prevent damage to the system and a gain scheduler to change low current behaviour.

7.15.5.2 Gain Parameters

The Gain Parameters allow for modification of the current controllers response. These parameters will vary depending on the motor being used. Refer to [5.2.2.6 Initial Current Loop Configuration](#) for information on how to determine initial values for a new motor and [7.27 Tuning – Current Control](#) for information on how to fine tune a current controller.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0106 / 106	Current Controller Q Axis Proportional Gain	Unsigned Integer (2 bytes)	V/A	3	3
S-0-0107 / 107	Current Controller Q Axis Integral Time	Unsigned Integer (2 bytes)	μs	3750	3750
S-0-0119 / 119	Current Controller D Axis Proportional Gain	Unsigned Integer (2 bytes)	V/A	3	3
S-0-0120 / 120	Current Controller D Axis Integral Time	Unsigned Integer (2 bytes)	μs	3750	3750
P-0-0231 / 32999	Current Control D and Q axes Gain Scale	Signed Integer (4 bytes)	2 ⁻²¹	2097152	2097152

7.15.5.3 Applying Current Control Loop Parameters

Tuning parameters for the current control loops may be configured at any time; however such configuration is not applied until either the drive is re-enabled via or manually triggered by the user. To manually trigger a current control loop re-tune, the user may execute the *Motor Control Tuning Procedure Command* (P-0-510 / 33278) by changing its default value from 0 to 3. Doing this 'execution' will also re-tune the torque control loop.⁴ The user must then complete the procedure command by setting *Motor Control Tuning Procedure Command* back to 0.



NOTE: A "re-tune" is an internally executed recalibration and calculation of the necessary control variables for efficient execution of the desired control behaviour. It does NOT mean the drive changes any variables or gains in the control loops as set by the user.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-0510 / 33278	Motor Control Tuning Procedure Command	Unsigned Integer (2 bytes)	Procedure Command	0	0

7.15.5.4 Current and Torque Limits

7.15.5.4.1 Description

The AMD servo drive contains a number of current and torque limits that can be applied. Limits can be configured for the motor and/or the application or due to the limitations of the servo drive hardware. All limits are combined to determine the final peak current and torque commanded of the motor with the lowest value taking precedence. All limits described in this section apply to both positive and negative current and torque directions.

⁴ Note that a similar execution command exists for the position and velocity control loops, called the Servo Control Tuning Procedure Command (P-0-187 / 32955). The user should be made aware that if the Servo Control Tuning Procedure Command is executed, it ALSO re-tunes the current and torque control loops at the same time.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0109 / 109	Motor Peak Current	Signed Integer (4 bytes)	Standard Current	0	0
P-0-0232 / 33000	Q-axis Current Limit	Signed Integer (4 bytes)	Standard Current	4000 A	4000A
P-0-0518 / 33286	Current Limit Active	Unsigned Integer (2 bytes)	None	N/A	N/A
P-0-0519 / 33287	Active Current Limit Source	Unsigned Integer (2 bytes)	Source	N/A	N/A
P-0-0926 / 33694	D-axis Current Limit	Signed Integer (4 bytes)	Standard Current	0	0

7.15.5.4.2 Definitions

7.15.5.4.2.1 Motor Peak Current

The *Motor Peak Current* defines the maximum allowable current that can be applied to the motor. This will be defined by the motor manufacturer.

7.15.5.4.2.2 Q-axis Current Limit

The *Q-Axis Current Limit* defines the maximum allowable torque producing current that is applied to the motor. This is generally used to limit current/torque for a specific application⁵.

7.15.5.4.2.3 Current Limit Active

Current Limit Active can be used by the control unit/device to determine if the motor motion is being restricted due to a current limit. This is separated into positive and negative current directions.

Value	Description
0	No Current Limit Active
1	Minimum Current Limit Active
2	Maximum Current Limit Active

7.15.5.4.2.4 Active Current Limit Source

The *Active Current Limit Source* indicates the effective current limit applied to the motor i.e the lowest current limit of all possible sources. This can be used in conjunction with the *Current Limit Active* IDN for diagnostic purposes.

Value	Description
0	Q axis Current Limit
1	Motor Current Limit
2	Torque Limit
3	Power Limit
4	Amplifier Current Limit
5	Field Weakening Current Limit

⁵ Note that SoE treats the basic units for specifying current and torque quite separately, so the user needs to be aware that current and torque may not simply be value equivalent, even if $K_t=1$. For example, torque may be represented in basic units of 0.01 N or N/m whereas current is in A. If this is the situation, then with $K_t=1$, a value of 100 reported for torque over SoE would represent a value of 1 for quadrature current since the torque is being represented in units of cN/m or equivalently N/hm.

6

Dynamic Torque Limit

7.15.5.4.2.5 D-axis Current Limit

The *D-Axis Current Limit* defines the maximum allowable current that can be applied to the motor along the flux producing axis for *Field Weakening*. This should be set to '0' if the Field Weakening feature is not required.

7.15.5.5 Current Loop Integral Gain Scheduler

7.15.5.5.1 Description

A current loop integral gain scheduler is provided in the AMD servo drive to allow increased current controller gain during periods of small current command, where the gain characteristic decays drastically as signal frequency increases.

The current loop integral gain scheduler is enabled by setting *Current Control Q-axis Low Current Boost Enable* to a value of '1'. The parameter *Current Control Q-axis Low Current Boost Integral Time* specifies the current loop integral time constant that will be used at zero current command in place of the normal Q-axis Integral Time. A smaller time constant results in a larger integral gain, K_i , and vice-versa. The resulting gain is then interpolated adjusted linearly when the current command hovers between the symmetric +/- values of the *Current Control Q-axis Low Current Boost Threshold*. See *Figure 7-28* for a graphical representation of the function. Note that



NOTE: The scheduler does not have to result in an increased gain near zero current, it could be just as easily configured to decrease the gain near zero current; however, in most normal applications the desire will be to increase the integral gain in such circumstance.

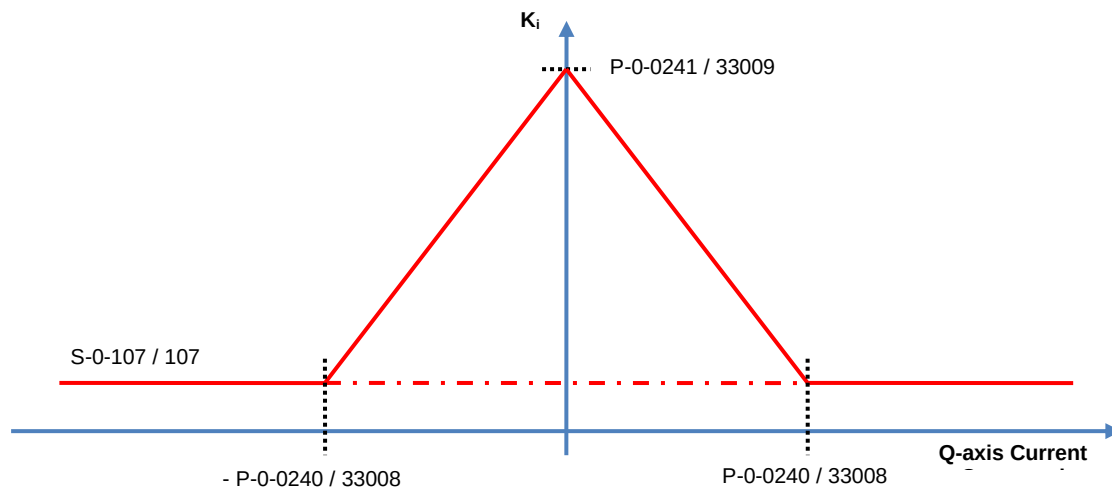


Figure 7-28 Integral Gain Boost

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-0239 / 33007	Current Control Q-axis Low Current Boost Enable	Unsigned Integer (2 bytes)	Boolean	0	0
P-0-0240 / 33008	Current Control Q-axis Low Current Boost Threshold	Signed Integer (4 bytes)	Standard Current	0	0
P-0-0241 / 33009	Current Control Q-axis Low Current Boost Integral Time	Unsigned Integer (2 bytes)	μ s	0	0

7.15.5.5.2 Definitions

7.15.5.5.2.1 Current Control Q-axis Low Current Boost Enable

The *Current Control Q-axis Low Current Boost Enable* allows the Current Loop Integral Gain Scheduler to be enabled or disabled. Set to '1' to enable this feature.

7.15.5.5.2.2 Current Control Q-axis Low Current Boost Threshold

The *Current Control Q-axis Low Current Boost Threshold* determines the current command below which the *Current Control Q-axis Low Current Boost Integral Time* will begin to be applied.

7.15.5.5.2.3 Current Control Q-axis Low Current Boost Integral Time

The *Current Control Q-axis Low Current Boost Integral Time* defines the current loop integral time to be used at 0 commanded current.

7.15.6 Variable Torque Control

7.15.6.1 Description

Variable Torque Control is a feature which allows the applied motor torque to be independently varied in the "positive" and "negative" direction. It is intended to be modified during operation as required for the application.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-0225 / 32993	Variable Torque Control Word	Unsigned Integer (2 bytes)	Binary	0	0
P-0-0226 / 3994	Variable Torque Limit – Maximum	Signed Integer (2 bytes)	Standard Torque	4 Nm	4 Nm
P-0-0227 / 32995	Variable Torque Limit - Minimum	Signed Integer (2 bytes)	Standard Torque	-4 Nm	-4 Nm

7.15.6.2 Definitions

7.15.6.2.1 Variable Torque Control Word

The *Variable Torque Control Word* allows the *Variable Torque Limit - Maximum* and *Variable Torque Limit Minimum* to be enabled and disabled independently.

Bit	Value	Definition
0	0	Disable <i>Variable Torque Limit - Maximum</i>
	1	Enable <i>Variable Torque Limit - Maximum</i>
1	0	Disable <i>Variable Torque Limit - Minimum</i>
	1	Enable <i>Variable Torque Limit - Minimum</i>

Table 7-31 – Variable Torque Control Word definition

7.15.6.2.2 Variable Torque Limit - Maximum

The *Variable Torque Limit - Maximum* defines the maximum allowable torque/current that can be applied to the motor. This limit can contain a positive or negative value depending on the direction in which it is to be applied.

7.15.6.2.3 Variable Torque Limit - Minimum

The *Variable Torque Limit - Minimum* defines the minimum allowable torque/current that can be applied to the motor. This limit can contain a positive or negative value depending on the direction in which it is to be applied.

7.15.6.3 Operation Example

Figure 7-29 below shows how different configurations of the *Variable Torque Control Word* influence the torque limits that are applied. We define a Unified Torque Limit as the smallest current/torque limit that is found from comparing all other current/torque limits current configured in the drive.

Some examples of torque/current limit configurations are:

- **Case 1:** Variable Torque Limits disabled. Unified Torque Limit is applied.
- **Case 2:** Maximum Variable Torque Limit enabled. Unified Torque Limit used for minimum limit.
- **Case 3:** Minimum Variable Torque Limit enabled. Unified Torque Limit used for maximum limit.
- **Case 4:** Maximum and Minimum Variable Torque Limits enabled.
- **Case 5:** Minimum Variable Torque Limit enabled. Minimum limit set to a value larger than zero, which means the drive will be unable to produce zero torque. This configuration is useful for axes which have a non-zero static load applied to them, for example a vertical axis subject to gravity.
- **Case 6:** Maximum Variable Torque Limit enabled. Maximum limit is set to a value which is larger than the Unified Torque Limit; hence the Unified Torque Limit is used.



NOTE: The minimum limit need not necessarily be negative and the maximum limit need not necessarily be positive, as highlighted in Case 5. However the maximum limit must be larger than the minimum limit, otherwise the drive will report a Class 1 Diagnostic (C1D) error (E080).

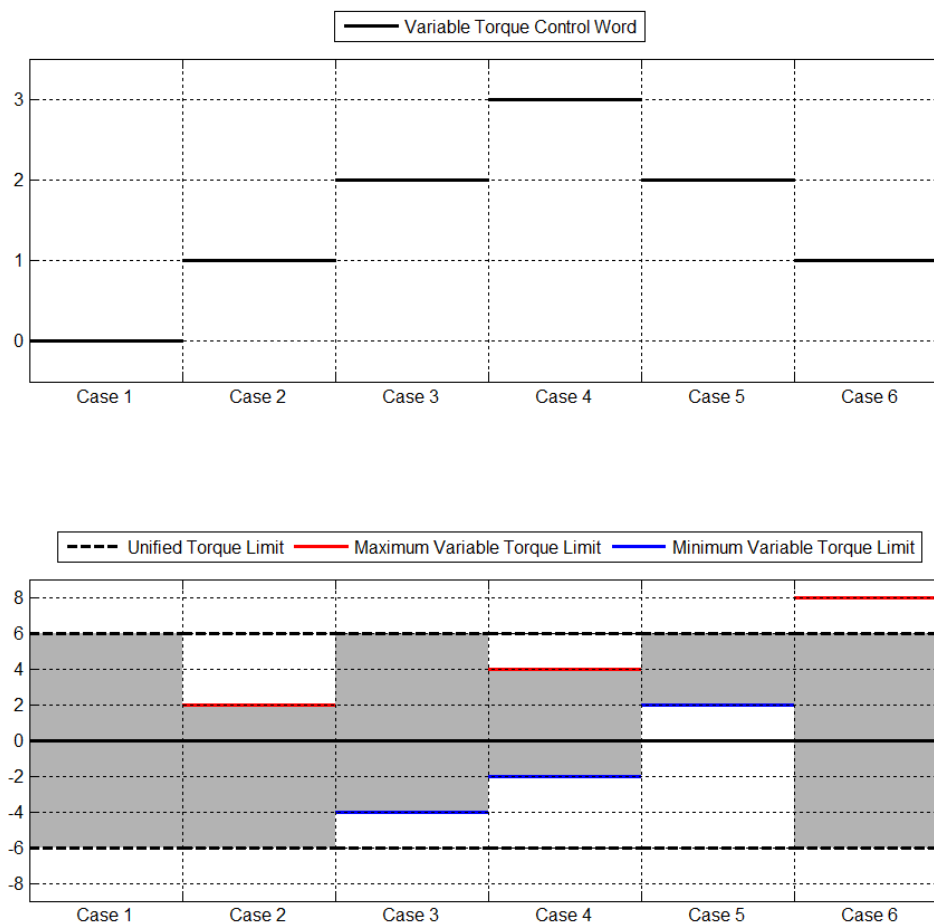


Figure 7-29 Variable Torque Control

7.15.7 Motor and Amplifier Temperature Current Limits

7.15.7.1 Description



NOTE: Motor temperature monitoring is only available on the AMD5x series servo drives.

The AMD servo drives support current limiting based on both motor and servo drive amplifier temperature. To utilise the motor temperature based current limits, motor temperature monitoring must be enabled. Amplifier temperature monitoring is always enabled. Refer to [7.25 Temperature Monitoring](#) for more information.

Both motor and amplifier temperature current limits are configured via two variables; the temperature threshold and the decay rate. The threshold defines the temperature at which the current limit is reduced to zero, while the decay rate determines the current per degree at which to reduce the limit.

The upper limit of current given a sensed temperature can be calculated by:

$$I_{limit} = (Temp_{zero\ current} - Temp_{sensed}) \times Current\ decay\ rate$$

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-1242 / 34010	Temperature Monitor Control Word	Unsigned Integer (2 bytes)	Binary	0001 0001	0000 0001

P-0-1243 / 34011	Amplifier Temperature Current Limiting - Temp Threshold	Unsigned Integer (2 bytes)	Standard Temperature	90 °C	90 °C
P-0-1244 / 34012	Motor Temperature Current Limiting - Temp Threshold	Unsigned Integer (2 bytes)	Standard Temperature	90 °C	90 °C
P-0-1245 / 34013	Amplifier Temperature Current Limiting - Decay Rate	Unsigned Integer (2 bytes)	2^{-11} A/°C	0.75 A/°C	0.75 A/°C
P-0-1246 / 34014	Motor Temperature Current Limiting - Decay Rate	Unsigned Integer (2 bytes)	2^{-11} A/°C	0.75 A/°C	0.75 A/°C

7.15.7.2 Definitions

7.15.7.2.1 Temperature Monitoring Control Word

The *Temperature Monitoring Control Word* allows enabling of temperature current limiting for the servo drive amplifier and/or motor via bits 8 and 9 respectively. Temperature measurement of the amplifier and/or motor must also be enabled via bits 0 and 1 respectively. Refer to [7.25 Temperature Monitoring](#) for more information.

7.15.7.2.2 Amplifier Temperature Current Limiting - Temp Threshold

The *Amplifier Temperature Current Limiting - Temp Threshold* defines the servo drive amplifier temperature at which the current limit is set to 0.

7.15.7.2.3 Motor Temperature Current Limiting - Temp Threshold

The *Motor Temperature Current Limiting - Temp Threshold* defines the motor temperature at which the current limit is set to 0.

7.15.7.2.4 Amplifier Temperature Current Limiting - Decay Rate

The *Amplifier Temperature Current Limiting - Decay Rate* defines the rate per degree of amplifier temperature at which the current limit is reduced to 0.

7.15.7.2.5 Motor Temperature Current Limiting - Decay Rate

The *Motor Temperature Current Limiting - Decay Rate* defines the rate per degree of motor temperature at which the current limit is reduced to 0.

7.15.7.3 Operation Example

[Figure 7-30](#) illustrates example configurations for Amplifier / Motor Temperature Based Current Limiting.

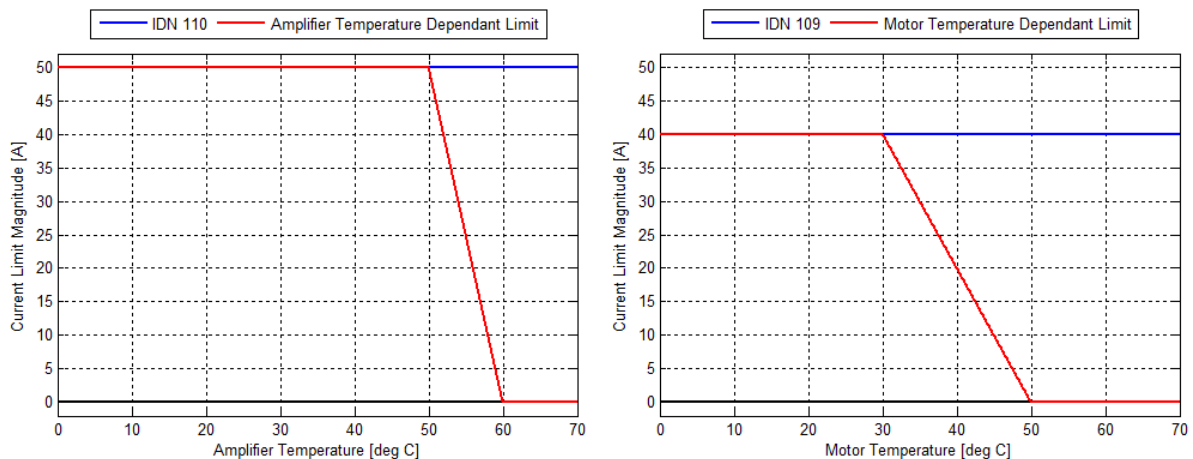


Figure 7-30 Amplifier / Motor Temperature Based Current Limiting

Parameter	Value
Amplifier Temperature Current Limiting - Temp Threshold	60 °C
Motor Temperature Current Limiting - Temp Threshold	50 °C
Amplifier Temperature Current Limiting - Decay Rate	5 A/°C
Motor Temperature Current Limiting - Decay Rate	2 A/°C

Table 7-32 - Amplifier / Motor Temperature Based Current Limiting

7.16 Non-Volatile Parameters

7.16.1 Description



NOTE: This feature is only available on the AMD2000 series servo drives.

The ANCA Motion servo drive provides an EEPROM so that parameters can be retained even if the drive is switched off and then on again (ie. power cycled). Non-Volatile Parameters (NVP) provide the user with flexibility to modify parameter values and store such changes to the drive's EEPROM. Drives may also be tailored with a specific set of parameter values saved to the drive at the end of the production line or for any specific application. These saved parameters may be automatically loaded upon start-up without the intervention of an external controller device / unit.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-263 / 263	NV Parameters - Read	Unsigned Integer (2 bytes)	Procedure Command	0	0
S-0-264 / 264	NV Parameters - Write	Unsigned Integer (2 bytes)	Procedure Command	0	0
P-0-0352 / 33120	Non-volatile Parameters Auto-load on Start-up	Unsigned Integer (2 bytes)	Boolean	1	1
P-0-0353 / 33121	Full NV Parameter(s) List (200 elements)	Unsigned Integer (2 bytes) Array	IDN	N/A	N/A
P-0-0354 / 33122	Selected NV Parameter(s) List (50 elements)	Unsigned Integer (2 bytes) Array	IDN	0,0,...,0	0,0,...,0
P-0-0355 / 33123	Non-Volatile Parameter Control Word	Unsigned Integer (2 bytes)	Binary	0000 0100	0000 0100
P-0-0357 / 33125	Non-Volatile Parameters Last Error Code	Unsigned Integer (2 bytes)	Error Code	N/A	N/A

7.16.2 Definitions

7.16.2.1 NV Parameters - Read

The *NV Parameters - Read* procedure command allows the control device / unit to trigger a write to the servo drive's Non-Volatile Memory. To perform the write operation:

1. Set *NV Parameters - Read* to '3' to trigger the read procedure command.
2. The servo drive will read parameters according to the configuration defined in *Non-Volatile Parameter Control Word*. Any errors encountered during this process will abort the procedure command with the error code stored in *Non-Volatile Parameters Last Error Code*.
3. Set *NV Parameters - Read* to '0' to cancel the procedure command.

7.16.2.2 NV Parameters - Write

The *NV Parameters - Write* procedure command allows the control device / unit to trigger a write to the servo drive's Non-Volatile Memory. To perform the write operation:

1. If required, modify *Selected NV Parameter(s) List*.
2. Set *NV Parameters - Write* to '3' to trigger the save procedure command.
3. The servo drive will save the parameters. Any errors encountered during this process will abort the procedure command with the error code stored in *Non-Volatile Parameters Last Error Code*.
4. Set *NV Parameters - Write* to '0' to cancel the procedure command.

7.16.2.3 Non-Volatile Parameters Auto-load on Start-up

The servo drive can be configured to automatically perform restoration of last saved Non-Volatile Parameters upon power up. To enable this feature, set *Non-Volatile Parameters Auto-load on Start-up* to '1'. This parameter is stored by default whenever a write operation is triggered. However explicit write procedure is required to make new value effective.

7.16.2.4 Full NV Parameter(s) List

The *Full NV Parameter(s) List* is a read only IDN that lists all the servo drive IDNs that can be saved into Non-Volatile memory.

7.16.2.5 Selected NV Parameter(s) List

The *Selected NV Parameter(s) List* defines the IDNs that will be saved into the drive's EEPROM. A maximum of 50 IDNs may be selected including arrays.

7.16.2.6 Non-Volatile Parameter Control Word

The *Non-Volatile Parameter Control Word* specifies the behaviour of non-volatile parameters loading.

Bit	Value	Definition
0	0	Do not allow partial load. The servo drive returns an error if any invalid IDN is found in the saved NVP list. An invalid IDN is one that does not exist in the current profile. This scenario is likely to occur after a firmware upgrade where an IDN may have been deleted or changed in the new firmware
	1	Allow partial load. The servo drive ignores invalid IDNs found in the saved NVP list and continues to load valid IDNs.
1	0	Load all saved parameters.
	1	Only load the saved parameters currently in the <i>Selected NV Parameter(s) List</i> .
2	0	<i>Selected NV Parameter(s) List</i> will not be modified upon loading from EEPROM.
	1	Upon loading from EEPROM, the <i>Selected NV Parameter(s) List</i> will be updated with the IDNs that were successfully loaded.

Table 7-33 – Non-Volatile Parameter Control Word bit definition

7.16.2.7 Non-Volatile Parameters Last Error Code

If any error occurs whilst executing NVP save/load operation, the error code is lodged in *Non-Volatile Parameters Last Error Code*. A list of the possible error codes is shown below.

Code	Description
20	Error on Save - Invalid IDN

21	Error on Save - Invalid Parameter Array Length
22	Error on Save - Insufficient Space
23	Error on Read - Invalid IDN
24	Error on Read - Header Checksum Mismatch (last NVP save may be corrupted)
25	Error on Read - Header Information Mismatch (last NVP save may be corrupted)
26	Error on Read - Header Status Bit Mismatch (last NVP save may be corrupted)
27	Error on Read - Too Many IDNs to Load
28	Error on Read - Invalid IDN and Allow Partial Load Disabled
29	Error on Read - Attempt to Load Volatile Parameter
30	Error on Read - Invalid IDN Data Length
31	Error on Read - NULL IDN Data

Table 7-34 - Non-Volatile Parameter Error Codes

7.16.3 Operation Examples

7.16.3.1 Load Selected IDNs Only

1. The previous save had *Selected NV Parameter(s) List* = {1,2,3,4,5} and the contents were saved.
2. The *Selected NV Parameter(s) List* is updated to {1,2,3,5,6}.
3. Bit 1 of the *Non-Volatile Parameter Control Word* is set to '1'.
4. Upon NVP read either via *NV Parameters – Read* or *Non-Volatile Parameters Auto-load on Start-up*, only {1,2,3,5} will be loaded from memory.

7.16.3.2 Replacing Selected IDN List

1. The previous save had *Selected NV Parameter(s) List* = {1,2,3,4,5} and the contents were saved.
2. The *Selected NV Parameter(s) List* is updated to {6,7,8,9,10}.
3. Bit 1 & 2 of the *Non-Volatile Parameter Control Word* is set to '0' & '1' respectively.
4. Upon NVP read either via *NV Parameters – Read* or *Non-Volatile Parameters Auto-load on Start-up*, {1,2,3,4,5} will be loaded. *Selected NV Parameter(s) List* will also be updated to {1,2,3,4,5}.



NOTE: The updated *Selected NV Parameter(s) List* only contains IDNs that were successfully loaded.

7.16.3.3 Erasing Non-Volatile Memory

To erase the servo drive's Non-Volatile Memory and restore default parameters:

1. Set *Non-Volatile Parameters Auto-load on Start-up* to 0 (if it is not already set to 0).
2. Save using the *NV Parameters - Write* procedure command.
3. Power cycle the drive.
4. Default values are restored to all IDNs.

7.17 Probing

7.17.1 Description

Probing provides a measuring function by latching the current position value when the probe input changes (configurable rising and/or falling edge). A single servo drive can latch the axis position based on a connected input. Multiple servos can also latch positions at the same moment with one drive setup as the probe 'master', performing the same functions as a single drive, while the other drives are setup as probe 'slaves' that are triggered from a timestamp transmitted by the EtherCAT master from the 'master' to the 'slaves'. The use of distributed clocks allows the position of multiple servo drives to be latched to a single signal accurately. Probing may be used in various applications, such as detecting axis boundary limit, or measuring contour and geometry.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0130 / 130	Probe Value 1 - Rising Edge	Signed Integer (4 bytes)	Standard Position	N/A	N/A
S-0-0131 / 131	Probe Value 1 - Falling Edge	Signed Integer (4 bytes)	Standard Position	N/A	N/A
S-0-0169 / 169	Probe Control Parameter	Unsigned Integer (2 bytes)	Binary	0	0
S-0-0170 / 170	Probing Procedure Command	Unsigned Integer (2 bytes)	Procedure Command	0	0
S-0-0179 / 179	Probe Status	Unsigned Integer (2 bytes)	Binary	N/A	N/A
S-0-0401 / 401	Probe 1 Active	Unsigned Integer (2 bytes)	Boolean	N/A	N/A
S-0-0405 / 405	Probe 1 Enable	Unsigned Integer (2 bytes)	Boolean	0	0
S-0-0409 / 409	Probe 1 Rising Latched	Unsigned Integer (2 bytes)	Boolean	N/A	N/A
S-0-0410 / 410	Probe 1 Falling Latched	Unsigned Integer (2 bytes)	Boolean	N/A	N/A
P-0-0382 / 33150	Joint Control Event Capture Finish Status	Unsigned Integer (2 bytes)	Binary	N/A	N/A
P-0-0383 / 33151	Event Capture Global Time Out - Ch 0	Signed Integer (4 bytes)	$10^{-3}\mu\text{s}$	N/A	N/A
P-0-0387 / 33155	Joint Control Event Capture Commence Status	Unsigned Integer (2 bytes)	Binary	0	0
P-0-0388 / 33156	Event Capture Global Time In - Ch0	Signed Integer (4 bytes)	$10^{-3}\mu\text{s}$	0	0
P-0-0562 / 33330	Probing Device	Unsigned Integer (2 bytes)	Binary	1	1
P-0-0565 / 33333	Probe Digital Input Source	Unsigned Integer (2 bytes)	None	0	0

7.17.2 Definitions

7.17.2.1 Probe Value 1 - Rising Edge

The *Probe Value 1 - Rising Edge* contains the axis position that was latched when probe 1 rising edge was detected. Rising edge must be enabled via the *Probe Control Parameter*.

7.17.2.2 Probe Value 1 - Falling Edge

The *Probe Value 1 - Falling Edge* contains the axis position that was latched when probe 1 falling edge was detected. Falling edge must be enabled via the *Probe Control Parameter*.

7.17.2.3 Probe Control Parameter

The *Probe Control Parameter* can be used to enable different probe latching behaviour.

Bit	Value	Definition
0	0	Position will not be latched on probe 1 rising edge.
	1	Position will be latched on probe 1 rising edge.
1	0	Position will not be latched on probe 1 falling edge.
	1	Position will be latched on probe 1 falling edge.

Table 7-35 Probe Control Parameter bit definition

7.17.2.4 Probing Procedure Command

The *Probing Procedure Command* is used in conjunction with *Probe 1 Enable* to start and stop the probing process. The procedure command follows the standard syntax; set to 3 to run and 0 to stop.

7.17.2.5 Probe Status

The *Probe Status* is a bit field that combines the individual IDNs *Probe 1 Rising Latched* and *Probe 1 Falling Latched* to indicate when the probe has latched a position.

Bit	Value	Definition
0	0	Probe 1 rising edge has not been latched. Reflects <i>Probe 1 Rising Latched</i> .
	1	Probe 1 rising edge has been latched. Reflects <i>Probe 1 Rising Latched</i> .
1	0	Probe 1 falling edge has not been latched. Reflects <i>Probe 1 Falling Latched</i> .
	1	Probe 1 falling edge has been latched. Reflects <i>Probe 1 Falling Latched</i> .

Table 7-36 - Probe Status bit definition

7.17.2.6 Probe 1 Active

The *Probe 1 Active* signal indicates that probe 1 is armed and awaiting a probe signal trigger. The trigger will depend on the configuration of the *Probe Control Parameter*. Often utilised with *Real Time Status Bits* for monitoring via process (cyclic) data.

7.17.2.7 Probe 1 Enable

Probe 1 Enable is used in conjunction with the *Probing Procedure Command* to begin the probing process. Often used with *Real Time Control Bits* for control via process (cyclic) data.

7.17.2.8 Probe 1 Rising Latched

The *Probe 1 Rising Latched* signal is true when a rising edge has been latch on probe 1 and *Probe Value 1 – Rising Edge* has been populated. This is mirrored by bit 0 of *Probe Status*. Often utilised with *Real Time Status Bits* for monitoring via process (cyclic) data.

7.17.2.9 Probe 1 Falling Latched

The *Probe 1 Falling Latched* signal is true when a falling edge has been latch on probe 1 and *Probe Value 1 – Falling Edge* has been populated. This is mirrored by bit 1 of *Probe Status*. Often utilised with *Real Time Status Bits* for monitoring via process (cyclic) data.

7.17.2.10 Probing Device

The *Probing Device* parameter allows configuration of the probe trigger source. Available sources are shown below.

Value	Description
0	Motor Index Pulse
1	Probe Input. Dedicated input on AMD5x or digital input on AMD2000.
2	SoE Timestamp

Table 7-37 - Probing Device definition

7.17.2.11 Probe Digital Input Source



NOTE: This feature is only available on the AMD2000 series servo drives.

The *Probe Digital Input Source* allows a digital input to be assigned as the probe input as defined in the *Probing Device* parameter.

Value	Description
0	Digital Input 1
1	Digital Input 2
2	Digital Input 3
3	Digital Input 4
4	Digital Input 5
5	Digital Input 6
6	Digital Input 9 (High Speed Differential)
7	Digital Input 10 (High Speed Differential)

Table 7-38 – Probe Digital Input Source definition

7.17.3 General Operation

Operation of probing occurs as described below.

1. Probing is configured via the *Probe Control Parameter* and *Probing Device*.
2. The *Probing Procedure Command* is set to 3.
3. The probing procedure is started by setting *Probe 1 Enable* to 1. *Probe 1 Active* will be equal 1.
4. The probe will trigger depending on the configuration of *Probing Device*:
 - a. *Probing Device* = 0: The probe is triggered by the index pulse.
 - b. *Probing Device* = 1: The probe is triggered by a probe input (either dedicated or allocated digital input depending on servo drive model).
 - c. *Probing Device* = 2: The probe is triggered by an SoE timestamp.
5. *Probe 1 Active* will be equal to 0.
6. Depending on the *Probe Control Parameter* configuration:
 - a. Rising edge enabled: *Probe 1 Rising Latched* and *Probe Status* bit 0 will equal 1. The latched position will be available from *Probe Value 1 – Rising Edge*.
 - b. Falling edge enabled: *Probe 1 Falling Latched* and *Probe Status* bit 1 will equal 1. The latched position will be available from *Probe Value 1 – Falling Edge*.
7. *Probe 1 Enable* is set to 0. This can be set to 1 to start a new probing cycle without modifying the *Probing Procedure Command*.
8. If no more probing is to be performed, the *Probing Procedure Command* is set to 0 to cancel the procedure.

7.18 Real Time Control Bits

7.18.1 Description

Real Time Control Bits allow for the allocation of up to two IDNs to be controlled via the [Master Control Word](#). This provides the ability to control a function via process data without the explicitly adding it to process data before the Safe-Op EtherCAT state. Only the Least Significant Bit (LSB) of the allocated IDN is controlled.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0300 / 300	Real Time Control Bit 1	Unsigned Integer (2 bytes)	Boolean	N/A	N/A
S-0-0301 / 301	Real Time Control Bit 1 Allocation	Unsigned Integer (2 bytes)	IDN	0	0
S-0-0302 / 302	Real Time Control Bit 2	Unsigned Integer (2 bytes)	Boolean	N/A	N/A
S-0-0303 / 303	Real Time Control Bit 2 Allocation	Unsigned Integer (2 bytes)	IDN	0	0

7.18.2 Definitions

7.18.2.1 Real Time Control Bit 1

The *Real Time Control Bit 1* reflects the state of the allocated IDN's bit 0 as set by bit 6 of the [Master Control Word](#).

7.18.2.2 Real Time Control Bit 1 Allocation

The *Real Time Control Bit 1 Allocation* defines the IDN that is mapped to *Real Time Control Bit 1*. Only the LSB (bit 0) is mapped.

7.18.2.3 Real Time Control Bit 2

The *Real Time Control Bit 2* reflects the state of the allocated IDN's bit 0 as set by bit 7 of the *Master Control Word*.

7.18.2.4 Real Time Control Bit 2 Allocation

The *Real Time Control Bit 2 Allocation* defines the IDN that is mapped to *Real Time Control Bit 2*. Only the LSB (bit 0) is mapped.

7.19 Real Time Status Bits

7.19.1 Description

Real Time Status Bits allow for the allocation of up to two IDNs to be monitored via the *Drive Status Word*. This provides the ability to monitor a function via process data without the explicitly adding it to process data before the Safe-Op EtherCAT state. Only the Least Significant Bit (LSB) of the allocated IDN is monitored.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0304 / 304	Real Time Status Bit 1	Unsigned Integer (2 bytes)	Boolean	N/A	N/A
S-0-0305 / 305	Real Time Status Bit 1 Allocation	Unsigned Integer (2 bytes)	IDN	0	0
S-0-0306 / 306	Real Time Status Bit 2	Unsigned Integer (2 bytes)	Boolean	N/A	N/A
S-0-0307 / 307	Real Time Status Bit 2 Allocation	Unsigned Integer (2 bytes)	IDN	0	0

7.19.2 Definitions

7.19.2.1 Real Time Status Bit 1

The *Real Time Status Bit 1* is a duplicate of bit 6 of the *Drive Status Word* and reflects the state of the allocated IDN's bit 0.

7.19.2.2 Real Time Status Bit 1 Allocation

The *Real Time Status Bit 1 Allocation* defines the IDN that is mapped to *Real Time Status Bit 1*. Only the LSB (bit 0) is mapped.

7.19.2.3 Real Time Status Bit 2

The *Real Time Status Bit 2* is a duplicate of bit 7 of the *Drive Status Word* and reflects the state of the allocated IDN's bit 0.

7.19.2.4 Real Time Status Bit 2 Allocation

The *Real Time Status Bit 2 Allocation* defines the IDN that is mapped to *Real Time Status Bit 2*. Only the LSB (bit 0) is mapped.

7.20 Reversal Compensation

7.20.1 Description

During axis reversal, non-linear dynamic friction can cause an increase in position following error. This error may be reduced by a high-gain velocity controller and/or use of the [Velocity Feed-forward Gain Factor](#) but this must be balanced with system stability. The AMD servo drive supports Reversal Compensation, which compensates for the behaviour of friction at low speed without affecting system stability.

Reversal Compensation operates by inserting up to two pulses of additional velocity command into the velocity controller to overcome the predicted friction of the axis. This feature requires calibration under typical application conditions for best results.

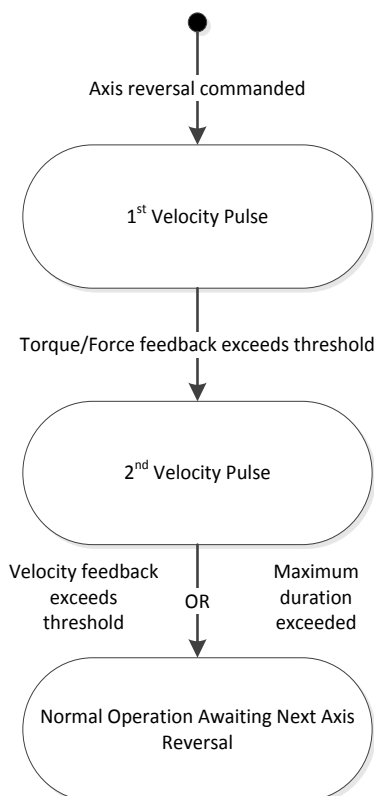


Figure 7-31 - Reversal Compensation algorithm

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-0642 / 33410	Reversal Compensation – Enable	Unsigned Integer (2 bytes)	Boolean	0	0
P-0-0643 / 33411	Reversal Compensation – 1 st Pulse Amplitude	Signed Integer (4 bytes)	Standard Velocity	0	0
P-0-0644 / 33412	Reversal Compensation – Torque Threshold	Signed Integer (4 bytes)	Standard Torque	0	0
P-0-0645 / 33413	Reversal Compensation – 2 nd Pulse Amplitude	Signed Integer (4 bytes)	Standard Velocity	0	0
P-0-0646 / 33414	Reversal Compensation – 2 nd Pulse Max Duration	Unsigned Integer (2 bytes)	250 μ s	0	0
P-0-0647 / 33415	Reversal Compensation – Velocity Threshold	Signed Integer (4 bytes)	Standard Velocity	0	0

7.20.2 Definitions

7.20.2.1 Reversal Compensation – Enable

The *Reversal Compensation – Enable* parameter is used to enable or disable the Reversal Compensation feature. Set to 1 to enable.

7.20.2.2 Reversal Compensation – 1st Pulse Amplitude

The *Reversal Compensation – 1st Pulse Amplitude* is the amplitude of the of additive velocity to be added to the output of the position controller until the measured torque reaches the *Reversal Compensation – Torque Threshold*. This velocity is added in the direction of travel.

7.20.2.3 Reversal Compensation – Torque Threshold

The *Reversal Compensation – Torque Threshold* specifies the torque threshold above which the *Reversal Compensation – 1st Pulse Amplitude* additive velocity will cease. This torque threshold is in the direction of travel.

7.20.2.4 Reversal Compensation – 2nd Pulse Amplitude

The *Reversal Compensation – 2nd Pulse Amplitude* is the amplitude of the of additive velocity to be added to the output of the position controller until the measured velocity reaches the *Reversal Compensation – Velocity Threshold* or the duration reaches that specified in *Reversal Compensation – 2nd Pulse Max Duration*. This velocity is added in the direction of travel.

7.20.2.5 Reversal Compensation – 2nd Pulse Max Duration

The *Reversal Compensation – 2nd Pulse Max Duration* parameter defines the maximum allowable time the *Reversal Compensation – 2nd Pulse Amplitude* additive velocity is applied.

7.20.2.6 Reversal Compensation – Velocity Threshold

The *Reversal Compensation – Velocity Threshold* specifies the velocity threshold above which the *Reversal Compensation – 2nd Pulse Amplitude* additive velocity will cease. This velocity threshold is in the direction of travel.

7.20.3 Operation Example

The following example demonstrates how Reversal Compensation can be used to overcome friction effects upon axis reversal.

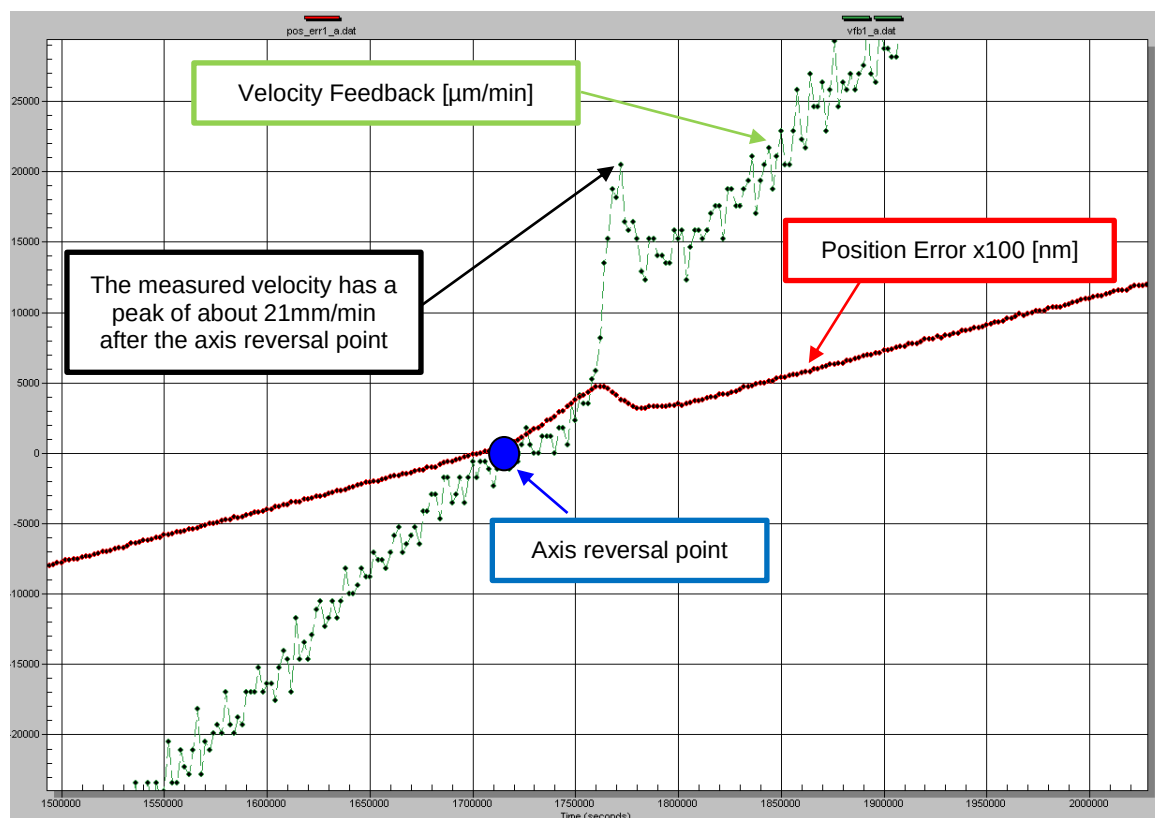


Figure 7-32 - Effect of friction at axis reversal

Figure 7-32 shows an axis that demonstrates a noticeable position error increase when the velocity feedback changes polarity. It can also be seen that the position controller responds to this increase in position error and causes a velocity overshoot when the friction behaviour changes at higher velocities.

To use Reversal Compensation to remove this behaviour:

1. Ensure the servo drive is disabled.
2. The value of *Reversal Compensation – 1st Pulse Amplitude* is chosen based upon the measured peak of the velocity overshoot after axis reversal. *Reversal Compensation – 1st Pulse Amplitude* = 21000 µm/min.
3. The *Reversal Compensation – Torque Threshold* is chosen to be 0. This is the recommended initial and can be increased later if required.
4. The *Reversal Compensation – 2nd Pulse Amplitude* is chosen using the 'rule of thumb' that it is equal to 50% of the *Reversal Compensation – 1st Pulse Amplitude*. *Reversal Compensation – 2nd Pulse Amplitude* = 10500 µm/min.
5. The *Reversal Compensation – 2nd Pulse Max Duration* is set to 1 second as an initial estimate.
6. The *Reversal Compensation – Velocity Threshold* is chosen using the 'rule of thumb' that it is chosen to be 30% of the *Reversal Compensation – 1st Pulse Amplitude*. *Reversal Compensation – Velocity Threshold* = 7000 µm/min.
7. Enable Reversal Compensation by setting *Reversal Compensation – Enable* to 1.

The same test is performed again with the above configuration, the results are shown in Figure 7-33 below.

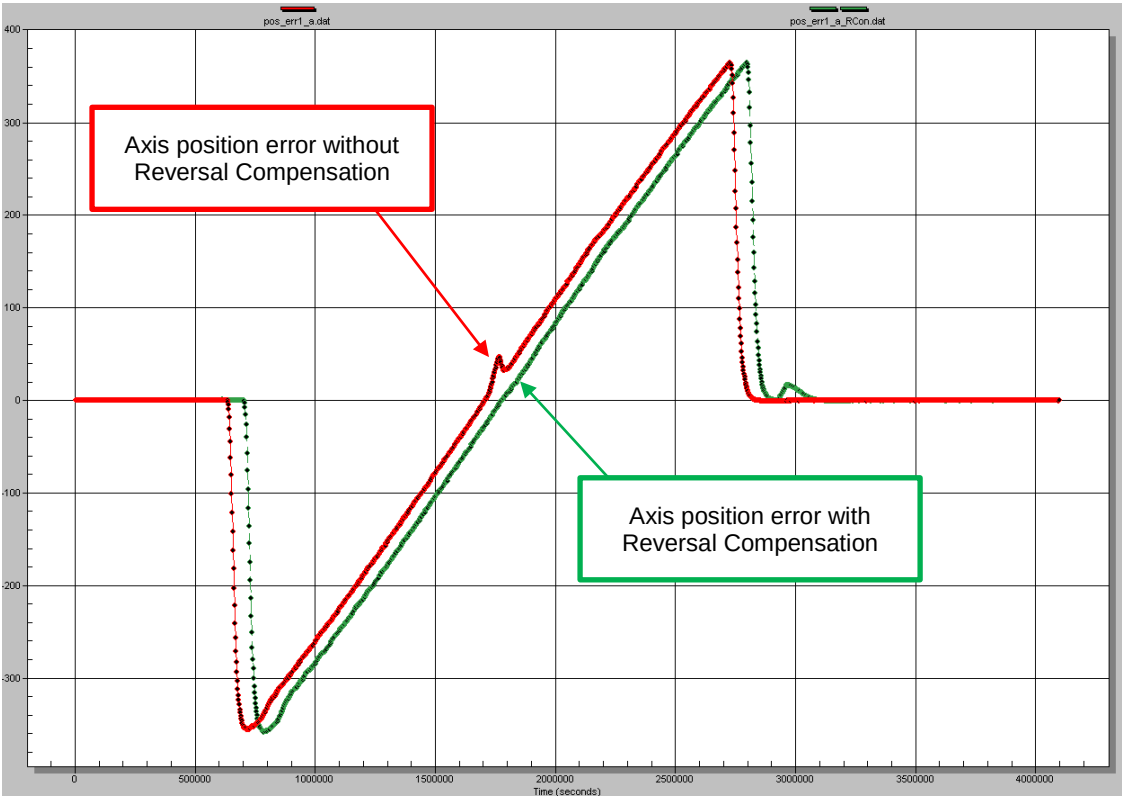


Figure 7-33 - Example Reversal Compensation position error difference

7.21 Safe Torque Off

7.21.1 Description



NOTE: This feature is only available on the AMD2000 series D21xx servo drives.

The AMD servo drives with Safe Torque Off (STO) feature diagnostics to aid in identifying the cause of a STO being active or the cause of a STO related fault.

IDN	Description	Data Type	Units	Default
				AMD2000
P-0-3242 / 36010	STO Module Signals	Unsigned Integer (2 bytes)	Binary	0,0,0,...,0

Please refer to the [AMD2000 Series D21xx User Guide](#) for more information on the Safe Torque Off feature.

7.21.2 Definitions

7.21.2.1 STO Module Signals

STO Module Signals reflects the diagnostic signals from the Safe Torque Off (STO) module as given in Table 7-38.

Bit	Value	Definition	C1D Error
0	0	STO Channel 1 Inactive	No Error
	1	STO Channel 1 Asserted	70
10	0	STO Channel 2 Voltage Normal	No Error
	1	STO Channel 2 Voltage Fault	73
11	0	STO Timer Normal	No Error
	1	STO Timer Fault	71
14	0	STO Channel 2 Inactive	No Error
	1	STO Channel 2 Asserted	70
15	0	STO Channel 1 Voltage Normal	No Error
	1	STO Channel 1 Voltage Fault	72

Table 7-39 - Velocity Data Scaling Type bit definition

7.22 Servo Control

7.22.1 Overview

The AMD servo drive control loops utilise the architecture for position, velocity and torque/current control displayed by [Figure 7-34](#). On the AMD servo drive, the term Servo Control refers to the position controller, velocity controller and related functions.

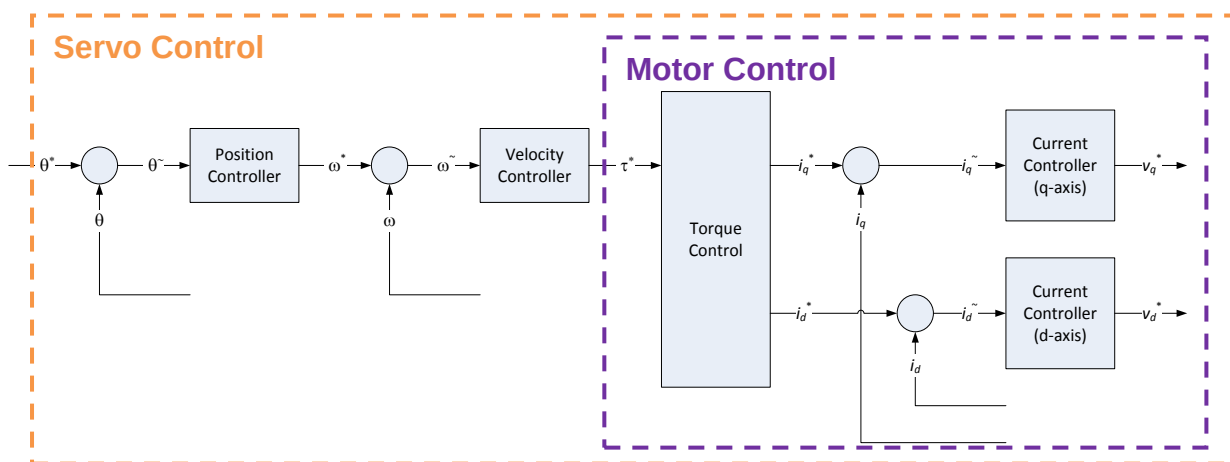


Figure 7-34 - Cascaded control architecture found in IEC 61491

7.22.2 Velocity Controller

7.22.2.1 Description

The AMD servo drive's velocity control is performed by a Proportional-Integral controller with acceleration feedforward. This controller contains gain parameters that allow the user to 'tune' the response of the system. For more information on determining controller parameter values refer to [7.28 Tuning – Velocity Control](#).

Additionally, the position controller provides position monitoring capabilities providing the control device unit additional high level information such as when the axis is moving and if the difference between the velocity command (output of position controller or control device/unit command) and the velocity feedback exceeds a configurable threshold.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0100 / 100	Velocity Controller Proportional Gain	Unsigned Integer (2 bytes)	1/s	100	100
S-0-0101 / 101	Velocity Controller Integral Time	Unsigned Integer (2 bytes)	10 ⁻¹ ms	100	100
S-0-0124 / 124	Standstill Window	Signed Integer (4 bytes)	Standard Velocity	0	0
S-0-0331 / 331	Status 'At Standstill'	Unsigned Integer (2 bytes)	Boolean	N/A	N/A
P-0-0202 / 32970	Acceleration Feed-forward Gain Factor	Signed Integer (2 bytes)	2 ⁻¹⁴	0	0
P-0-0203 / 32971	Velocity Controller Proportional Gain Scale	Signed Integer (2 bytes)	bits	0	0
P-0-0210 / 32978	Velocity Following Error Threshold	Signed Integer (4 bytes)	Standard Velocity	0	0

7.22.2.2 Definitions

7.22.2.2.1 Velocity Controller Proportional Gain

The *Velocity Controller Proportional Gain* determines the responsiveness of the velocity controller. A higher value will result in a faster response to a command or disturbance but may cause instability. The default units for this parameter are 1/s but can be modified by the *Velocity Controller Proportional Gain Scale*.

7.22.2.2.2 Velocity Controller Integral Time

The *Velocity Controller Integral Time* determines the low frequency responsiveness of the velocity controller. A higher value will result in a stiffer axis at standstill but may cause instability. The integral component will be disabled if this value is set to 6553.5 ms.

7.22.2.2.3 Standstill Window

The *Standstill Window* defines the absolute velocity below which the servo drive will consider the axis 'at standstill'. This is indicated by *Status 'At Standstill'* set to 1. This value should be set above any noise or quantisation artefacts from the feedback device.

7.22.2.2.4 Status 'At Standstill'

The *Status 'At Standstill'* will be set to 1 when the absolute velocity feedback of the axis is measured to be below the *Standstill Window*. The value will be set to 0 otherwise.

7.22.2.2.5 Acceleration Feed-forward Gain Factor

The *Acceleration Feed-forward Gain Factor* is applied to the commanded acceleration i.e. the derivative of the velocity command and added to the output of the velocity controller. As such, this gain is not affected by the value of the velocity feedback and will cause a non-zero velocity controller output before any velocity error is measured.

7.22.2.2.6 Velocity Controller Proportional Gain Scale

The *Velocity Controller Proportional Gain Scale* allows modification to the value represented by the *Velocity Controller Proportional Gain*. Modification of this value allows for much higher or lower gains than would otherwise be possible. The effective gain is calculated by

$$\text{Effective Gain} = \text{Velocity Controller Proportional Gain} \times 2^{\text{Velocity Controller Proportional Gain Scale}}$$

7.22.2.2.7 Velocity Following Error Threshold

The *Velocity Following Error Threshold* determines the maximum allowable difference between velocity command and feedback i.e. error before a C1D error (E0024) is asserted. The threshold is independent of velocity direction. Setting the threshold to 0 will disable the velocity error monitoring.

7.22.3 Position Controller

7.22.3.1 Description

The AMD servo drive's position control is performed by a Proportional-Integral controller with velocity feedforward. This controller contains gain parameters that allow the user to 'tune' the response of the system. In most applications, only the proportional term is recommended.

Additionally, the position controller provides position monitoring capabilities providing the control device unit additional high level information such as when the target position has been reached and if the difference between the position command and feedback has exceeded a configurable threshold.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0057 / 57	Position Window	Signed Integer (4 bytes)	Standard Position	1 mm	1 mm
S-0-0104 / 104	Position Controller Proportional Gain	Unsigned Integer (2 bytes)	10^{-2} (m/min)/mm	2.4	2.4
S-0-0105 / 105	Position Controller Integral Time	Unsigned Integer (2 bytes)	10^{-1} ms	0	0
S-0-0159 / 159	Monitoring Window	Signed Integer (4 bytes)	Standard Position	1000 mm	1000 mm
S-0-0336 / 336	Status 'In Position'	Unsigned Integer (2 bytes)	Boolean	N/A	N/A
P-0-0192 / 32960	Velocity Feed-forward Gain Factor	Signed Integer (2 bytes)	%	0	0

7.22.3.2 Definitions

7.22.3.2.1 Position Window

The *Position Window* defines the allowable difference between the commanded position and position feedback i.e. position error, within which the servo drive will consider the axis 'in position'. This is indicated by *Status 'In Position'* set to 1. The *Position Window* value applies equally to both positive and negative error, and can therefore be thought of as half of the allowable 'in position' window around the target position.

7.22.3.2.2 Position Controller Proportional Gain

The *Position Controller Proportional Gain* determines the responsiveness of the position controller. A higher value will result in a faster response to a command or disturbance but may cause instability.

7.22.3.2.3 Position Controller Integral Time

The *Position Controller Integral Time* determines the low frequency responsiveness of the position controller. A higher value will result in a stiffer axis at standstill but may cause instability. The integral component will be disabled if this value is set to below 200 ms.

7.22.3.2.4 Monitoring Window

The *Monitoring Window* determines the maximum allowable absolute difference between position command and feedback i.e. error before a C1D error (E0023) is asserted. Setting the threshold to 0 will disable the position error monitoring.

7.22.3.2.5 Status 'In Position'

The *Status 'In Position'* signal is set to 1 when the absolute value of the difference between the commanded position and position feedback is less than the value defined in *Position Window*. The value will be set to 0 otherwise.

7.22.3.2.6 Velocity Feed-forward Gain Factor

The *Velocity Feed-forward Gain Factor* is applied to the commanded velocity i.e. the derivative of the position command and added to the output of the position controller. As such, this gain is not affected by the value of the position feedback and will cause a non-zero position controller output before any position error is measured.

7.22.4 Applying Servo Control Parameters

Tuning parameters for servo control may be configured at any time; however such configuration is not applied until either the drive is re-enabled via or manually triggered by the user. To manually trigger a servo control re-tune, the user may execute the *Servo Control Tuning Procedure Command* by changing its default value from 0 to 3. The user must then complete the procedure command by setting *Servo Control Tuning Procedure Command* back to 0. Execution of the *Servo Control Tuning Procedure Command* will also execute the *Motor Control Tuning Procedure Command*. Refer to [7.15.5.3 Applying Current Control Loop Parameters](#) for more information on the *Motor Control Tuning Procedure Command*.



NOTE: A “re-tune” is an internal calculation of the necessary control parameters. It does NOT mean the drive changes any configuration parameters or gains in the control loops as set by the user.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-0187 / 32955	Servo Control Tuning Procedure Command	Unsigned Integer (2 bytes)	Procedure Command	0	0

7.23 SoE Data Scaling

7.23.1 Overview



WARNING: Modification of data scaling on the servo drive must be done collaboratively with the control unit / device. Updating of data scaling on the servo drive alone may result in unintended behaviour.

The AMD servo drive allows for configurable scaling of standard data types over SoE. This allows for conversion of data into units that are more meaningful to the application e.g. configuring the position feedback to be linear or rotary.

Each datatype can be configured with a unit, a factor and an exponent. Additionally, position and velocity have a dynamic shift factor that determines the maximum representable position and velocity in the drive. A change to scaling will only take affect once the *Parameter Rescaling Procedure Command* is run.

7.23.2 How Scaling is Calculated

7.23.2.1 Standard

As a general rule, the weight of the Least Significant Bit (LSB) shall be derived from the multiplication of the scaling factor and the scaling exponent (base 10):

$$LSB\ weight = \alpha \times factor \times 10^{exponent}$$

Where α varies depending on the data to be scaled.

When the *Parameter Rescaling Procedure Command* is run, both the *exponent* and *factor* will be recalculated depending on the chosen scaling method.

7.23.2.2 Including the Dynamic Shift

A dynamic shift exists for position and velocity that allows configuration of the maximum representable position and velocity in the drive. The shifts will be taken into account when executing the *Parameter Rescaling Procedure Command* as shown below:

$$LSB\ weight = \alpha \times factor \times 10^{exponent} \times 2^{dynamic\ exponent}$$

Example: The *Velocity Scaling Shift Factor* allows you to change the maximum velocity that can be obtained internally in the drive. *Velocity Scaling Shift Factor* = 0 allows representation of 1875 RPM. Furthermore, 1: 3750RPM, 2: 7500, 3: 15,000RPM, 4: 30,000RPM, 5: 60,000RPM, 6: 120,000RPM etc.

The value of the dynamic exponent is a trade-off between the resolution of the velocity/position and the maximum value that can be represented. For peak performance, the lowest possible *dynamic exponent* that provides a representation of the entire working range of the application should be chosen.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-0852 / 33620	Velocity Scaling Shift Factor	Unsigned Integer (2 bytes)	None	0	0
P-0-0853 / 33621	Position Scaling Shift Factor	Unsigned Integer (2 bytes)	None	0	0

7.23.3 Parameter Rescaling Procedure Command

The *Parameter Rescaling Procedure Command* will activate all scaling as configured with the parameters listed in this section. To run the procedure command, set *Parameter Rescaling Procedure Command* to '3'. Cancel the Procedure Command by setting *Parameter Rescaling Procedure Command* to '0'.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-0682 / 33450	Parameter Rescaling Procedure Command	Unsigned Integer (2 bytes)	Procedure Command	0	0

7.23.4 Position Scaling

7.23.4.1 Description



NOTE: Position data on the AMD servo drive is always with respect to the load or end effector i.e. after the gear box and feed constant calculations.

The SoE position scaling can be broadly split into linear or rotary. Both linear and rotary have a 'preferred' scaling or fully customisable parameter scaling option. Additionally, the axis can be configured as modulo. This allows a position to be defined where the position 'wraps', allowing rotary axes to perform multiple rotations without the accumulation of position.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0076 / 76	SoE Position Scaling - Type	Unsigned Integer (2 bytes)	Binary	65	65

S-0-0077 / 77	SoE Position Scaling - Linear Factor	Unsigned Integer (2 bytes)	None	1	1
S-0-0078 / 78	SoE Position Scaling - Linear Exponent	Signed Integer (2 bytes)	None	-4	-4
S-0-0079 / 79	SoE Position Scaling - Rotational Resolution	Unsigned Integer (4 bytes)	10^{-4} degrees	3,600,000°	3,600,000°

7.23.4.2 Position Data Scaling Type

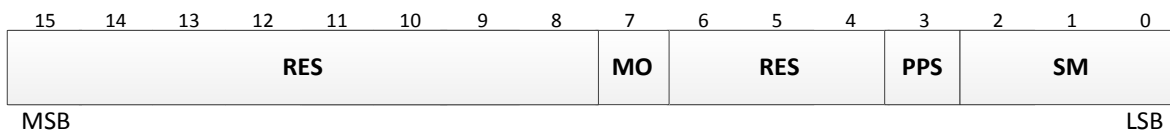


Figure 7-35 - Position Data Scaling Type definition

Bit	Key	Definition	Value	Description
0-2	SM	Scaling Method	000	No scaling
			001	Linear scaling
			010	Rotational scaling
3	PPS	Preferred/Parameter Scaling	0	Preferred scaling
			1	Parameter scaling
4-6	RES	Reserved	X	-
7	MO	Modulo Operation	0	Absolute format
			1	Modulo format
8-15	RES	Reserved	X	-

Table 7-40 - Position Data Scaling Type bit definition

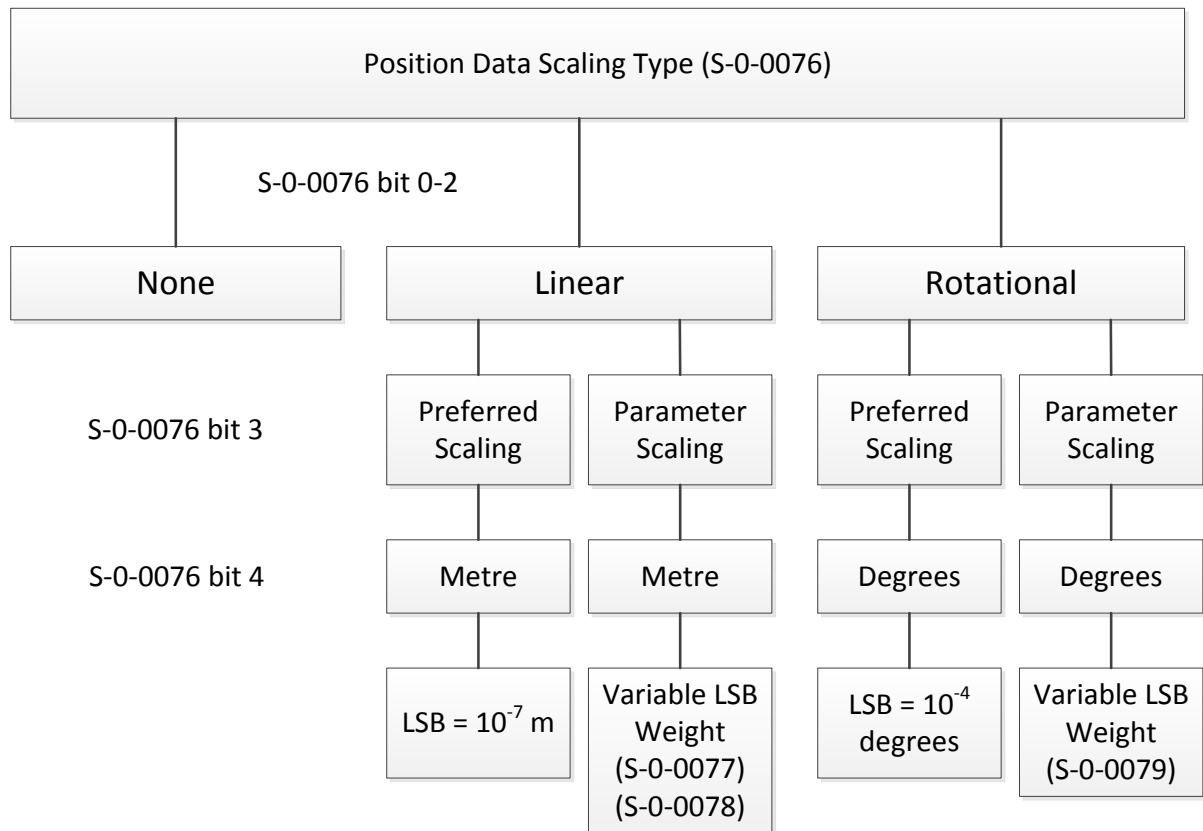


Figure 7-36 - Position Data Scaling Type guide

7.23.4.3 None (S-0-0076, bits0-2 = 0)

$$LSB\ weight = \alpha \times factor \times 10^{exponent}$$

$$\alpha = 2^{-24}$$

$$factor = 1, exponent = 0$$

There are no further available options when this scaling method is chosen. No unit string is displayed.

7.23.4.4 Linear (S-0-0076, bits0-2 = 1)

$$LSB\ weight = \alpha \times factor \times 10^{exponent}$$

$$\alpha = unit$$

$$factor = 1, exponent = -4\ (default)$$

If parameter scaling is chosen (*SoE Position Scaling - Type* bit3 = '1') then the *SoE Position Scaling - Linear Factor* will be used as "factor" and *SoE Position Scaling - Linear Exponent* as "exponent". If preferred scaling is chosen (*SoE Position Scaling - Type* bit3 = '0'), default values as shown in [Figure 7-36](#) will be used.

Typical application use in linear mode will set *SoE Position Scaling - Type* = 0x41, this uses the preferred scaling of 0.1um per LSB.

7.23.4.5 Rotational (S-0-0076, bits0-2 = 2)

$$LSB\ weight = \alpha \times factor \times 10^{exponent}$$

$$\alpha = \frac{1\ revolution}{Rotational\ position\ resolution}$$

$$factor = 1, exponent = 0\ (default)$$

If parameter scaling is chosen (*SoE Position Scaling - Type* bit3 = '1') then *SoE Position Scaling - Linear Factor* will be used as "factor" and *SoE Position Scaling - Linear Exponent* as "exponent". *SoE Position Scaling - Rotational Resolution* is used to calculate α as shown above. If preferred scaling is chosen (*SoE Position Scaling - Type* bit3 = '0'), default values as shown in [Figure 7-36](#) will be used.

Typical application use in rotational mode will set *SoE Position Scaling - Type* = 0xc2, this uses the preferred scaling of 0.0001 degrees per LSB and sets the axis to be modulo. For more information on modulo, refer to [7.23.4.6 Modulo Operation](#).



NOTE: Preferred linear position scaling on the AMD servo drive is $\text{LSB} = 10^{-3} \text{ mm}$.

7.23.4.6 Modulo Operation

Modulo operation is a feature that can be applied only to position feedback. Modulo operations are almost always applied to rotational axes where one revolution, or a sector of one revolution, results in the axis returning to a reference starting position (e.g. one complete revolution of a shaft or gear). The axis position as it is represented by the drive can be wrapped upon a specific proportion of a revolution (usually a full revolution). For example, a 500° motion from an initial reference starting point will be shown as 140° if the modulo operation is configured to 'wrap' or 'modulo' the motion every 360°.

To enable modulo operation, set bit 7 of the *SoE Position Scaling - Type* to 1. The *Modulo Value* must then be set to the value at which "wrapping" occurs. The modulo operation is applied to the axis position feedback regardless of if it is from the motor or external encoder.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0103 / 103	Modulo Value	Signed Integer (4 bytes)	Standard Position	360°	360°

7.23.5 Velocity Scaling

7.23.5.1 Description



NOTE: Velocity data on the AMD servo drive is always with respect to the load or end effector i.e. after the gear box and feed constant calculations.

The SoE velocity scaling can be broadly split into linear or rotary. Both linear and rotary have a 'preferred' scaling or fully customisable parameter scaling option.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0044 / 44	SoE Velocity Scaling - Type	Unsigned Integer (2 bytes)	Binary	65	65
S-0-0045 / 45	SoE Velocity Scaling - Factor	Unsigned Integer (2 bytes)	None	1	1
S-0-0046 / 46	SoE Velocity Scaling - Exponent	Signed Integer (2 bytes)	None	-3	-3

7.23.5.2 Velocity Data Scaling Type

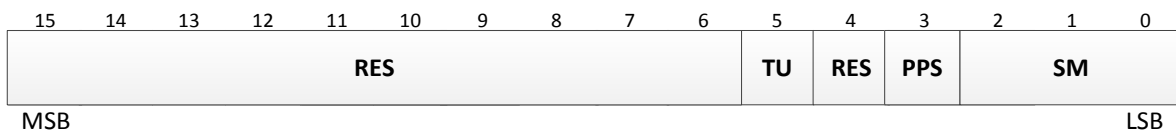


Figure 7-37 - Velocity Data Scaling Type definition

Bit	Key	Definition	Value	Description
0-2	SM	Scaling Method	000	No scaling
			001	Linear scaling
			010	Rotational scaling
3	PPS	Preferred/Parameter Scaling	0	Preferred scaling
			1	Parameter scaling
4	RES	Reserved	X	-
5	TU	Time Units	0	Minutes
			1	Seconds
6-15	RES	Reserved	X	-

Table 7-41 - Velocity Data Scaling Type bit definition

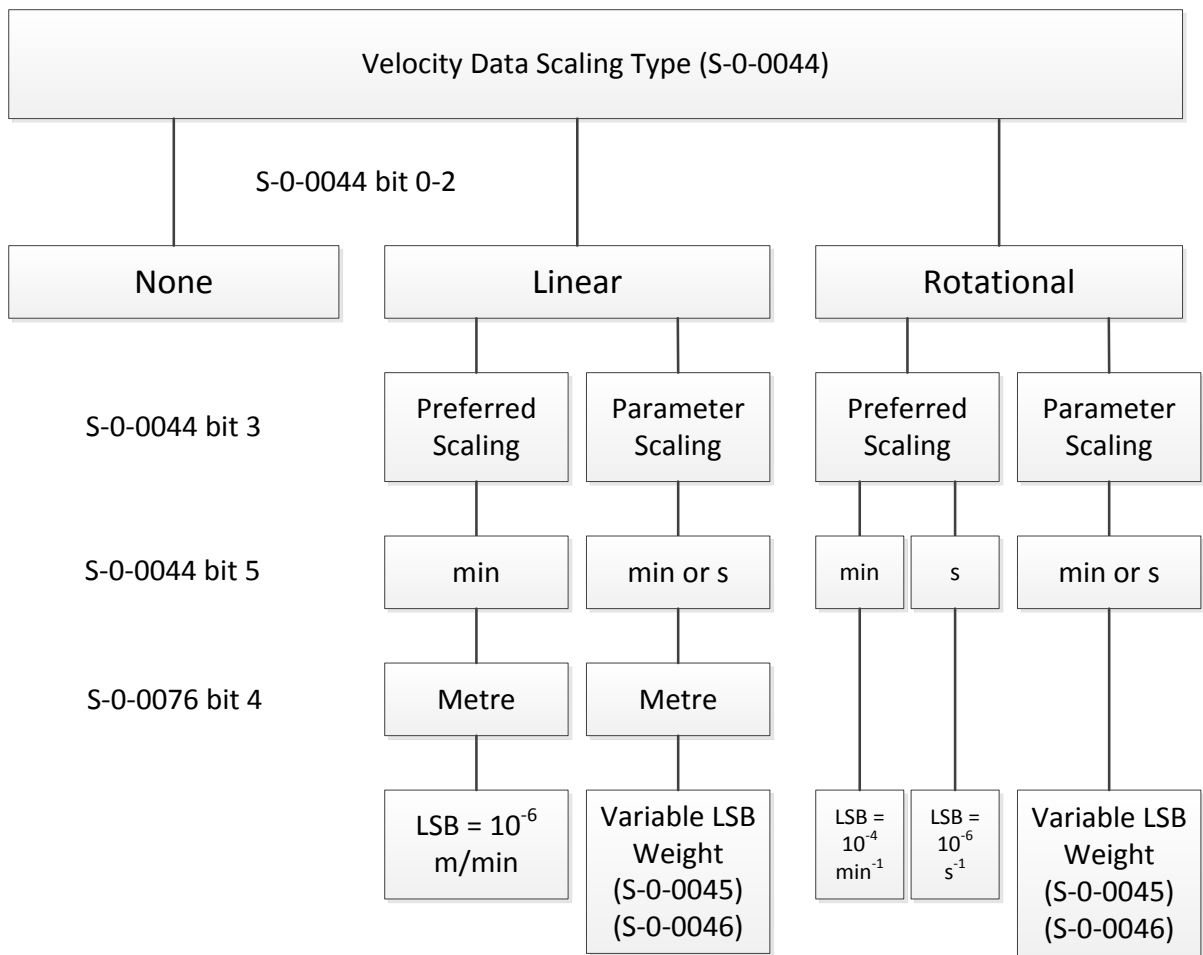


Figure 7-38 - Velocity Data Scaling Type guide

7.23.5.3 None (S-0-0044, bits0-2 = 0)

$$LSB\ weight = \alpha \times factor \times 10^{exponent}$$

$$\alpha = \frac{2^{-38}}{250\mu s}$$

$factor = 1, exponent = 0$

There are no further available options when this scaling method is chosen. No unit string is displayed.

7.23.5.4 Linear (S-0-0044, bits0-2 = 1)

$$LSB\ weight = \alpha \times factor \times 10^{exponent}$$

$$\alpha = \frac{unit}{time\ unit}$$

$factor = 1, exponent = -6\ (default)$

If parameter scaling is chosen (SoE Velocity Scaling - Type bit3 = '1') then SoE Velocity Scaling - Factor will be used as "factor" and SoE Velocity Scaling - Exponent as "exponent". If preferred scaling is chosen (SoE Velocity Scaling - Type bit3 = '0') then default values as shown in [Figure 7-38](#) will be used.

Typical application use in linear mode will set SoE Velocity Scaling - Type = 0x41, this uses the preferred scaling of 0.001 mm/min per LSB.

7.23.5.5 Rotational (S-0-0044, bits0-2 = 2)

$$LSB\ weight = \alpha \times factor \times 10^{exponent}$$

$$\alpha = \frac{unit}{time\ unit}$$

$factor = 1, exponent = -6$ (default)

If parameter scaling is chosen (SoE Velocity Scaling - Type bit3 = '1') then SoE Velocity Scaling - Factor will be used as "factor" and SoE Velocity Scaling - Exponent as "exponent". If preferred scaling is chosen (SoE Velocity Scaling - Type bit3 = '0') then default values as shown in [Figure 7-38](#) will be used.

Typical application use in rotational mode will set SoE Velocity Scaling - Type = 0x42, this uses the preferred scaling of 0.0001 revolutions per minute (RPM) per LSB. Alternatively, it is also common to set SoE Velocity Scaling - Type = 0x62. This uses the preferred scaling of 0.000001 revolutions per second (rev/s).

7.23.6 Acceleration Scaling

7.23.6.1 Description



NOTE: Acceleration data on the AMD servo drive is always with respect to the load or end effector i.e. after the gear box and feed constant calculations.

The SoE acceleration scaling can be broadly split into linear or rotary. Both linear and rotary have a 'preferred' scaling or fully customisable parameter scaling option.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0160 / 160	SoE Acceleration Scaling - Type	Unsigned Integer (2 bytes)	Binary	65	65
S-0-0161 / 161	SoE Acceleration Scaling - Factor	Unsigned Integer (2 bytes)	None	1	1
S-0-0162 / 162	SoE Acceleration Scaling - Exponent	Signed Integer (2 bytes)	None	-6	-6

7.23.6.2 Acceleration Data Scaling Type

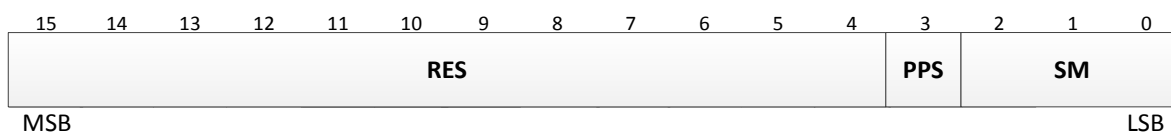


Figure 7-39 - Acceleration Data Scaling Type definition

Bit	Key	Definition	Value	Description
0-2	SM	Scaling Method	000	No scaling
			001	Linear scaling
			010	Rotational scaling
3	PPS	Preferred/Parameter Scaling	0	Preferred scaling
			1	Parameter scaling
6-15	RES	Reserved	X	-

Table 7-42 - Acceleration Data Scaling Type bit definition

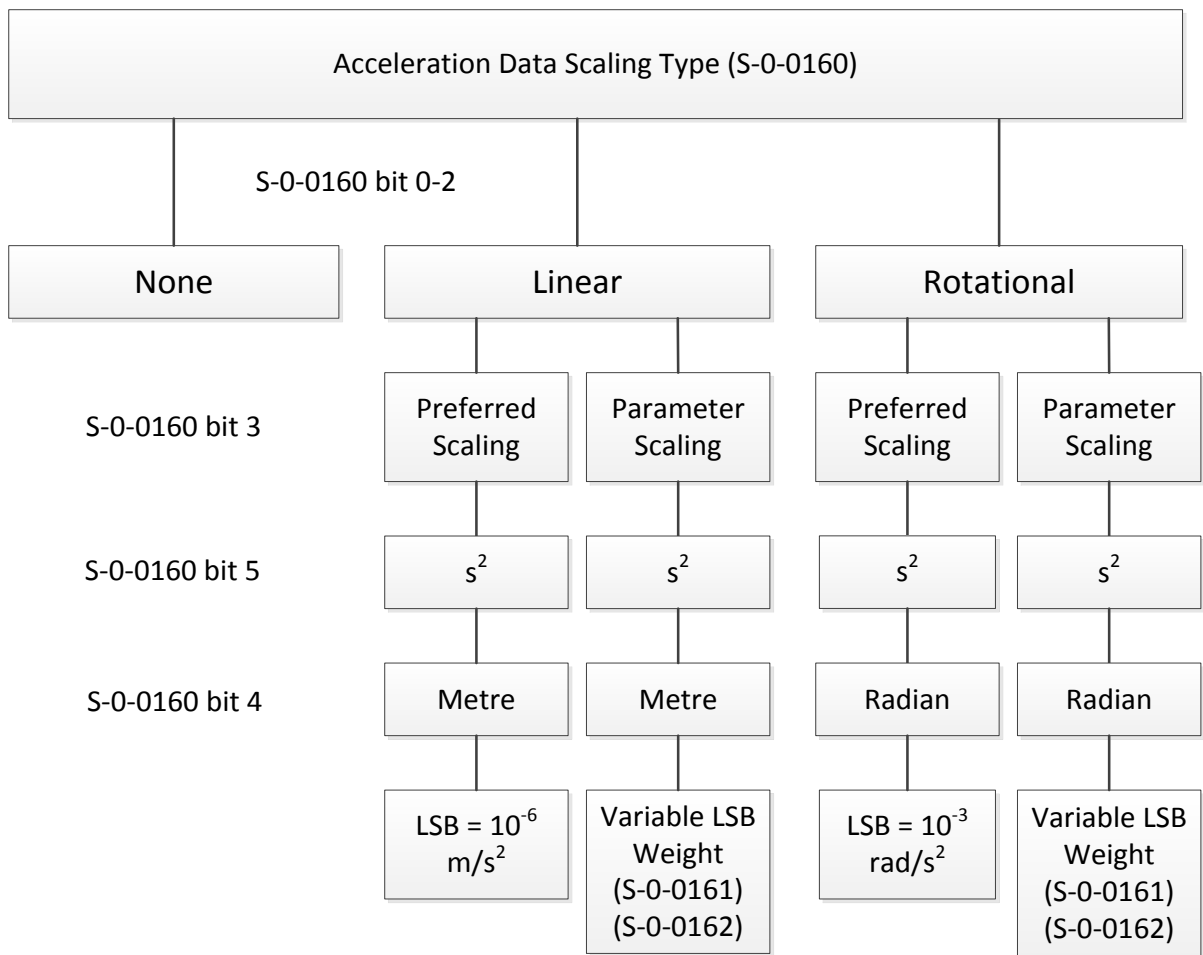


Figure 7-40 - Acceleration Data Scaling Type guide

7.23.6.3 None (S-0-0160, bits0-2 = 0)

$$LSB\ weight = \alpha \times factor \times 10^{exponent}$$

$$\alpha = \frac{2^{-46}}{(250\mu s)^2}$$

$factor = 1, exponent = 0$

There are no further available options when this scaling method is chosen. No unit string is displayed.

7.23.6.4 Linear (S-0-0160, bits0-2 = 1)

$$LSB\ weight = \alpha \times factor \times 10^{exponent}$$

$$\alpha = \frac{unit}{time\ unit^2}$$

$factor = 1, exponent = -6\ (default)$

If parameter scaling is chosen (SoE Acceleration Scaling - Type bit3 = 1) then SoE Acceleration Scaling - Factor will be used as “factor” and SoE Acceleration Scaling - Exponent as “exponent”. If preferred scaling is chosen (SoE Acceleration Scaling - Type bit3 = 0) then default values as shown in [Figure 7-40](#) will be used.

Typical application use in linear mode will set SoE Acceleration Scaling - Type = 0x41, this uses the preferred scaling of 0.001 mm/s² per LSB.

7.23.6.5 Rotational (S-0-0160, bits0-2 = 2)

$$LSB\ weight = \alpha \times factor \times 10^{exponent}$$

$$\alpha = \frac{unit}{time\ unit^2}$$

$factor = 1, exponent = -3$ (default)

If parameter scaling is chosen (SoE Acceleration Scaling - Type bit3 = 1) then SoE Acceleration Scaling - Factor will be used as “factor” and SoE Acceleration Scaling - Exponent as “exponent”. If preferred scaling is chosen (SoE Acceleration Scaling - Type bit3 = 1) then default values as shown in [Figure 7-40](#) will be used.

Typical application use in rotary mode will set SoE Acceleration Scaling - Type = 0x41, this uses the preferred scaling of 0.001 rad/s² per LSB.

7.23.7 Torque / Force Scaling

7.23.7.1 Description



NOTE: Torque / Force data on the AMD servo drive is always with respect to the load or end effector i.e. after the gear box and feed constant calculations.

The SoE torque / force scaling can be broadly split into linear or rotary. Both linear and rotary have a ‘preferred’ scaling or fully customisable parameter scaling option.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0086 / 86	SoE Torque/Force Scaling - Type	Unsigned Integer (2 bytes)	Binary	65	65
S-0-0093 / 93	SoE Torque/Force Scaling - Factor	Unsigned Integer (2 bytes)	None	1	1
S-0-0094 / 94	SoE Torque/Force Scaling - Exponent	Signed Integer (2 bytes)	None	0	0
S-0-0111 / 111	Motor Continuous Current Rating	Signed Integer (4 bytes)	Standard Current	1.25 A	1.25 A
S-0-0196 / 196	Motor Rated Current	Signed Integer (4 bytes)	Standard Current	4 A	4 A

7.23.7.2 Torque/Force Data Scaling Type

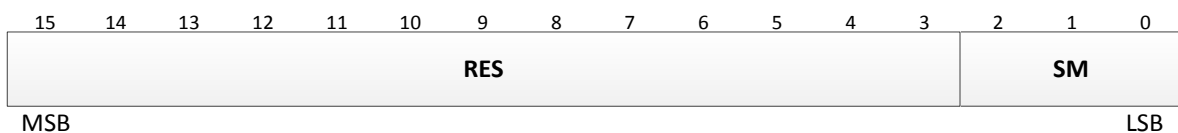


Figure 7-41 - Torque/Force Data Scaling Type definition

Bit	Key	Definition	Value	Description
0-2	SM	Scaling Method	000	Percentage scaling
			001	Linear scaling
			010	Rotational scaling
3	PPS	Preferred/Parameter Scaling	0	Preferred scaling
			1	Parameter scaling
6-15	RES	Reserved	X	-

Table 7-43 - Torque/Force Data Scaling Type bit definition

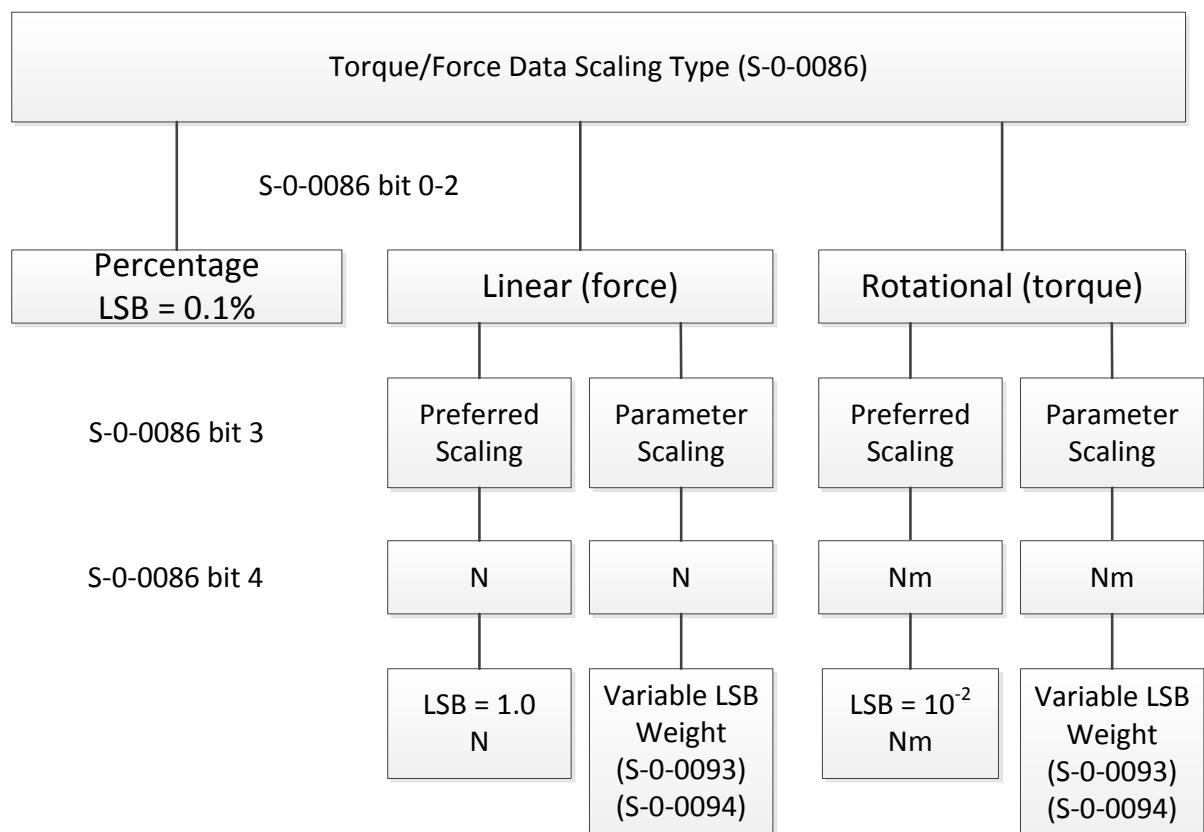


Figure 7-42 - Torque/Force Data Scaling Type guide

7.23.7.3 Percentage (S-0-0086, bits0-2 = 0)

Percentage scaling is configured by the *SoE Torque/Force Scaling - Type*. No further scaling parameters are required. With percentage torque scaling, the LSB weight of the torque data is based on 0.1% of the *Motor Continuous Current Rating* if the motor is synchronous or 0.1% of the *Motor Rated Current* if the motor is asynchronous.

7.23.7.4 Linear (S-0-0086, bits0-2 = 1)

$$LSB\ weight = \alpha \times factor \times 10^{exponent}$$

$\alpha = \text{unit}$ $\text{factor} = 1, \text{exponent} = 0$ (default)

If parameter scaling is chosen (*SoE Torque/Force Scaling - Type* bit3 = '1') then *SoE Torque/Force Scaling - Factor* will be used as "factor" and *SoE Torque/Force Scaling - Exponent* as "exponent". If preferred scaling is chosen (*SoE Torque/Force Scaling - Type* bit3 = '0') then default values as shown in [Figure 7-42](#) will be used.

Typical application use in linear mode will set *SoE Torque/Force Scaling - Type* = 0x41, this uses the preferred scaling of 1N per LSB. Alternatively, it is also common to set *SoE Torque/Force Scaling - Type* = 0x49, S-0-0093/93 = 1 and S-0-0093/94 = -3 to use scaling of 0.001N per LSB.

7.23.7.5 Rotational (S-0-0086, bits0-2 = 2)

$$\text{LSB weight} = \alpha \times \text{factor} \times 10^{\text{exponent}}$$

 $\alpha = \text{unit}$ $\text{factor} = 1, \text{exponent} = -3$ (default)

If parameter scaling is chosen (*SoE Torque/Force Scaling - Type* bit3 = '1') then *SoE Torque/Force Scaling - Factor* will be used as "factor" and *SoE Torque/Force Scaling - Exponent* as "exponent". If preferred scaling is chosen (*SoE Torque/Force Scaling - Type* bit3 = '0') then default values as shown in [Figure 7-42](#) will be used.

Typical application use in rotary mode will set *SoE Torque/Force Scaling - Type* = 0x42, this uses the preferred scaling of 0.01Nm per LSB.

7.23.8 Temperature Scaling

The AMD servo drive has no configuration options for temperature scaling; the unit of 0.1° C per LSB will be used.

7.23.9 Current Scaling

The AMD servo drive has no configuration options for current scaling; the unit of milliamps per LSB will be used.

7.23.10 Voltage Scaling

The AMD servo drive has no configuration options for voltage scaling; the units of volts per LSB will be used for 16 bit values and millivolts per LSB for 32 bit values.

7.23.11 Power Scaling

The AMD servo drive has no configuration options for power scaling; the unit of watts per LSB will be used.

7.24 Strobing

7.24.1 Description



NOTE: This feature is only available on the AMD2000 series servo drives.

The strobing feature allows a digital input of the AMD servo drive to latch the current count of a pulse or encoder input. The count value is latched into separate IDNs on the rising and falling edge of the digital input signal. This count may contain an offset to pulse / encoder counts found elsewhere in the system; therefore, it is recommended that the *Strobe Pulse / Encoder Count* is used as a reference.

IDN	Description	Data Type	Units	Default
				AMD2000
P-0-0672 / 33440	Strobing Falling Edge Latched Count	Unsigned Integer (4 bytes)	Count	N/A
P-0-0673 / 33441	Strobing Rising Edge Latched Count	Unsigned Integer (4 bytes)	Count	N/A
P-0-0674 / 33442	Strobing Pulse / Encoder Count	Unsigned Integer (4 bytes)	Count	N/A
P-0-0675 / 33443	Strobing Digital Input Select	Unsigned Integer (2 bytes)	None	0
P-0-0676 / 33444	Strobing Pulse / Encoder Source	Unsigned Integer (2 bytes)	None	0
P-0-0677 / 33445	Strobing Enable	Unsigned Integer (2 bytes)	Boolean	0
P-0-0678 / 33446	Strobing Status	Unsigned Integer (2 bytes)	None	N/A

7.24.2 Definitions

7.24.2.1 Strobing Digital Input Select

The *Strobing Digital Input Select* is defined as shown below.

Value	Description
0	Digital Input 1
1	Digital Input 2
2	Digital Input 3
3	Digital Input 4
4	Digital Input 5
5	Digital Input 6
6	Digital Input 9 (High Speed Differential)
7	Digital Input 10 (High Speed Differential)

Table 7-44 - Strobing Digital Input Select Definition

7.24.2.2 Strobing Pulse / Encoder Source

The *Strobing Status* is defined as shown below.

Value	Description
0	Pulse Input (Digital Inputs 9 & 10)
1	Encoder Channel 1
2	Encoder Channel 2

Table 7-45 - Strobing Status definition

7.24.2.3 Strobing Status

The *Strobing Status* is defined as shown below.

Value	Description
0	Idle
1	Probing Active (Unavailable for Strobing)
2	Strobing Active
4	Error

Table 7-46 - Strobing Status definition

7.24.3 Operation Example



NOTE: The Strobing feature cannot be used at the same time as the Probing feature. If both features are requested at the same time, an error will be shown in the *Strobing Status*. Refer to [7.17 Probing](#).



NOTE: The Strobing feature cannot be used at the same time as the [Pulse / Stepper Position Control](#) feature.

In the example below, the strobing function has been assigned digital input 1 to trigger the latching function via *Strobing Digital Input Select*. The motor encoder has been assigned to the strobing counter via *Strobing Pulse / Encoder Source*. Once *Strobing Enable* is set to 1, *Strobe Status* is 2 to indicate strobing is active. The motor moves forward causing the counter to count forward, the digital input is then pulsed. This causes the strobing latching to occur. Latched encoder counts are stored in *Strobing Rising Edge Latched Count* and *Strobing Falling Edge Latched Count*.

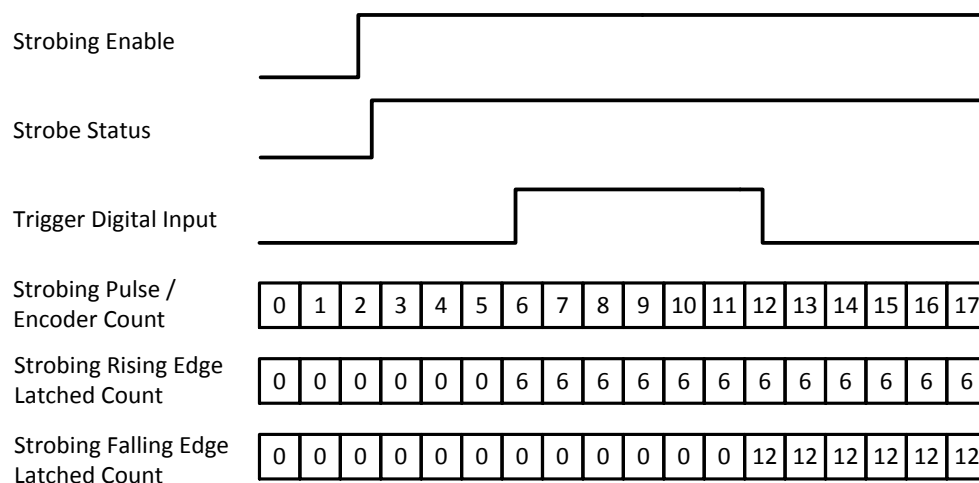


Figure 7-43 - Strobing operation example

7.25 Temperature Monitoring

7.25.1 Description



NOTE: Motor temperature monitoring is only available on the AMD5x series servo drives.

The AMD servo drive supports temperature monitoring of both it and the connected motor provided that the motor has a compatible temperature sensor. Refer to the AMD servo drive User Guide for more information on compatible motor temperature sensors.

Temperature monitoring provides temperature measurements for the control unit/device to monitor and configurable thresholds at which to trigger Class 1 Diagnostic (C1D) errors.

Configuration of temperature monitoring is available via the *Temperature Monitor Control Word* allowing enabling/disabling of individual components. To use any of the motor temperature monitoring features, motor temperature measurement must be enabled in the *Temperature Monitor Control Word*.



NOTE: All drive's temperature related parameters are read only and they can vary by drive rating and variants. Table below is only typical values for presentation.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0203 / 203	Amplifier Shut-Down Temperature	Signed Integer (2 bytes)	Standard Temperature	80°C	80°C
S-0-0204 / 204	Motor Shut-Down Temperature	Signed Integer (2 bytes)	Standard Temperature	120°C	120°C
P-0-1236 / 34004	Motor Shut-Down Temperature with PWM On	Signed Integer (2 bytes)	Standard Temperature	120°C	120°C
P-0-1237 / 34005	Amplifier Shut-Down Temperature with PWM On	Signed Integer (2 bytes)	Standard Temperature	80°C	80°C
P-0-1240 / 34008	Sensed Motor Temperature	Signed Integer (2 bytes)	Standard Temperature	N/A	N/A
P-0-1241 / 34009	Sensed Amplifier Temperature	Signed Integer (2 bytes)	Standard Temperature	N/A	N/A
P-0-1242 / 34010	Temperature Monitor Control Word	Unsigned Integer (2 bytes)	Binary	0001 0001	0000 0001

7.25.2 Definitions

7.25.2.1 Amplifier Shut-Down Temperature

The *Amplifier Shut-Down Temperature* defines the servo drive amplifier temperature at which the C1D error, E104, is asserted. This is property of the servo drive hardware and cannot be modified.

7.25.2.2 Motor Shut-Down Temperature

The *Motor Shut-Down Temperature* defines the servo drive amplifier temperature at which the C1D error, E114, is asserted. This is a user configurable parameter dependent on the motor rating and normal application operating range.

7.25.2.3 Motor Shut-Down Temperature with PWM On

The *Motor Shut-Down Temperature with PWM On* defines the servo drive amplifier temperature at which the C1D error, E115, is asserted when power is applied to the motor. This is a user configurable parameter for use in specific applications where the measured temperature is acceptable to enable to the drive's controller but not the drive's power module. This error is commonly used to ensure that external cooling systems are active in applications that require it.

7.25.2.4 Amplifier Shut-Down Temperature with PWM On

The *Amplifier Shut-Down Temperature with PWM On* defines the servo drive amplifier temperature at which the C1D error, E105, is asserted when power is applied to the motor. This is a user configurable parameter for use in specific applications where the measured temperature is acceptable to enable to the drive's controller but not the drive's power module. This error is commonly used to ensure that external cooling systems are active in applications that require it.

7.25.2.5 Sensed Motor Temperature

The *Sensed Motor Temperature* contains the currently measured temperature from the motor temperature sensor.

7.25.2.6 Sensed Amplifier Temperature

The *Sensed Amplifier Temperature* contains the currently measured temperature from the servo drive amplifier temperature sensor.

7.25.2.7 Temperature Monitor Control Word

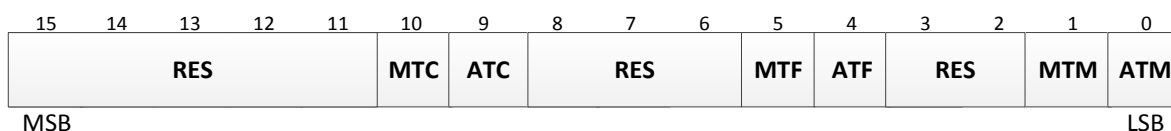


Figure 7-44 - Temperature Monitor Control Word definition

Bit	Key	Definition	Value	Description
0	ATM	Amplifier Temperature Measurement	0	Disabled
			1	Enabled
1	MTM	Motor Temperature Measurement	0	Disabled
			1	Enabled
2	TM	Ambient Temperature Measurement	0	Disabled
			1	Enabled
3	RES	Reserved	X	-
4	ATF	Amplifier Temperature Fault Monitoring	0	Disabled
			1	Enabled. Amplifier Temperature Measurement must be enabled.
5	MTF	Motor Temperature Fault Monitoring	0	Disabled
			1	Enabled. Motor Temperature Measurement must be enabled.
6-8	RES	Reserved	X	-
9	ATC	Amplifier Temperature Current Limiting	0	Disabled
			1	Enabled. Amplifier Temperature Measurement must be enabled.
10	MTC	Motor Temperature Current Limiting	0	Disabled
			1	Enabled. Motor Temperature Measurement must be enabled.
11-15	RES	Reserved	X	-

Table 7-47 - Temperature Monitor Control Word bit definition

7.26 Torque Command Filters

7.26.1 Description

The AMD servo drive features of torque command low-pass and notch filters. These filters are used to remove unwanted high frequency signals or defined frequency bands (such as regions of mechanical resonance) in the torque used to drive the machine. Removal of resonant frequency bands is often desirable to reduce machine noise and vibration, improving lifespan by reducing fatigue or improving quality in finished products from the machine. The drive can be configured for up to 1 low-pass and 5 notch filters. Only a corner frequency parameter need be set by the user for the low-pass filter, whereas the notch filters require a centre frequency and a Q factor.

The parameters can be calculated or estimated as follows:

Low-pass filter corner frequency – signals above this base frequency will be attenuated by more than -3 dB,

Notch filter centre frequency, f_0 – centre frequency of the band needing to be attenuated with the notch, and

Q factor – a dimension-less parameter which characterises a resonator's bandwidth relative to its centre frequency. Q factor is determined from centre frequency f_0 and filter bandwidth f_Δ by the relationship,

$$Q = \frac{f_0}{f_\Delta}, \text{ and } f_\Delta = f_{high} - f_{low}$$

The recommended procedure for setting the filter configuration is

1. Ensure the drive is disabled.
2. Set the associated filter frequency parameter to the desired frequency (ON) or 0 (OFF).
3. Set the QFactor parameter if required.
4. Enable the drive to enable the new filter settings.

Alternatively, the *Motor Control Tuning Procedure Command* can be used to apply Torque Command Filter changes. Refer to [7.15.5.3 Applying Current Control Loop Parameters](#) for more information.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-0012 / 32780	Torque Cmd Low Pass Filter Freq	Unsigned Integer (2 bytes)	Hz	0	0
P-0-0013 / 32781	Torque Cmd Notch 1 Filter Freq	Unsigned Integer (2 bytes)	Hz	0	0
P-0-0014 / 32782	Torque Cmd Notch 1 QFactor	Signed Integer (2 bytes)	2^{-10}	973	973
P-0-0015 / 32783	Torque Cmd Notch 2 Filter Freq	Unsigned Integer (2 bytes)	Hz	0	0
P-0-0016 / 32784	Torque Cmd Notch 2 QFactor	Signed Integer (2 bytes)	2^{-10}	973	973
P-0-0017 / 32785	Torque Cmd Notch 3 Filter Freq	Unsigned Integer (2 bytes)	Hz	0	0
P-0-0018 / 32786	Torque Cmd Notch 3 QFactor	Signed Integer (2 bytes)	2^{-10}	973	973
P-0-0019 / 32787	Torque Cmd Notch 4 Filter Freq	Unsigned Integer (2 bytes)	Hz	0	0
P-0-0020 / 32788	Torque Cmd Notch 4 QFactor	Signed Integer (2 bytes)	2^{-10}	973	973
P-0-0021 / 32789	Torque Cmd Notch 5 Filter Freq	Unsigned Integer (2 bytes)	Hz	0	0
P-0-0022 / 32790	Torque Cmd Notch 5 QFactor	Signed Integer (2 bytes)	2^{-10}	973	973

7.27 Tuning – Current Control

7.27.1 Description

The current control loop is responsible for supplying current to the motor windings which in turn are responsible for supplying torque via magnetic fields to turn the motor rotor. A well-tuned current loop will have a fast response to changes in current demand, with little or no oscillation.

The physical parameters affecting current loop tuning are primarily the resistance and inductance of the winding circuits. However, the supply voltage and PWM update rate can also have an impact. These parameters are usually fixed for any particular motor model, thus the current loop tuning is determined primarily by the motor model (usually marked on the name plate for the motor). If the same motor model is used in any application, it is usually the case that the same current loop parameters may be applied regardless of the load or other mechanical changes.

The current controller included in the AMD servo drive is a Proportional-Integral (PI) controller applied to two independent current loops, one for quadrature current (q), and one for direct current (d). A more complete description of these currents and their relationship to driving torques is discussed elsewhere in this manual, in the section entitled 'Field Orientation Initialisation.' For a description of how the torque demand from the velocity controller is resolved into current demands for the current controller the interested reader should refer to the section entitled 'Motor Control.'

The frequency domain Laplace transfer function representation of the current controllers has the form:

$$V_d(s) = K_d \left(1 + \frac{1}{T_d s} \right) (I_d^*(s) - I_d(s))$$
$$V_q(s) = K_q \left(1 + \frac{1}{T_q s} \right) (I_q^*(s) - I_q(s))$$

This is implemented in a discrete time version within the AMD servo drive where the parameters are discussed elsewhere in this manual under the title 'Motor Control.' They can be briefly summarised as follows;

K_d is the current loop proportional d -axis gain
 K_q is the current loop proportional q -axis gain
 T_d is the current loop d -axis integral action time
 T_q is the current loop q -axis integral action time
 I_d^* is the d -axis current reference
 I_q^* is the q -axis current reference
 I_d is the d -axis estimated current
 I_q is the q -axis estimated current
 V_d is the d -axis voltage command
 V_q is the q -axis voltage command

7.27.2 Tuning Procedure



Warning: It is recommended that the motor be disconnected from any load for this procedure to prevent possible damage to equipment.

In general, the procedure for setting the values velocity control loop's K and T involves the following steps performed in sequence.

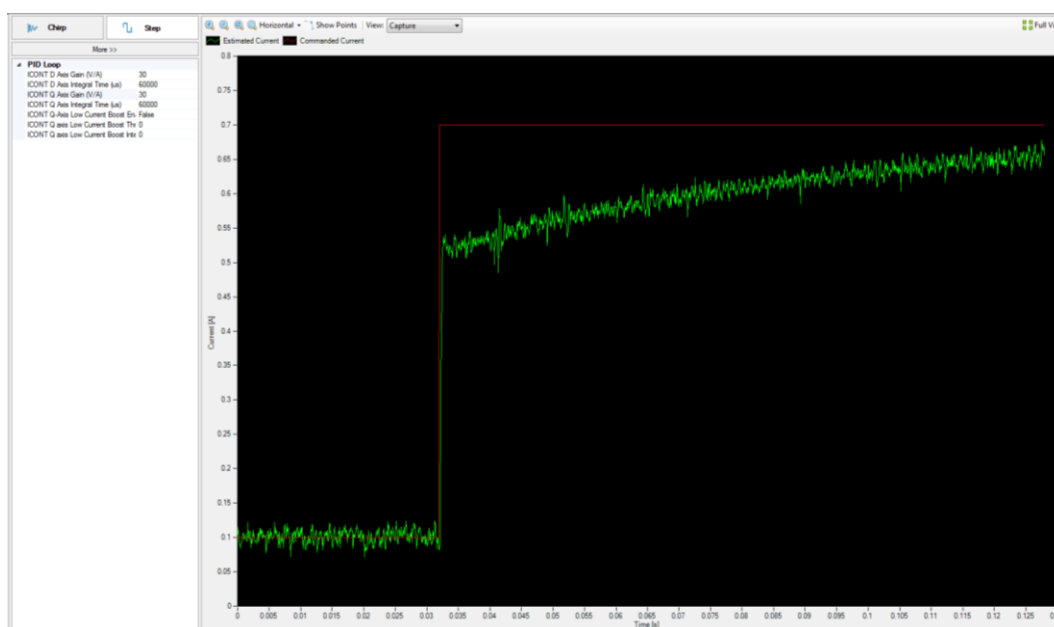
The tuning procedure involves repeatedly testing the control loop's response to a step change in the command input and altering the proportional gain and integral time towards the desired characteristics. Both D and Q axis parameters will be set to the same value.

1. Start with a low proportional gain e.g. 5 and a very high integral time value e.g. 60000. This essentially eliminates the integral action allowing purely the proportional gain to be tuned.
2. Perform iterations of increasing the proportional gain and testing the step response until oscillations are seen in the response.
3. Reduce the proportional gain by ~30%.
4. Perform iterations of reducing the integral time to increase the speed of the response until a maximum overshoot of 5-10% is experienced.
5. Procedure complete.

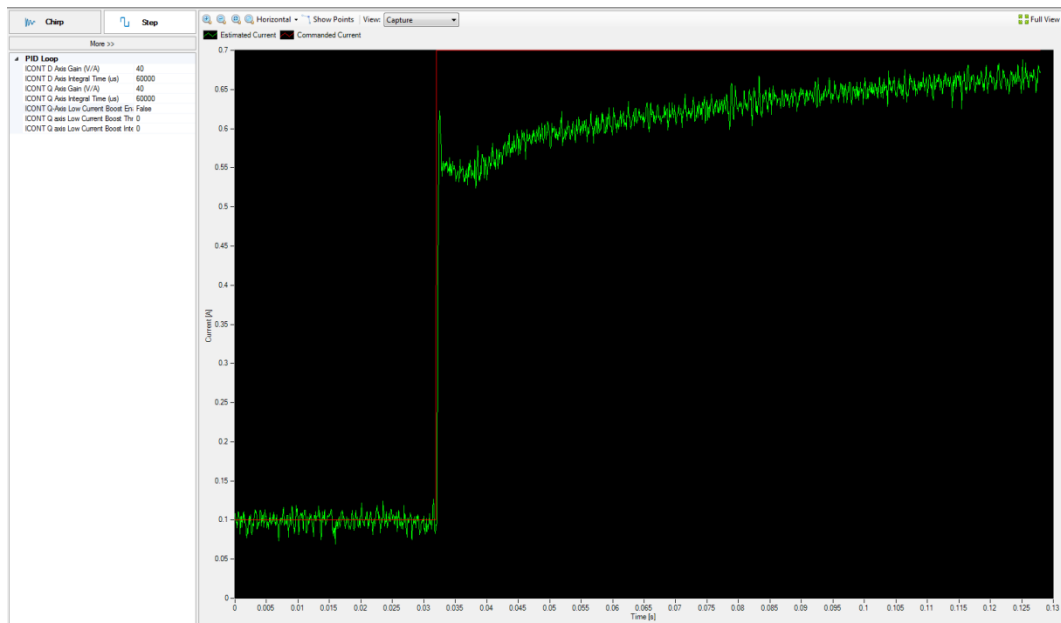
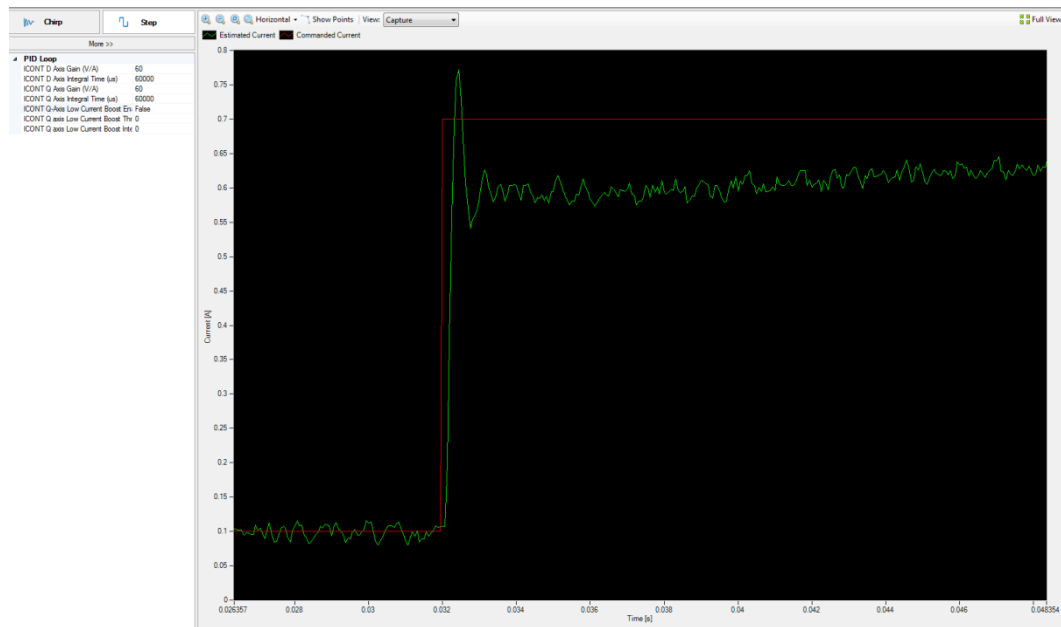
7.27.3 Example

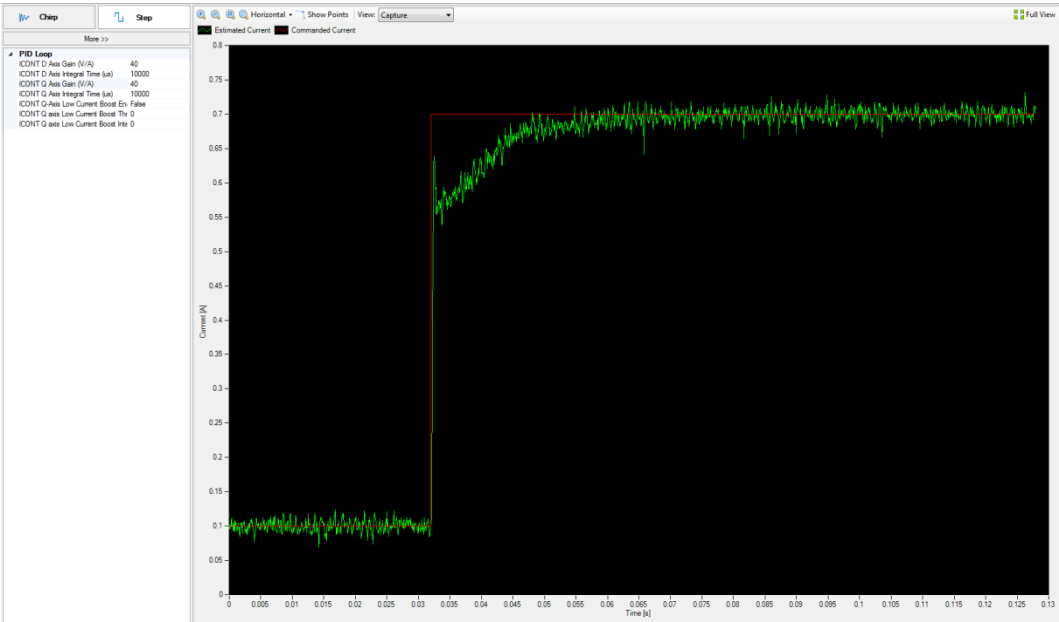
The following example demonstrates basic current control tuning using the step response page in MotionBench.

1. Starting values with low proportional gain and high integral time.

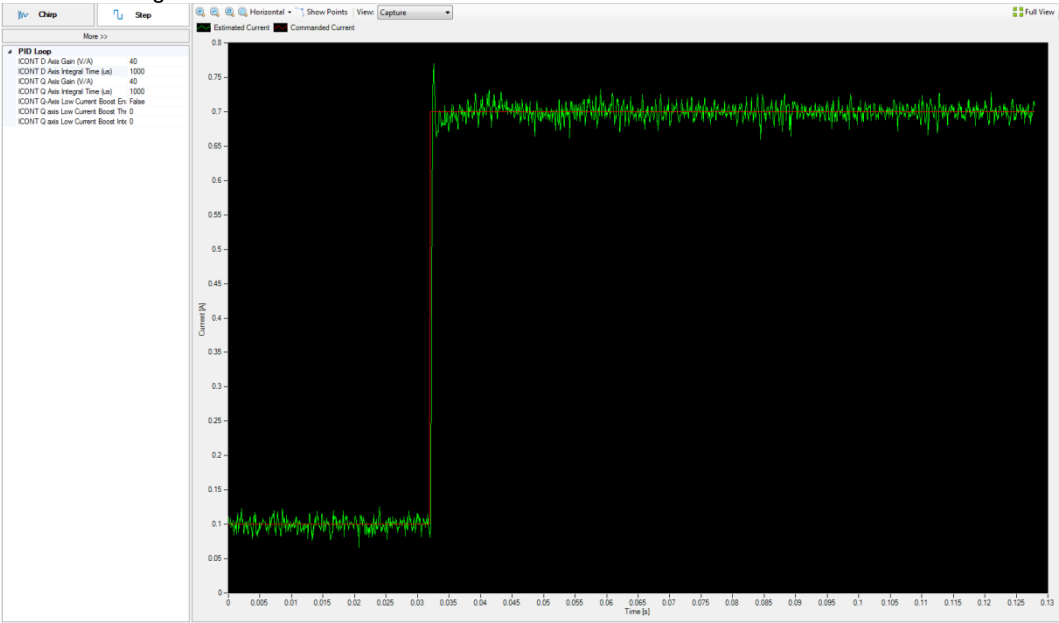


2. Increase in proportional gain resulting mild oscillations

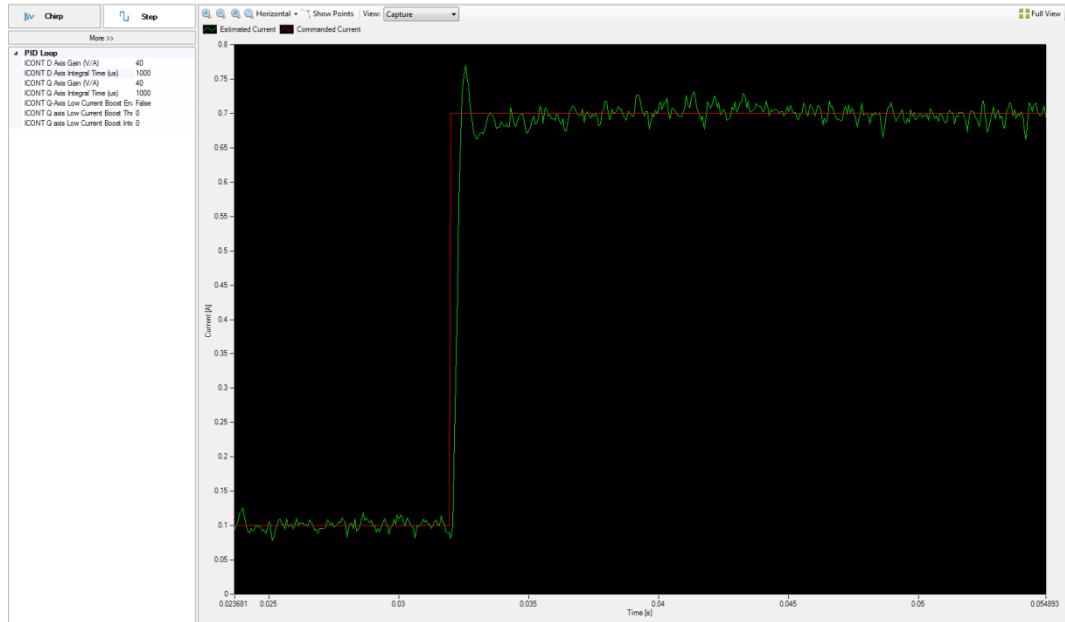




5. Reduce integral time further.



6. Tuning complete.



7.28 Tuning – Velocity Control

7.28.1 Description

The velocity control loop is responsible for tracking velocity demands, either issuing from the position control loop, or from direct external demands. In addition, it is responsible for rejecting disturbances that may result in deviations from the demand levels. In control systems, tracking the demand is commonly known as the servo task, while rejecting disturbances is known as the regulation task.

A well-tuned velocity loop will result in a stiff and stable axis. The physical parameters affecting the velocity loop tuning are primarily the inertia and friction of the moving parts. However, the compliance of any shafts or belts, the torque constant of the motor, and the velocity loop update rate also have an impact. The velocity tuning is greatly affected by any gearbox or pulley ratio between the motor and the moving part(s). Differences in manufacture between machines can also have an impact on velocity loop tuning. It is consequently possible that similar machines can require different tunings to achieve similar levels of performance.

Due to the effects of mechanical resonance, tuning the velocity controller may be more complex than tuning the current controller. Resonance comes from various parts of the mechanical structure, including the bearing, belt and pulley. Changes in resonant frequency may result from structural re-design, in which case the axes need to be re-tuned. It is possible for the user to specify a number of filters that may help deal with such resonant response, and these are addressed elsewhere in the manual under the title 'Torque Command Filters.'

The velocity controller included in the AMD servo drive is a Proportional-Integral (PI) controller. The frequency domain Laplace transfer function representation of this controller has the form:

$$Tor_q^*(s) = K \left(1 + \frac{1}{Ts} \right) (\omega^*(s) - \omega(s))$$

This is implemented in a discrete time version within the AMD servo drive where the parameters are discussed elsewhere in this manual under the title 'Servo Control.' They can be briefly summarised as follows;

K is the velocity loop proportional gain,
 T is the velocity loop integral action time,
 ω^* is the velocity loop demand (or reference),
 ω is the velocity loop sensed or estimated feedback velocity, and
 Tor_q^* is the torque command.

'Velocity Tuning' is the process whereby the user selects appropriate K and T settings for their particular control application.

7.28.2 Tuning Procedure

In general, the procedure for setting the values velocity control loop's K and T involves the following steps performed in sequence. The section details the typical way to tune the velocity loop of a basic axis. The addition of torque command filters to mitigate vibration is not included in this section.

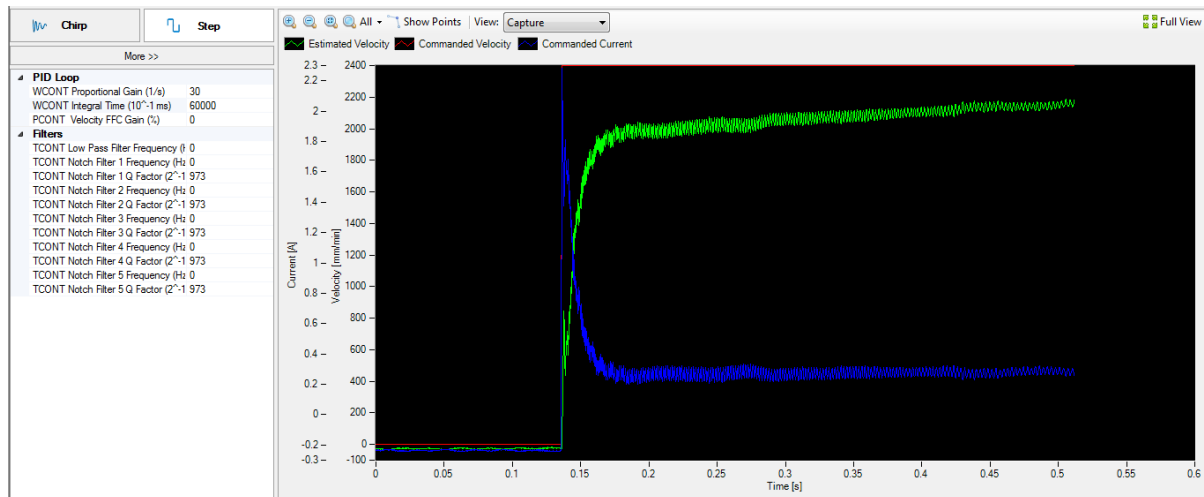
The tuning procedure involves repeatedly testing the control loop's response to a step change in the command input and altering the proportional gain and integral time towards the desired characteristics.

1. Start with a low proportional gain e.g. 5 and a very high integral time value e.g. 60000. This essentially eliminates the integral action allowing purely the proportional gain to be tuned.
2. Perform iterations of increasing the proportional gain and testing the step response until oscillations are seen in the response.
3. Reduce the proportional gain by ~30%.
4. Perform iterations of reducing the integral time to increase the speed of the response until a maximum overshoot of 5-10% is experienced.
5. Procedure complete.

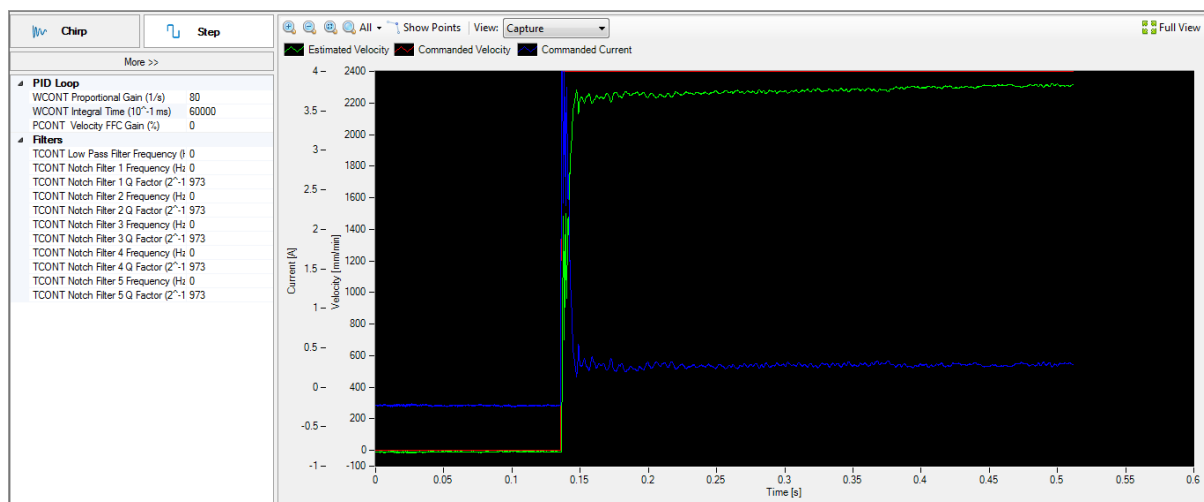
7.28.3 Example

The following example demonstrates basic velocity control tuning using the step response page in MotionBench.

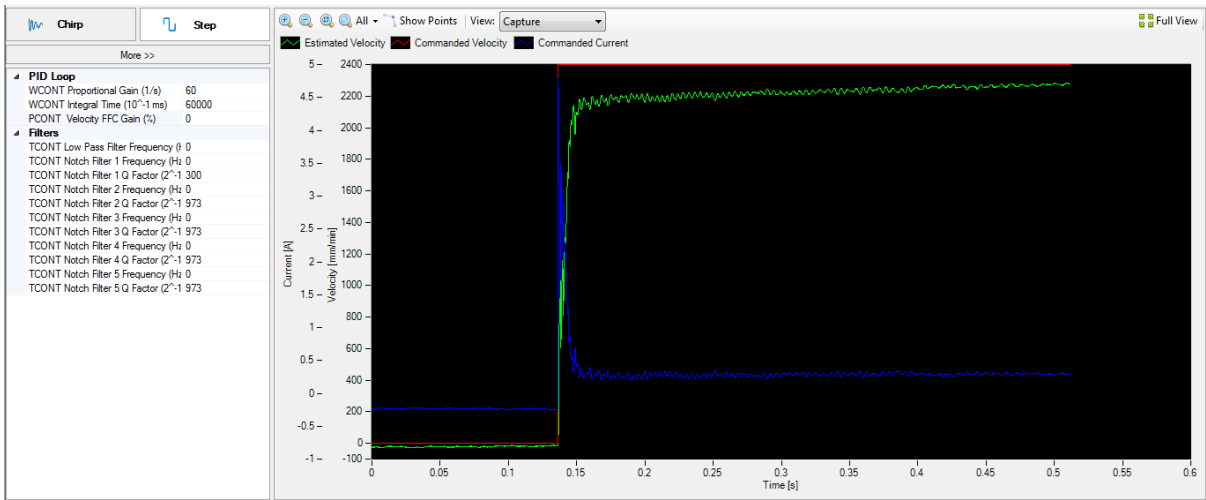
1. Starting values with low proportional gain and high integral time.



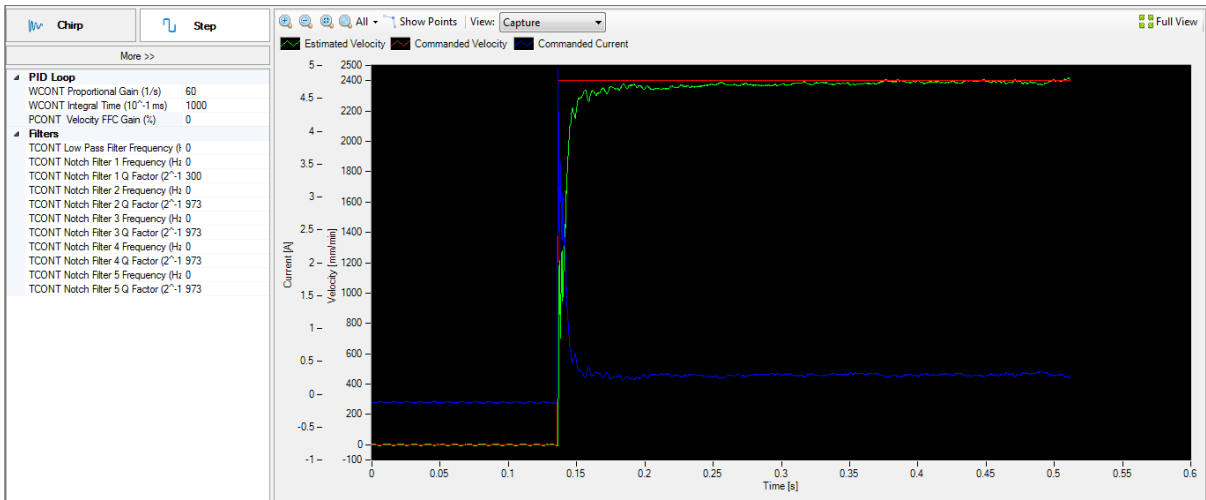
2. Increase in proportional gain resulting mild oscillations



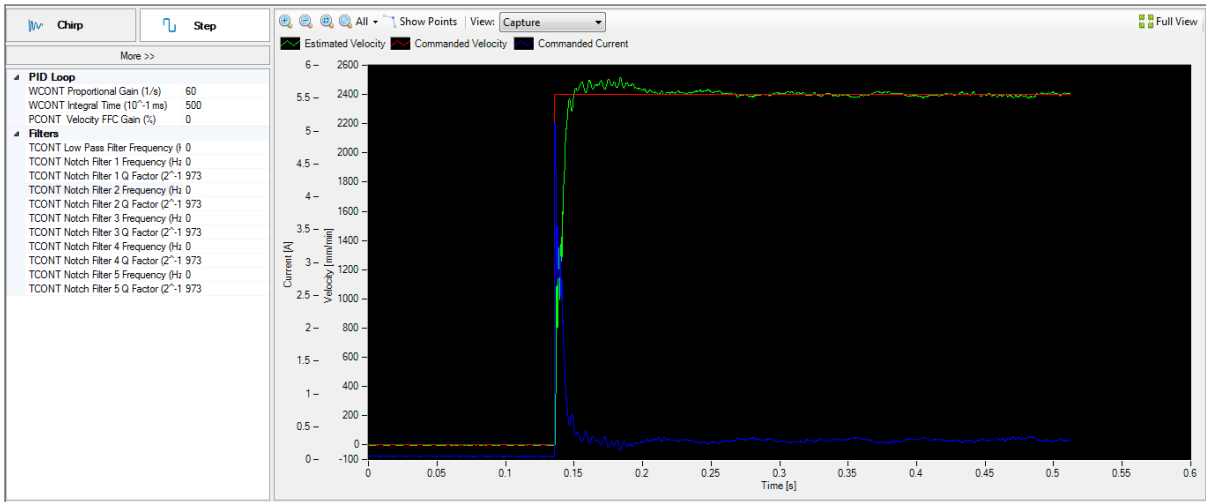
3. Reduce proportional gain by $\sim 1/3$.



4. Reduce integral time.



5. Reduce integral time further. Tuning complete.



8 Communication

8.1 What this Chapter Contains

This chapter contains information about communicating with the AMD servo drive:

- Servo Drive Profile over EtherCAT (SoE)
- EtherCAT state machine
- EtherCAT Process Data
- Distributed Clocks over EtherCAT

8.2 EtherCAT

8.2.1 Servo Drive Profile over EtherCAT (SoE) Device Architecture

The AMD servo drive communications is composed of EtherCAT communication in the physical and data link layers, while Servo drive profile over EtherCAT (SoE), FoE and EoE are utilised in the application layer. The Identification Number (IDN) dictionary includes all parameters, application data and process data mapping information between the process data interface and application data. The process data IDNs contain IDNs from the IDN dictionary which are mapped to process data. Process data updates cyclically, reading and writing to the mapped IDNs. Service data uses the EtherCAT mailbox to non-cyclically perform send and receive requests of IDN data.

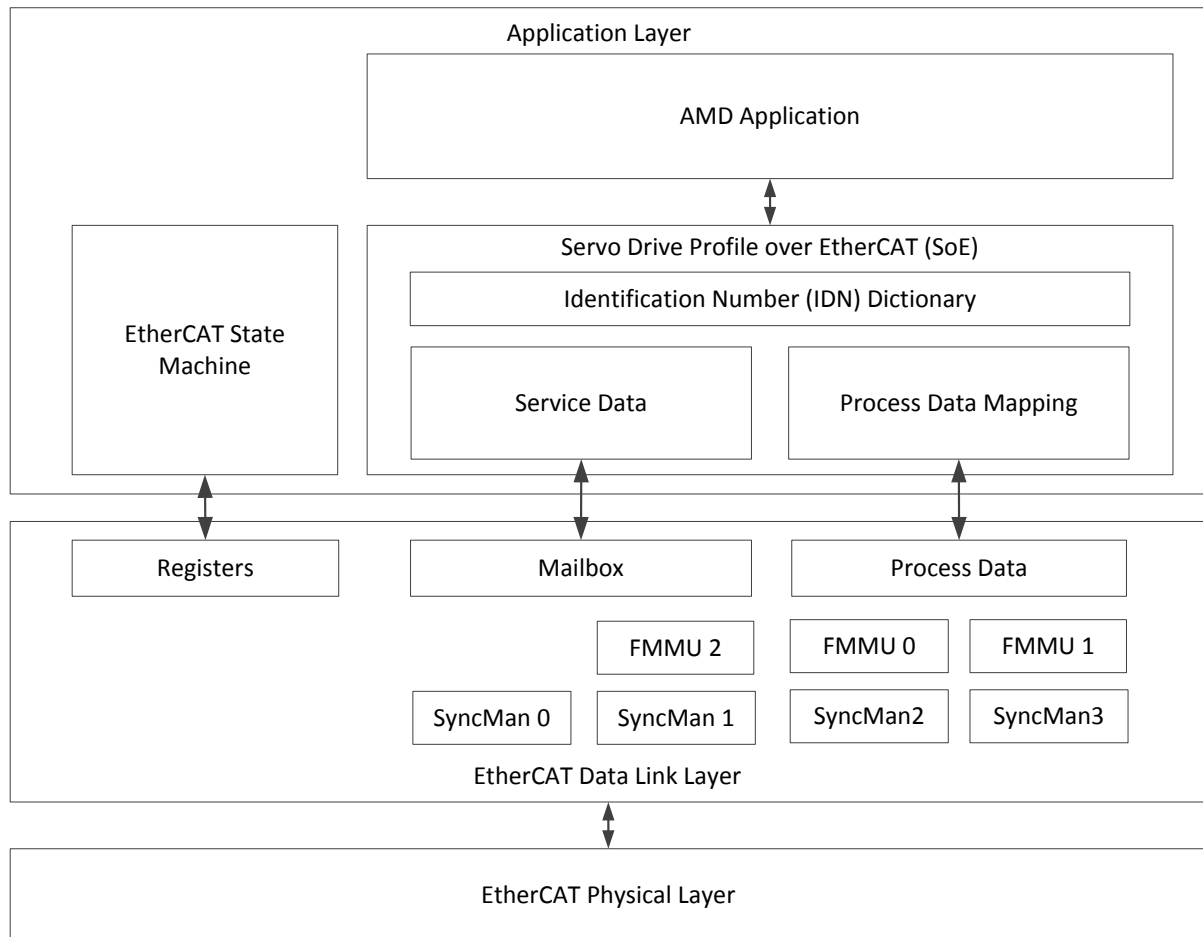


Figure 8-1 - SoE device architecture

8.2.2 EtherCAT Slave Information

The EtherCAT Slave Information (ESI) XML file describes the EtherCAT slave device to an EtherCAT master. The ESI file contains the communication configuration required to utilise the device. Each AMD servo drive model has a unique ESI file that is provided as part of the firmware bundle.

8.2.3 EtherCAT State Machine

The EtherCAT State Machine (ESM) is shown below in [Figure 8-2](#). Under normal operation conditions, each state transition is initiated by the EtherCAT master with the master responsible for performing prerequisite operations before transition to the next state as defined in the ESI file.

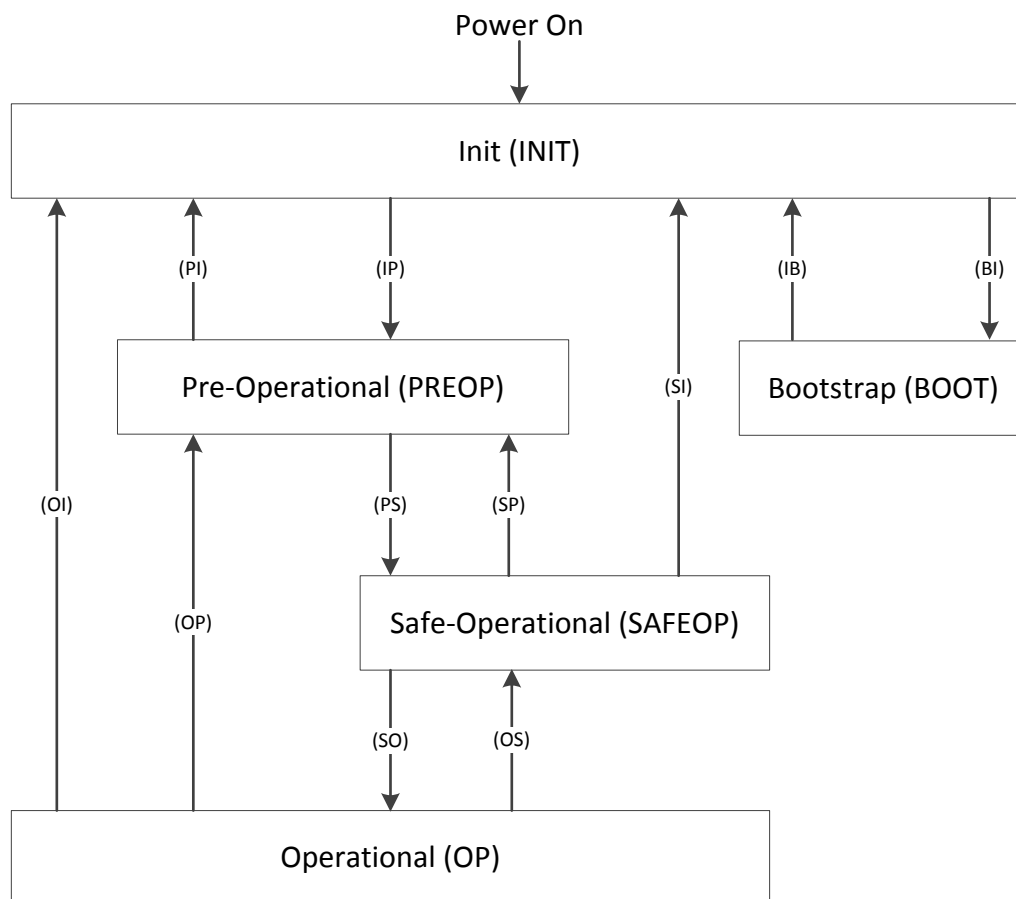


Figure 8-2 - EtherCAT State Machine

State	Description
INIT	The initial power up state. No mailbox or process data communication is possible.
PREOP	Only mailbox communication is available.
SAFEOP	Mailbox communication is available. Only process data communication from the drive is available. Process data to the drive remains in a 'safe state'.
OP	Full mailbox and process data communication available.
BOOT	The boot state unloads the current firmware and operates only the bootloader. This state is a prerequisite before FoE transfer of new firmware can be achieved.

Table 8-1 - EtherCAT State Machine state definitions

8.2.4 Service Data

8.2.5 Process Data Mapping

8.2.5.1 Description

Process data, also known as cyclic data, is data that is transferred to and from an EtherCAT slave device once every communications cycle. Process data is used for operational data that must be delivered in real-time such as motor commands and feedback or IO.

AT process data consists of data sent from the servo drive to the EtherCAT master. The data will always be sent with the *Drive Status Word* followed by the data from the IDNs listed in *AT (transmit) Configuration List*.

MDT process data consists of data sent from the EtherCAT master to the servo drive. The data will always be received with the *Master Control Word* followed by the data from the IDNs listed in *MDT (receive) Configuration List*.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0016 / 16	AT (transmit) Configuration List (32 elements)	Unsigned Integer (2 bytes) Array	IDN	0,0,...,0	0,0,...,0
S-0-0024 / 24	MDT (receive) Configuration List (32 elements)	Unsigned Integer (2 bytes) Array	IDN	0,0,...,0	0,0,...,0
S-0-0185 / 185	AT List Maximum Size	Unsigned Integer (2 bytes)	Bytes	256	256
S-0-0186 / 186	MDT List Maximum Size	Unsigned Integer (2 bytes)	Bytes	256	256
S-0-0187 / 187	All AT (transmit) IDNs (192 elements)	Unsigned Integer (2 bytes) Array	IDN	N/A	N/A
S-0-0187 / 187	All MDT (receive) IDNs (192 elements)	Unsigned Integer (2 bytes) Array	IDN	N/A	N/A

8.2.5.2 Definitions

8.2.5.2.1 AT (transmit) Configuration List

The *AT (transmit) Configuration List* allows configuration of the data to be cyclically transferred from the drive to the EtherCAT master. Data from each IDN that is entered will be mapped to cyclic data and transmitted in the order that they are placed in this list.

8.2.5.2.2 MDT (receive) Configuration List

The *MDT (receive) Configuration List* allows configuration of the data to be cyclically transferred from the EtherCAT master to the drive. Data from each IDN that is entered will be mapped to cyclic data and received in the order that they are placed in this list.

8.2.5.2.3 AT List Maximum Size

This parameter lists the total size, in bytes, that is available in the AT List. This value is independent of the number of IDNs in the list.

8.2.5.2.4 MDT List Maximum Size

This parameter lists the total size, in bytes, that is available in the MDT List. This value is independent of the number of IDNs in the list.

8.2.5.2.5 All AT (transmit) IDNs

This parameter lists all the IDNs that are permitted in the AT list. Only IDNs in this list can be added to the *AT (transmit) Configuration List*.

8.2.5.2.6 All MDT (receive) IDNs

This parameter lists all the IDNs that are permitted in the MDT list. Only IDNs in this list can be added to the *MDT (receive) Configuration List*.

8.2.6 Cyclic Data Update Rates

8.2.6.1 Description

The AMD servo drive can be configured to process cyclic data at 4ms and 1ms update rates. This option can be broken down into two configuration parameters, the *Communications Cycle Time* and the *Control Unit Cycle Time*.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0001 / 1	Control Unit Cycle Time	Unsigned Integer (2 bytes)	us	4000	4000
S-0-0002 / 2	Communications Cycle Time	Unsigned Integer (2 bytes)	us	4000	4000

8.2.6.2 Definitions

8.2.6.2.1 Control Unit Cycle Time

The *Control Unit Cycle Time* defines the rate at which the control unit consumes new command data. The parameter is only utilised when following NC set points. This parameter must be a multiple of the *Communications Cycle Time*. Valid values are 1000us and 4000us. If an invalid value is set the default of 4000us will be used.

8.2.6.2.2 Communications Cycle Time

The *Communications Cycle Time* defines the rate at which the cyclic data is transferred to and from the drive over EtherCAT. Valid values are 1000us and 4000us. If an invalid value is set the default 4000us will be used.

8.2.6.3 Configuration

By default, the AMD servo drive is configured to run at 4ms cycle time for both the control unit and communications. If it is desired to run in NC mode at 1ms then set both IDN1 and IDN2 to 1000. If the drive is not in NC mode then the control unit cycle time will be at 4ms regardless of the value of IDN1.

8.2.7 Synchronisation with Distributed Clocks

Synchronisation of EtherCAT communications is performed via the Distributed Clock mechanism. With an EtherCAT master that supports distributed clocks, all EtherCAT devices synchronise to a reference clock (system time). Propagation delay between devices is also taken into account resulting in precise timing amongst numerous devices.

The use of a distributed clock also unburdens the EtherCAT master from precise timing of data. Since actions such as position measurement are triggered by the local clock instead of when the frame is received, the master device doesn't have such strict requirements for sending frames. The EtherCAT master needs only to ensure that the EtherCAT telegram is sent before the DC signal in the slave devices triggers the output.

The Distributed Clock must be setup and configured before the drive transitions to the SAFEOP state. Refer to the device's [EtherCAT Slave Information](#) file for more detail on the use of distributed clocks on the AMD servo drive.

The AMD servo drive includes a diagnostic that monitors the error between the drive's local clock to the system clock. If the drive cannot lock onto the system clock or the measured *Distributed Clock Time Error* is found to be greater than *Distributed Clock Time Error Threshold* then a C1D (E0221) will be asserted.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
P-0-0338 / 33106	Distributed Clock Time Error	Signed Integer (4 bytes)	$10^{-3} \mu\text{s}$	N/A	N/A
P-0-0341 / 33109	Distributed Clock Time Error Monitoring	Unsigned Integer (2 bytes)	Boolean	1	1
P-0-0342 / 33110	Distributed Clock Time Error Threshold	Signed Integer (4 bytes)	$10^{-3} \mu\text{s}$	10 ms	10 ms

8.2.8 Ethernet over EtherCAT (EoE)

EtherCAT makes use of the physical layers of Ethernet and the Ethernet frame to provide its real-time behaviour. The Ethernet over EtherCAT (EoE) protocol allows standard IP protocols to be tunnelled over EtherCAT. The EtherCAT master acts as a switch with only standard Ethernet IP visible to applications requiring it.

While EoE is not required to operate the AMD servo drive, support for ANCA MotionBench requires the EoE protocol.

8.2.9 File access over EtherCAT (FoE)

The File access over EtherCAT (FoE) protocol allows for simple file transfer to the EtherCAT device. The AMD servo drive utilises FoE to enable simple firmware upgrades. The EtherCAT master can enforce a firmware version of all EtherCAT slaves and modify the firmware as required.

9 Fault Tracing

9.1 What this Chapter Contains

This chapter contains information that will guide the user in trouble shooting AMD Servo Drive:

- Diagnostic indicators on the drive
- Error recovery
- Base Firmware Error Codes and Possible Causes
- Firmware Upgrade Errors

9.2 Problem Diagnosis

9.2.1 AMD2000 Display Indicators



The 7 segment LED display on the AMD2000 serves three functions. It is used to report errors, to indicate the state of the EtherCAT communications and to indicate the state of the drive.

9.2.1.1 Error state

In an error condition, the display will read **E-####** where **####** refers to the relevant error code. See [9.4 Supported Error Codes](#) for the description and cause of each. When no error has been reported, the display will provide information on both the drive state and the communications state.

9.2.1.2 Communication State

To indicate the state of the EtherCAT communications, the leftmost digits of the display will read **C#**, where **#** refers to the current communications condition as shown in the following table:

C#	Communications State
C0	None
C1	Initialization
C2	Pre-operational
C4	Safe-operational
C8	Operational

9.2.1.3 Drive State

To indicate the state of the drive, the rightmost digits of the display will read d#, where # refers to the current drive condition as shown in the following table:

d#	Drive State
d0	Off
d2	Ready to operate
d3	Enabling
d4	Enabled

9.2.2 Error Reporting

If the AMD servo drive encounters an error, this will be communicated back to the control unit / device via 3 mechanisms.

The primary indicator of an error is bit 13 of the drive status word. This will equal 1 when there is a Class 1 Diagnostics Error on the drive. For more information on the drive status word refer to [6.2.1.2 Drive Status Word](#). Once the error occurring has been registered by the SoE master, the diagnostic message and diagnostic message code can be accessed via IDNs S-0-0095 / 95 and P-0-0489 / 33257 respectively.

See [9.4 Supported Error Codes](#) for the description and cause of each.

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0095 / 95	Diagnostic Message (32 elements)	Unsigned Integer (1 byte) Array	String	N/A	N/A
S-0-0135 / 135	Drive Status Word	Unsigned Integer (2 bytes)	Binary	N/A	N/A
P-0-0489 / 33257	Diagnostic Message Code	Unsigned Integer (2 bytes)	None	N/A	N/A

9.3 Resetting From Errors



NOTE: If the drive has a persistent error then the reset command may appear not to work. Ensure the cause of the error is removed before attempting to reset.

In the case that the AMD servo drive has experienced a Class 1 Diagnostic Error, the drive will remain in an error state until the error has been cleared.

To clear the error:

1. Solve the cause of the error based on the error message
2. If not already done so, set IDN S-0-0099 / 99 to 0
3. Set IDN S-0-0099 / 99 to 3 to trigger the procedure command
4. Set IDN S-0-0099 / 99 to 0

IDN	Description	Data Type	Units	Default	
				AMD2000	AMD5x
S-0-0099 / 99	Reset Class 1 Diagnostic	Unsigned Integer (2 bytes)	Procedure Command	0	0

9.4 Supported Error Codes

Error codes are displayed on the seven segment display in the format of a prefix followed by a number

9.4.1 Error Code Prefixes

Prefix	Severity	Description
E	Error	Critical faults which disables the drive. (Class 1 Diagnostics).

9.4.2 Error Codes

A summary of the AMD servo drive error codes is shown in [Table 9-1](#). Please note that error codes are firmware dependent and a complete and up-to-date error listing will be delivered as part of a firmware package ("Digital_Servo_Drive_Error_Code_Reference.pdf").

Code	Severity	Label
E0004	Error	Powerstage General Fault
E0007	Error	Current Offset Adapt Error
E0013	Error	Encoder Amplitude Low - Motor Commutation Track
E0014	Error	Encoder Amplitude High - Motor Commutation Track
E0015	Error	Encoder Amplitude Low - Motor
E0016	Error	Encoder Amplitude High - Motor
E0017	Error	Encoder Amplitude Low - Auxiliary
E0018	Error	Encoder Amplitude High - Auxiliary
E0023	Error	Excess Servo Position Error
E0024	Error	Excess Servo Velocity Error
E0027	Error	Encoder Amplitude Low - External
E0028	Error	Encoder Amplitude High - External
E0033	Error	Excess Difference in Position Feedback
E0050	Error	Serial Encoder Timeout Error - Motor
E0051	Error	Serial Encoder CRC Error - Motor
E0052	Error	Serial Encoder Head Error - Motor
E0054	Error	Serial Encoder Timeout Error - External
E0055	Error	Serial Encoder CRC Error - External
E0056	Error	Serial Encoder Head Error - External
E0060	Error	Fan Fault
E0070	Error	Drive Enabled with STO Active
E0071	Error	STO Input Signal Mismatch
E0072	Error	STO Channel 1 Voltage Fault
E0073	Error	STO Channel 2 Voltage Fault
E0080	Error	Variable Torque Limit Error
E0100	Error	Drive Not Configured
E0101	Error	Amplifier Temperature Sensor Error - High
E0102	Error	Amplifier Temperature Sensor Error - Low
E0104	Error	Amplifier Temperature High Error

E0105	Error	Amplifier Temperature High Error with Drive Enabled
E0111	Error	Motor Temperature Sensor Error - High
E0112	Error	Motor Temperature Sensor Error - Low
E0114	Error	Motor Temperature High Error
E0115	Error	Motor Temperature High Error when Drive Enabled
E0120	Error	Motor Not Standstill during Enable
E0150	Error	Home Switch Error
E0205	Error	EtherCAT Watchdog Timeout
E0206	Error	Configuration Mode Watchdog Timeout
E0210	Error	PSU Main Power Start Timeout Error
E0211	Error	PSU Error
E0215	Error	Encoder Adjusted Amplitude Low - Motor
E0216	Error	Encoder Adjusted Amplitude High - Motor
E0217	Error	Encoder Adjusted Amplitude Low - Auxiliary
E0218	Error	Encoder Adjusted Amplitude High - Auxiliary
E0220	Error	Control Unit Synchronization Bit Toggle Missing
E0221	Error	Distributed Clocks Error
E0227	Error	Encoder Adjusted Amplitude Low - External
E0228	Error	Encoder Adjusted Amplitude High - External
E0302	Error	DC Bus Voltage High
E0303	Error	DC Bus Voltage Low
E0304	Error	Positive Position Soft Limit
E0305	Error	Negative Position Soft Limit
E0306	Error	Positive Position Dead Stop
E0307	Error	Negative Position Dead Stop
E0308	Error	Instantaneous Current Limit Exceeded
E0309	Error	Amplifier I2R Overload
E0320	Error	Motor I2T Overload
E0325	Error	Motor I2R Overload
E0330	Error	Motor Temperature High Error when Drive Enabled
E0331	Error	Motor Not Standstill during Enable
E0332	Error	Home Switch Error
E0333	Error	Current Limit Active
E0340	Error	Homing Invalid
E0380	Error	EtherCAT Watchdog Timeout
E0402	Error	Configuration Mode Watchdog Timeout
E0403	Error	PSU Main Power Start Timeout Error
E0404	Error	PSU Error
E0405	Error	Encoder Adjusted Amplitude Low - Motor
E0406	Error	Encoder Adjusted Amplitude High - Motor
E0407	Error	Encoder Adjusted Amplitude Low - Auxiliary
E0411	Error	Control Unit Synchronization Bit Toggle Missing
E0412	Error	Distributed Clocks Error
E0413	Error	SoE Scaling Has Reduced Range
E0414	Error	Encoder Adjusted Amplitude Low - External

E0415	Error	Encoder Adjusted Amplitude High - External
E0416	Error	DC Bus Voltage High
E0417	Error	DC Bus Voltage Low

Table 9-1 - Error / Warning Code summary

9.4.3 Error Codes Detailed Descriptions

Current Offset Adapt Error

E0007

Description	Calibrated current offset value has exceeded the specified tolerance. Possible causes for this error are: 1. Fault in the current measurement system. 2. Incorrect current scaling parameters configured.
Severity	Error

Encoder Amplitude Low - Motor

E0015

Description	The magnitude of the signals coming from the motor analogue encoder is too low. Possible causes for this error are: 1. Encoder cable is disconnected. 2. Encoder cable is wired incorrectly. 3. Encoder is not analogue. 4. Encoder is not outputting the correct voltage. 5. Encoder is faulty. 6. Drive is faulty. Please contact ANCA Motion for support.
Severity	Error

Encoder Amplitude High - Motor

E0016

Description	The magnitude of the signals coming from the motor analogue encoder is too high. Possible causes for this error are: 1. Encoder cable is wired incorrectly. 2. Encoder is not analogue. 3. Encoder is not outputting the correct voltage. 4. Encoder is faulty. 5. Drive is faulty. Please contact ANCA Motion for support.
Severity	Error

Excess Servo Position Error

E0023

Description	Position following error exceeded the configured threshold. Possible causes for this error are: 1. Contouring commands too demanding. 2. Insufficient DC bus voltage. 3. Poor controller tuning. 4. Axis has crashed or jammed. 5. Field orientation alignment is inaccurate (possible encoder fault).
Severity	Error

Excess Servo Velocity Error

E0024

Description	Velocity following error exceeded the configured threshold. Possible causes for this error are: 1. Contouring commands too demanding. 2. Insufficient DC bus voltage.
-------------	---

	- Poor controller tuning. - Axis has crashed or jammed. - Field orientation alignment is inaccurate (possible encoder fault).
Severity	Error

Encoder Amplitude Low - External

E0027

Description	The magnitude of the signals coming from the external analogue encoder is too low. Possible causes for this error are:
-------------	--

- External encoder configured where no encoder exists.
- Encoder cable is disconnected.
- Encoder cable is wired incorrectly.
- Encoder is not analogue.
- Encoder is not outputting the correct voltage.
- Encoder is faulty.
- Drive is faulty. Please contact ANCA Motion for support.

Severity	Error
----------	-------

Encoder Amplitude High - External

E0028

Description	The magnitude of the signals coming from the external analogue encoder is too high. Possible causes for this error are:
-------------	---

- Encoder cable is wired incorrectly.
- Encoder is not analogue.
- Encoder is not outputting the correct voltage.
- Encoder is faulty.
- Drive is faulty. Please contact ANCA Motion for support.

Severity	Error
----------	-------

Excess Difference in Position Feedback

E0033

Description	The difference between motor and external feedback is greater than the configured threshold. Possible causes for this error are:
-------------	--

- Incorrect encoder line count.
- Incorrect gear box ratio and/or feed constant.
- Faulty drive mechanism (e.g. loss coupling).
- Faulty encoder.
- Fault drive. Please contact ANCA Motion for support.

Severity	Error
----------	-------

Variable Torque Limit Error

E0080

Description	Configuration of the variable torque limit is invalid. The minimum torque limit is configured larger than the maximum torque limit.
-------------	---

Severity	Error
----------	-------

Drive Not Configured

E0100

Description	The drive has been enabled before being configured
-------------	--

Severity	Error
----------	-------

Amplifier Temperature Sensor Error - High

E0101

Description	Drive power stage temperature sensor is reading a value higher than the drive's sensor operating range. Drive is faulty. Please contact ANCA Motion for support.
-------------	--

Severity	Error
----------	-------

Amplifier Temperature Sensor Error - Low**E0102**

Description	Drive power stage temperature sensor is reading a value lower than the drive's sensor operating range. Drive is faulty. Please contact ANCA Motion for support.
Severity	Error

Amplifier Temperature High Warning**W0103**

Description	The drive amplifier (power stage) temperature exceeds the hardware's warning level. Possible causes for this error are: 1. Operating environment is outside specification. 2. Drive ventilation is insufficient. 3. Application is too demanding. 4. Cooling fan is faulty. Please contact ANCA Motion for support.
Severity	Warning

Amplifier Temperature High Error**E0104**

Description	The drive amplifier (power stage) temperature exceeds the hardware's physical operating threshold. Possible causes for this error are: 1. Operating environment is outside specification. 2. Drive ventilation is insufficient. 3. Application is too demanding. 4. Cooling fan is faulty. Please contact ANCA Motion for support.
Severity	Error

Amplifier Temperature High Error with Drive Enabled**E0105**

Description	With the drive enabled the drive amplifier (power stage) temperature exceeds the hardware's physical operating threshold. Possible causes for this error are: 1. Operating environment is outside specification. 2. Drive ventilation is insufficient. 3. Application is too demanding. 4. Cooling fan is faulty. Please contact ANCA Motion for support.
Severity	Error

Motor Not Standstill During Enable**E0120**

Description	When the drive is enabled the motor must be stationary. This error is triggered if the motor moves during initialisation. Possible causes for this error are: 1. Motor is moving via some external interaction during initialisation. 2. Analogue encoder feedback is excessively noisy (if applicable). 3. Standstill threshold is set below the analogue encoder feedback noise floor (if applicable). 4. Encoder is faulty. 5. Drive is faulty. Please contact ANCA Motion for support.
Severity	Error

EtherCAT Watchdog Timeout**E0205**

Description	This error indicates a problem with communications between the EtherCAT Master (eg. CNC) and the drive. Please contact ANCA Motion for support.
Severity	Error

Configuration Mode Watchdog Timeout**E0206**

Description This error indicates a problem with communications between ANCA MotionBench software and the drive. Please refer to [4 Start-up](#)

Severity Error

PSU Main Power Start Timeout Error**E0210**

Description The main power supply failed to enable within the configured time. Please contact ANCA Motion for support.

Severity Error

PSU Error**E0211**

Description The main power supply has reported an error. Please contact ANCA Motion for support.

Severity Error

Encoder Adjusted Amplitude Low - Motor**E0215**

Description The magnitude of the adjusted signals for the motor analogue encoder is too low. Incorrect gain and/or offset values have been configured.

Severity Error

Encoder Adjusted Amplitude High - Motor**E0216**

Description The magnitude of the adjusted signals for the motor analogue encoder is too high. Incorrect gain and/or offset values have been configured.

Severity Error

Control Unit Synchronization Bit Toggle Missing**E0220**

Description This error indicates a problem with communications between the EtherCAT Master (e.g. CNC) and the drive. Please contact ANCA Motion for support.

Severity Error

Distributed Clocks Error**E0221**

Description This error indicates a problem with communications between the EtherCAT Master (e.g. CNC) and the drive. Please contact ANCA Motion for support.

Severity Error

Encoder Adjusted Amplitude Low - External**E0227**

Description The magnitude of the adjusted signals for the external analogue encoder is too low. Incorrect gain and/or offset values have been configured.

Severity Error

Encoder Adjusted Amplitude High - External**E0228**

Description The magnitude of the adjusted signals for the external analogue encoder is too high. Incorrect gain and/or offset values have been configured.

Severity	Error	
	DC Bus Voltage High	E0302
Description	DC bus voltage in the power stage exceeded the hardware maximum limit. Possible causes for this error are: 1. Regenerative load is outside the specification for the drive. 2. Mains supply voltage is too high. 3. Regeneration resistor or drive is faulty. Please contact ANCA Motion for support.	
Severity	Error	
	DC Bus Voltage Low	E0303
Description	DC bus voltage in the power stage is below the hardware minimum limit. Possible causes for this error are: 1. Mains supply is not connected. 2. Connector for external inductor, missing inductor or link across P1, P2. 3. Mains supply voltage is too low. 4. Power requirements for the application are outside the specification for the drive. 5. Drive is faulty. Please contact ANCA Motion for support.	
Severity	Error	
	Positive Position Soft Limit	E0304
Description	Position soft limit in the positive direction has been exceeded. Possible causes for this error are: 1. Error limit is enabled before the axis has been successfully homed. 2. Master or other High Level Function have commanded the drive to a state (position & velocity) where it will be unable to decelerate before exceeding the positive position limit. 3. Drive has experienced a fault which has resulted in a runaway event.	
Severity	Error	
	Negative Position Soft Limit	E0305
Description	Position soft limit in the negative direction has been exceeded. Possible causes for this error are: 1. Error limit is enabled before the axis has been successfully homed. 2. Master or other High Level Function have commanded the drive to a state (position & velocity) where it will be unable to decelerate before exceeding the negative position limit. 3. Drive has experienced a fault which has resulted in a runaway event.	
Severity	Error	
	Positive Position Dead stop	E0306
Description	Position positive dead stop input is active.	
Severity	Error	
	Negative Position Dead stop	E0307
Description	Position negative dead stop input is active.	
Severity	Error	
	Instantaneous Current Limit Exceeded	E0308

Description	One or more of the motor phase currents has exceeded the instantaneous limit. Possible causes for this error are: 1. Instantaneous current limit is configured too low given the unified current limit. 2. Current loop is poorly tuned. 3. Motor is faulty. 4. Drive is faulty. Please contact ANCA Motion for support.
Severity	Error

Amplifier I2R Overload

E0309

Description	Residual heat within the drive power stage (amplifier) exceeds the thermal limit. Possible causes for this error are: 1. Application is outside the specification for the drive: too demanding. 2. Current controller is poorly tuned. 3. Field orientation alignment is inaccurate (possible encoder fault). 4. Drive is faulty. Please contact ANCA Motion for support.
Severity	Error

Motor I2T Overload

E0320

Description	Motor current is consistently higher than the specified load threshold. Possible causes for this error are: 1. Load on axis is above configured threshold. 2. Axis has crashed or jammed.
Severity	Error

Amplifier I2R Warning

W0321

Description	Residual heat within the drive power stage (amplifier) exceeds the warning level. Possible causes for this error are: 1. Application is outside the specification for the drive: too demanding. 2. Current controller is poorly tuned. 3. Field orientation alignment is inaccurate (possible encoder fault). 4. Drive is faulty. Please contact ANCA Motion for support.
Severity	Warning

Motor I2T Warning

W0322

Description	Motor current is higher than the specified load threshold. Possible causes for this error are: 1. Load on axis is above configured threshold. 2. Axis has crashed or jammed.
Severity	Warning

Motor I2R Warning

W0324

Description	Residual heat within the motor exceeds the warning level. Possible causes for this error are: 1. Application is outside the specification for the motor: too demanding. 2. Current controller is poorly tuned. 3. Field orientation alignment is inaccurate (possible encoder fault). 4. Motor is faulty.
Severity	Warning

Motor I2R Overload

E0325

Description	Residual heat within the motor exceeds the thermal limit. Possible causes for this error are: 1. Application is outside the specification for the motor: too demanding. 2. Current controller is poorly tuned. 3. Field orientation alignment is inaccurate (possible encoder fault). 4. Motor is faulty.
Severity	Error

Positive Position Hard Limit**E0330**

Description	Position hard limit in the positive direction has been exceeded. Possible causes for this error are: 1. Error limit is enabled before the axis has been successfully homed. 2. Master or other High Level Function has commanded the drive to a position that exceeds the positive position limit. 3. Drive has experienced a fault which has resulted in a runaway event.
Severity	Error

Negative Position Hard Limit**E0331**

Description	Position hard limit in the negative direction has been exceeded. Possible causes for this error are: 1. Error limit is enabled before the axis has been successfully homed. 2. Master or other High Level Function have commanded the drive to a position that exceeds the negative position limit. 3. Drive has experienced a fault which has resulted in a runaway event.
Severity	Error

Positive Velocity Hard Limit**E0332**

Description	Positive velocity hard limit has been exceeded. Possible causes for this error are: 1. Master or other High Level Function has commanded the drive to a velocity that exceeds the positive velocity limit. 2. Drive has experienced a fault which has resulted in a runaway event.
Severity	Error

Negative Velocity Hard Limit**E0333**

Description	Negative velocity hard limit has been exceeded. Possible causes for this error are: 1. Master or other High Level Function have commanded the drive to a velocity that exceeds the negative velocity limit. 2. Drive has experienced a fault which has resulted in a runaway event.
Severity	Error

Invalid Command Reference Frame**E0340**

Description	The set point command is in referenced coordinates, that is zeroed or homed, but the feedback is not in reference coordinates. Handshaking between the master and the drive while attempting to change referenced coordinates has failed.
Severity	Error

Event Detection Error**E0380**

Description	The event detection module has reported an error while attempting to latch a position during homing or probing. Please contact ANCA Motion for support.
Severity	Error

DQA Invalid Movement Detected**E0402**

Description	During DQ alignment an invalid movement was detected. Possible causes for this error are: 1. Incorrect motor poles configured. 2. Incorrect motor phase sequence. 3. Incorrect motor encoder line count configured. 4. Incorrect motor encoder polarity configured. 5. The configured alignment current is too low to drive the motor. 6. Motor/axis is jammed.
Severity	Error

Alignment Off Index Pulse Error**E0403**

Description	The difference between the configured and estimated field orientation alignment offset is larger than the configured threshold. Possible causes for this error are: 1. Incorrect encoder configuration (i.e. UVW hexant binary). 2. The relative position between the motor and encoder has been modified.
Severity	Error

DQA Current Magnitude Error**E0404**

Description	The configured alignment current exceeds the unified current limit.
Severity	Error

DQA Current Control Error**E0405**

Description	Sensed motor current is not following the DQ alignment current with sufficient accuracy, Possible causes for this error are: 1. Poorly tuned current loop. 2. DC bus voltage too low. 3. Motor armature cable is disconnected. 4. Motor is faulty. 5. Drive is faulty. Please contact ANCA Motion for support.
Severity	Error

Absolute Encoder Alignment Error**E0406**

Description	Absolute encoder used for field orientation initialisation has failed to latch an alignment angle. Possible causes for this error are: 1. Encoder is faulty. Power cycling the drive/encoder may resolve the issue. 2. Drive is faulty. Please contact ANCA Motion for support.
Severity	Error

Field Orientation Alignment in Progress**I0409**

Description	Field Orientation Alignment is in progress - active stimulus signal (current) has been commanded to the motor.
Severity	Info

Acceleration Observer Excessive Movement**E0411**

Description	Motor movement exceeded tolerance while executing acceleration observer field orientation. Possible causes for this error are: 1. Incorrect motor poles configured. 2. Incorrect motor phase sequence. 3. Incorrect motor encoder line count configured. 4. Incorrect motor encoder
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Severity polarity configured.
Error

Acceleration Observer Current Control Error

E0412

Description Sensed motor current is not following the Acceleration Observer alignment current with sufficient accuracy. Possible causes for this error are:

1. Poorly tuned current loop.
2. DC bus voltage too low.
3. Motor armature cable is disconnected.
4. Motor is faulty.
5. Drive is faulty. Please contact ANCA Motion for support.

Severity Error

Acceleration Observer Excessive Velocity

E0413

Description Excessive velocity detected while acceleration observer is executing. Possible causes for this error are:

1. Incorrect motor poles configured.
2. Incorrect motor encoder line count configured.
3. Stimulus frequency is too low.
4. Stimulus current is too high.

Severity Error

Acceleration Observer Torque Response Amplitude Low

E0414

Description The fundamental frequency component of the torque response is below the minimum amplitude threshold. Possible causes for this error are:

1. Stimulus frequency is too high.
2. Stimulus current is too low.

Severity Error

Acceleration Observer Torque Response Amplitude High

E0415

Description The fundamental frequency component of the torque response is above the maximum amplitude threshold. Possible causes for this error are:

1. Stimulus frequency is too low.
2. Stimulus current is too high.

Severity Error

Acceleration Observer Validation Failed

E0416

Description The axis moved in the wrong direction after Acceleration Observer completed. Possible causes for this error are:

1. Incorrect motor poles configured.
2. Incorrect motor phase sequence.
3. Incorrect motor encoder line count configured.
4. Incorrect motor encoder polarity configured.
5. The configured stimulus current is too low.
6. The configured stimulus frequency is too high.

Severity Error

Acceleration Observer Current Magnitude Error

E0417

Description Stimulus current magnitude exceeds the unified current limit.

9.4.4 Firmware Upgrade Errors



NOTE: Display state is only supported in AMD2000 drive series.

Displayed state	Description
BOOT1	Boot loader started
BOOT2	Boot loader finished
BLUP0	Boot loader Updater: Firmware processing state idle
BLUP1	Boot loader Updater: Firmware processing state validate after write
BLUP2	Boot loader Updater: Firmware processing state finished
E0001	EFW Streaming error: unexpected flash programming in progress
E0002	EFW Streaming error in receiving the file header
E0003	EFW Streaming error in validating the file header (CRC)
E0004	EFW Streaming error in initializing flash programming
E0005	EFW Streaming error in receiving the image block header
E0006	EFW Streaming error while decrypting the image block header
E0007	EFW Streaming error in validating the image block header (CRC)
E0008	EFW Streaming error: unexpected flash erasing or writing in progress while receiving the image block header
E0009	EFW Streaming error: software image size is larger than the allocated receiving buffer
E0010	EFW Streaming error in receiving the software image header
E0011	EFW Streaming error in decrypting the software image header
E0012	EFW Streaming error in validating the software image header (CRC)
E0013	EFW Streaming error: unexpected flash erasing or writing in progress while receiving the software image header
E0014	EFW Streaming error in receiving the software image data
E0015	EFW Streaming error in decrypting the software image data
E0016	EFW Streaming error: flash interface is not enabled while finalizing image stage 1
E0017	EFW Streaming error in validating the image block (CRC)
E0018	EFW Streaming in validating the software image

E0019	EFW Streaming error: flash interface is not enabled while finalizing image stage 2
E0020	EFW Streaming error in validating the file (CRC)
E0021	Boot loader Updater error: boot loader CRC check failed
E0022	Boot loader Updater Firmware Processing error: Flash controller interface error while preparing to write
E0023	Boot loader Updater Firmware Processing error: Flash controller interface error while submitting a block to be written
E0024	Boot loader Updater Firmware Processing error: Flash controller interface programming error
E0025	Boot loader Updater Firmware Processing error: Flash has attempted to write to the boot loader sector in flash but was unsuccessful, wipe the boot loader sector so the blcheck can jump straight to the BLUPG code again; failed to erase.
E0026	Boot loader Updater error: The attempt to erase the first sector of the application section (this has the jump instruction and header) failed.

9.5 Servo over EtherCAT® IDN Listing



NOTE: Refer to 'SoE Parameter Reference' associated to firmware release for an updated list.

IDN	Name	Data Type
1	Control Unit Cycle Time	Unsigned Integer (2 bytes)
2	Communications Cycle Time	Unsigned Integer (2 bytes)
11	Class 1 Diagnostics (C1D)	Unsigned Integer (2 bytes)
12	Class 2 Diagnostics (C2D)	Unsigned Integer (2 bytes)
13	Class 3 Diagnostics (C3D)	Unsigned Integer (2 bytes)
15	Telegram Type	Unsigned Integer (2 bytes)
16	AT (Transmit) Configuration List	Unsigned Integer (2 bytes)
17	IDN list of All Operational Data	Unsigned Integer (2 bytes)
21	CP2 Invalid Data IDN List	Unsigned Integer (2 bytes)
22	CP3 Invalid Data IDN List	Unsigned Integer (2 bytes)
24	MDT (receive) Configuration List	Unsigned Integer (2 bytes)
25	All procedure commands	Unsigned Integer (2 bytes)
30	Firmware Version Label	Unsigned Integer (1 bytes)
32	Primary Operating Mode	Unsigned Integer (2 bytes)
33	Operating Mode Secondary1	Unsigned Integer (2 bytes)
34	Operating Mode Secondary2	Unsigned Integer (2 bytes)
35	Operating Mode Secondary3	Unsigned Integer (2 bytes)
36	NC Velocity Setpoint Command	Signed Integer (4 bytes)
37	Velocity Loop Additive Velocity Command	Signed Integer (4 bytes)
40	NC Velocity Feedback	Signed Integer (4 bytes)
41	Homing Velocity to locate home switch	Signed Integer (4 bytes)
42	DCH Max Acceleration	Signed Integer (4 bytes)
43	NC Velocity Polarity Configuration	Unsigned Integer (2 bytes)
44	SoE Velocity Scaling - Type	Unsigned Integer (2 bytes)
45	SoE Velocity Scaling - Factor	Unsigned Integer (2 bytes)
46	SoE Velocity Scaling - Exponent	Signed Integer (2 bytes)
47	NC Position Setpoint Command	Signed Integer (4 bytes)
48	Position Loop Additive Position Command	Signed Integer (4 bytes)
51	NC Motor Position Feedback	Signed Integer (4 bytes)
52	Reference Distance 1	Signed Integer (4 bytes)
53	NC External Position Feedback	Signed Integer (4 bytes)
54	Reference Distance 2	Signed Integer (4 bytes)
55	NC Position polarity configuration	Unsigned Integer (2 bytes)
56	Auxilliary Sample Time	Unsigned Integer (4 bytes)
57	Position Window	Signed Integer (4 bytes)
58	Backlash Compensation Distance	Signed Integer (4 bytes)
76	SoE Position Scaling - Type	Unsigned Integer (2 bytes)
77	SoE Position Scaling - Linear Factor	Unsigned Integer (2 bytes)

78	SoE Position Scaling - Linear Exponent	Signed Integer (2 bytes)
79	SoE Position Scaling - Rotational Resolution	Unsigned Integer (4 bytes)
80	NC Torque/Force Setpoint Command	Signed Integer (2 bytes)
81	Torque Loop Additive Torque Command	Signed Integer (2 bytes)
84	TCONT Torque Feedback	Signed Integer (2 bytes)
85	Torque polarity configuration	Unsigned Integer (2 bytes)
86	SoE Torque Scaling - Type	Unsigned Integer (2 bytes)
93	SoE Torque Scaling - Factor	Unsigned Integer (2 bytes)
94	SoE Torque Scaling - Exponent	Signed Integer (2 bytes)
95	Diagnostic message	Unsigned Integer (1 bytes)
99	Reset C1D	Unsigned Integer (2 bytes)
100	WCONT Proportional Gain	Unsigned Integer (2 bytes)
101	WCONT Integral Time	Unsigned Integer (2 bytes)
102	Velocity Control Differential Time	Unsigned Integer (2 bytes)
103	Modulo Value	Signed Integer (4 bytes)
104	Position Controller Proportional Gain Kv	Unsigned Integer (2 bytes)
105	Position Loop Integral Time Constant	Unsigned Integer (2 bytes)
106	ICONT Q Axis Gain	Unsigned Integer (2 bytes)
107	ICONT Q Axis Integral Time	Unsigned Integer (2 bytes)
109	ICONT Motor Peak Current	Signed Integer (4 bytes)
110	ICONT Amplifier Peak Current	Signed Integer (4 bytes)
111	Motor Continuous Current Rating	Signed Integer (4 bytes)
112	Power Stage Continuous Current Rating	Signed Integer (4 bytes)
113	Maximum Motor Velocity	Signed Integer (4 bytes)
114	Motor Load Limit	Unsigned Integer (2 bytes)
115	Position Feedback 2 Type	Unsigned Integer (2 bytes)
116	Encoder 1 Pulse Per Revolution	Unsigned Integer (4 bytes)
117	Encoder2 PPR	Unsigned Integer (4 bytes)
118	External Encoder Linear Resolution	Unsigned Integer (4 bytes)
119	ICONT D Axis Gain	Unsigned Integer (2 bytes)
120	ICONT D Axis Integral Time	Unsigned Integer (2 bytes)
121	Input Revolutions	Unsigned Integer (4 bytes)
122	Output Revolutions	Unsigned Integer (4 bytes)
123	Feed Constant	Signed Integer (4 bytes)
124	Half Width for Zero Velocity Detection	Signed Integer (4 bytes)
129	Manufacturer-specific Class 1 Diagnostics	Unsigned Integer (2 bytes)
130	Probe 1 - Rising Edge	Signed Integer (4 bytes)
131	Probe 1 - Falling Edge	Signed Integer (4 bytes)
132	Probe 2 - Rising Edge	Signed Integer (4 bytes)
133	Probe 2 - Falling Edge	Signed Integer (4 bytes)
134	Master Control Word	Unsigned Integer (2 bytes)
135	Drive Status Word	Unsigned Integer (2 bytes)
140	Manufacturer Product Label	Unsigned Integer (1 bytes)

142	Application Label	Unsigned Integer (1 bytes)
143	SoE Version Label	Unsigned Integer (1 bytes)
146	CUCH Proc Cmd	Unsigned Integer (2 bytes)
147	Homing Parameters	Unsigned Integer (2 bytes)
148	Drive Controlled Homing Procedure Command	Unsigned Integer (2 bytes)
150	Reference Offset 1	Signed Integer (4 bytes)
151	Ref Offset 2	Signed Integer (4 bytes)
159	Position Following Error Window Half Width	Signed Integer (4 bytes)
160	SoE Acceleration Scaling - Type	Unsigned Integer (2 bytes)
161	SoE Acceleration Scaling - Factor	Unsigned Integer (2 bytes)
162	SoE Acceleration Scaling - Exponent	Signed Integer (2 bytes)
169	Probe Control Parameter	Unsigned Integer (2 bytes)
170	Probing Procedure Command	Unsigned Integer (2 bytes)
171	Calculate Displacement Procedure Command	Unsigned Integer (2 bytes)
172	Set Displacement to the reference system	Unsigned Integer (2 bytes)
173	Marker Position A	Signed Integer (4 bytes)
174	Marker Position B	Signed Integer (4 bytes)
175	Displacement Parameter 1	Signed Integer (4 bytes)
176	Displacement Parameter 2	Signed Integer (4 bytes)
179	Probe Status	Unsigned Integer (2 bytes)
181	Manufacturer-specific Class 2 Diagnostics	Unsigned Integer (2 bytes)
182	Manufacturer-specific Class 3 Diagnostics	Unsigned Integer (2 bytes)
185	AT maximum size	Unsigned Integer (2 bytes)
186	MDT maximum size	Unsigned Integer (2 bytes)
187	All AT (transmit) IDNs	Unsigned Integer (2 bytes)
188	All MDT (receive) IDNs	Unsigned Integer (2 bytes)
189	Position Controller Error Term	Signed Integer (4 bytes)
191	Reference Point Cancel Procedure Command	Unsigned Integer (2 bytes)
194	Acceleration setpoint	Signed Integer (4 bytes)
196	Motor Rated Current	Signed Integer (4 bytes)
200	Amplifier Warning Temperature	Signed Integer (2 bytes)
201	Motor Warning Temperature	Signed Integer (2 bytes)
203	Amplifier Shut-Down Temperature	Signed Integer (2 bytes)
204	Motor Shut-Down Temperature	Signed Integer (2 bytes)
206	Drive on delay time	Unsigned Integer (2 bytes)
207	Drive off delay time	Unsigned Integer (2 bytes)
208	Temperature Scaling Type	Unsigned Integer (2 bytes)
263	Non-Volatile Parameters - Read	Unsigned Integer (2 bytes)
264	Non-Volatile Parameters - Write	Unsigned Integer (2 bytes)
273	Drive off maximum delay time	Unsigned Integer (2 bytes)
277	Position Feedback 1 Type	Unsigned Integer (2 bytes)
284	Operating Mode Secondary 4	Unsigned Integer (2 bytes)
285	Operating Mode Secondary 5	Unsigned Integer (2 bytes)

286	Operating Mode Secondary 6	Unsigned Integer (2 bytes)
287	Operating Mode Secondary 7	Unsigned Integer (2 bytes)
292	List of Supported Operation Modes	Unsigned Integer (2 bytes)
295	Drive on delay time	Unsigned Integer (2 bytes)
300	Real Time Control Bit 1	Unsigned Integer (2 bytes)
301	Real Time Control Bit 1 Allocation	Unsigned Integer (2 bytes)
302	Real Time Control Bit 2	Unsigned Integer (2 bytes)
303	Real Time Control Bit 2 Allocation	Unsigned Integer (2 bytes)
304	Real Time Status Bit 1	Unsigned Integer (2 bytes)
305	Real Time Status Bit 1 Allocation	Unsigned Integer (2 bytes)
306	Real Time Status Bit 2	Unsigned Integer (2 bytes)
307	Real Time Status Bit 2 Allocation	Unsigned Integer (2 bytes)
331	Standstill	Unsigned Integer (2 bytes)
336	In position	Unsigned Integer (2 bytes)
372	Drive Halt Acceleration Bipolar	Signed Integer (4 bytes)
380	Filtered Sensed DC Bus Voltage	Signed Integer (2 bytes)
400	Home Switch	Unsigned Integer (2 bytes)
401	Probe 1	Unsigned Integer (2 bytes)
402	Probe 2	Unsigned Integer (2 bytes)
403	Position Feedback Value Status	Unsigned Integer (2 bytes)
404	Position Command Value Status	Unsigned Integer (2 bytes)
405	Probe 1 Enable	Unsigned Integer (2 bytes)
406	Probe 2 Enable	Unsigned Integer (2 bytes)
407	Control Unit Controlled Homing Enable	Unsigned Integer (2 bytes)
408	CUCH Ref Marker Registered	Unsigned Integer (2 bytes)
409	Probe 1 Rising Latched	Unsigned Integer (2 bytes)
410	Probe 1 Falling Latched	Unsigned Integer (2 bytes)
411	Probe 2 Rising Latched	Unsigned Integer (2 bytes)
412	Probe 2 Falling Latched	Unsigned Integer (2 bytes)
32771	Encoder Error	Unsigned Integer (2 bytes)
32772	Motor Encoder Control	Unsigned Integer (2 bytes)
32773	External Encoder Control	Unsigned Integer (2 bytes)
32774	Motor poles	Unsigned Integer (2 bytes)
32775	Motor Encoder Linear Resolution	Unsigned Integer (4 bytes)
32776	Motor Velocity Estimation Method	Unsigned Integer (2 bytes)
32777	Motor Velocity Feedback Pulse Width - Maximum Pulse Width	Signed Integer (4 bytes)
32778	Motor Velocity Feedback Auto Mode - Hysteresis Half Span	Signed Integer (4 bytes)
32780	TCONT Low Pass Filter Frequency	Unsigned Integer (2 bytes)
32781	TCONT Notch Filter 1 Frequency	Unsigned Integer (2 bytes)
32782	TCONT Notch Filter 1 Q Factor	Signed Integer (2 bytes)
32783	TCONT Notch Filter 2 Frequency	Unsigned Integer (2 bytes)
32784	TCONT Notch Filter 2 Q Factor	Signed Integer (2 bytes)

32785	TCONT Notch Filter 3 Frequency	Unsigned Integer (2 bytes)
32786	TCONT Notch Filter 3 Q Factor	Signed Integer (2 bytes)
32787	TCONT Notch Filter 4 Frequency	Unsigned Integer (2 bytes)
32788	TCONT Notch Filter 4 Q Factor	Signed Integer (2 bytes)
32789	TCONT Notch Filter 5 Frequency	Unsigned Integer (2 bytes)
32790	TCONT Notch Filter 5 Q Factor	Signed Integer (2 bytes)
32796	Internal Estimated U, V & W Phase Currents	Signed Integer (4 bytes)
32797	ISENSE Iuvw Used in Calculating Idq Feedback	Signed Integer (4 bytes)
32798	Current ADC Sensors Gain	Signed Integer (2 bytes)
32799	ISENSE Instantaneous Phase Current Limit	Signed Integer (4 bytes)
32800	Current Sensor Offset Adaption Rate	Signed Integer (2 bytes)
32801	Current Offset Calibration Time	Unsigned Integer (2 bytes)
32802	Phase U Current Sensor Offset - EOL Calibrated	Unsigned Integer (2 bytes)
32803	Phase W Current Sensor Offset - EOL Calibrated	Unsigned Integer (2 bytes)
32807	Calibrate PWM off	Unsigned Integer (2 bytes)
32809	Enable Absolute Current Feedback	Unsigned Integer (2 bytes)
32810	ISENSE Ripple Compensation Ctrl	Unsigned Integer (2 bytes)
32811	ISENSE Iuvw With Ripple Compensation	Signed Integer (4 bytes)
32820	Menu Display List	Unsigned Integer (2 bytes)
32850	In Position Threshold - Command	Signed Integer (4 bytes)
32851	Position Near Threshold - Command	Signed Integer (4 bytes)
32852	In Position Threshold - Actual	Signed Integer (4 bytes)
32853	Position Near Threshold - Actual	Signed Integer (4 bytes)
32854	In Position Flags	Unsigned Integer (2 bytes)
32867	Global Limits - Enable Flag	Unsigned Integer (2 bytes)
32868	Global Maximum Position Limit	Signed Integer (4 bytes)
32869	Global Minimum Position Limit	Signed Integer (4 bytes)
32870	Global Maximum Velocity Limit	Signed Integer (4 bytes)
32871	Global Minimum Velocity Limit	Signed Integer (4 bytes)
32872	Global Acceleration Limit - Positive Velocity Region	Signed Integer (4 bytes)
32873	Global Acceleration Limit - Negative Velocity Region	Signed Integer (4 bytes)
32874	Global Deceleration Limit - Positive Velocity Region	Signed Integer (4 bytes)
32875	Global Deceleration Limit - Negative Velocity Region	Signed Integer (4 bytes)
32876	Global Maximum Force Limit	Signed Integer (4 bytes)
32877	Global Minimum Force Limit	Signed Integer (4 bytes)
32882	Joint Motion Profiler Safety Limit Master Enable	Unsigned Integer (2 bytes)
32883	Joint Motion Profiler Safety Limits Enable Flags	Unsigned Integer (2 bytes)
32884	Safety Maximum Position Limit	Signed Integer (4 bytes)
32885	Safety Minimum Position Limit	Signed Integer (4 bytes)
32886	Safety Maximum Velocity Limit	Signed Integer (4 bytes)
32887	Safety Minimum Velocity Limit	Signed Integer (4 bytes)
32888	Safety Acceleration Limit - Positive Velocity Region	Signed Integer (4 bytes)
32889	Safety Acceleration Limit - Negative Velocity Region	Signed Integer (4 bytes)

32890	Safety Deceleration Limit - Positive Velocity Region	Signed Integer (4 bytes)
32891	Safety Deceleration Limit - Negative Velocity Region	Signed Integer (4 bytes)
32892	Safety Maximum Force Limit	Signed Integer (4 bytes)
32893	Safety Minimum Force Limit	Signed Integer (4 bytes)
32894	Motion Profiler Error Limits Master Enable	Unsigned Integer (2 bytes)
32895	Error Limits Enable Flags	Unsigned Integer (2 bytes)
32896	Position Hard Min and Max	Signed Integer (4 bytes)
32897	Position Soft Min and Max	Signed Integer (4 bytes)
32898	Motion Profiler Error Limits Tripped Flags	Unsigned Integer (2 bytes)
32899	Rotary Joint With Limited Stroke	Unsigned Integer (2 bytes)
32900	Location of End Stops on a Rotary Joint	Signed Integer (4 bytes)
32901	Velocity Hard Min and Max	Signed Integer (4 bytes)
32902	JMP Error Limit Soft Limit Max Deceleration	Signed Integer (4 bytes)
32903	Positive Limit Switch	Unsigned Integer (2 bytes)
32904	Negative Limit Switch	Unsigned Integer (2 bytes)
32908	Analog Input 1	Signed Integer (4 bytes)
32909	Analog Input 2	Signed Integer (4 bytes)
32910	Analog Input Bias	Signed Integer (2 bytes)
32911	Analog Input Dead Zone	Signed Integer (4 bytes)
32912	Analog Input filter coefficient	Unsigned Integer (2 bytes)
32913	Analog Input Span	Signed Integer (4 bytes)
32914	Analog Input Min	Signed Integer (4 bytes)
32915	Analog Input Max	Signed Integer (4 bytes)
32916	General Purpose Analogue Inputs - Dead Zone Type	Unsigned Integer (2 bytes)
32917	Analogue Input Zero - Channel 1	Unsigned Integer (2 bytes)
32918	Analogue Input Zero - Channel 2	Unsigned Integer (2 bytes)
32919	Analogue Output 1 Desired Voltage	Signed Integer (2 bytes)
32920	Analogue Output 2 Desired Voltage	Signed Integer (2 bytes)
32921	Analogue Output Voltage Offset	Signed Integer (2 bytes)
32922	Analogue Output Gain	Unsigned Integer (2 bytes)
32923	Analogue Output Safe State Enable	Unsigned Integer (2 bytes)
32924	Analogue Output Safe State Value	Signed Integer (2 bytes)
32928	Synchronous Memory Control Word	Unsigned Integer (2 bytes)
32929	Synchronous Memory Status Word	Unsigned Integer (2 bytes)
32930	Synchronous Memory Transmit Data	Unsigned Integer (2 bytes)
32931	Sync Mem RxData	Unsigned Integer (2 bytes)
32938	FPGA Proxy Period Count - Cycle Count	Unsigned Integer (4 bytes)
32939	FPGA Proxy Period Count - Previous Complete Cycle	Unsigned Integer (4 bytes)
32955	Servo Control Tuning Procedure Command	Unsigned Integer (2 bytes)
32960	PCONT Velocity FFC Gain	Signed Integer (2 bytes)
32961	Position Command	Signed Integer (4 bytes)
32962	PCONT Velocity Cmd	Signed Integer (4 bytes)
32963	Interpolation - external	Unsigned Integer (2 bytes)

32969	Position Feedback for Linear Scale Compensation	Signed Integer (4 bytes)
32970	Feed-forward Acceleration Gain Factor	Signed Integer (2 bytes)
32971	WCONT Proportional Gain Scale (2*ID32971)	Signed Integer (2 bytes)
32972	Velocity Control State Saturation Gain	Signed Integer (2 bytes)
32973	Velocity Control Saturation Recovery Gain	Signed Integer (2 bytes)
32974	Velocity Controller Error Term	Signed Integer (4 bytes)
32975	Linear Scale Compensation Control	Unsigned Integer (2 bytes)
32976	Minimum Velocity in Linear Scale Compensation	Signed Integer (4 bytes)
32977	Minimum Velocity in Linear Scale Compensation	Signed Integer (4 bytes)
32978	Velocity Following Error Threshold	Signed Integer (4 bytes)
32979	LSCOMP_s32Maximum_Position_Difference	Signed Integer (4 bytes)
32980	VF Curve - Break Points Program Length	Unsigned Integer (2 bytes)
32981	VF Curve - Velocity Break Points	Signed Integer (4 bytes)
32982	VF Curve - Voltage Break Points	Signed Integer (2 bytes)
32983	VF Control - Max Current	Signed Integer (4 bytes)
32984	VF Control - Stop Current	Signed Integer (4 bytes)
32985	VF Control - Stop Voltage	Signed Integer (2 bytes)
32986	VF Control - Stop Time	Unsigned Integer (2 bytes)
32987	VF Control - Minimum Velocity Command	Signed Integer (4 bytes)
32988	VF Control - Velocity Command Scale Factor	Signed Integer (2 bytes)
32989	Induction Motor VF Control State	Unsigned Integer (2 bytes)
32990	Torque Setpoint Switch	Unsigned Integer (2 bytes)
32991	TCONT Torque Cmd PreFilters	Signed Integer (4 bytes)
32992	TCONT MotorTorque Constant	Signed Integer (4 bytes)
32993	Dynamic Torque Min-Max Control Flags	Unsigned Integer (2 bytes)
32994	Dynamic Maximum Torque Limit	Signed Integer (2 bytes)
32995	Dynamic Minimum Torque Limit	Signed Integer (2 bytes)
32996	Torque Setpoint Switch	Unsigned Integer (2 bytes)
32998	Current Controller Tuning Axis Selection	Unsigned Integer (2 bytes)
32999	ICONT D- and Q- Axis Gain Scale	Signed Integer (4 bytes)
33000	ICONT Iq Limit	Signed Integer (4 bytes)
33001	ICONT Tuning Mode ProcCmd	Unsigned Integer (2 bytes)
33002	ICONT StateSaturationGain	Signed Integer (2 bytes)
33003	ICONT Vdq Overmodulation Limit Factor	Signed Integer (2 bytes)
33004	ICONT Additive Idq Command	Signed Integer (4 bytes)
33005	DQ Current Command Test Injection	Signed Integer (4 bytes)
33006	ICONT Idq Cmd	Signed Integer (4 bytes)
33007	ICONT Q-Axis Low Current Boost Enable	Unsigned Integer (2 bytes)
33008	ICONT Q axis Low Current Boost Threshold	Signed Integer (4 bytes)
33009	ICONT Q axis Low Current Boost Integral Time	Unsigned Integer (2 bytes)
33010	Nominal DC Voltage	Signed Integer (2 bytes)
33011	Maximum DC Voltage	Signed Integer (2 bytes)
33012	Minimum DC Voltage	Signed Integer (2 bytes)

33013	Brake Chopper Turn-On Voltage	Signed Integer (2 bytes)
33014	Brake Chopper Turn-Off Voltage	Signed Integer (2 bytes)
33016	VCONT Voltage Saturation Factor	Signed Integer (2 bytes)
33017	VCONT PWM Vector Time	Signed Integer (2 bytes)
33018	HWPWM DBC Value	Signed Integer (2 bytes)
33019	HWPWM DBC Current Hysteresis Threshold	Signed Integer (4 bytes)
33020	HWPWM DBC Polarity	Signed Integer (2 bytes)
33024	Drive Controlled Moves Procedure Command	Unsigned Integer (2 bytes)
33025	Drive Controlled Move Status Word	Unsigned Integer (2 bytes)
33027	Drive Controlled Move Types	Unsigned Integer (2 bytes)
33028	Drive Controlled Move Target Position	Signed Integer (4 bytes)
33029	DCM Segment Absolute Maximum Velocity	Signed Integer (4 bytes)
33030	DCM Segment Start Acceleration	Signed Integer (4 bytes)
33031	DCM Segment End Acceleration	Signed Integer (4 bytes)
33032	DCM Segment End Delay Time	Unsigned Integer (2 bytes)
33033	Drive Control Move Next Segment ID	Unsigned Integer (2 bytes)
33034	ID of the segment from which to start the move	Unsigned Integer (2 bytes)
33035	DCM Halt Acceleration	Signed Integer (4 bytes)
33036	Drive Controlled Moves - Control Word	Unsigned Integer (2 bytes)
33037	DCM Range Detection Limits	Signed Integer (4 bytes)
33038	DCM Range Detection State	Unsigned Integer (2 bytes)
33039	Drive Controlled Moves Cycle Counter	Unsigned Integer (4 bytes)
33040	Drive Controlled Homing Procedure Command Acknowledge	Unsigned Integer (2 bytes)
33049	CMNT EAC Enable	Unsigned Integer (2 bytes)
33050	ICONT Idq Feedback	Signed Integer (4 bytes)
33051	Reverse Phase Sequence Enable	Unsigned Integer (2 bytes)
33052	Manual Commutation Speed	Signed Integer (4 bytes)
33053	CMNT Electrical Commutation Angle	Signed Integer (4 bytes)
33055	Motor Field Alignment State	Unsigned Integer (2 bytes)
33057	Evaluation of Index Pulse Offset Complete	Unsigned Integer (2 bytes)
33058	FOI AIP ThetaElec Difference Threshold	Signed Integer (4 bytes)
33059	FOI Negative Electrical Angle Offset	Signed Integer (4 bytes)
33060	FOI Ctrl	Unsigned Integer (2 bytes)
33061	Alignment Commutation Angle Command	Signed Integer (4 bytes)
33062	FOI AIP Ctrl	Unsigned Integer (2 bytes)
33063	FOI AIP Measured ThetaElec at Index Pulse	Signed Integer (4 bytes)
33064	FOI AIP ThetaElec at Index Pulse	Signed Integer (4 bytes)
33065	FOI Type	Unsigned Integer (2 bytes)
33066	FOI PO ThetaElec Preset Offset	Signed Integer (4 bytes)
33067	Motor Control DQ Axis Alignment Repeat Count Test Points	Unsigned Integer (2 bytes)
33068	Motor Control DQ Axis Alignment Time	Unsigned Integer (2 bytes)
33069	FOI DQA Alignment Current	Signed Integer (4 bytes)

33070	FOI DQA Alignment Current Slew Rate	Signed Integer (4 bytes)
33071	Motor Control DQ Axis Alignment Current Tolerance	Signed Integer (4 bytes)
33072	DQ Axis Alignment Slew Angle Limit	Signed Integer (4 bytes)
33073	Motor Control DQ Axis Alignment Test Angles	Signed Integer (4 bytes)
33074	MCDQA Repeat offsets angles	Signed Integer (4 bytes)
33075	DQ Axis Alignment Init Angle	Signed Integer (4 bytes)
33076	Brake Engage Hold Time	Unsigned Integer (2 bytes)
33080	Absolute Feedback Type	Unsigned Integer (2 bytes)
33081	Motor Control Pulse Alignment Hexant Estimate	Signed Integer (2 bytes)
33083	Motor Control Pulse Alignment Hexant Input Map	Signed Integer (2 bytes)
33084	Motor Control Pulse Alignment Hexant Edge Angles	Signed Integer (4 bytes)
33085	Motor Control Pulse Alignment Angle	Signed Integer (4 bytes)
33086	Motor Control Pulse Alignment Encoder1 CosSin Offset	Unsigned Integer (2 bytes)
33087	Motor Control Pulse Alignment Scale Factor	Signed Integer (2 bytes)
33094	Missing Counts Threshold	Signed Integer (4 bytes)
33095	Missing Counts	Signed Integer (4 bytes)
33100	Distributed Clock PLL Proportional Gain	Signed Integer (2 bytes)
33101	Distributed Clock Minimum/Maximum Corection	Signed Integer (2 bytes)
33103	Next Distributed Clock Value	Signed Integer (4 bytes)
33104	Distributed Clock Current Time	Signed Integer (4 bytes)
33105	OSDC Time bias	Signed Integer (4 bytes)
33106	Distributed Clock Time Error	Signed Integer (4 bytes)
33107	Distributed Clocks Frequency Correction	Signed Integer (2 bytes)
33108	EtherCAT Slave Controller PLL Error	Signed Integer (4 bytes)
33109	Distributed Clocks Time Error Monitoring	Unsigned Integer (2 bytes)
33110	Distributed Clock Time Error Threshold	Signed Integer (4 bytes)
33111	Distributed Clocks Lock-on Threshold	Unsigned Integer (2 bytes)
33112	Distributed Clock PLL integral gain	Signed Integer (2 bytes)
33119	Features Data - Read	Unsigned Integer (2 bytes)
33120	Non-volatile Parameters Autoload on Startup	Unsigned Integer (2 bytes)
33121	Full NV Parameter(s) List	Unsigned Integer (2 bytes)
33122	Selected NV Parameter(s) list	Unsigned Integer (2 bytes)
33123	Non-Volatile Parameter Control Word	Unsigned Integer (2 bytes)
33124	EEPROM Control Word	Unsigned Integer (2 bytes)
33125	Non-Volatile Parameters Last Error Code	Unsigned Integer (2 bytes)
33126	NV Memory - Read	Unsigned Integer (2 bytes)
33127	Controller card EEPROM Read Word	Unsigned Integer (2 bytes)
33128	Controller card EEPROM Read Address	Unsigned Integer (4 bytes)
33129	ECSM Current State	Unsigned Integer (2 bytes)
33130	Drive Firmware Revision	Unsigned Integer (4 bytes)
33131	Drive Serial Number	Unsigned Integer (1 bytes)
33132	CPLD Version Number	Unsigned Integer (1 bytes)
33133	Bootloader Version Info	Unsigned Integer (1 bytes)

33135	Drive Identification	Unsigned Integer (1 bytes)
33136	Motor Identification	Unsigned Integer (1 bytes)
33140	Bootloader Customer ID	Unsigned Integer (2 bytes)
33141	Firmware Customer ID	Unsigned Integer (2 bytes)
33150	Joint Control Event Capture Finish Status	Unsigned Integer (2 bytes)
33151	Event Capture Global Time Out - Ch0	Signed Integer (4 bytes)
33152	Event Capture Global Time Out - Ch1	Signed Integer (4 bytes)
33153	Event Capture Global Time Out - Ch2	Signed Integer (4 bytes)
33154	Event Capture Global Time Out - Ch3	Signed Integer (4 bytes)
33155	Joint Control Event Capture Commence Status	Unsigned Integer (2 bytes)
33156	Event Capture Global Time In - Ch0	Signed Integer (4 bytes)
33157	Event Capture Global Time In - Ch1	Signed Integer (4 bytes)
33158	Event Capture Global Time In - Ch2	Signed Integer (4 bytes)
33159	Event Capture Global Time In - Ch3	Signed Integer (4 bytes)
33160	Use external feedback	Unsigned Integer (2 bytes)
33161	JCPED_boDisableBsplineSaturation	Unsigned Integer (2 bytes)
33190	Use a Digital Input In Place Of an Index Pulse for Homing	Unsigned Integer (2 bytes)
33200	Extended Homing Parameters	Unsigned Integer (2 bytes)
33201	Home Switch from Master	Unsigned Integer (2 bytes)
33202	DCH Home Switch Position	Signed Integer (4 bytes)
33203	Drive Controlled Homing Second Home Switch Position	Signed Integer (4 bytes)
33204	Drive Controlled Homing Latched Marker Position	Signed Integer (4 bytes)
33205	Homing Velocity to locate index pulse	Signed Integer (4 bytes)
33206	DCH Home Switch Active At Initialisation Runoff Distance	Signed Integer (4 bytes)
33207	Drive Controlled Homing Procedure Command Ack	Unsigned Integer (2 bytes)
33208	Homing Stall Detection Velocity Threshold	Signed Integer (4 bytes)
33209	Homing Stall Detection Torque Threshold	Signed Integer (4 bytes)
33210	DCH Stall Detect Debounce Time	Unsigned Integer (2 bytes)
33211	DCH Runoff Distance After Home Switch Detection	Signed Integer (4 bytes)
33212	Home Switch Input	Unsigned Integer (2 bytes)
33213	DCH Max Force (Torque)	Signed Integer (4 bytes)
33214	Homing Velocity Max	Signed Integer (4 bytes)
33250	Task Gross Worst Case Execution Times	Unsigned Integer (4 bytes)
33251	Task Execution Period	Unsigned Integer (4 bytes)
33255	Error state	Unsigned Integer (2 bytes)
33256	List of Currently Active Error Codes	Unsigned Integer (2 bytes)
33257	Diagnostic Message Code	Unsigned Integer (2 bytes)
33259	User Application Status String	Unsigned Integer (1 bytes)
33260	External Master Request Procedure Command	Unsigned Integer (2 bytes)
33261	External Master Request	Unsigned Integer (2 bytes)
33262	User Application Has Control	Unsigned Integer (2 bytes)
33265	Operation System Global Task Counter	Unsigned Integer (2 bytes)
33270	Current setpoint switch - FOI mode	Unsigned Integer (2 bytes)

33271	Current Setpoint Switch - Run Mode	Unsigned Integer (2 bytes)
33272	Voltage Setpoint Switch - Run Mode	Unsigned Integer (2 bytes)
33273	Commutation Angle Switch - FOI Mode	Unsigned Integer (2 bytes)
33274	Commutation Angle Switch - Run Mode	Unsigned Integer (2 bytes)
33275	Current Setpoint Switch - Run Secondary 1 mode	Unsigned Integer (2 bytes)
33278	Motor Control Tuning Procedure Command	Unsigned Integer (2 bytes)
33280	Motor Control Main State	Unsigned Integer (2 bytes)
33281	Motor Control Sub State	Unsigned Integer (2 bytes)
33286	Current Limit Active	Unsigned Integer (2 bytes)
33287	Active Current Limit Source	Unsigned Integer (2 bytes)
33290	Secondary Operation Mode - Control Word	Unsigned Integer (2 bytes)
33291	Secondary Op Mode	Unsigned Integer (2 bytes)
33295	Analogue Setpoint Source	Unsigned Integer (2 bytes)
33296	Analogue Input Velocity Command Gain	Signed Integer (4 bytes)
33299	Numerical Control Command Source	Unsigned Integer (2 bytes)
33300	Slave Mode Control Word	Unsigned Integer (2 bytes)
33301	Slave Position Command	Signed Integer (4 bytes)
33302	Slave Force / Torque Command	Signed Integer (4 bytes)
33304	Slave Velocity Command	Signed Integer (4 bytes)
33307	External Pulse Count - Relative Mode	Unsigned Integer (2 bytes)
33310	Interpolated Setpoint to Servo Loop	Signed Integer (4 bytes)
33320	SC Torque Cmd	Signed Integer (4 bytes)
33330	Probing Device	Unsigned Integer (2 bytes)
33331	Probe 1 Filtered	Unsigned Integer (2 bytes)
33333	Timestamp Trigger Source	Unsigned Integer (2 bytes)
33337	External Pulse Count Control Word	Unsigned Integer (2 bytes)
33340	Digital Input Event Capture Bit Mask	Unsigned Integer (4 bytes)
33342	Digital Input Polarity	Unsigned Integer (4 bytes)
33343	Digital Input Data	Unsigned Integer (4 bytes)
33344	Digital Output Polarity	Unsigned Integer (4 bytes)
33345	Digital Output Data	Unsigned Integer (4 bytes)
33346	Motor Brake Release	Unsigned Integer (2 bytes)
33348	HWDIO Hexant Inputs	Unsigned Integer (2 bytes)
33349	HWDIO_u16Latched_EncoderUVW_atPowerUp	Unsigned Integer (2 bytes)
33350	Digital Output Source IDN	Unsigned Integer (2 bytes)
33351	Digital Out Invert Mask	Unsigned Integer (2 bytes)
33352	Dig Out Source Bitmask	Unsigned Integer (2 bytes)
33353	Digital Output User Configurable Default Safe State	Unsigned Integer (2 bytes)
33354	Digital Output Default Safe State - Hardware Level	Unsigned Integer (4 bytes)
33355	Digital input to IDN mapping	Unsigned Integer (2 bytes)
33356	Dig In Map Bitmask	Unsigned Integer (2 bytes)
33361	ADC Measurement of DC Bus Voltage	Unsigned Integer (2 bytes)
33362	Filtered ADC measurement of DC bus voltage.	Unsigned Integer (2 bytes)

33363	ADC Sample - Cos1,Sin1,Cos2,Sin2	Unsigned Integer (2 bytes)
33364	COM Track ADC Sample - Cos1,Sin1,Cos2,Sin2	Unsigned Integer (2 bytes)
33365	Current ADC Low Pass Filter Coefficients	Unsigned Integer (2 bytes)
33370	Quadrature Encoder Phase Information	Unsigned Integer (2 bytes)
33371	Quadrature Encoder Input Qualification Time	Unsigned Integer (2 bytes)
33372	Quadrature Counts - Ch1,Ch2	Signed Integer (2 bytes)
33373	Encoder Pass Through Channel Selction	Unsigned Integer (2 bytes)
33380	Encoder Counter Sample - Ch1	Signed Integer (4 bytes)
33381	Encoder Counter Sample - Ch2	Signed Integer (4 bytes)
33390	Encoder Revolution Sample	Signed Integer (2 bytes)
33391	Encoder Counter Sample	Signed Integer (4 bytes)
33404	RCOMP State	Unsigned Integer (2 bytes)
33405	RCOMP Velocity Feedback	Signed Integer (4 bytes)
33406	RCOMP Force Feedback	Signed Integer (4 bytes)
33407	RCOMP Velocity Feedback Low-pass Filter Coefficient	Unsigned Integer (2 bytes)
33408	RCOMP Force Feedback Low-pass Filter Coefficient	Unsigned Integer (2 bytes)
33409	RCOMP Velocity Injection	Signed Integer (4 bytes)
33410	RCOMP Enable	Unsigned Integer (2 bytes)
33411	RCOMP 1st Pulse Amplitude	Signed Integer (4 bytes)
33412	RCOMP Torque Threshold	Signed Integer (4 bytes)
33413	RCOMP 2nd Pulse Amplitude	Signed Integer (4 bytes)
33414	RCOMP 2nd Pulse Max Duration	Unsigned Integer (2 bytes)
33415	RCOMP Velocity Threshold	Signed Integer (4 bytes)
33416	ESO Position Error	Signed Integer (4 bytes)
33417	ESO Torque Compensation Value	Signed Integer (4 bytes)
33418	ESO Control Word	Unsigned Integer (2 bytes)
33419	ESO Gain1	Unsigned Integer (2 bytes)
33420	ESO Gain2	Unsigned Integer (2 bytes)
33421	ESO Gain3	Unsigned Integer (2 bytes)
33422	ESO Axis Inertia	Unsigned Integer (4 bytes)
33429	TCONT PM Spindle Load LPF Coefficient	Unsigned Integer (2 bytes)
33431	ESO_s32Alpha_1	Signed Integer (4 bytes)
33432	ESO_s32Alpha_2	Signed Integer (4 bytes)
33433	ESO_u32Delta_1	Unsigned Integer (4 bytes)
33434	ESO_u32Delta_2	Unsigned Integer (4 bytes)
33440	Falling Edge Strobe QEP Count	Unsigned Integer (4 bytes)
33441	Rising Edge Strobe QEP Count	Unsigned Integer (4 bytes)
33442	QEP counter	Unsigned Integer (4 bytes)
33443	Strobe Digital Input Select	Unsigned Integer (2 bytes)
33444	Strobe Quadrature Input Select	Unsigned Integer (2 bytes)
33445	Enable Strobing	Unsigned Integer (2 bytes)
33446	Strobe Status	Unsigned Integer (2 bytes)
33450	Parameter Rescaling Procedure Command	Unsigned Integer (2 bytes)

33451	Task Reset Stats Procedure Command	Unsigned Integer (2 bytes)
33501	Main Power Enabled	Unsigned Integer (2 bytes)
33502	PSU Power Enable	Unsigned Integer (2 bytes)
33503	Main Power Enabled	Unsigned Integer (2 bytes)
33504	Main Power Enable Delay	Unsigned Integer (2 bytes)
33505	DSM Main Power En Timeout	Unsigned Integer (2 bytes)
33506	Standalone Mode Enable	Unsigned Integer (2 bytes)
33507	Bypass Mode Enable	Unsigned Integer (2 bytes)
33521	Toggle Fault Threshold	Unsigned Integer (2 bytes)
33522	Toggle Fault Count	Unsigned Integer (2 bytes)
33525	Servo Drive Reboot Required	Unsigned Integer (2 bytes)
33550	PWM Counter Period (Nominal)	Unsigned Integer (2 bytes)
33551	Tripzones - Global Enable/Disable	Unsigned Integer (2 bytes)
33570	NC Limit Enable Flags	Unsigned Integer (2 bytes)
33571	NC Maximum Position Limit	Signed Integer (4 bytes)
33572	NC Minimum Position Limit	Signed Integer (4 bytes)
33573	NC Maximum Velocity Limit	Signed Integer (4 bytes)
33574	NC Minimum Velocity Limit	Signed Integer (4 bytes)
33575	NC Acceleration Limit - Positive Velocity Region	Signed Integer (4 bytes)
33576	NC Acceleration Limit - Negative Velocity Region	Signed Integer (4 bytes)
33577	NC Deceleration Limit - Positive Velocity Region	Signed Integer (4 bytes)
33578	NC Deceleration Limit - Negative Velocity Region	Signed Integer (4 bytes)
33579	NC Maximum Force Limit	Signed Integer (4 bytes)
33580	NC Minimum Force Limit	Signed Integer (4 bytes)
33615	Backlash Comp min Speed	Signed Integer (4 bytes)
33617	Backlash Comp Slew Limit	Unsigned Integer (2 bytes)
33619	Invert Joint Direction	Unsigned Integer (2 bytes)
33620	WCONT Velocity Scaling Shift Factor	Unsigned Integer (2 bytes)
33621	Position Scaling Shift Factor	Unsigned Integer (2 bytes)
33625	Position Estimate	Signed Integer (4 bytes)
33626	WCONT Velocity Feedback	Signed Integer (4 bytes)
33627	Acceleration Estimate	Signed Integer (4 bytes)
33630	Calculate Absolute Offset Procedure Command	Unsigned Integer (2 bytes)
33680	VF Control Spindle Load	Unsigned Integer (2 bytes)
33681	VF control - Spindle Full Load Break Points	Signed Integer (4 bytes)
33682	VF Control - Spindle Full Load Slip Correction	Unsigned Integer (2 bytes)
33683	VF Control - Spindle Field Weakening Speed	Signed Integer (4 bytes)
33684	VF Control - Acceleration Limit	Signed Integer (4 bytes)
33691	FOC Control - Spindle Minimum Speed at Stop State	Signed Integer (4 bytes)
33692	Rated Magnetising Current for Induction Motor	Signed Integer (4 bytes)
33693	Induction Motor Rotor Time Constant	Unsigned Integer (4 bytes)
33694	ICONT Id Limit	Signed Integer (4 bytes)
33695	PMSMFW Enable	Unsigned Integer (2 bytes)

33696	PMSMFW D-Axis Current Polarity	Signed Integer (4 bytes)
33697	PMSMFW Curve - Break Points Number	Unsigned Integer (2 bytes)
33698	PMSMFW Curve - Velocity Break Points	Signed Integer (4 bytes)
33699	PMSMFW Curve - Current Break Points	Signed Integer (4 bytes)
33700	Reference Offset Valid	Unsigned Integer (2 bytes)
33701	Polynomial Model Update Disable	Unsigned Integer (2 bytes)
33708	Release Brake At FOI	Unsigned Integer (2 bytes)
33709	Release Brake Signal at FOI	Unsigned Integer (2 bytes)
33710	TCONT PMSM Spindle Load Update Enable	Unsigned Integer (2 bytes)
33711	Field Orientation Control D-Axis Polarity	Signed Integer (4 bytes)
33750	MB__en16BaudRate	Unsigned Integer (2 bytes)
33780	Adjusted ADC Squared-Motor Encoder	Signed Integer (4 bytes)
33781	Adjusted ADC Squared-External Encoder	Signed Integer (4 bytes)
33782	Adjusted Cos/Sin Signals - Motor Encoder	Signed Integer (2 bytes)
33783	Adjusted Cos/Sin Signals - External Encoder	Signed Integer (2 bytes)
33784	Cos/Sin Signals - External Encoder	Signed Integer (2 bytes)
33790	End Effect Encoder Cos Sine Check Enable	Unsigned Integer (2 bytes)
33791	ADC Min Value for External Encoder Sin and Cos	Unsigned Integer (2 bytes)
33792	ADC Max Value for External Encoder Sin and Cos	Unsigned Integer (2 bytes)
33793	ADC Squared-External Encoder ($\text{Cos}^2 + \text{Sin}^2$)	Signed Integer (4 bytes)
33796	Motor Encoder Source	Unsigned Integer (2 bytes)
33797	External Encoder Source	Unsigned Integer (2 bytes)
33800	ADC Squared-Motor Encoder ($\text{Cos}^2 + \text{Sin}^2$)	Signed Integer (4 bytes)
33801	ADC Min Value for Motor Encoder Sin and Cos	Unsigned Integer (2 bytes)
33802	ADC Max Value for Motor Encoder Sin and Cos	Unsigned Integer (2 bytes)
33803	Cos/Sin Gain - Motor Encoder	Signed Integer (2 bytes)
33804	Cos/Sin Offset - Motor Encoder	Signed Integer (2 bytes)
33805	Motor Mechanical Angle	Signed Integer (4 bytes)
33806	MCPOS Motor Rotary Speed	Signed Integer (4 bytes)
33807	MCPOS Motor Electrical Angle Including Offset	Signed Integer (4 bytes)
33808	Feedback Filter Type	Unsigned Integer (2 bytes)
33809	Cos/Sin Signals - Motor Encoder	Signed Integer (2 bytes)
33818	Motor Standstill Off Threshold	Signed Integer (4 bytes)
33819	Motor Standstill On Threshold	Signed Integer (4 bytes)
33820	MCPOS Velocity Feedback LPF Coefficient	Unsigned Integer (2 bytes)
33825	Velocity Estimate Proxy Sensor Edge per Rev	Unsigned Integer (2 bytes)
33826	MCPOS_s32OmegaMech_Estim_ProxySwitch_Minimum	Signed Integer (4 bytes)
33827	MCPOS EAC Gain	Signed Integer (4 bytes)
33843	Cos/Sin Gain - External Encoder	Signed Integer (2 bytes)
33844	Cos/Sin Offset - External Encoder	Signed Integer (2 bytes)
33860	Motor Velocity Filter Coefficient (End Effect Encoder)	Unsigned Integer (2 bytes)
33870	Drive Controlled Stroking Procedure Command	Unsigned Integer (2 bytes)
33871	Drive Controlled Stroking Control Word	Unsigned Integer (2 bytes)

33872	Drive Controlled Stroking Edge State	Unsigned Integer (2 bytes)
33873	Drive Controlled Stroking Velocity	Signed Integer (4 bytes)
33874	Drive Controlled Stroking Switching Position - Minimum	Signed Integer (4 bytes)
33875	Drive Controlled Stroking Switching Position - Maximum	Signed Integer (4 bytes)
33876	Drive Controlled Stroking Switching Position - Relative	Signed Integer (4 bytes)
33877	Drive Controlled Stroking Delay - Pre-Negative Velocity	Unsigned Integer (2 bytes)
33878	Drive Controlled Stroking Delay - Pre-Positive Velocity	Unsigned Integer (2 bytes)
33900	FOI AO Stimulus Amplitude	Signed Integer (4 bytes)
33901	FOI AO Stimulus Frequency	Signed Integer (4 bytes)
33902	MCFOAO_boFinalTestEn	Unsigned Integer (2 bytes)
33903	Motor Control Field Orientation State Machine Repeat Count	Unsigned Integer (2 bytes)
33904	Motor Control Field Orientation State Machine Stimulus Time	Unsigned Integer (2 bytes)
33905	FOI AO ThetaMech MaxDeviation	Signed Integer (4 bytes)
33906	Motor Control Field Orientation Theta Delay	Unsigned Integer (2 bytes)
33907	MCFOAO__s32PhaseEstimator_Phase	Signed Integer (4 bytes)
33908	PhaseEstimator_Amplitude	Signed Integer (4 bytes)
33909	MCFOAO_boPositiveVel_Detected	Unsigned Integer (2 bytes)
33910	Motor Control Field Orientation Estimated Torque	Unsigned Integer (2 bytes)
33911	MC FOI Torque Estimate Velocity Gain 1	Signed Integer (4 bytes)
33912	MC FOI Torque Estimate Velocity Gain 2	Signed Integer (4 bytes)
33913	FOI AO Axis Inertia	Signed Integer (4 bytes)
33914	MC FOI Torque Estimate Position Gain	Signed Integer (4 bytes)
33915	FOI AO LPF0 Coefficient	Unsigned Integer (2 bytes)
33916	FOI AO LPF1 Coefficient	Unsigned Integer (2 bytes)
33917	MC FOI Phase Estimator Low-Pass Filter 1 Minimum	Unsigned Integer (2 bytes)
33918	MC FOI Phase Estimator Low-Pass Filter 1 Decay Rate	Unsigned Integer (2 bytes)
33919	Motor Control Field Orientation Evaluator Low-Pass Filter	Unsigned Integer (2 bytes)
33920	Force Exit On Completion of Acceleration Observer Algorithm	Unsigned Integer (2 bytes)
33922	Motor Control Acceleration Observer Alignment Time	Unsigned Integer (2 bytes)
33923	Acceleration Observer Alignment Slew Current Rate	Signed Integer (4 bytes)
33924	Acceleration Observer Alignment Current Tolerance	Signed Integer (4 bytes)
33925	Excitation angle	Signed Integer (4 bytes)
33926	Torque prediction	Signed Integer (4 bytes)
33927	Torque estimate	Signed Integer (4 bytes)
33928	FOI AO OmegaMech Threshold	Signed Integer (4 bytes)
33970	Configuration Mode Watchdog Timeout	Unsigned Integer (2 bytes)
34000	Motor Thermal Rise Time	Signed Integer (2 bytes)
34001	Power Stage Thermal Time	Signed Integer (2 bytes)
34002	Motor I2R Overload Warn Level	Signed Integer (2 bytes)
34003	Power Stage I2R Overload Warn Level	Signed Integer (2 bytes)
34004	Motor Shut-Down Temperature with PWM On	Signed Integer (2 bytes)
34005	Amplifier Shut-Down Temperature with PWM On	Signed Integer (2 bytes)

34006	Cooling Fan Activation Amp Temperature Enable Threshold	Signed Integer (2 bytes)
34007	Cooling Fan Activation Amp Temperature Disable Threshold	Signed Integer (2 bytes)
34008	Sensed Motor Temperature	Signed Integer (2 bytes)
34009	Sensed Amplifier Temperature	Signed Integer (2 bytes)
34010	Temperature Monitor Control Word	Unsigned Integer (2 bytes)
34011	Zero Current - Amplifier Temperature Threshold	Unsigned Integer (2 bytes)
34012	Zero Current - Motor Temperature Threshold	Unsigned Integer (2 bytes)
34013	Current Decay Rate - Amplifier Temperature	Unsigned Integer (2 bytes)
34014	Current Decay Rate - Motor Temperature	Unsigned Integer (2 bytes)
34015	Amplifier Fan Control Word	Unsigned Integer (2 bytes)
34016	Motor Overload Threshold	Signed Integer (4 bytes)
34017	Sensed Ambient Temperature	Signed Integer (2 bytes)
34018	Amplifier Temperature Error Limit	Signed Integer (2 bytes)
34019	Motor I ² R Accumulator	Signed Integer (4 bytes)
34020	Power Limit	Signed Integer (4 bytes)
34021	Estimated Power (Electrical)	Signed Integer (4 bytes)
34040	Stimulus Input Procedure Command	Unsigned Integer (2 bytes)
34041	Stimulus Input Control Word	Unsigned Integer (2 bytes)
34042	Stimulus Input Status Word	Unsigned Integer (2 bytes)
34043	Stimulus Input Sample Delay	Unsigned Integer (2 bytes)
34044	Stimulus Input Length	Unsigned Integer (2 bytes)
34045	Stimulus Input Down Sample	Unsigned Integer (2 bytes)
34046	Stimulus Input Amplitude	Signed Integer (4 bytes)
34047	Stimulus Input Sine Frequency (Omega)	Signed Integer (4 bytes)
34048	Stimulus Input Sine Alpha	Signed Integer (4 bytes)
34049	OSSTIM Stimulus Scaled Amplitude	Signed Integer (4 bytes)
34050	Stimulus Error Word	Unsigned Integer (2 bytes)
34051	Stimulus Sample Index	Unsigned Integer (4 bytes)
34060	Data Logger Procedure Command	Unsigned Integer (2 bytes)
34061	Data Logger Variable List	Unsigned Integer (2 bytes)
34062	Data Logger Sample Period Factor	Unsigned Integer (2 bytes)
34063	Data Logger Pre-Trigger Samples	Unsigned Integer (2 bytes)
34064	Data Logger Trigger IDN	Unsigned Integer (2 bytes)
34065	Data Logger Trigger Mask	Unsigned Integer (4 bytes)
34066	Data Logger Trigger Value	Signed Integer (4 bytes)
34067	Data Logger Control Word	Unsigned Integer (2 bytes)
34068	Data Logger Variable List Indices	Unsigned Integer (2 bytes)
34069	Data Logger - Measured Signal - Channel 0	Unsigned Integer (4 bytes)
34070	Data Logger - Measured Signal - Channel 1	Unsigned Integer (4 bytes)
34071	Data Logger - Measured Signal - Channel 2	Unsigned Integer (4 bytes)
34072	Data Logger - Measured Signal - Channel 3	Unsigned Integer (4 bytes)
34073	Data Logger Error Word	Unsigned Integer (2 bytes)

34074	Data Logger Main State	Unsigned Integer (2 bytes)
34075	Data Logger Sample State	Unsigned Integer (2 bytes)
34076	Data Logger Trigger Flag	Unsigned Integer (2 bytes)
34100	Dual Axis Enabled	Unsigned Integer (2 bytes)
34101	Use Second Power Stage	Unsigned Integer (2 bytes)
34150	Height Following Command	Signed Integer (4 bytes)
34200	Encoder Type - Channel 1	Unsigned Integer (2 bytes)
34201	Encoder Type - Channel 2	Unsigned Integer (2 bytes)
34202	Encoder Type - Channel 3	Unsigned Integer (2 bytes)
34203	Encoder Type - Channel 4	Unsigned Integer (2 bytes)
34205	Latch Serial Encoder Errors - Motor	Unsigned Integer (2 bytes)
34206	Latch Serial Encoder Errors - External	Unsigned Integer (2 bytes)
34207	Encoder Index Count	Signed Integer (4 bytes)
34210	Encoder Index Pulse Status Word	Unsigned Integer (2 bytes)
34220	Pulse Count Feed Rate	Signed Integer (4 bytes)
34221	External Pulse Count	Signed Integer (4 bytes)
36010	STO Module Signals	Unsigned Integer (2 bytes)

10 Additional Information

10.1 What this Chapter Contains

This chapter contains information on product support and feedback:

- Contact Information
- Feedback on the manual

10.2 Product, Sales and Service Enquiries

If you require assistance for installation, training or other customer support issues, please contact the closest ANCA Motion Customer Service Office in your area for details.

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10.3 Feedback

This Manual is based on information available at the time of publication. Reasonable precautions have been taken in the preparation of this Manual, but the information contained herein does not purport to cover all details or variations in hardware and software configuration. Features may be described herein which are not present in all hardware and software systems. We would like to hear your feedback via our website:

www.ancamotion.com/Contact-Us